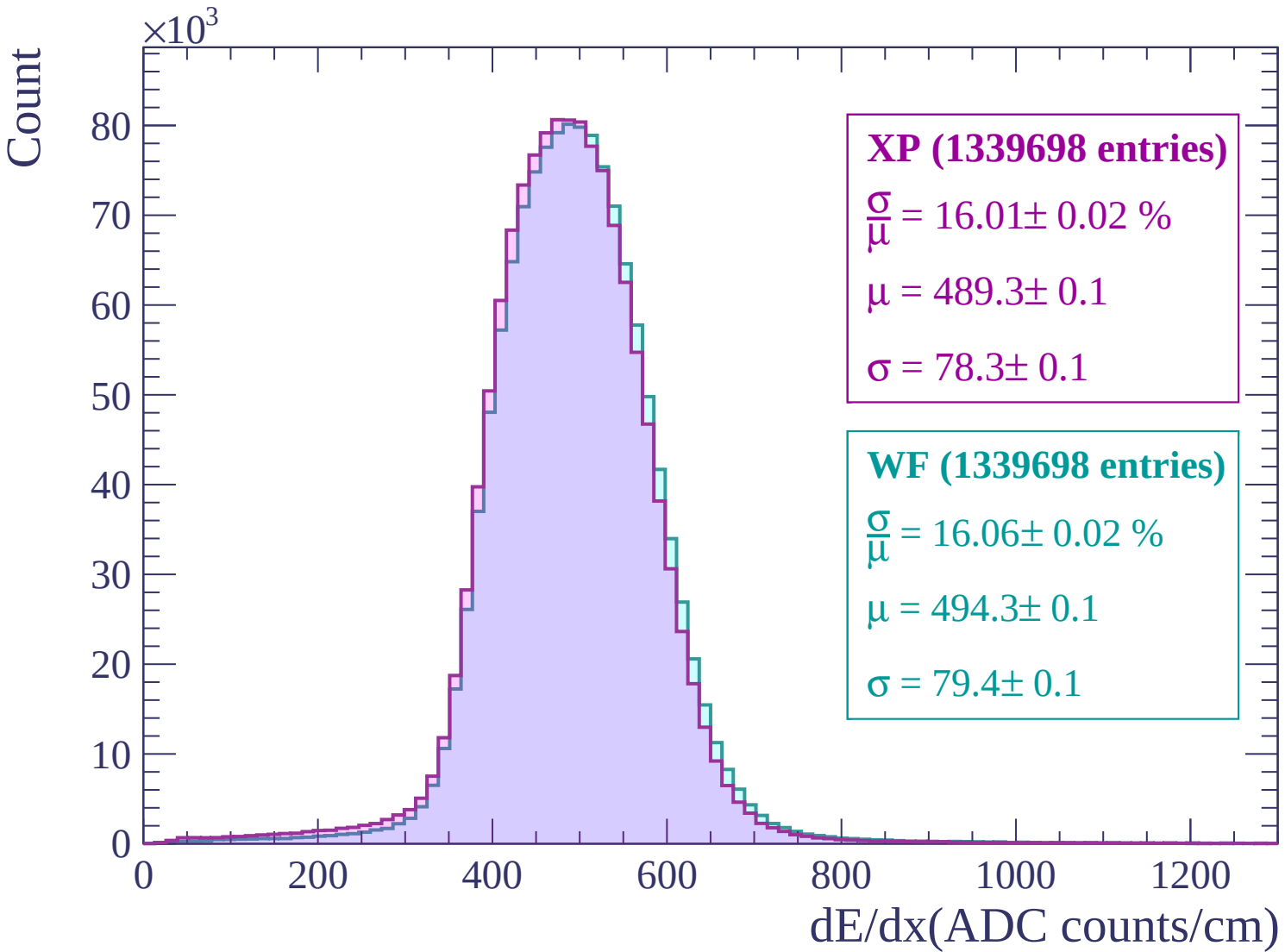
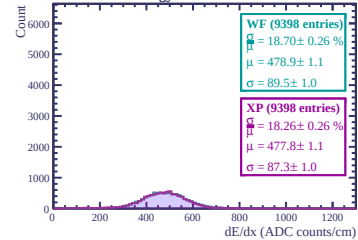
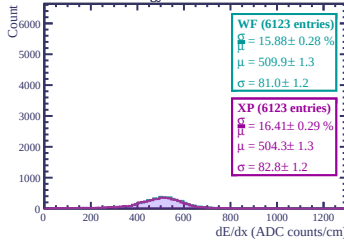




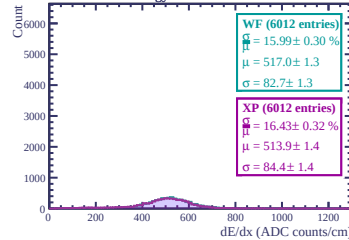
Energy loss in ERAM 24



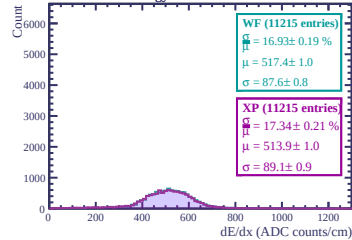
Energy loss in ERAM 30



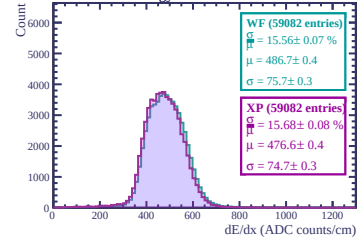
Energy loss in ERAM 36



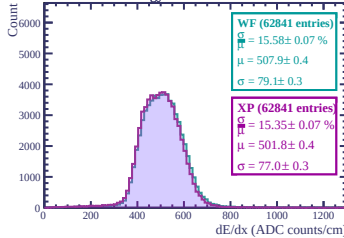
Energy loss in ERAM 42



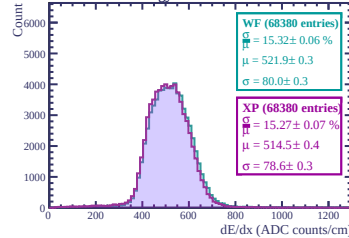
Energy loss in ERAM 21



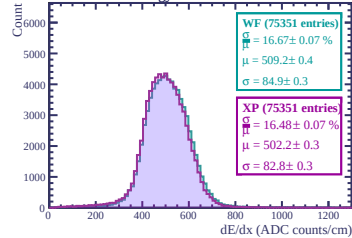
Energy loss in ERAM 13



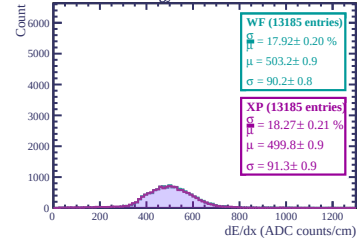
Energy loss in ERAM 9



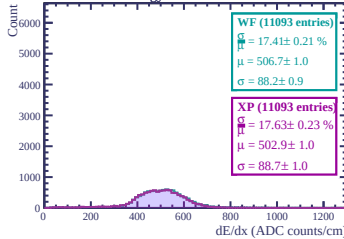
Energy loss in ERAM 2



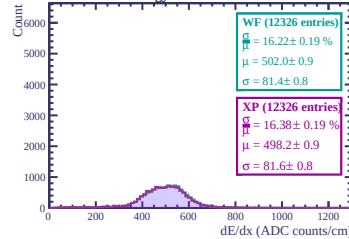
Energy loss in ERAM 26



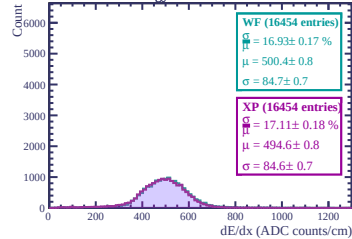
Energy loss in ERAM 17



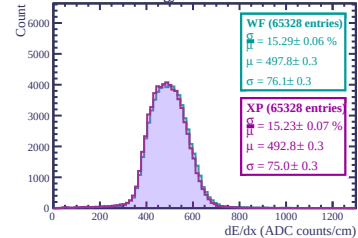
Energy loss in ERAM 23



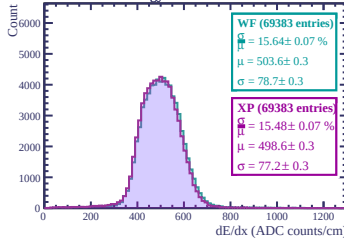
Energy loss in ERAM 29



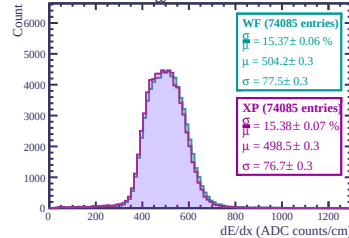
Energy loss in ERAM 1



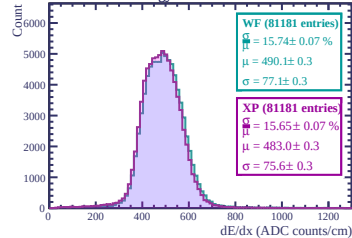
Energy loss in ERAM 10

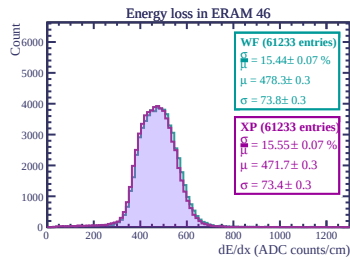
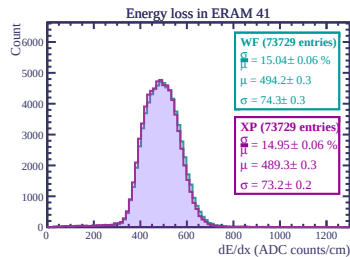
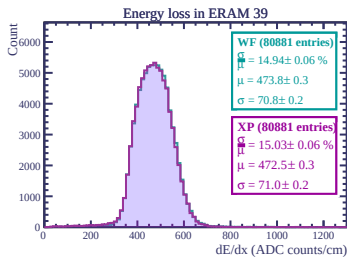
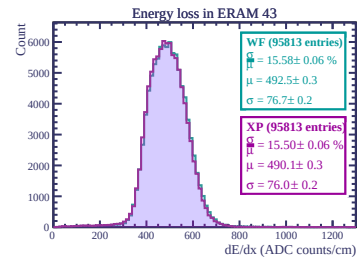
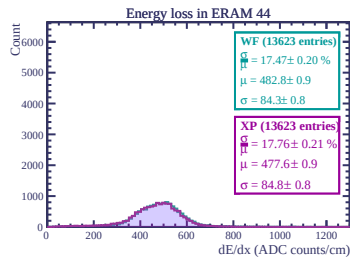
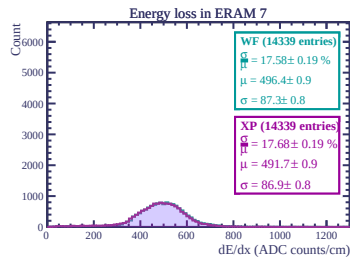
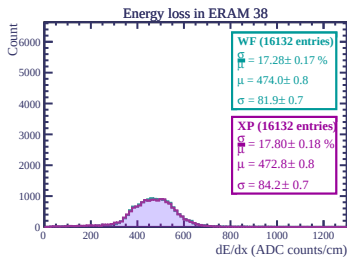
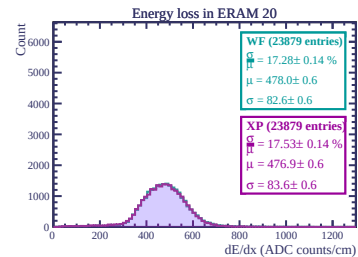
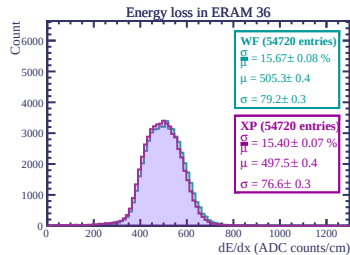
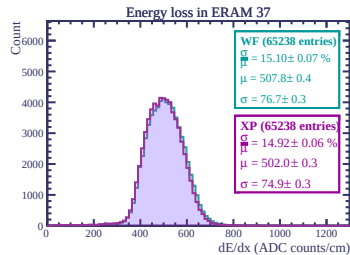
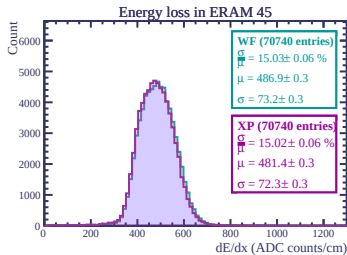
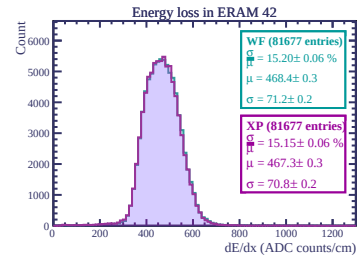
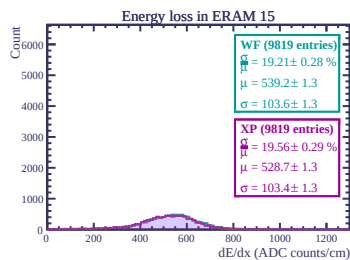
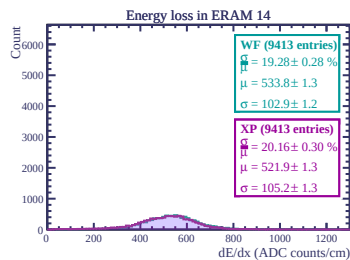
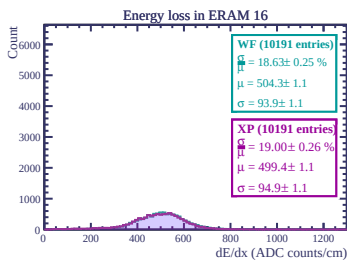
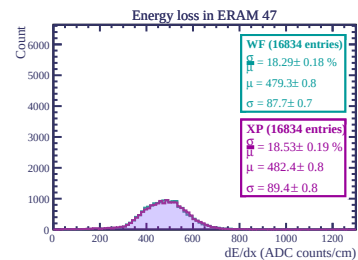


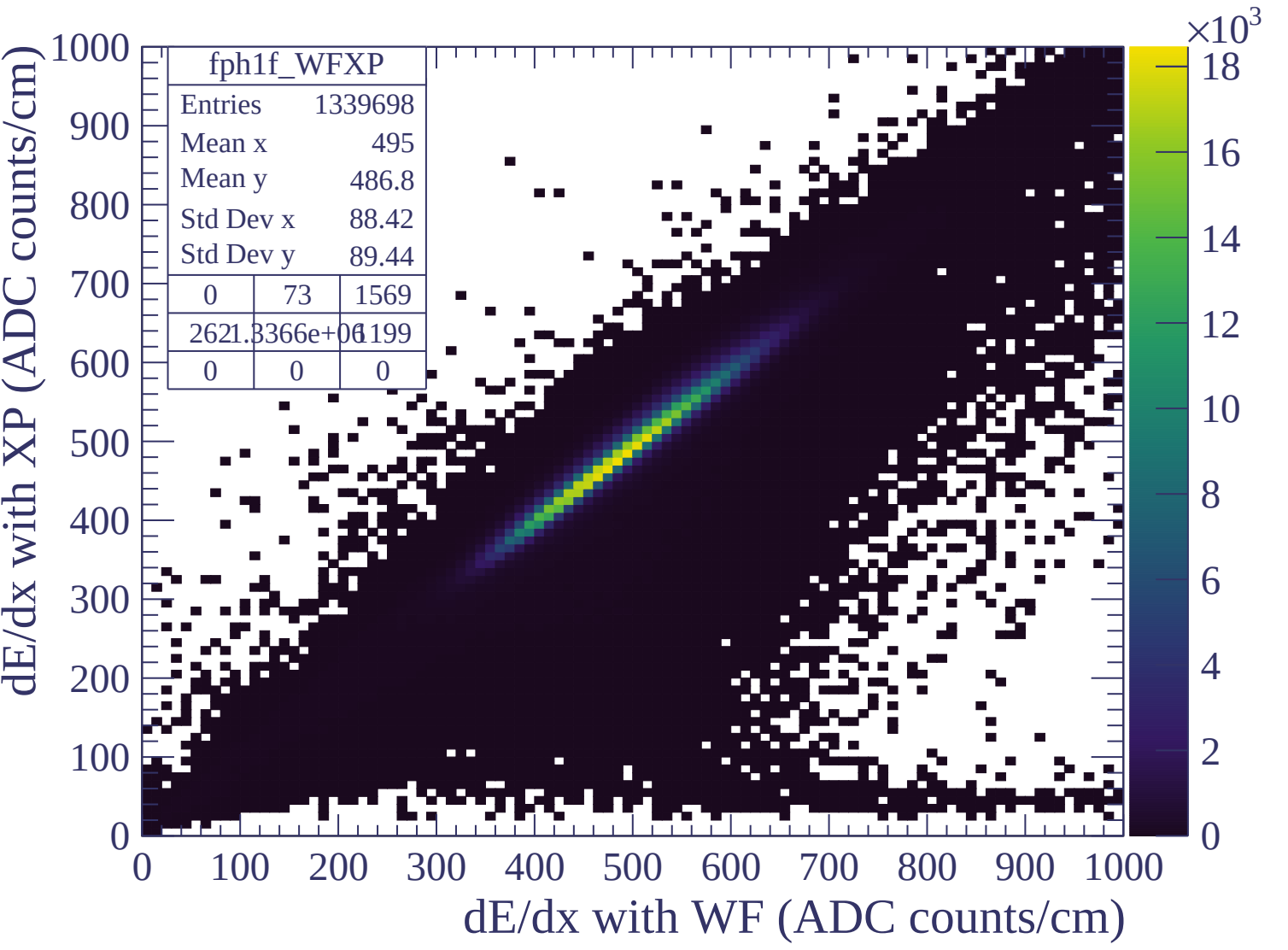
Energy loss in ERAM 11



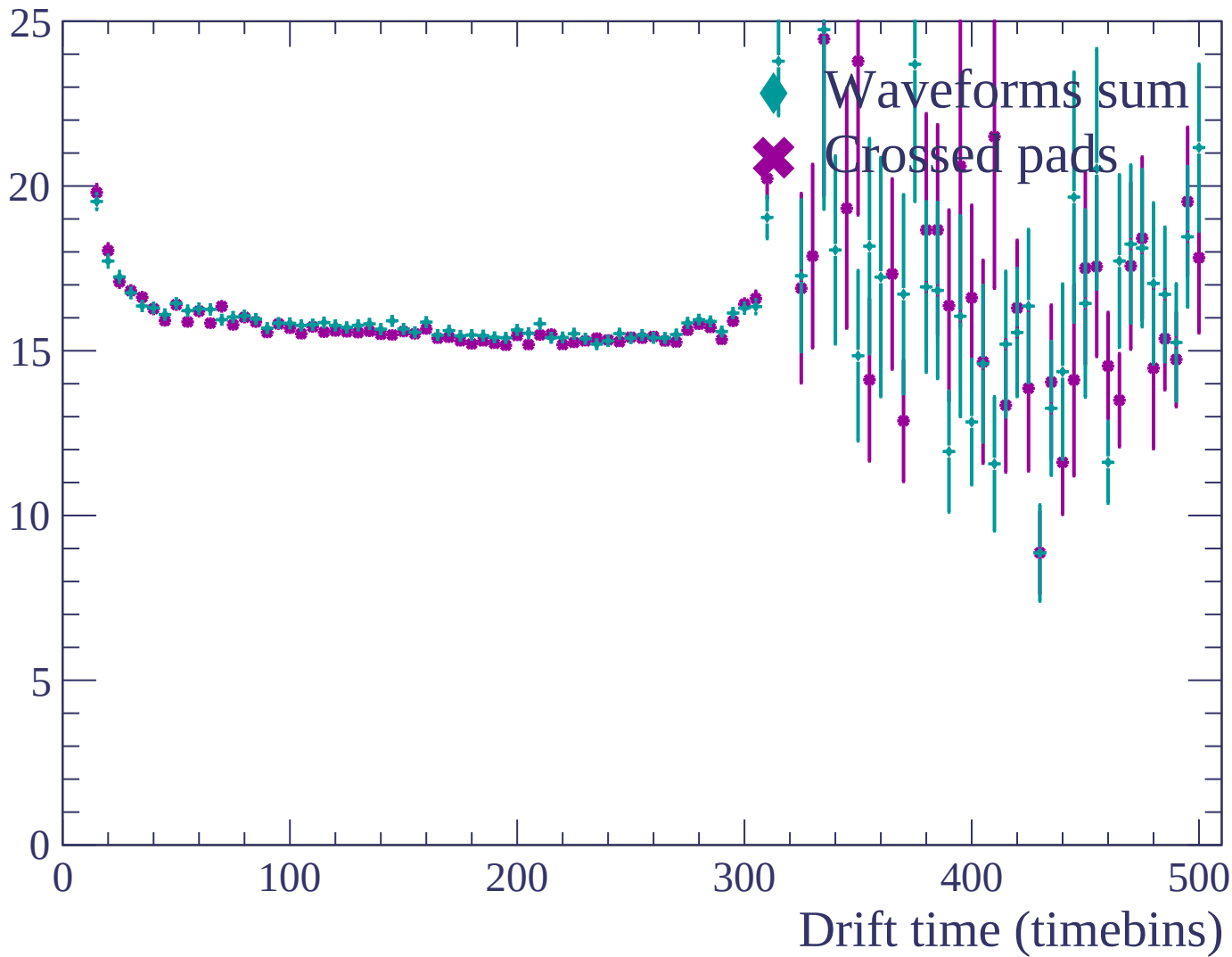
Energy loss in ERAM 3

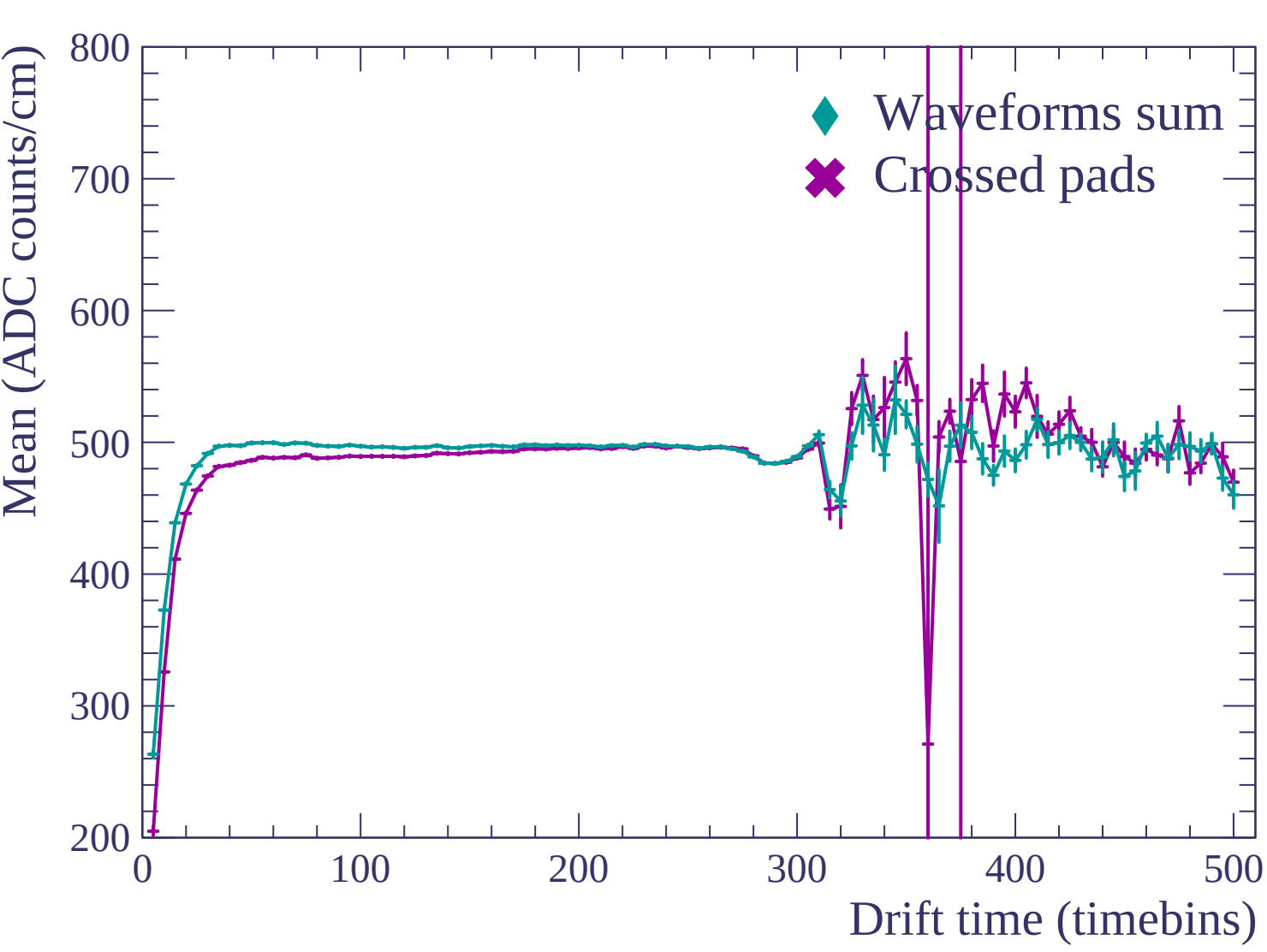


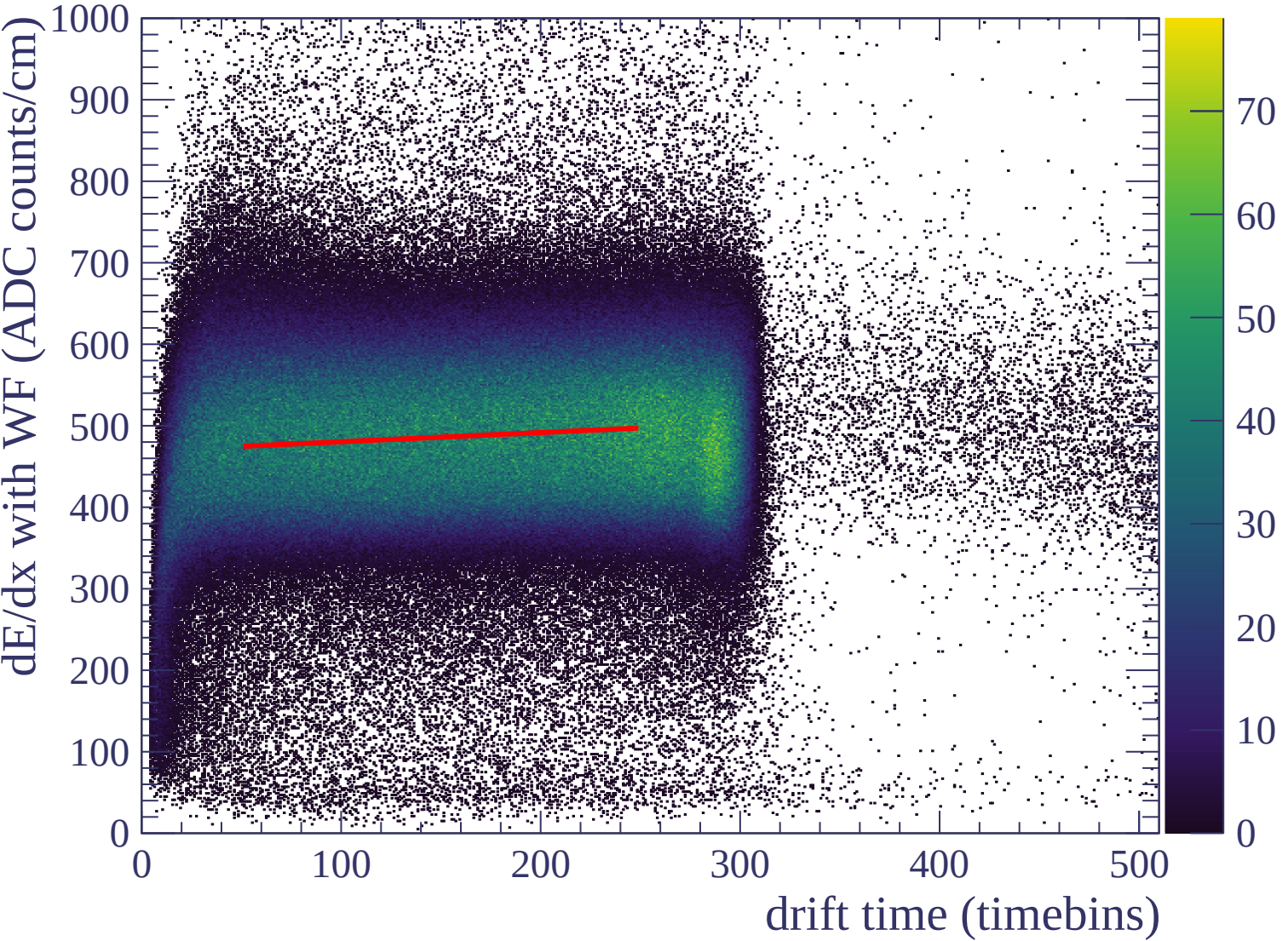


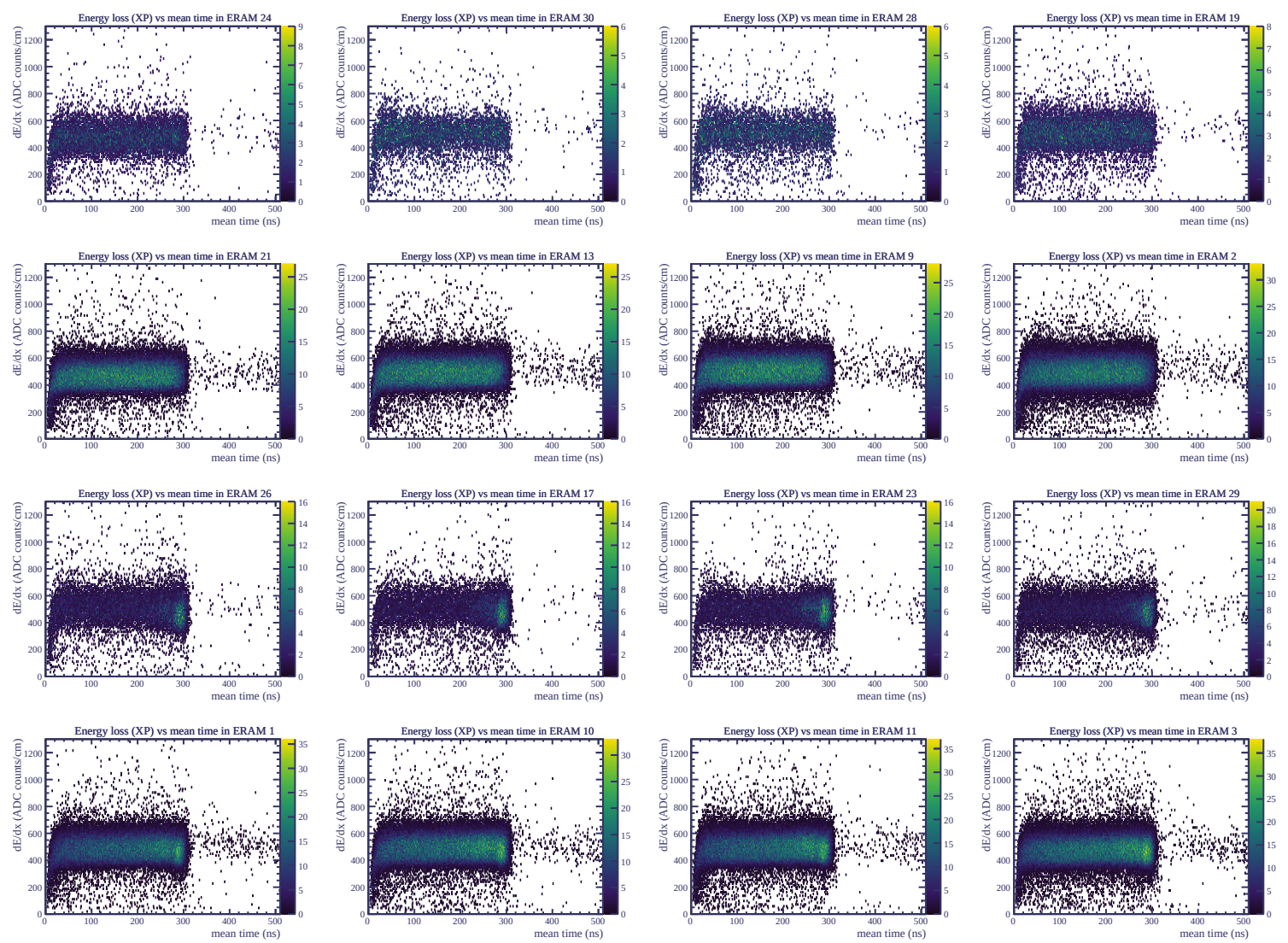


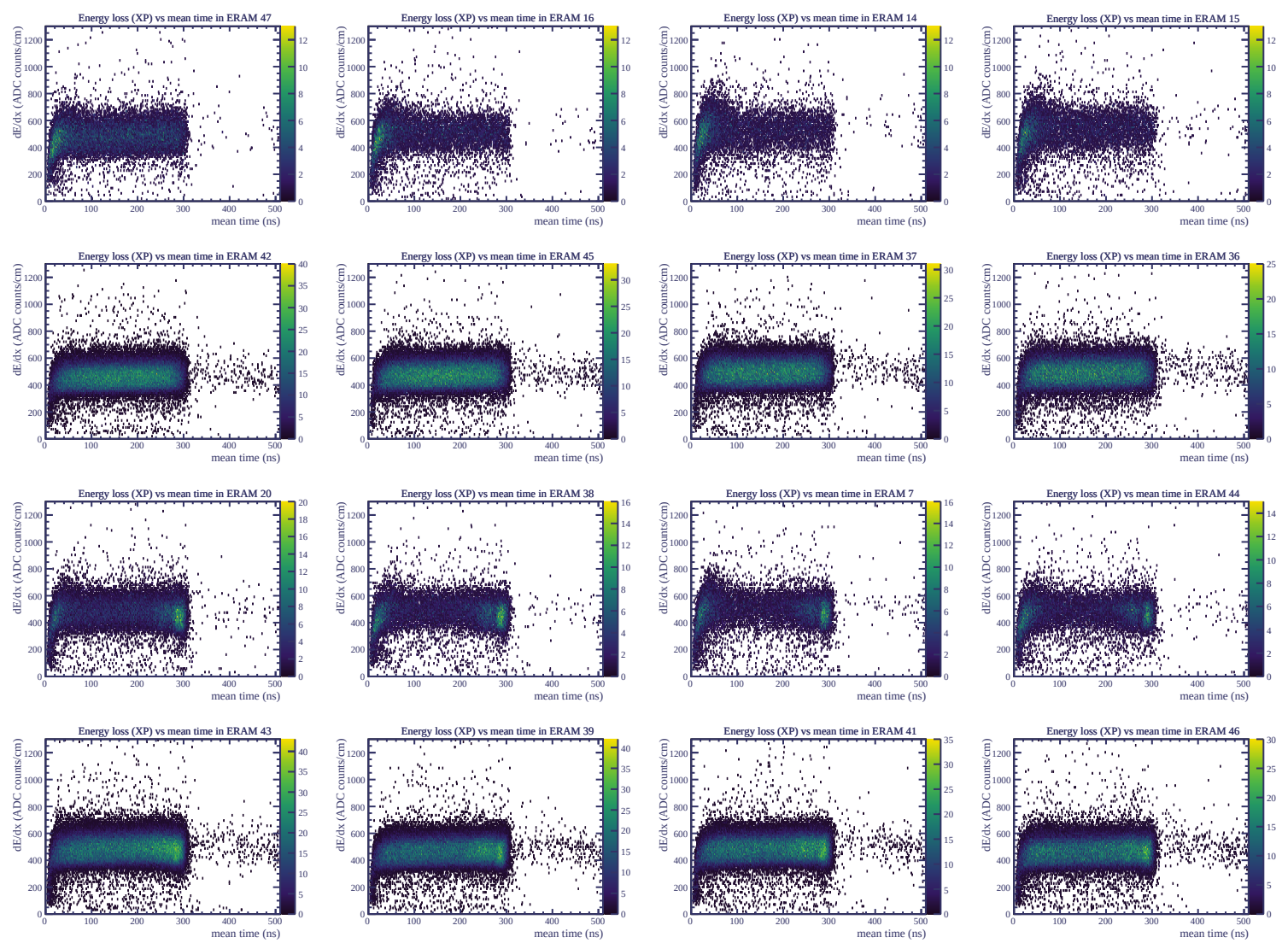
Resolution (%)

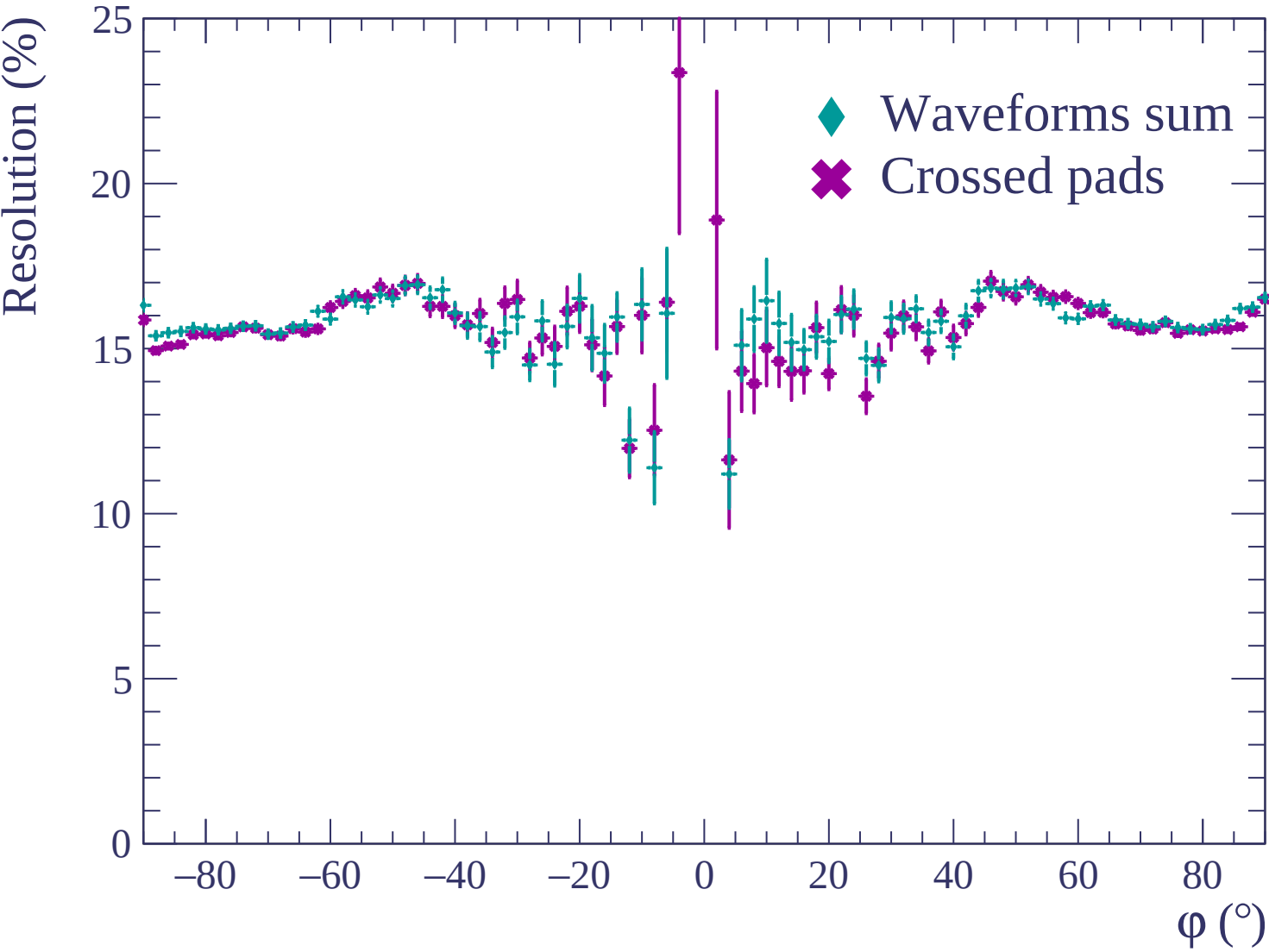


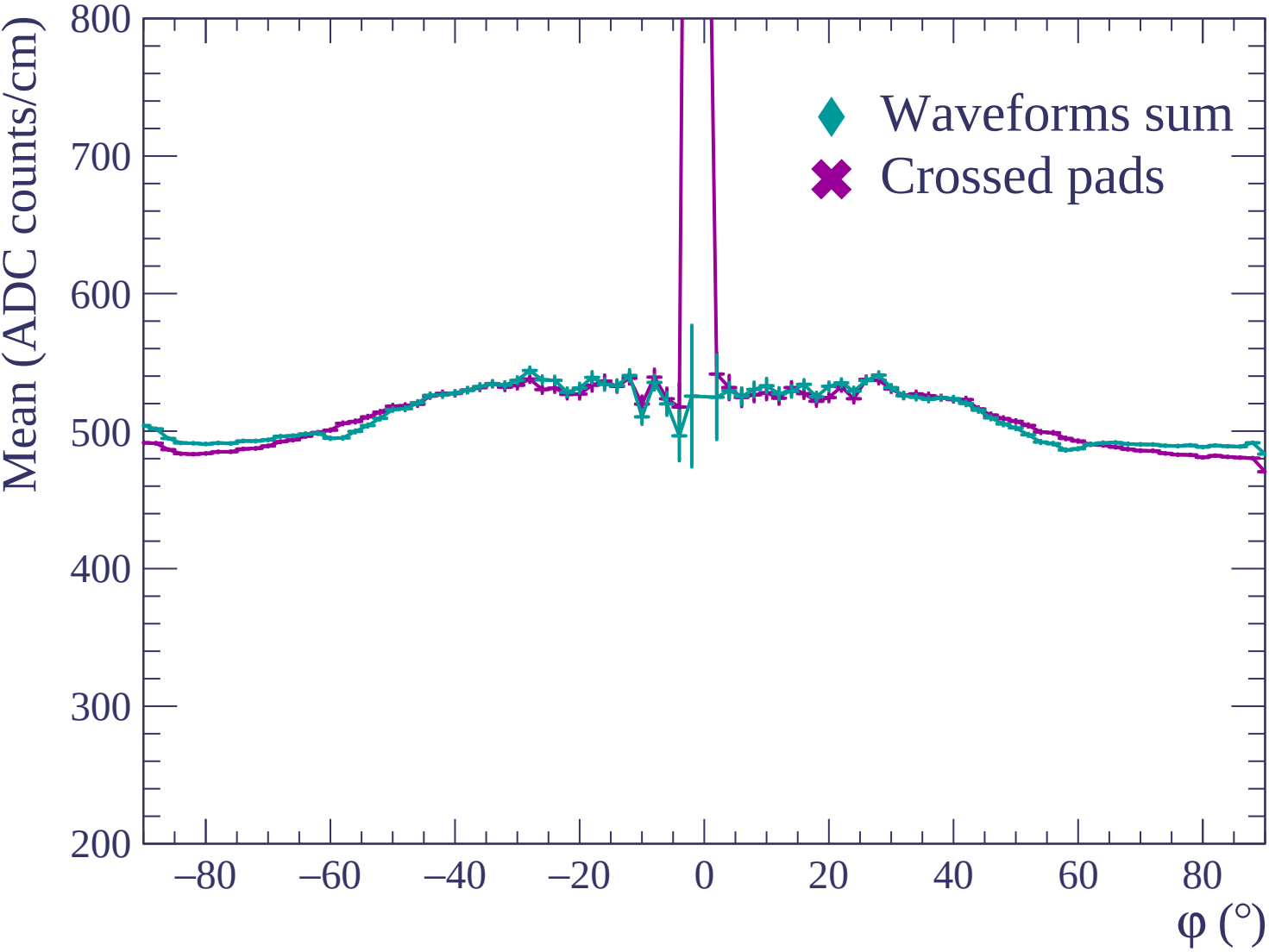


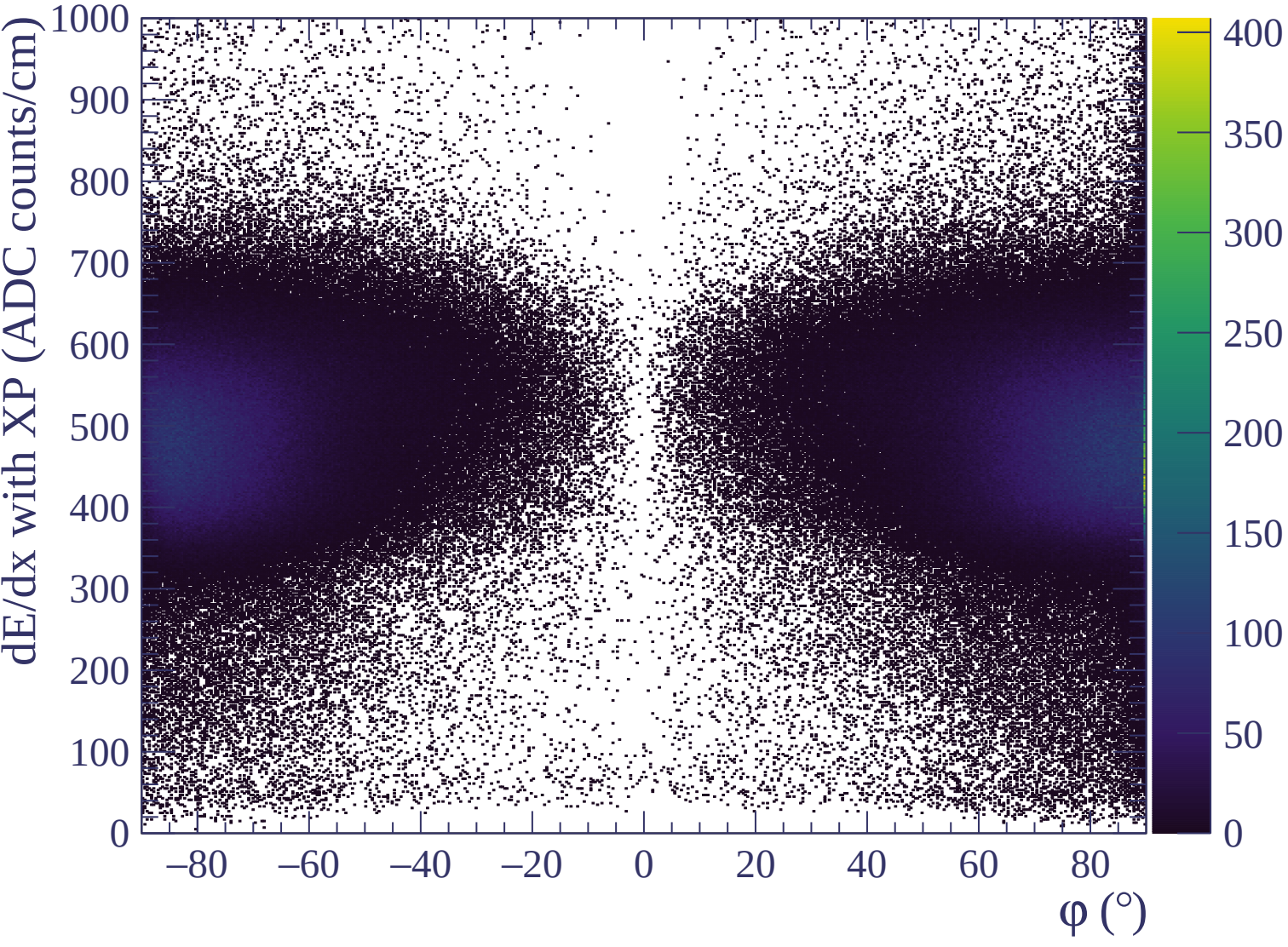


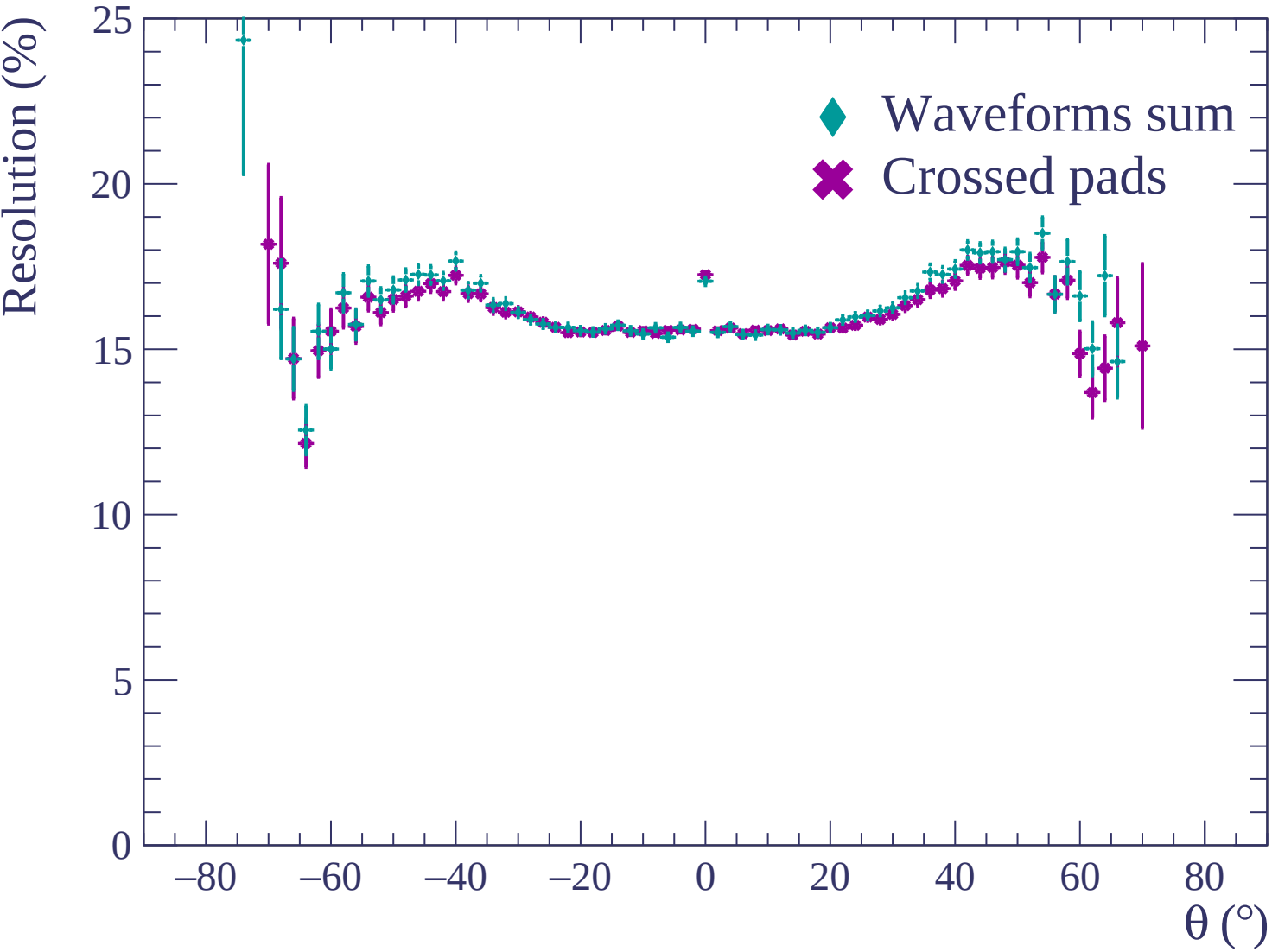


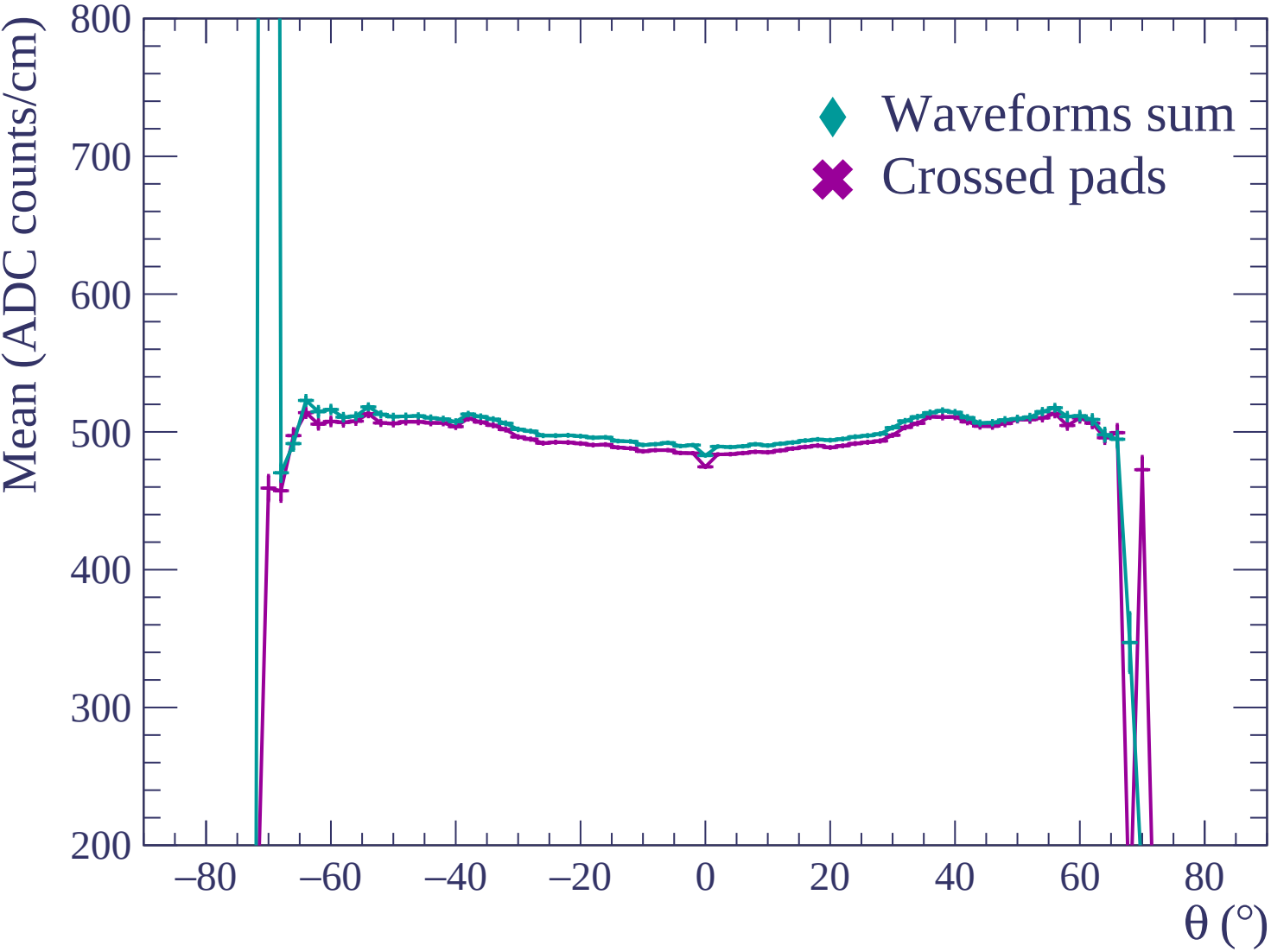


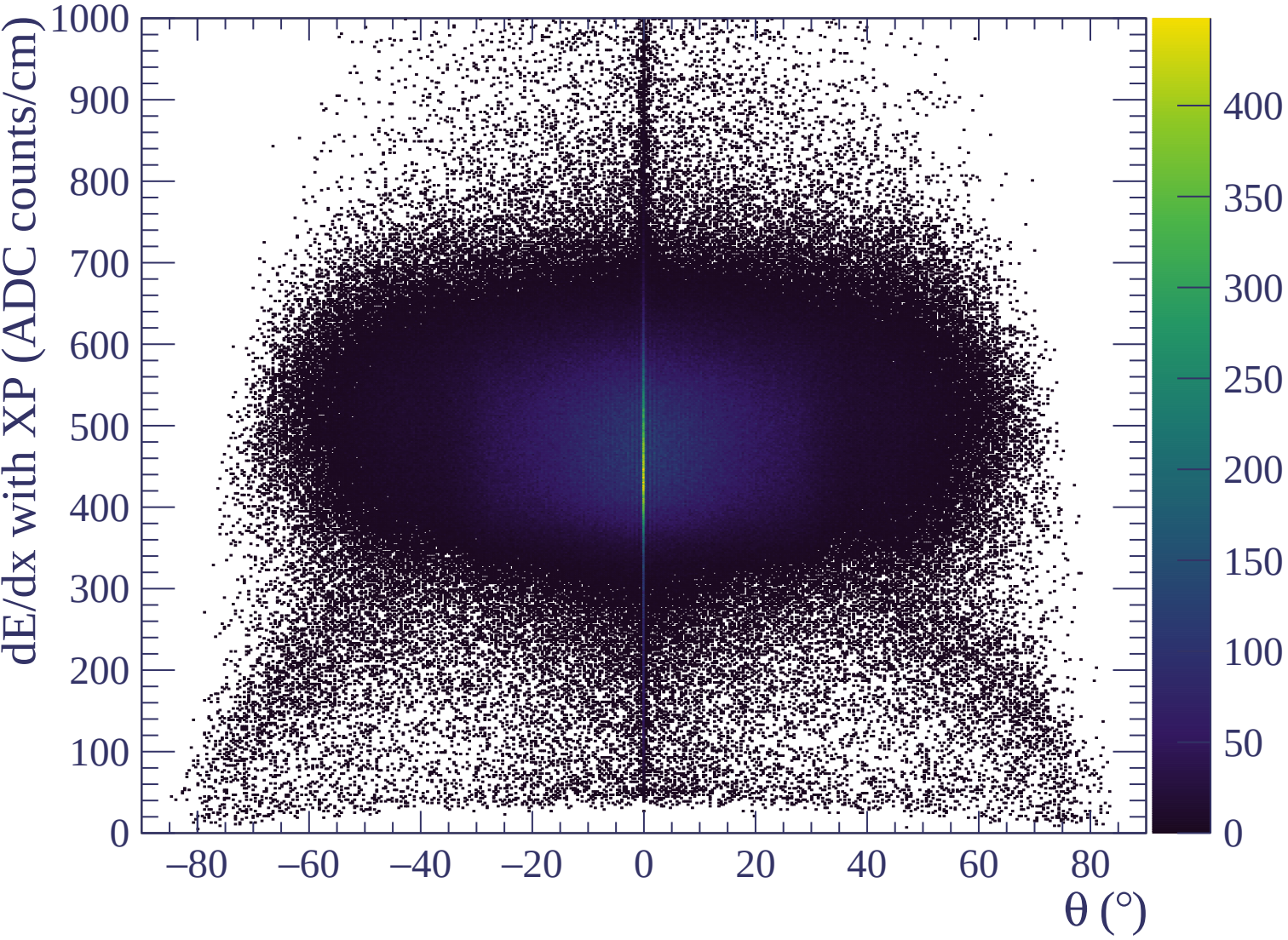


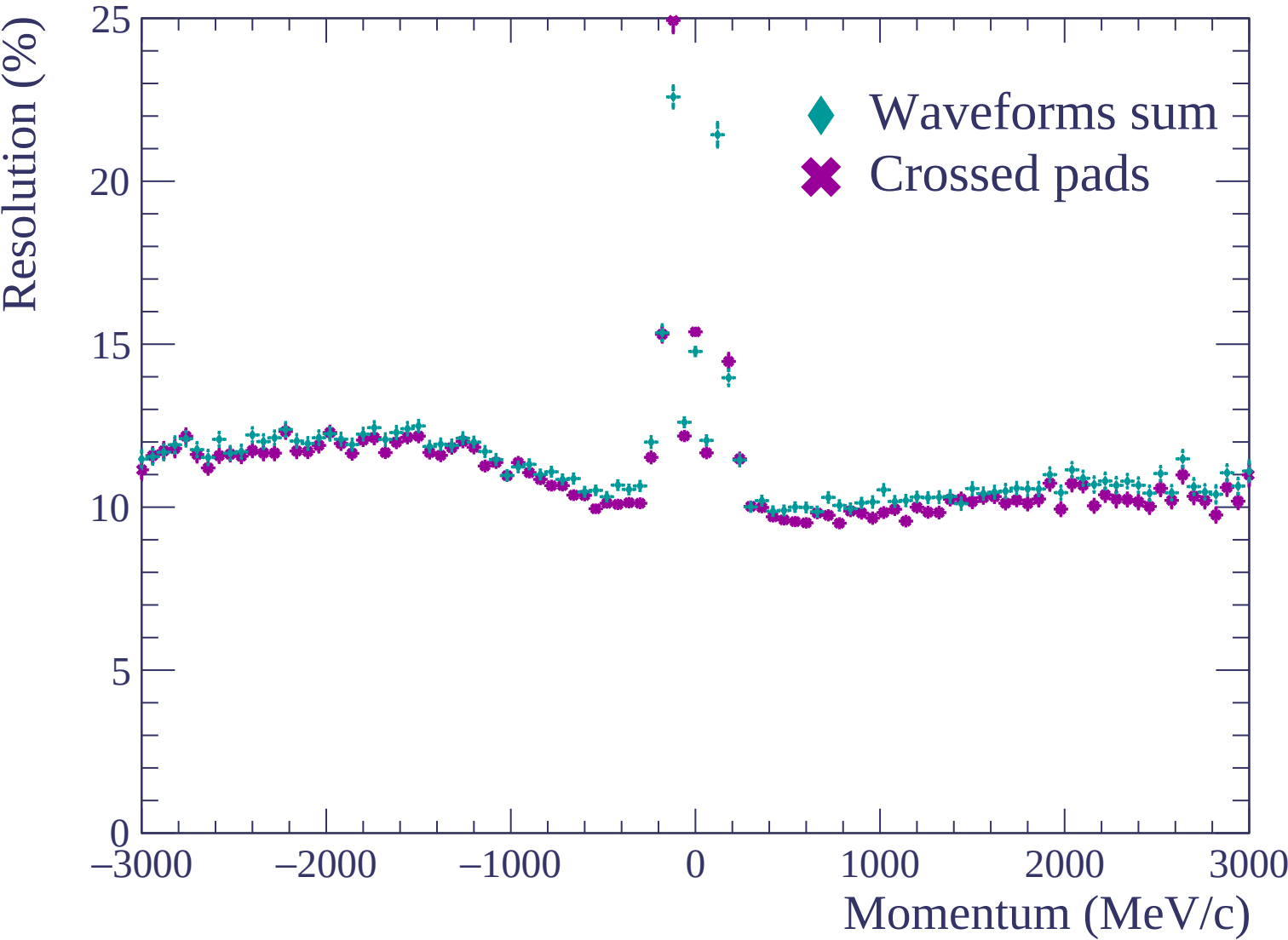


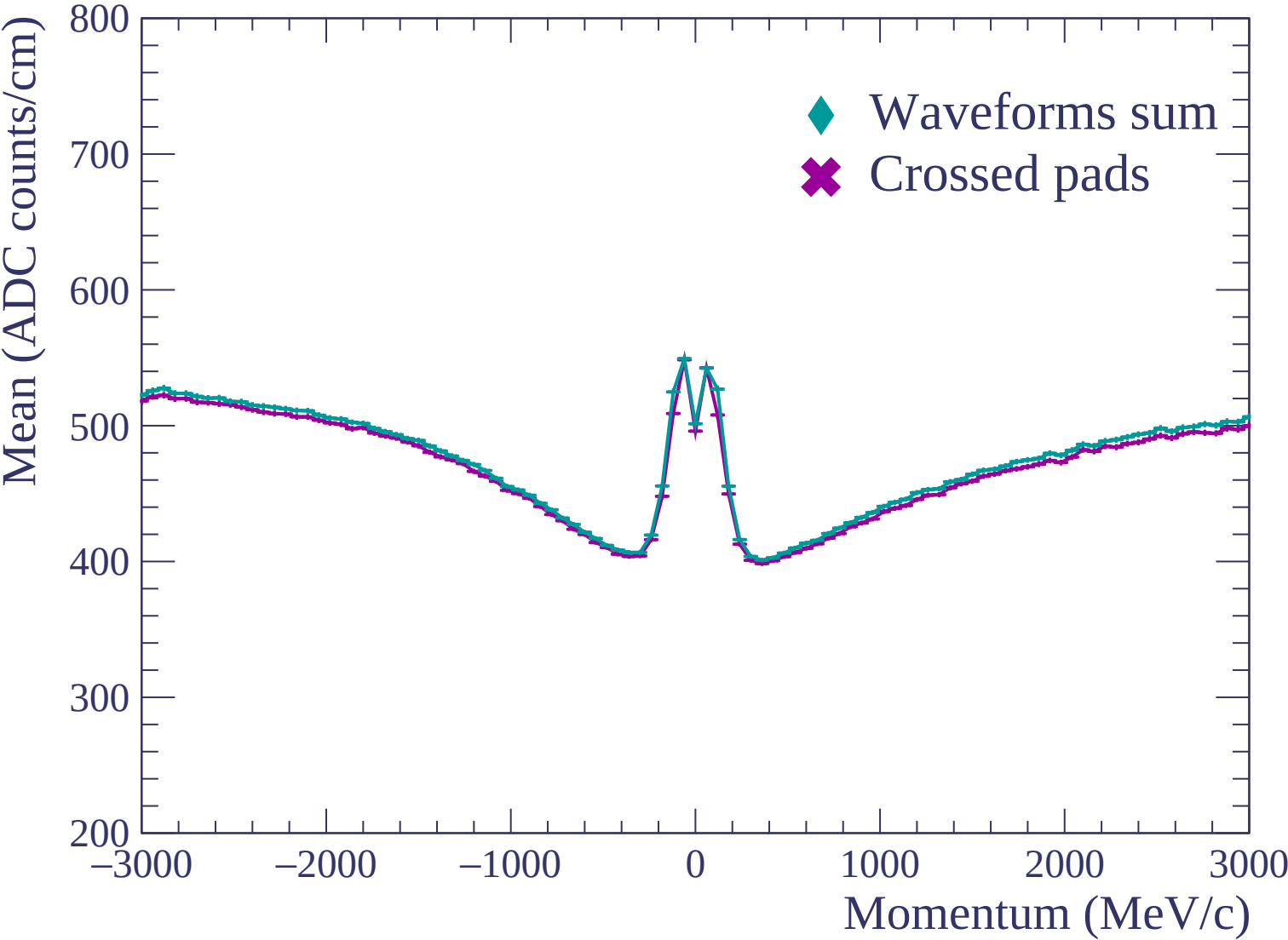


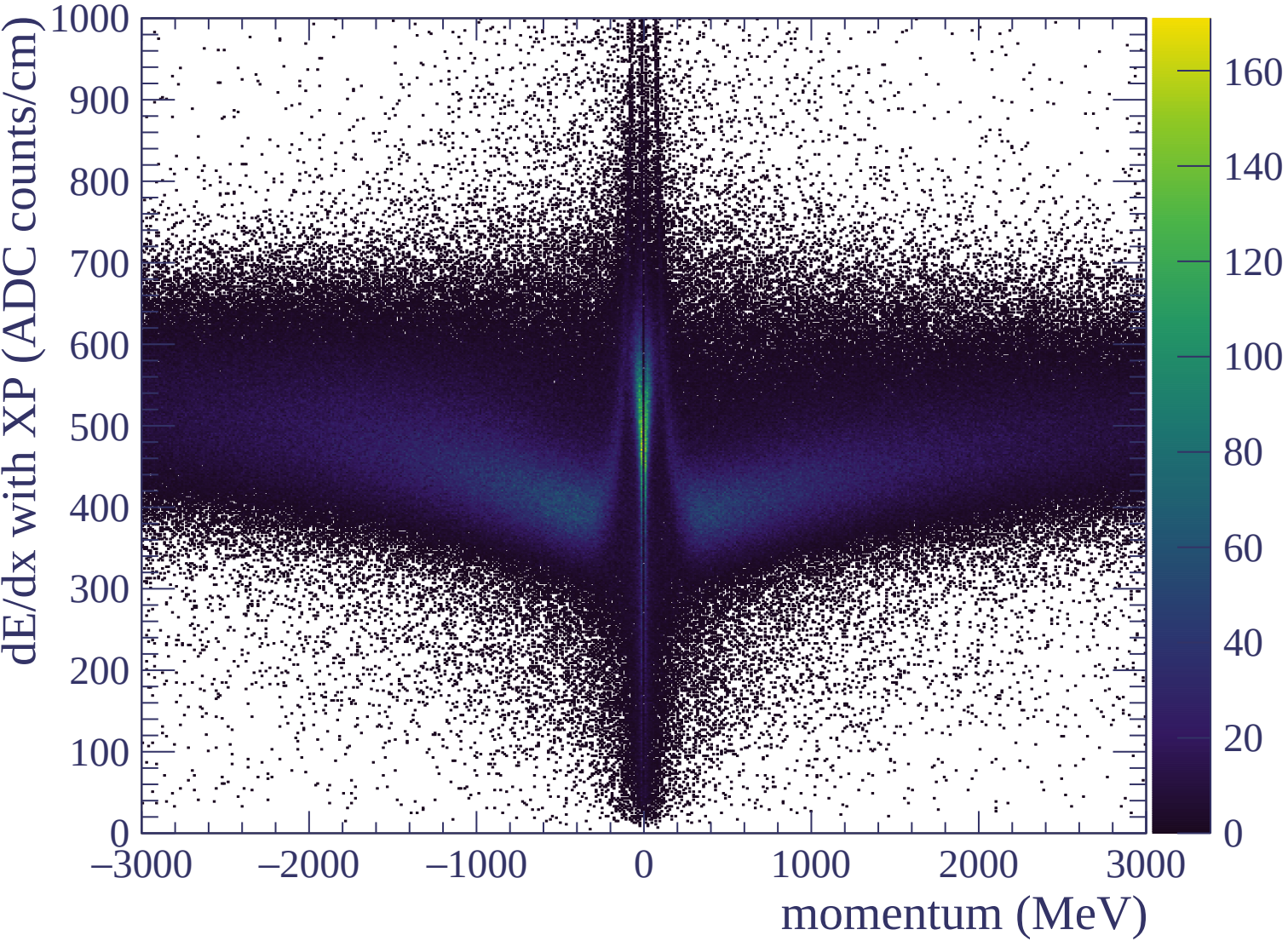


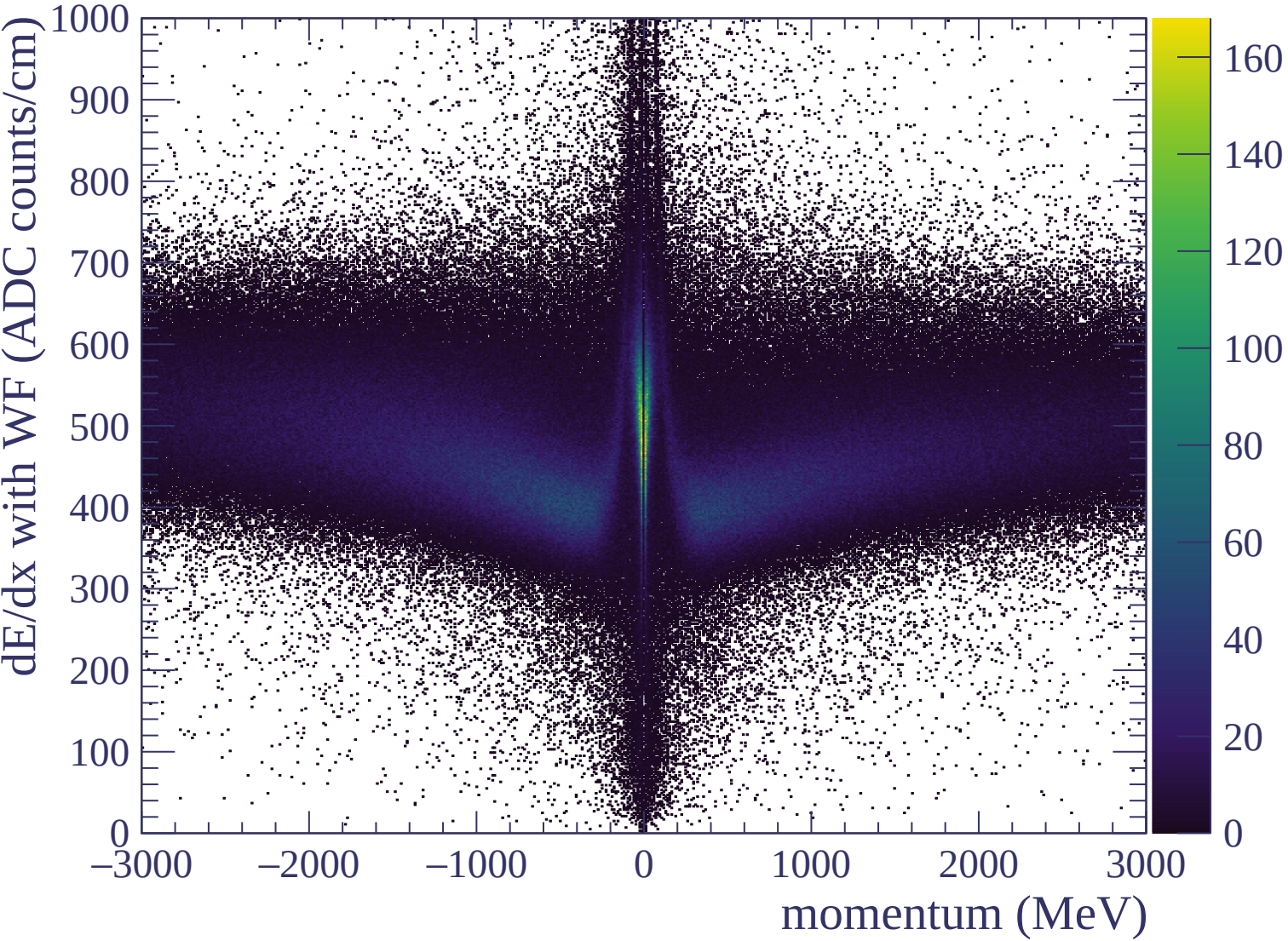


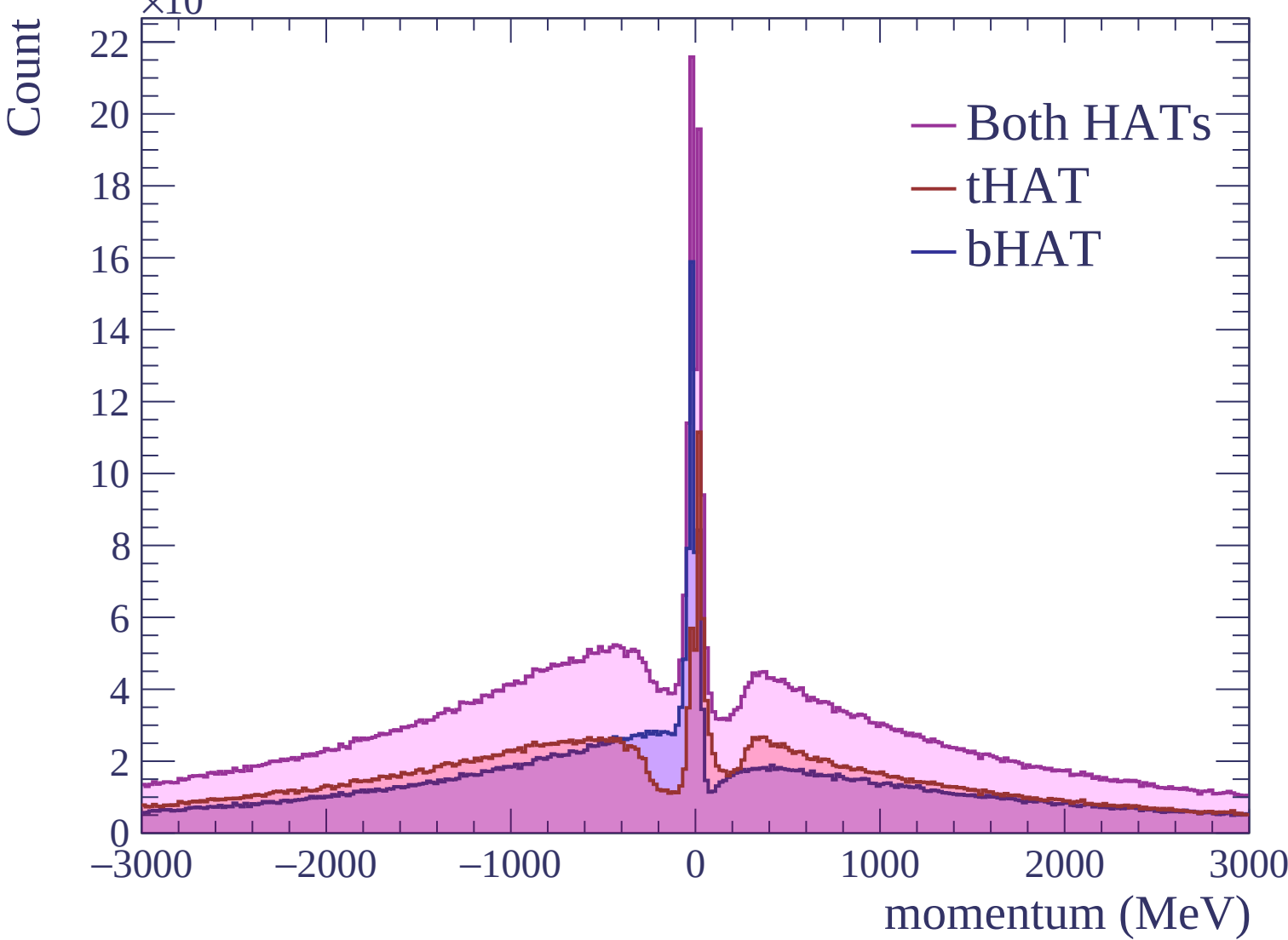


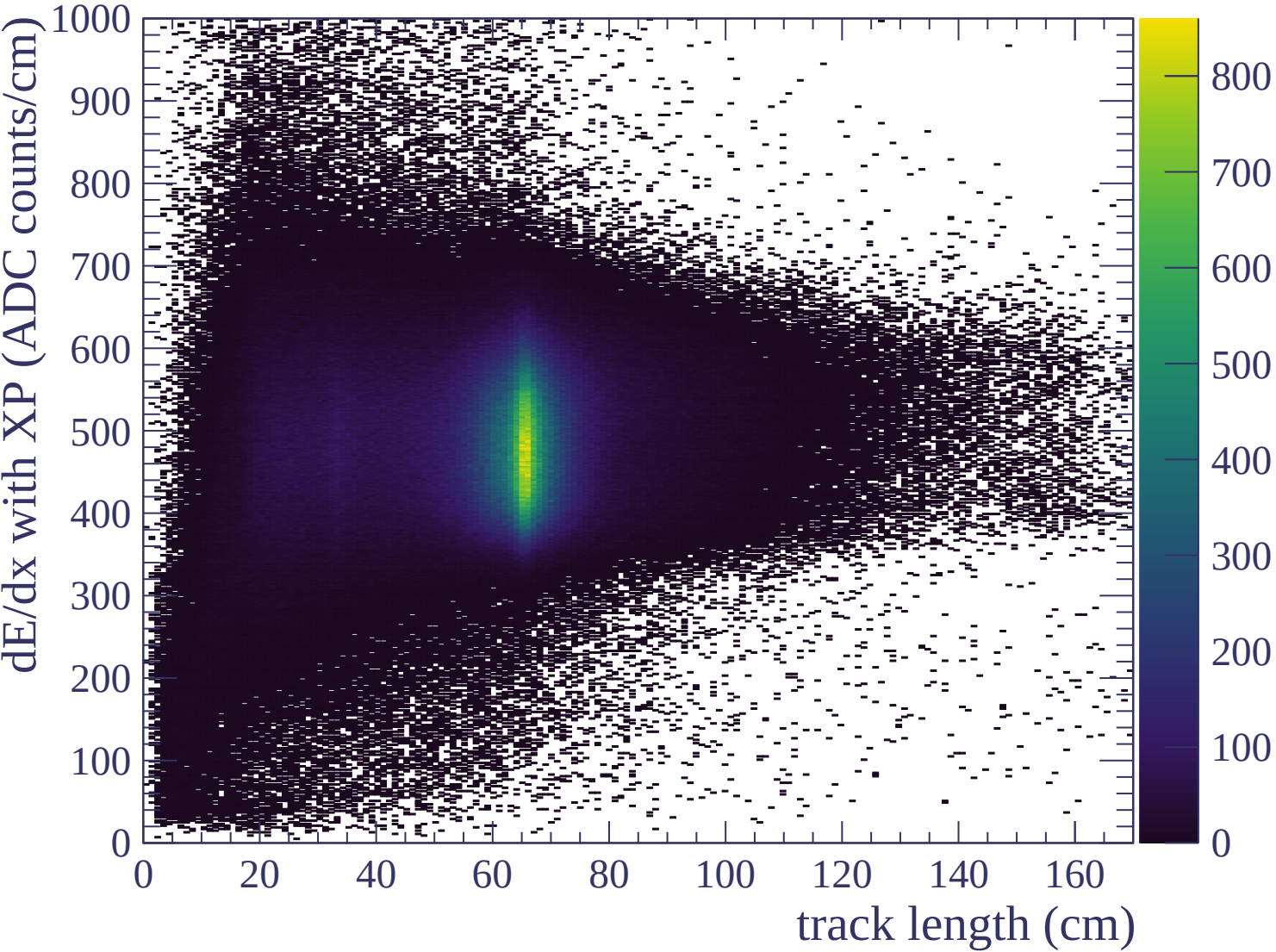


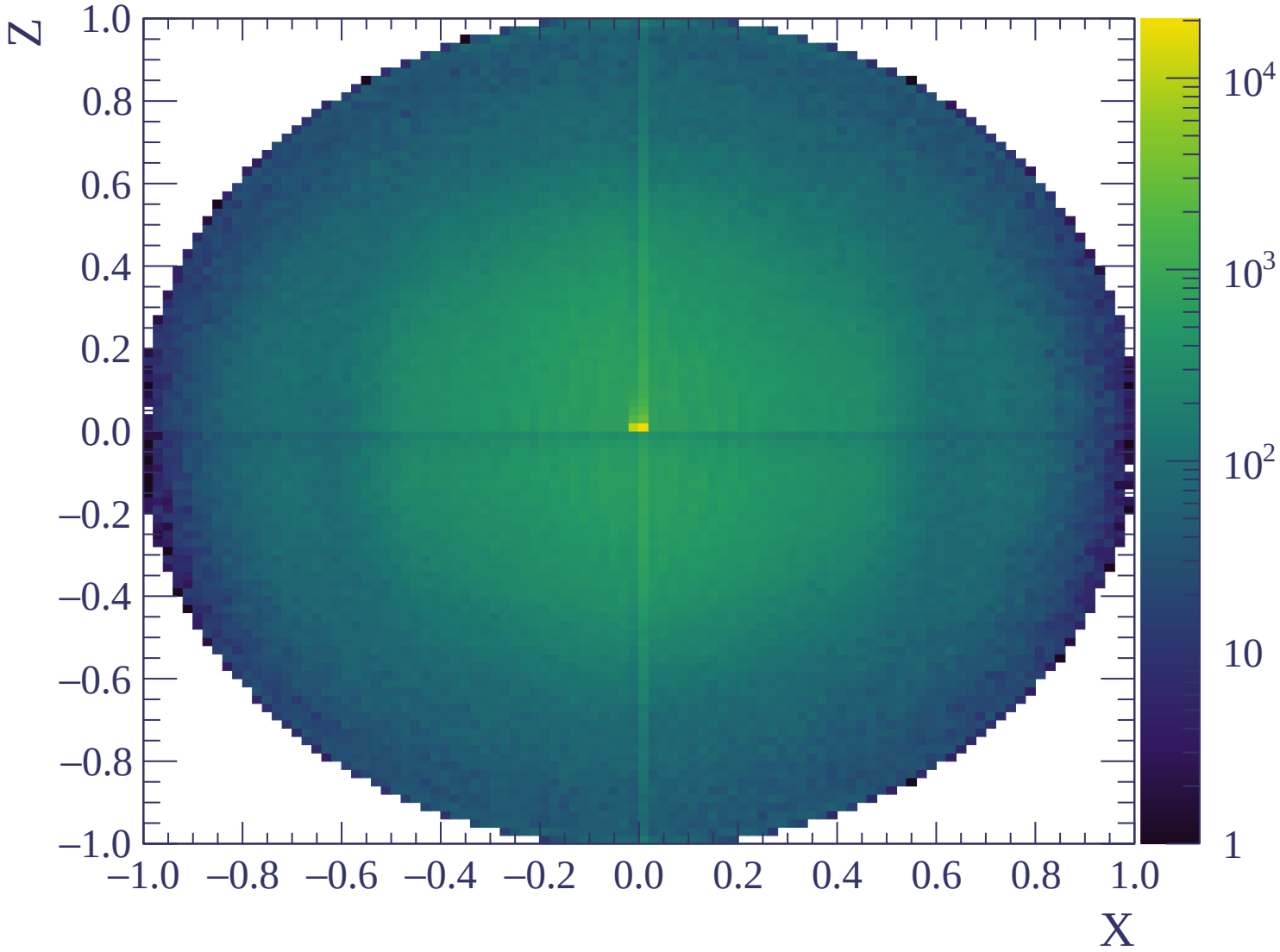


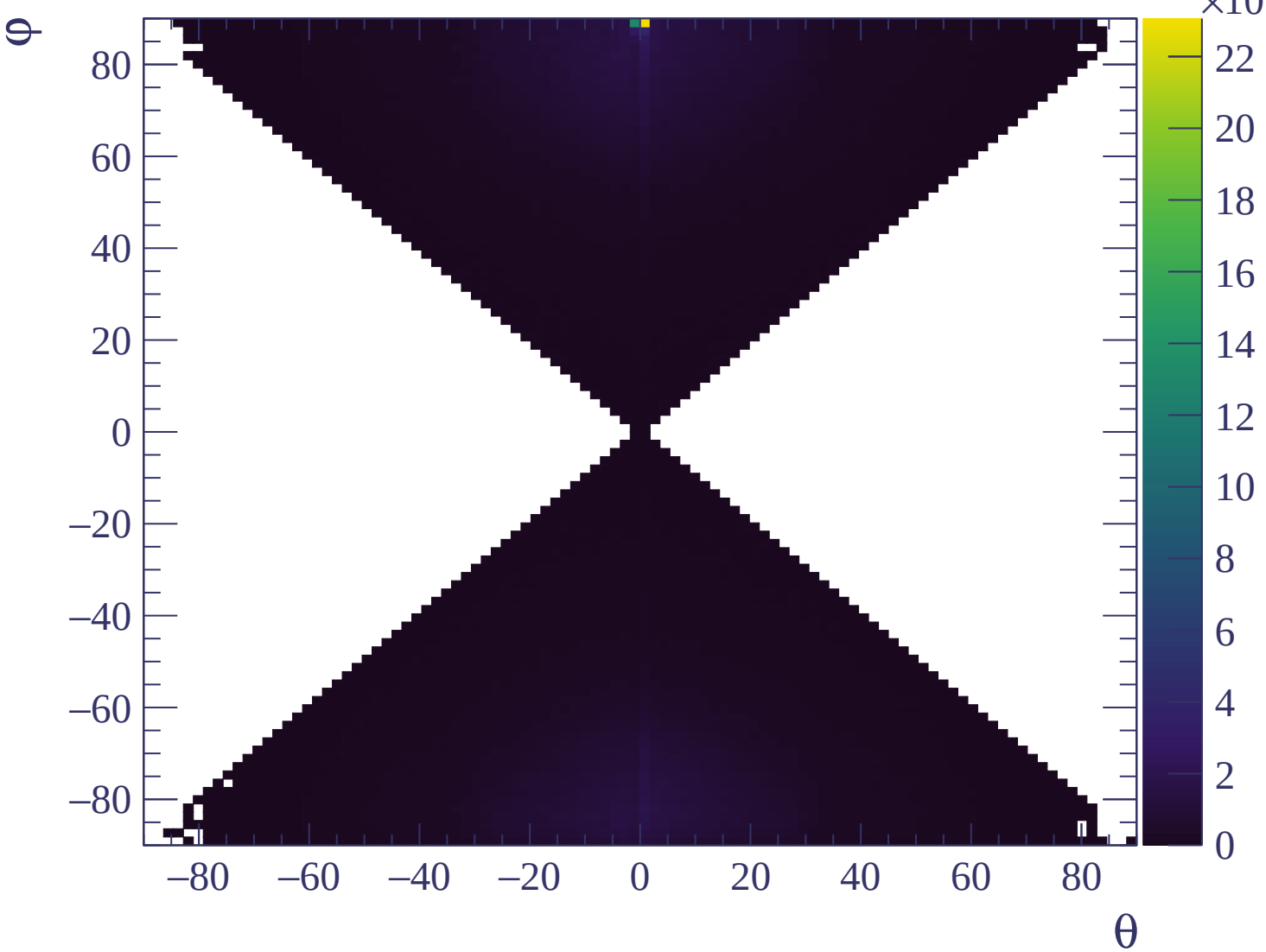


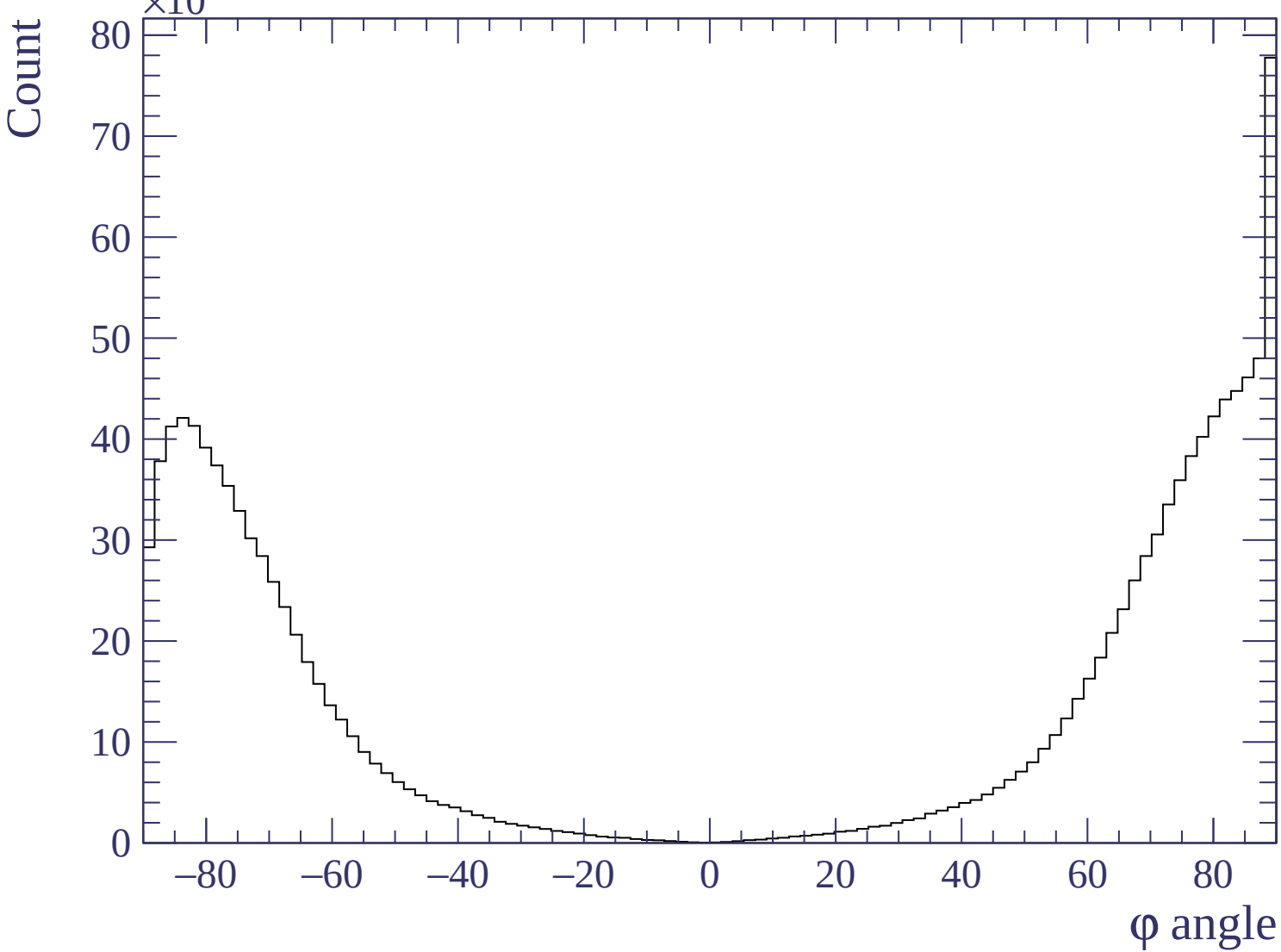


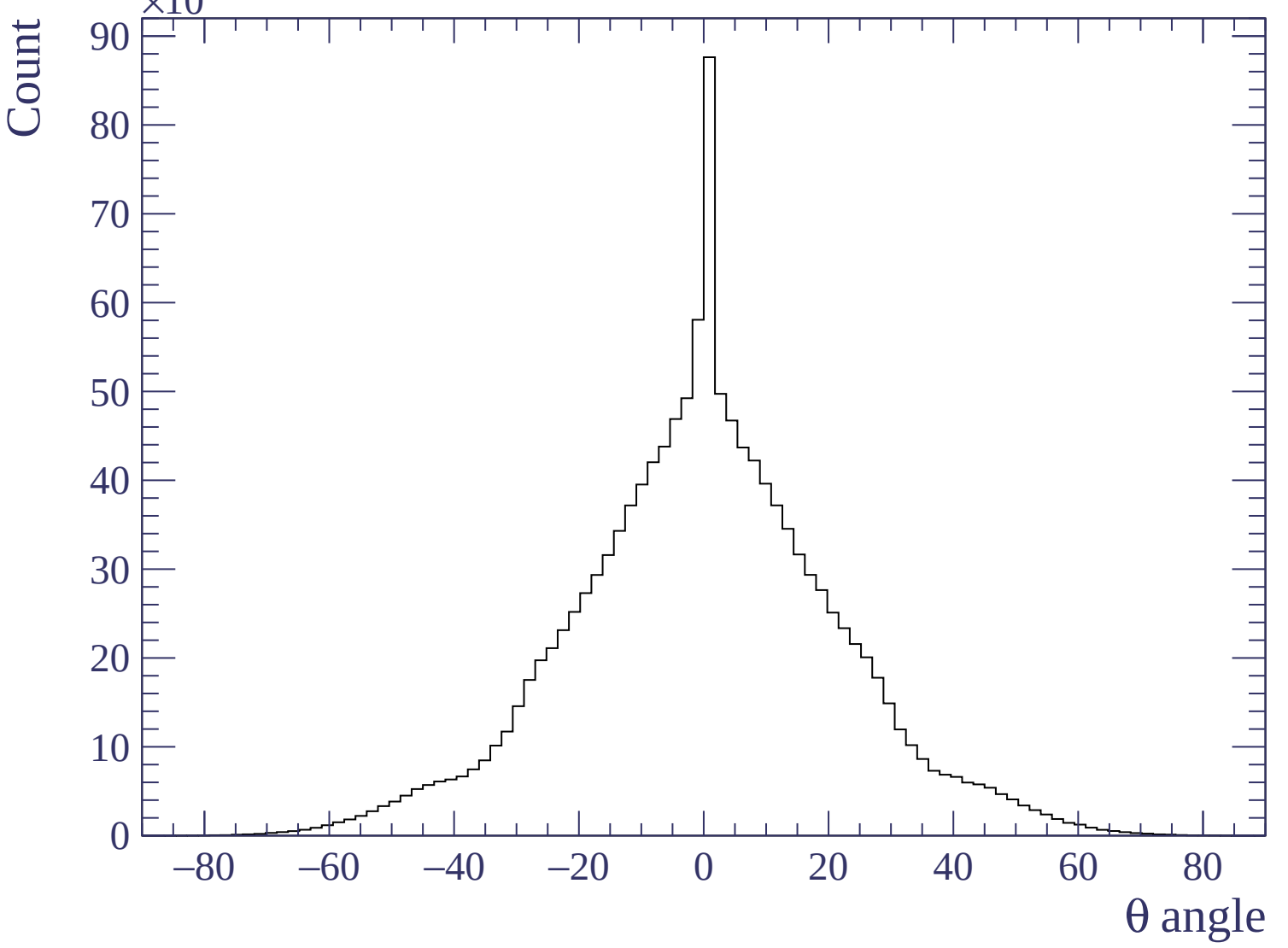


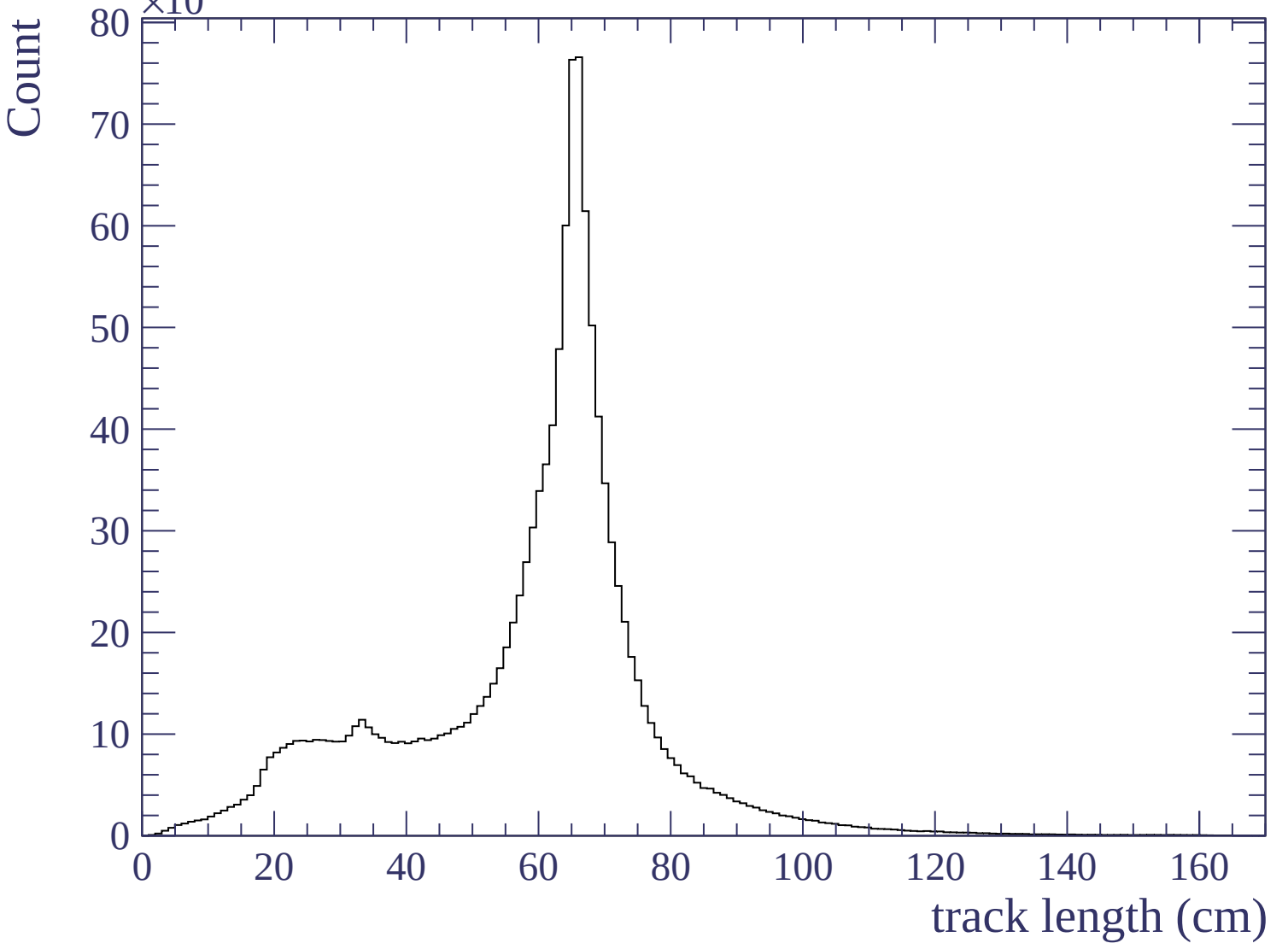


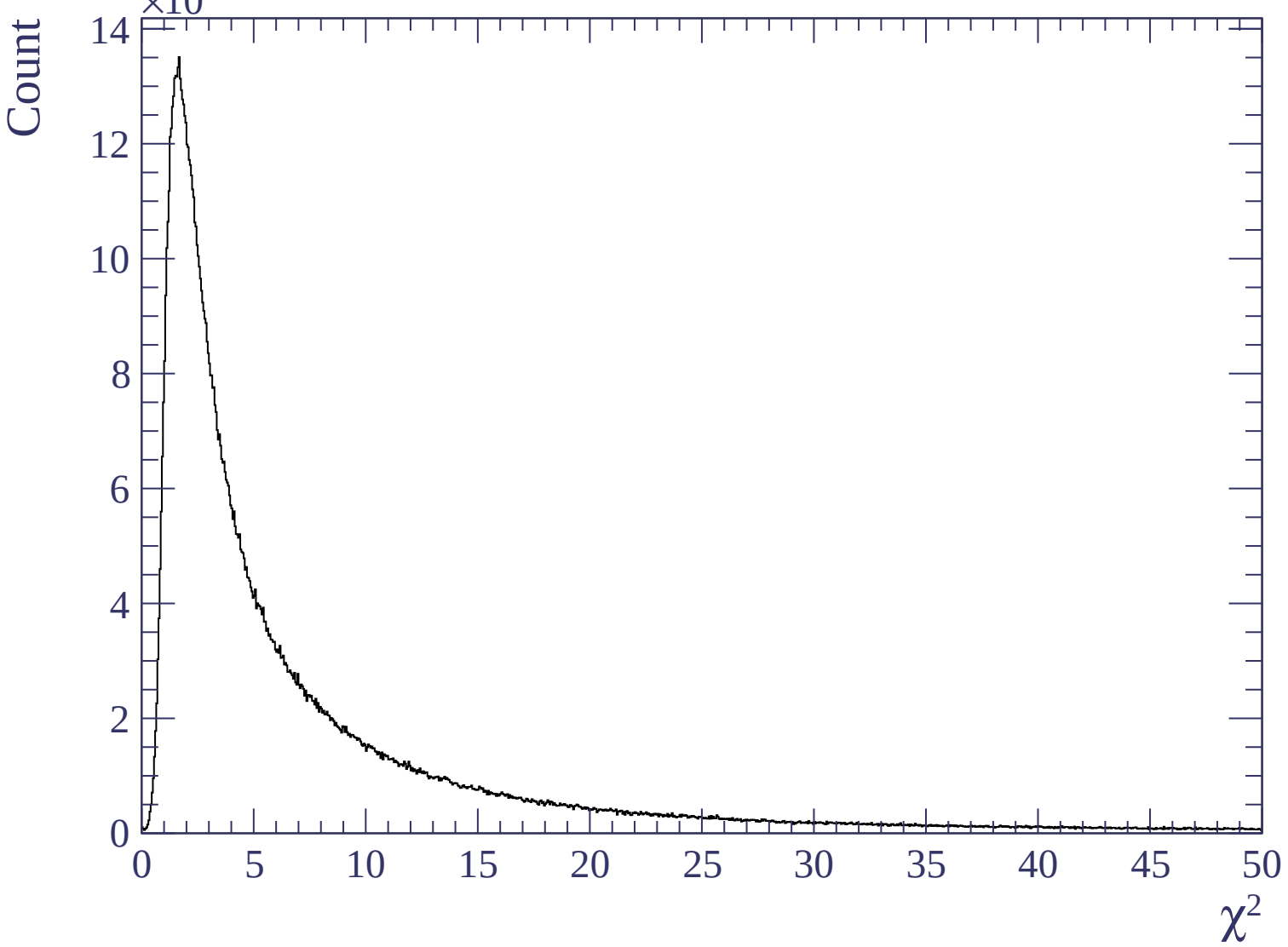


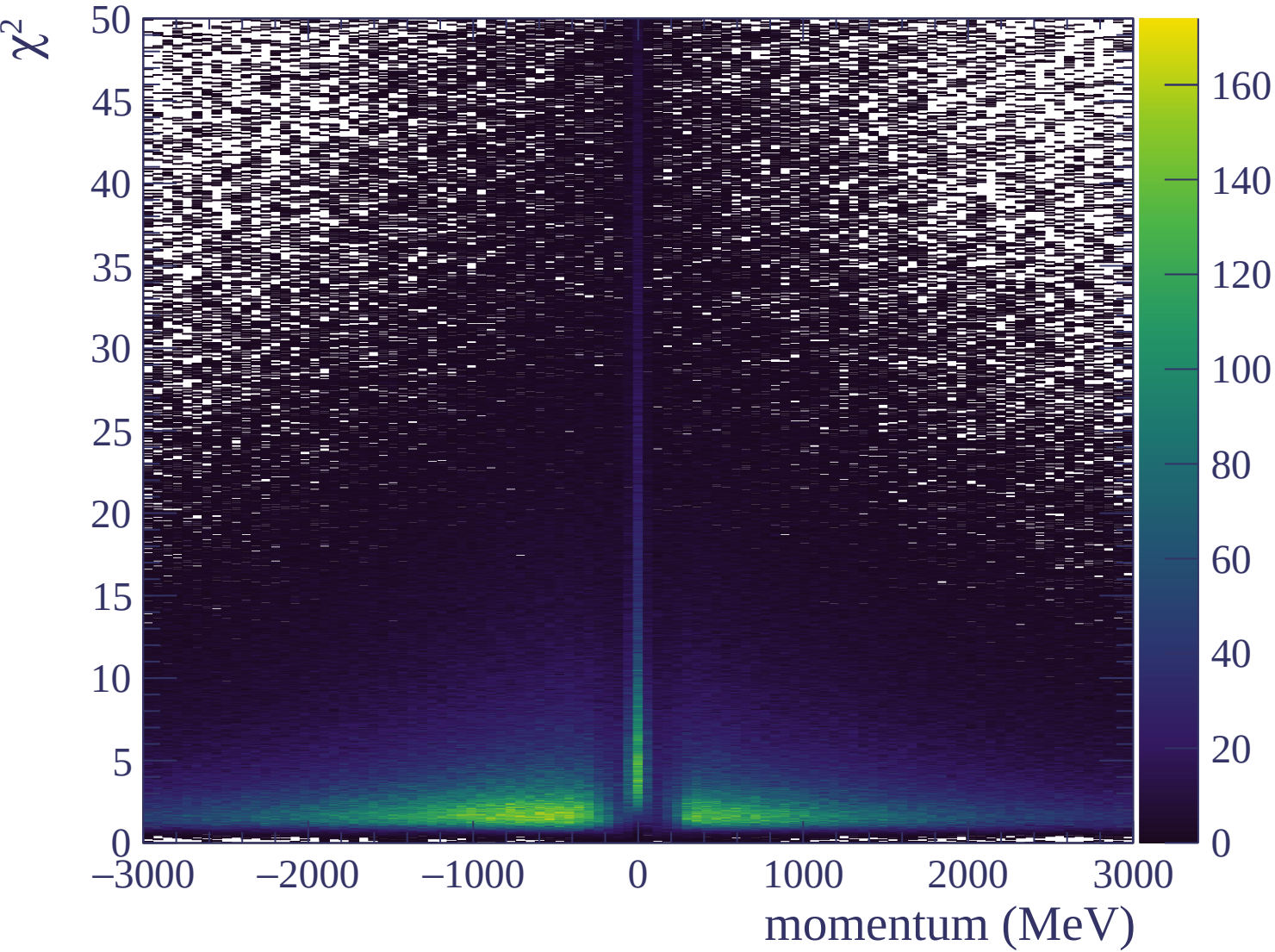


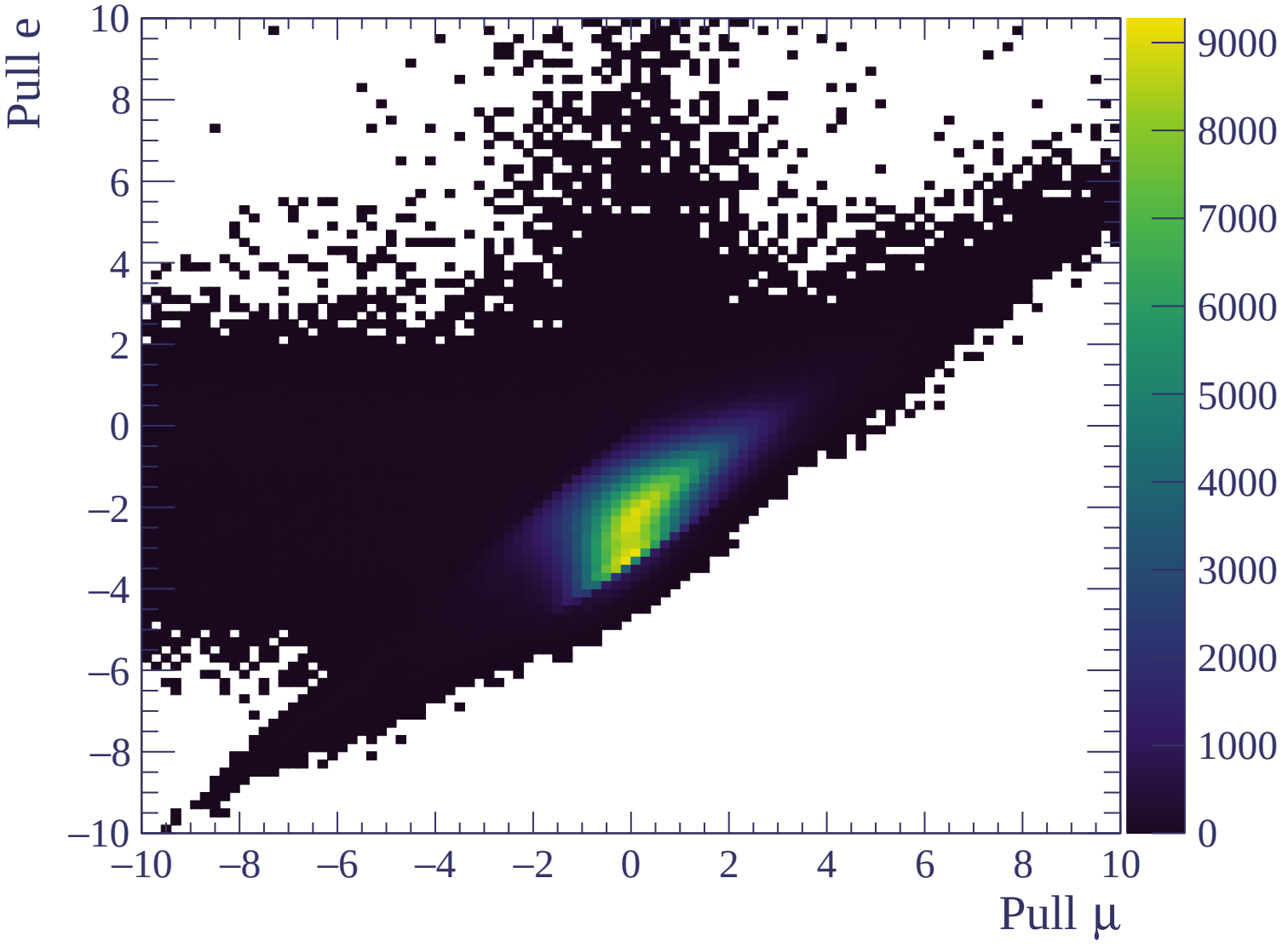


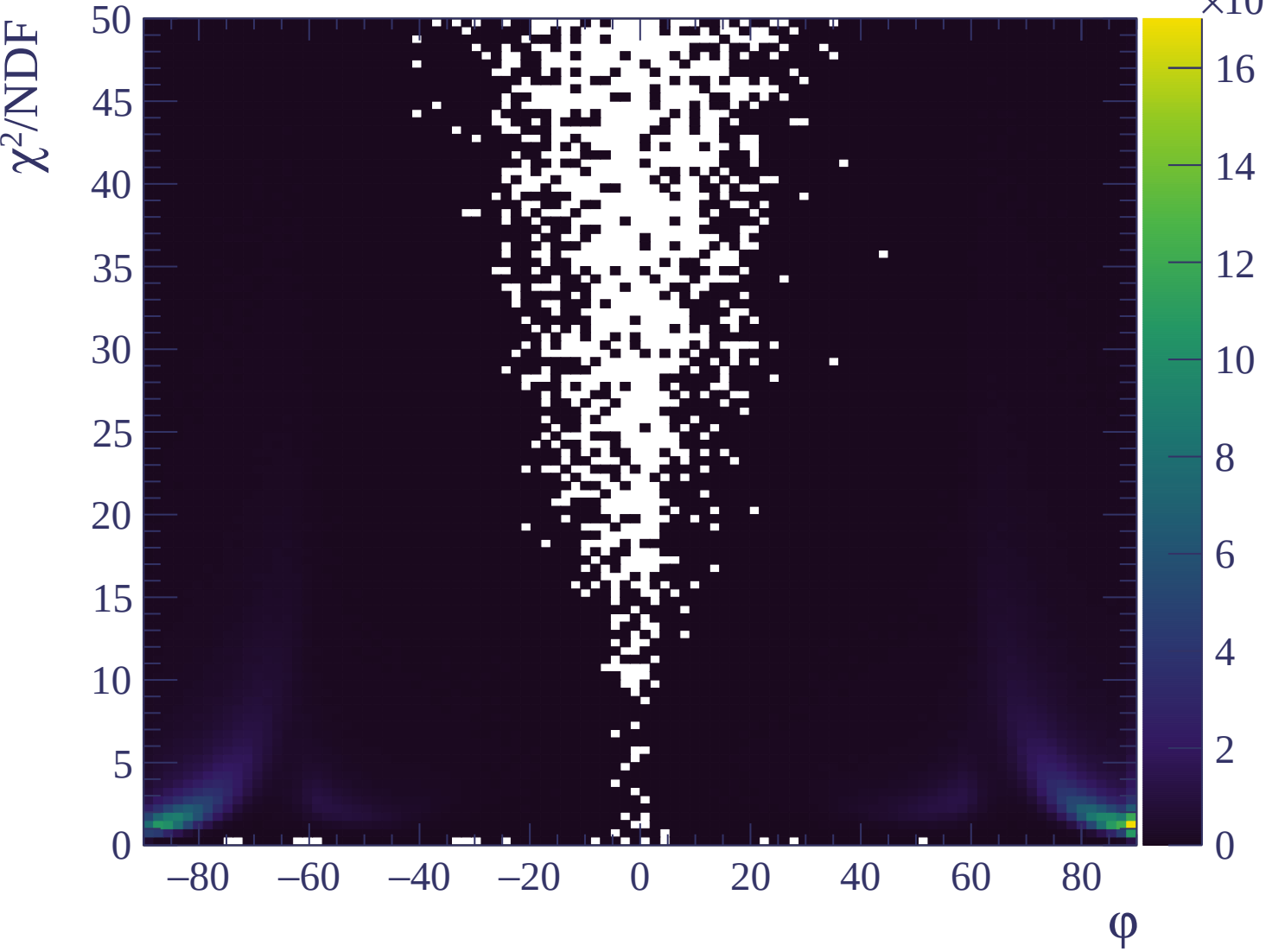


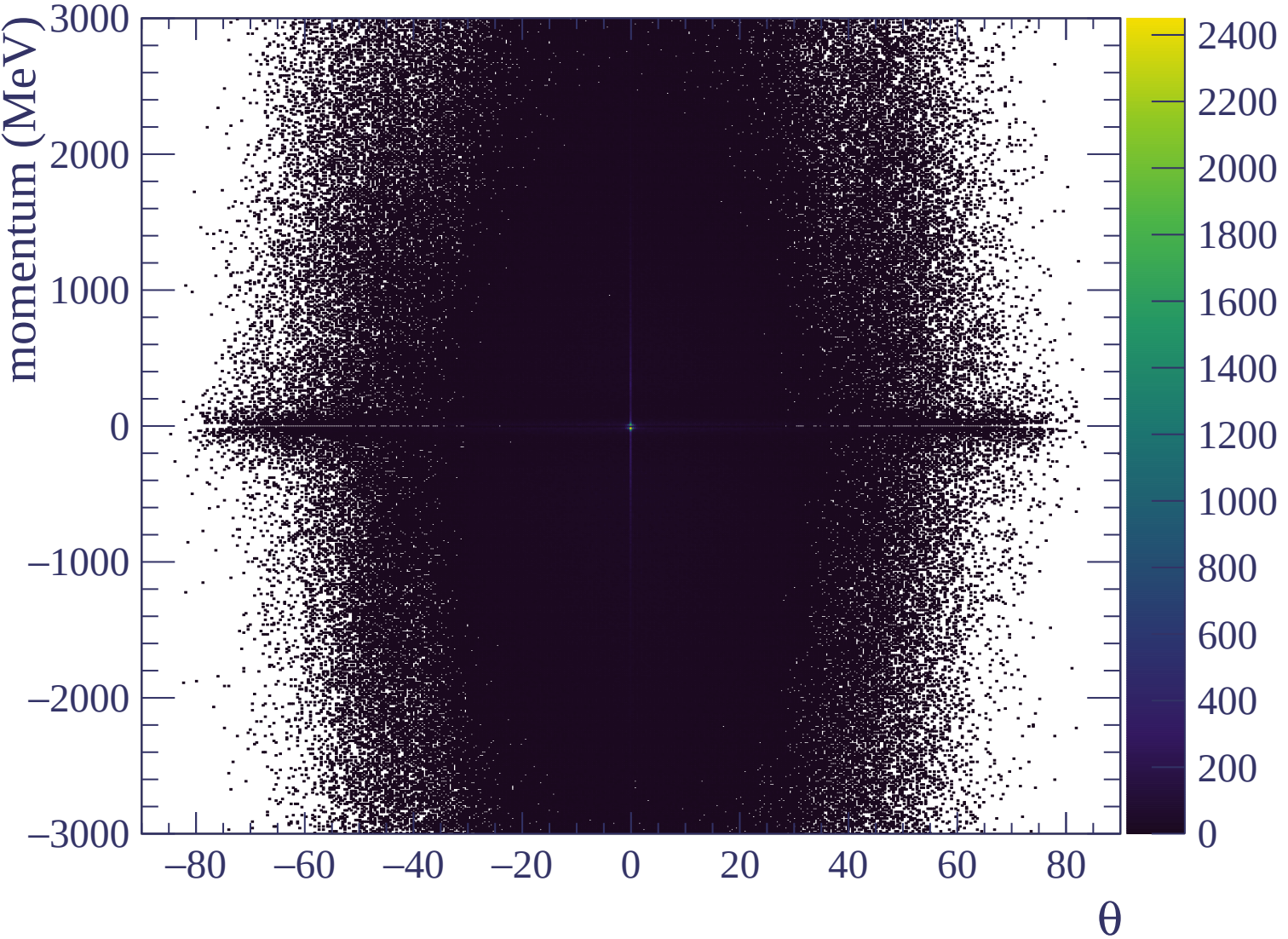


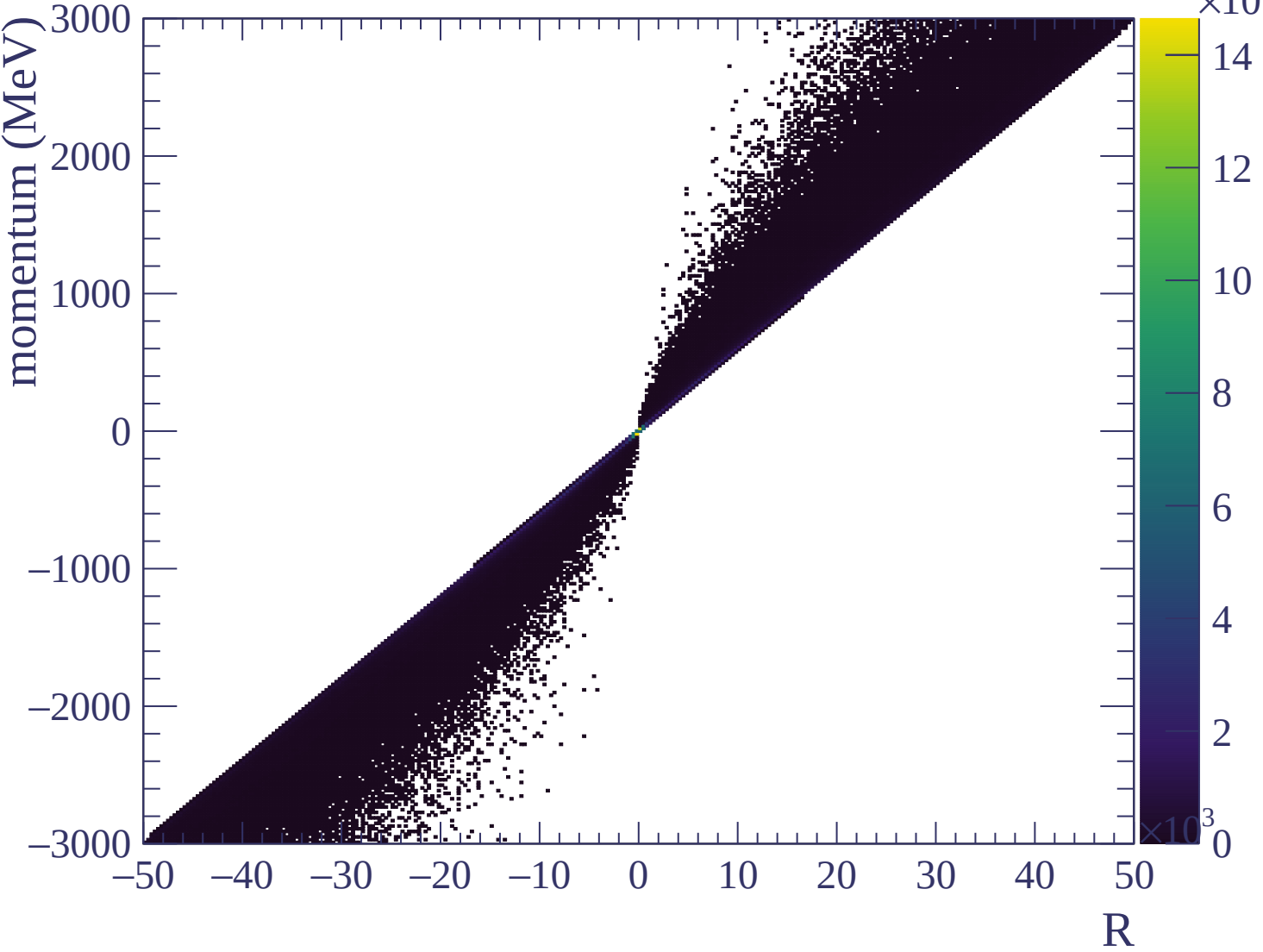


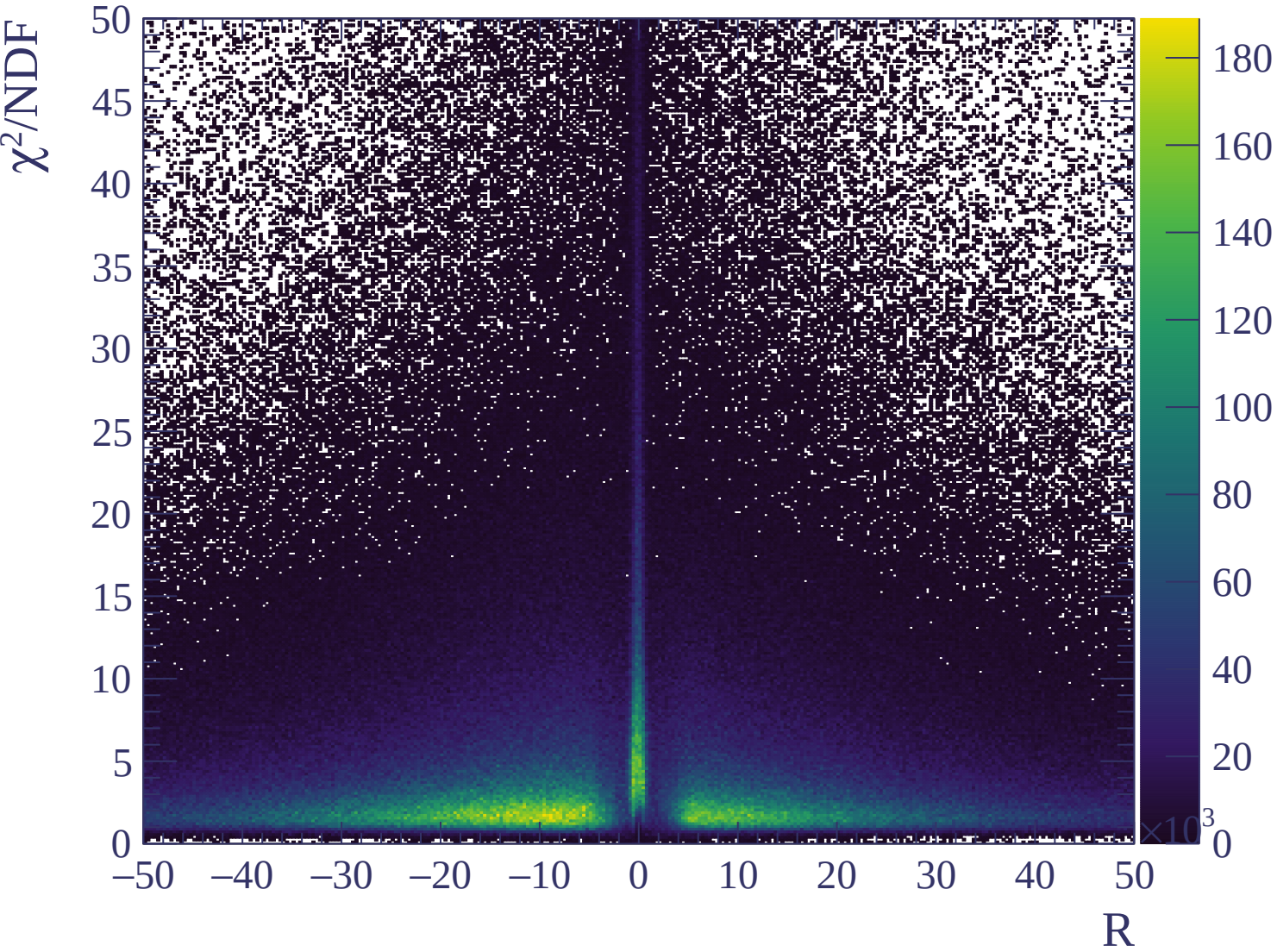


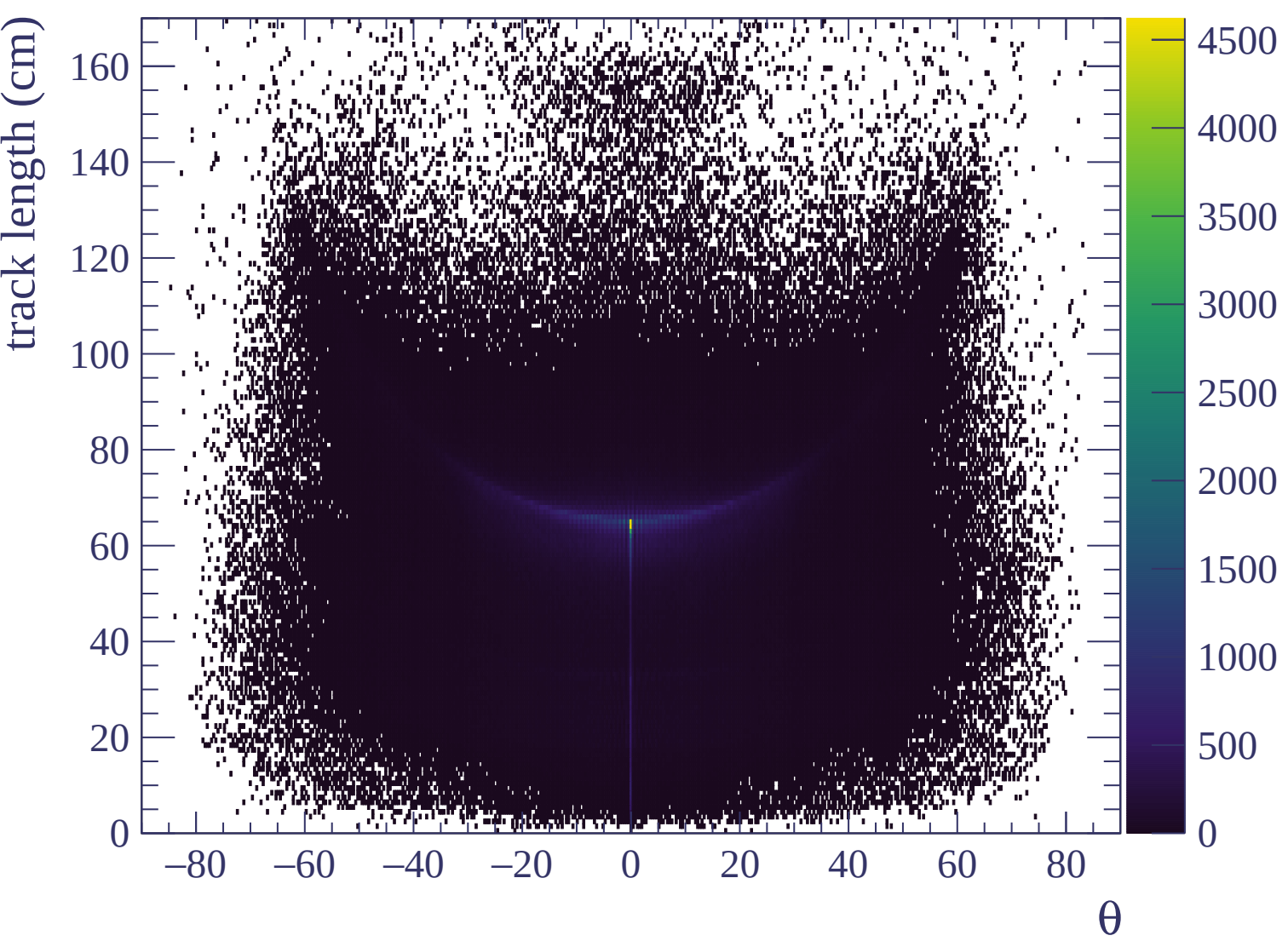


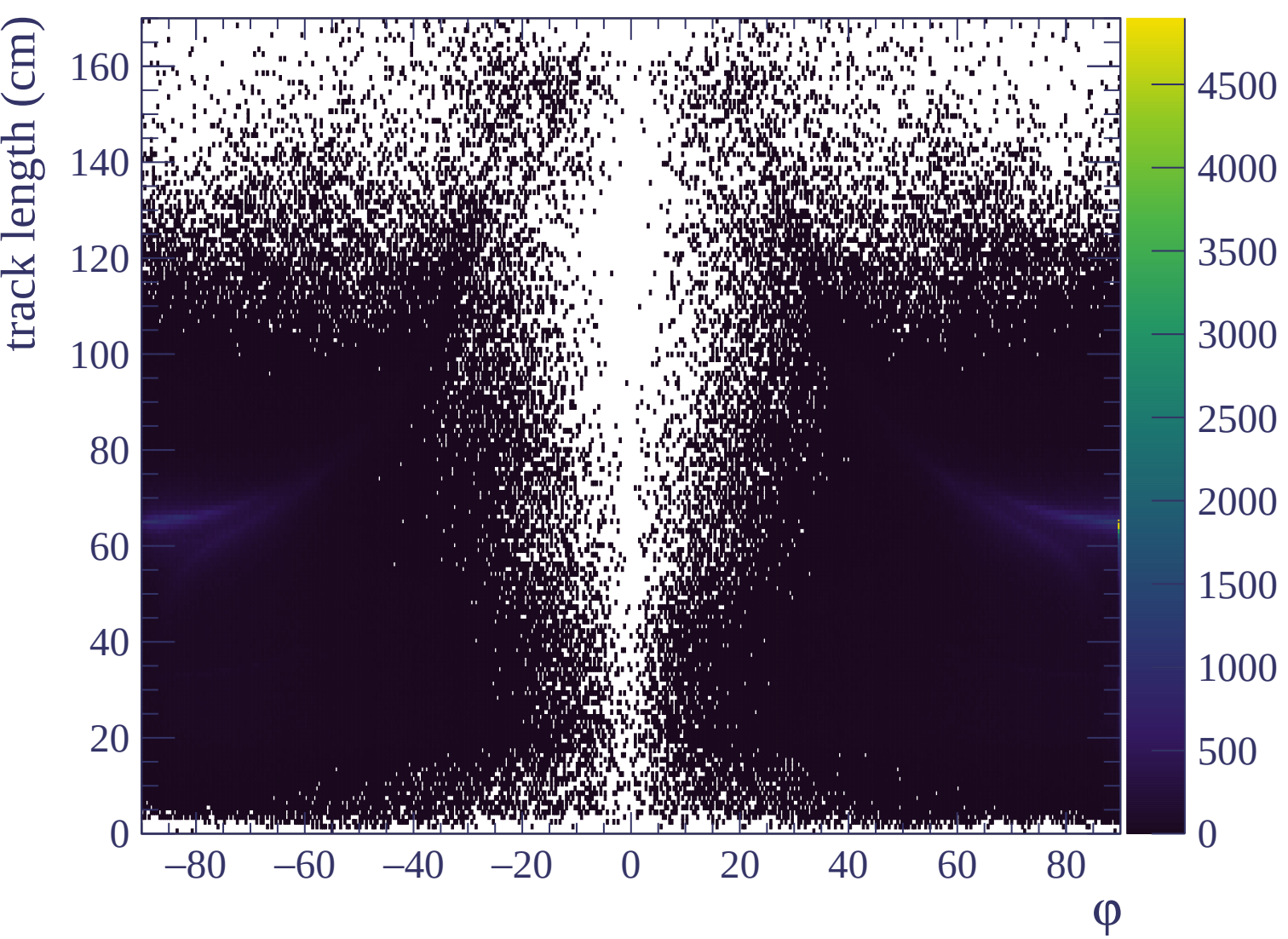






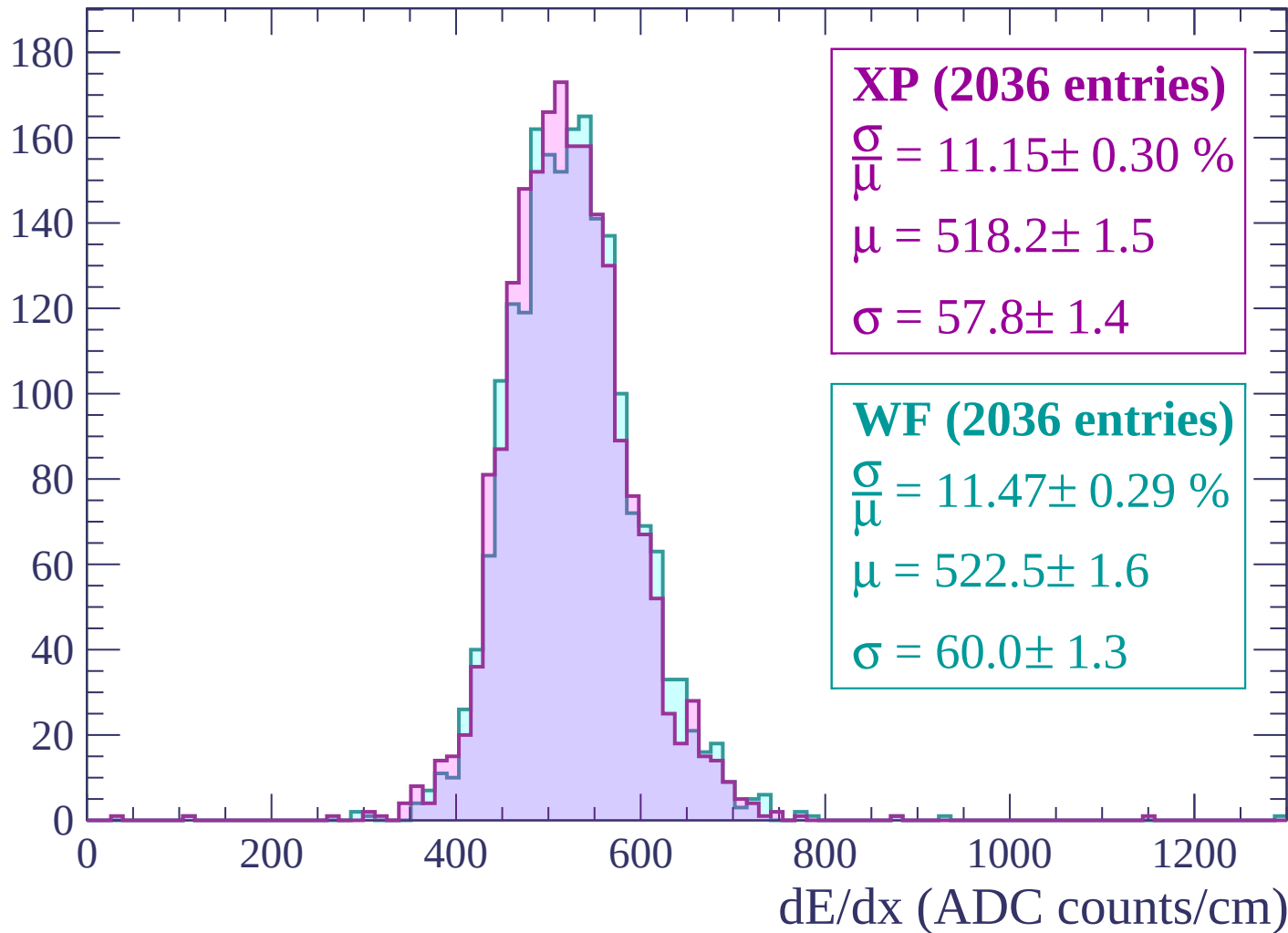






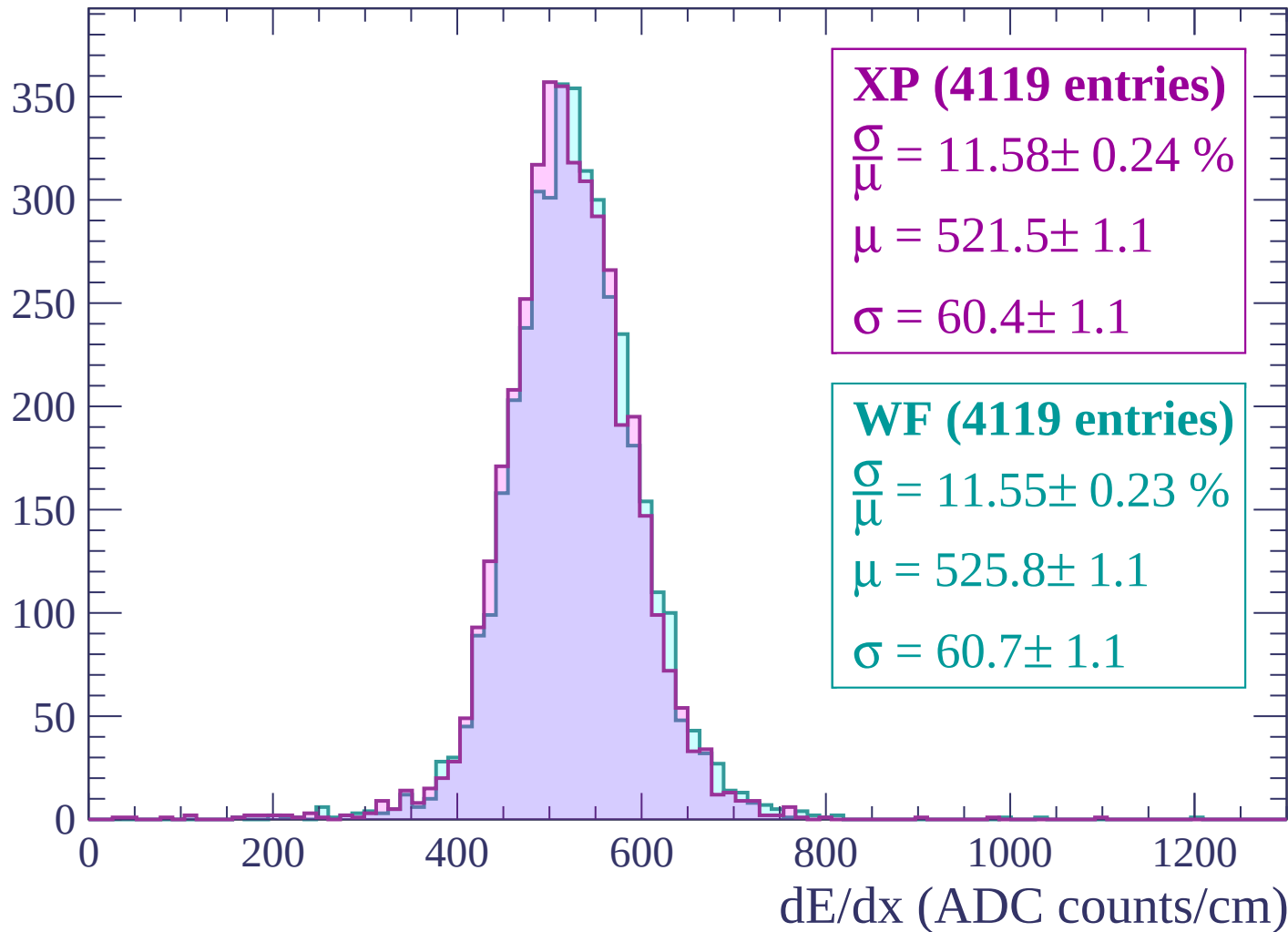
Energy loss | $-3000 < p < -2940$

Count



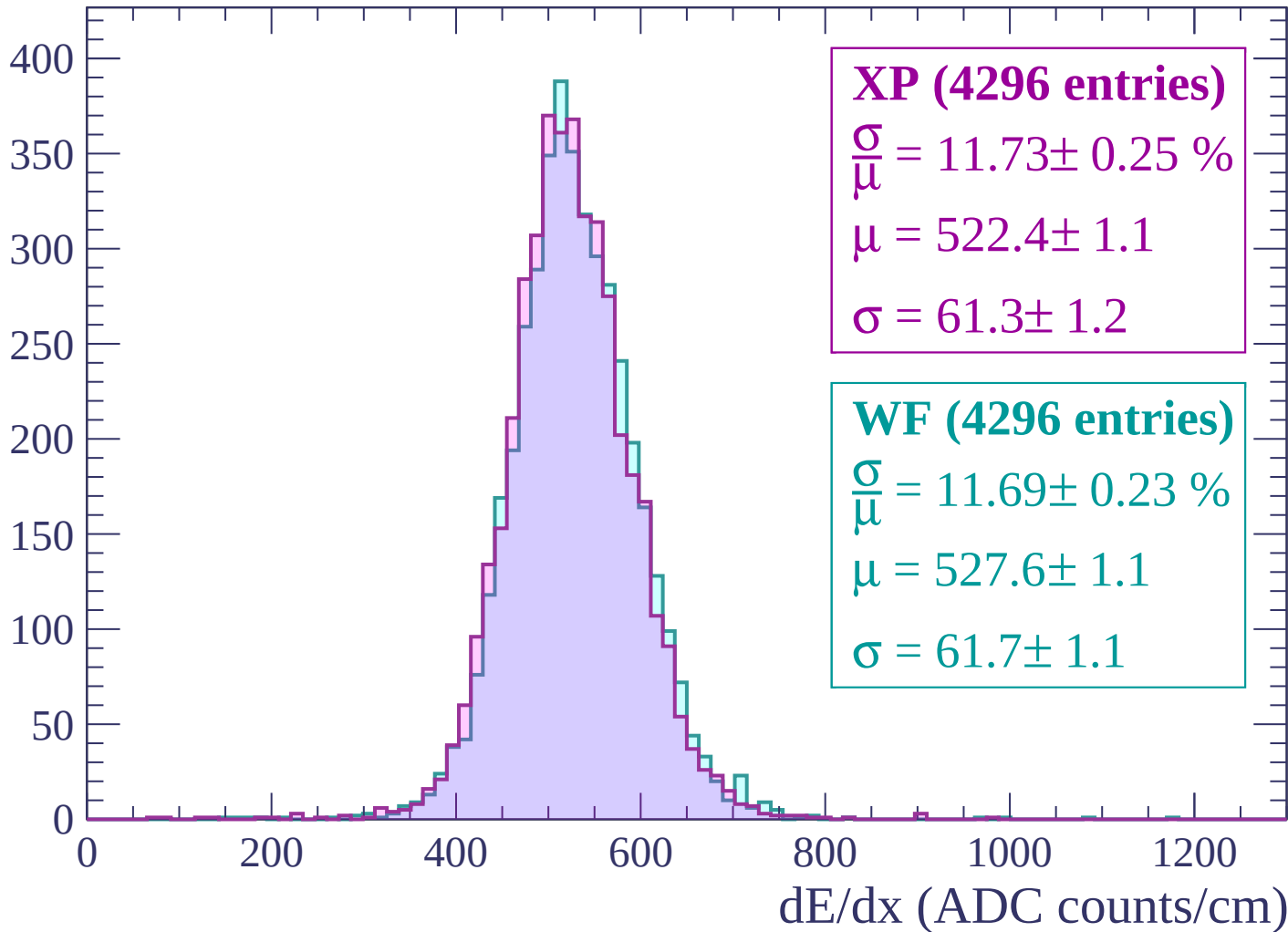
Energy loss | $-2940 < p < -2880$

Count



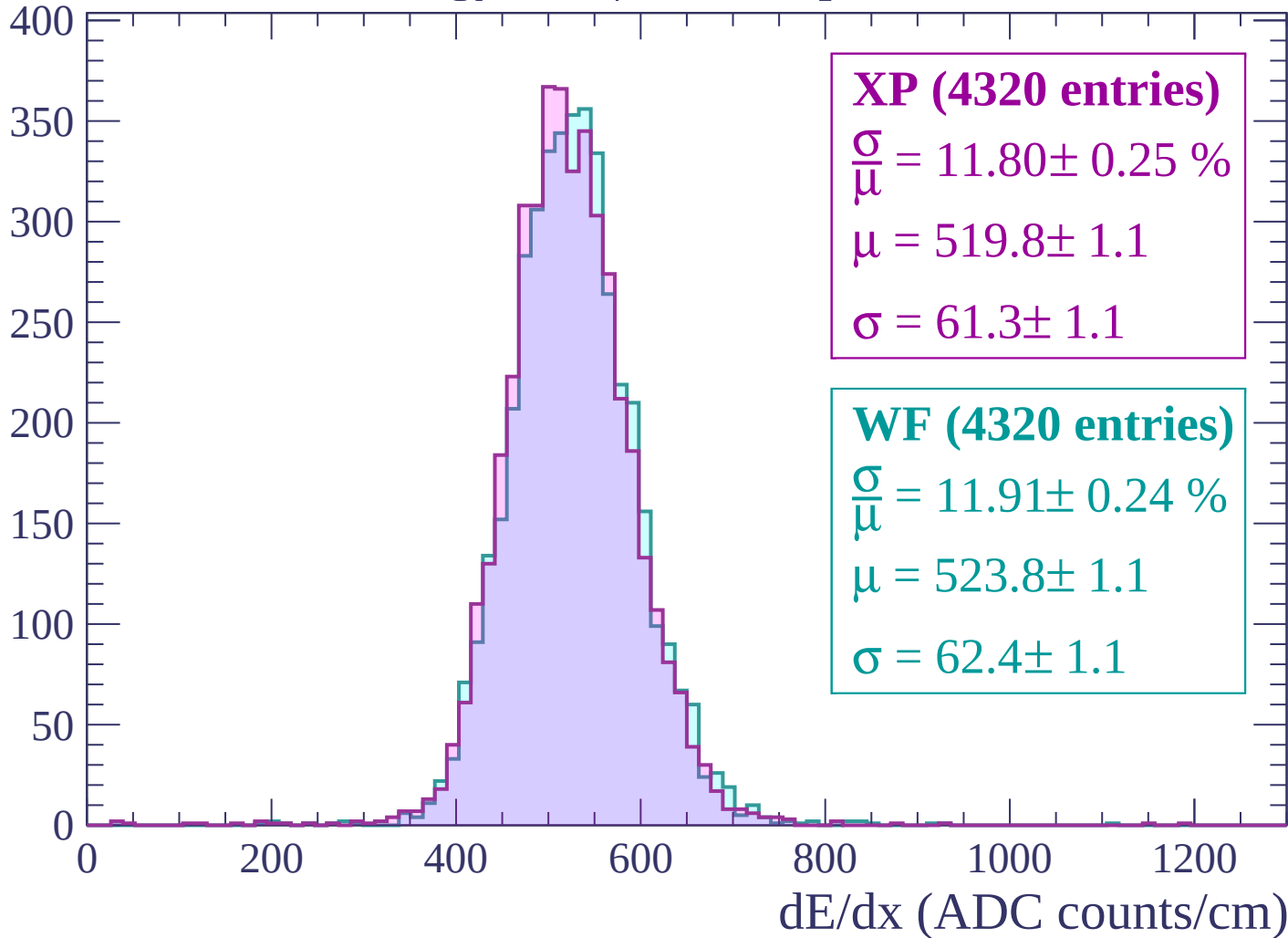
Energy loss | $-2880 < p < -2820$

Count



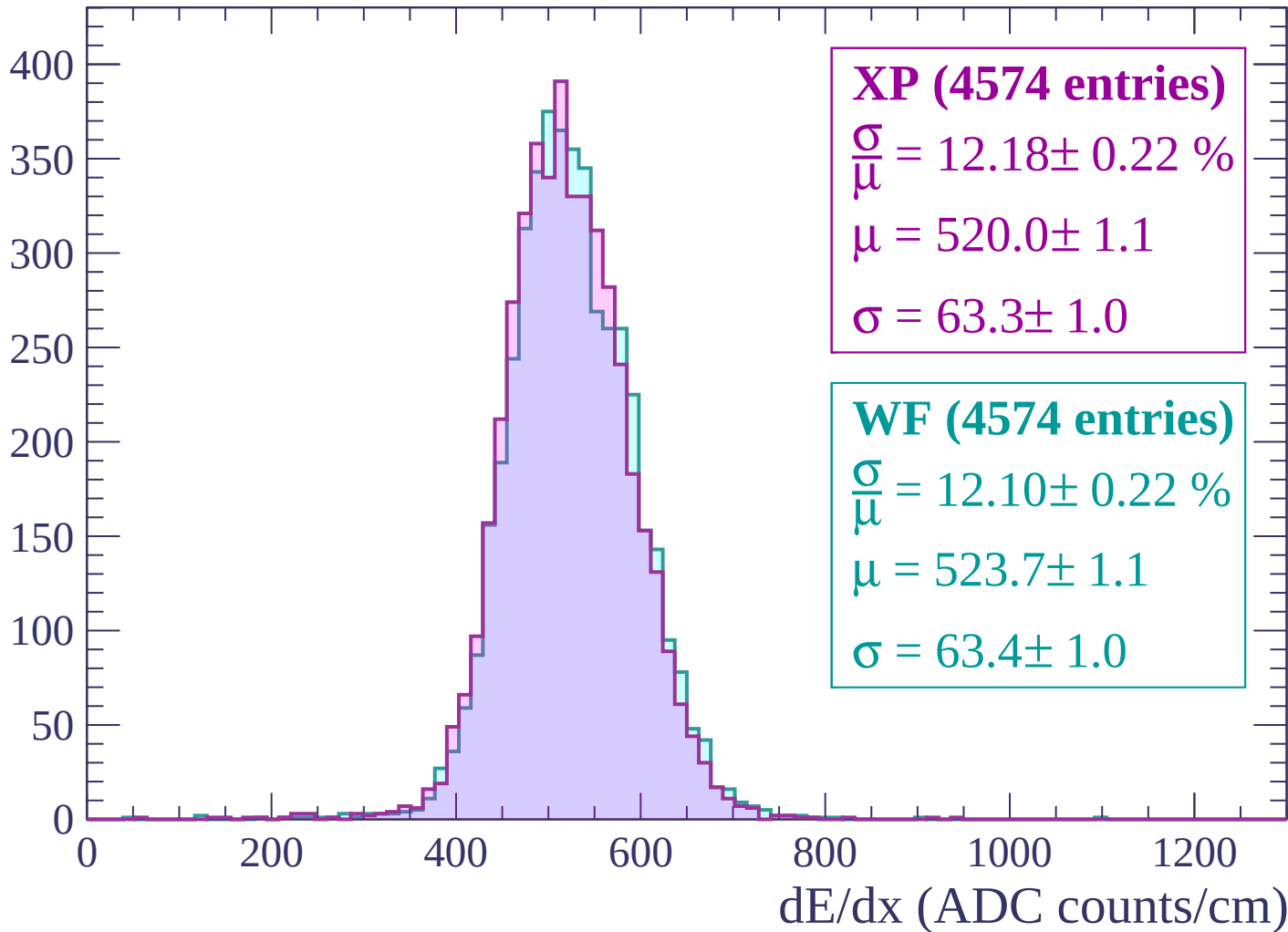
Energy loss | -2820 < p < -2760

Count



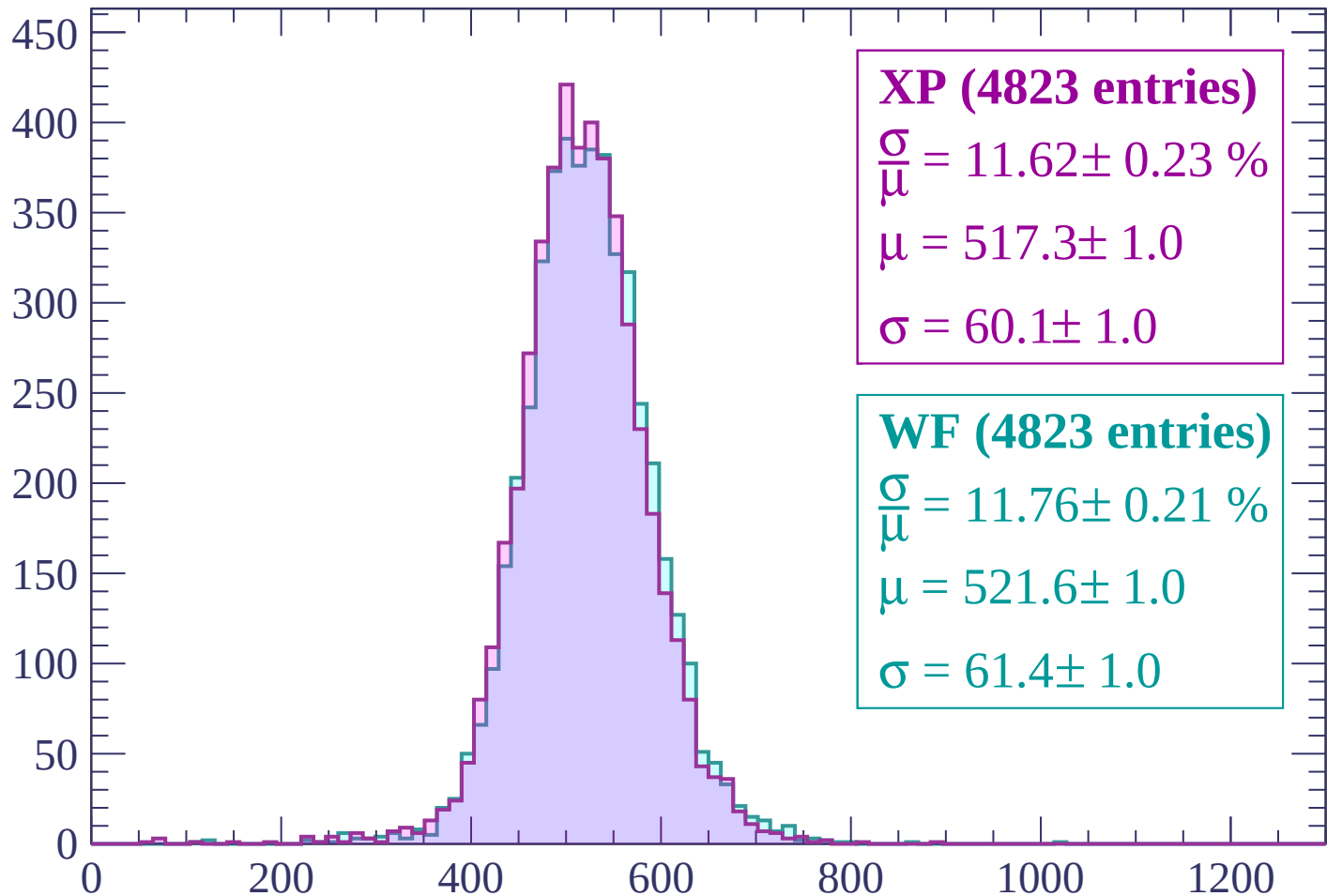
Energy loss | $-2760 < p < -2700$

Count



Energy loss | -2700 < p < -2640

Count



XP (4823 entries)

$$\frac{\sigma}{\mu} = 11.62 \pm 0.23 \%$$

$$\mu = 517.3 \pm 1.0$$

$$\sigma = 60.1 \pm 1.0$$

WF (4823 entries)

$$\frac{\sigma}{\mu} = 11.76 \pm 0.21 \%$$

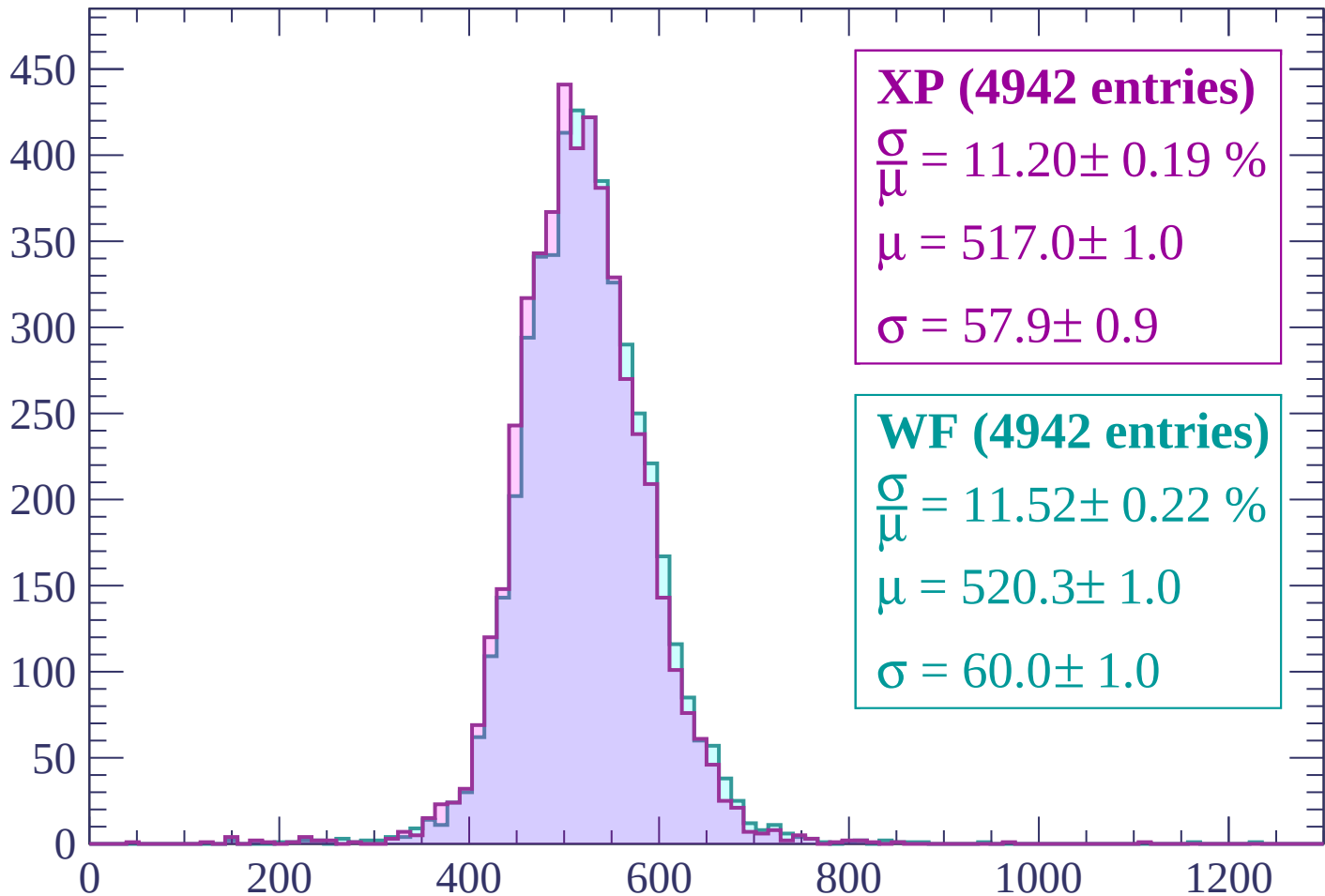
$$\mu = 521.6 \pm 1.0$$

$$\sigma = 61.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count



XP (4942 entries)

$\frac{\sigma}{\mu} = 11.20 \pm 0.19 \%$

$\mu = 517.0 \pm 1.0$

$\sigma = 57.9 \pm 0.9$

WF (4942 entries)

$\frac{\sigma}{\mu} = 11.52 \pm 0.22 \%$

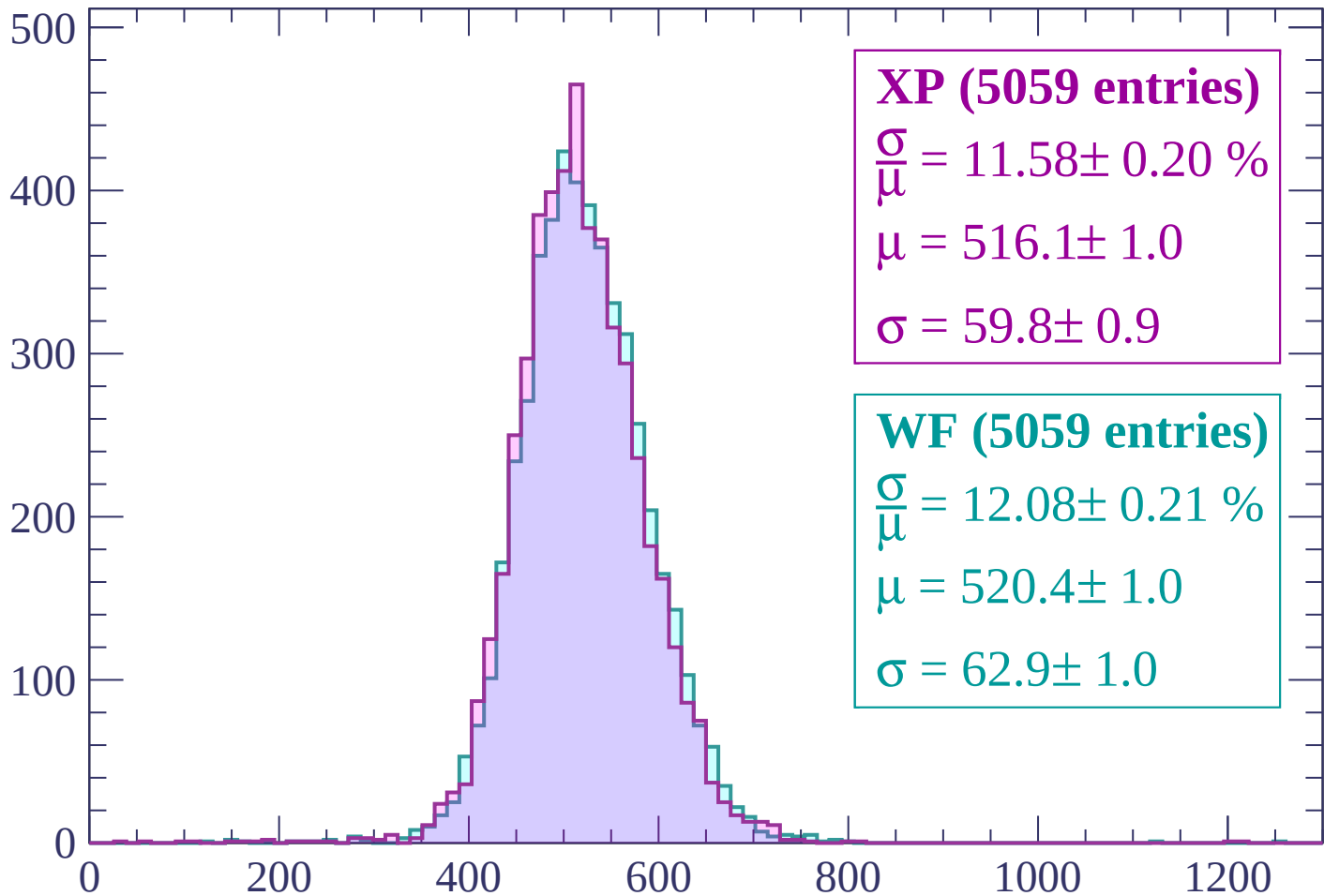
$\mu = 520.3 \pm 1.0$

$\sigma = 60.0 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $-2580 < p < -2520$

Count



XP (5059 entries)

$\frac{\sigma}{\mu} = 11.58 \pm 0.20 \%$

$\mu = 516.1 \pm 1.0$

$\sigma = 59.8 \pm 0.9$

WF (5059 entries)

$\frac{\sigma}{\mu} = 12.08 \pm 0.21 \%$

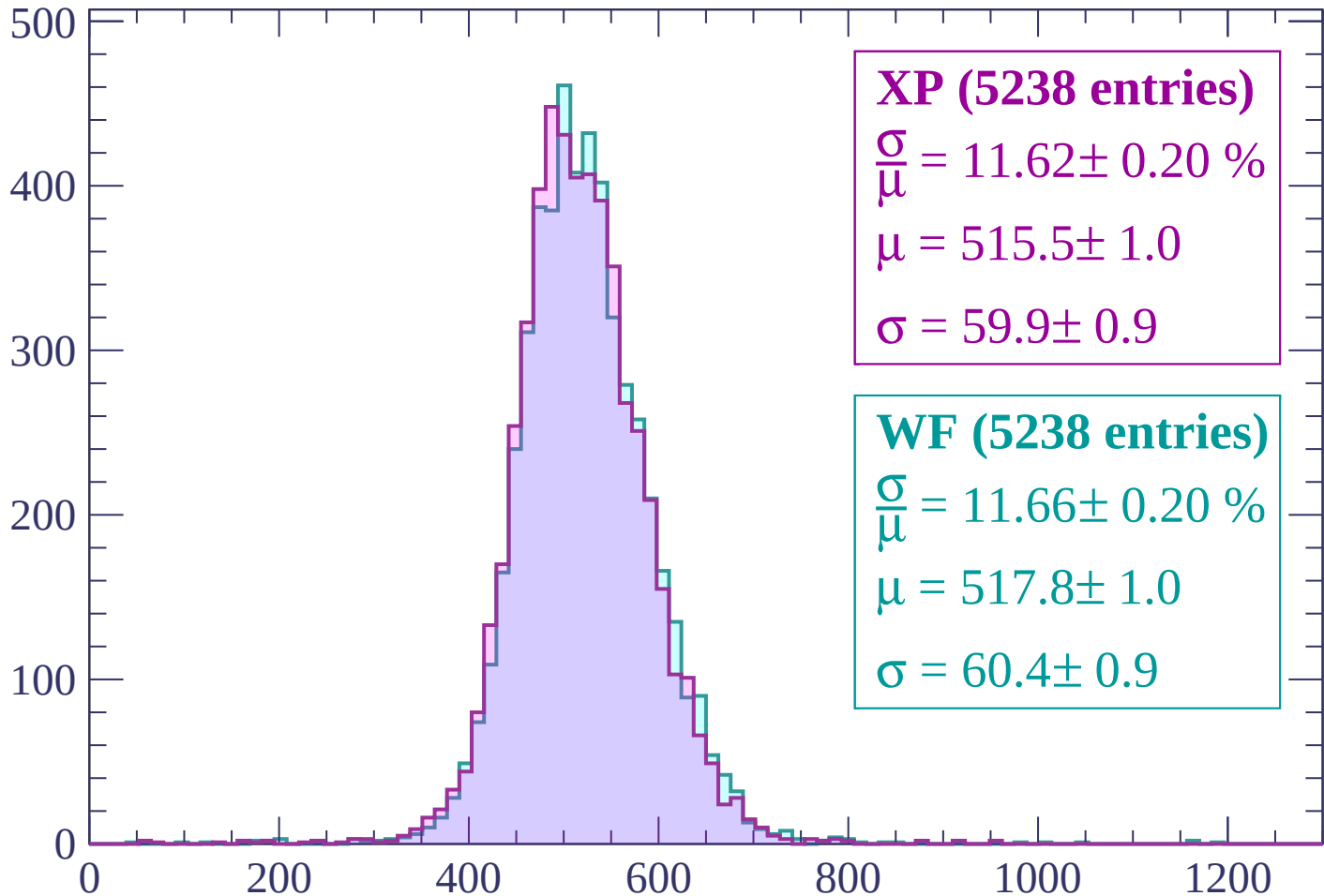
$\mu = 520.4 \pm 1.0$

$\sigma = 62.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -2520 < p < -2460

Count



XP (5238 entries)

$\frac{\sigma}{\mu} = 11.62 \pm 0.20 \%$

$\mu = 515.5 \pm 1.0$

$\sigma = 59.9 \pm 0.9$

WF (5238 entries)

$\frac{\sigma}{\mu} = 11.66 \pm 0.20 \%$

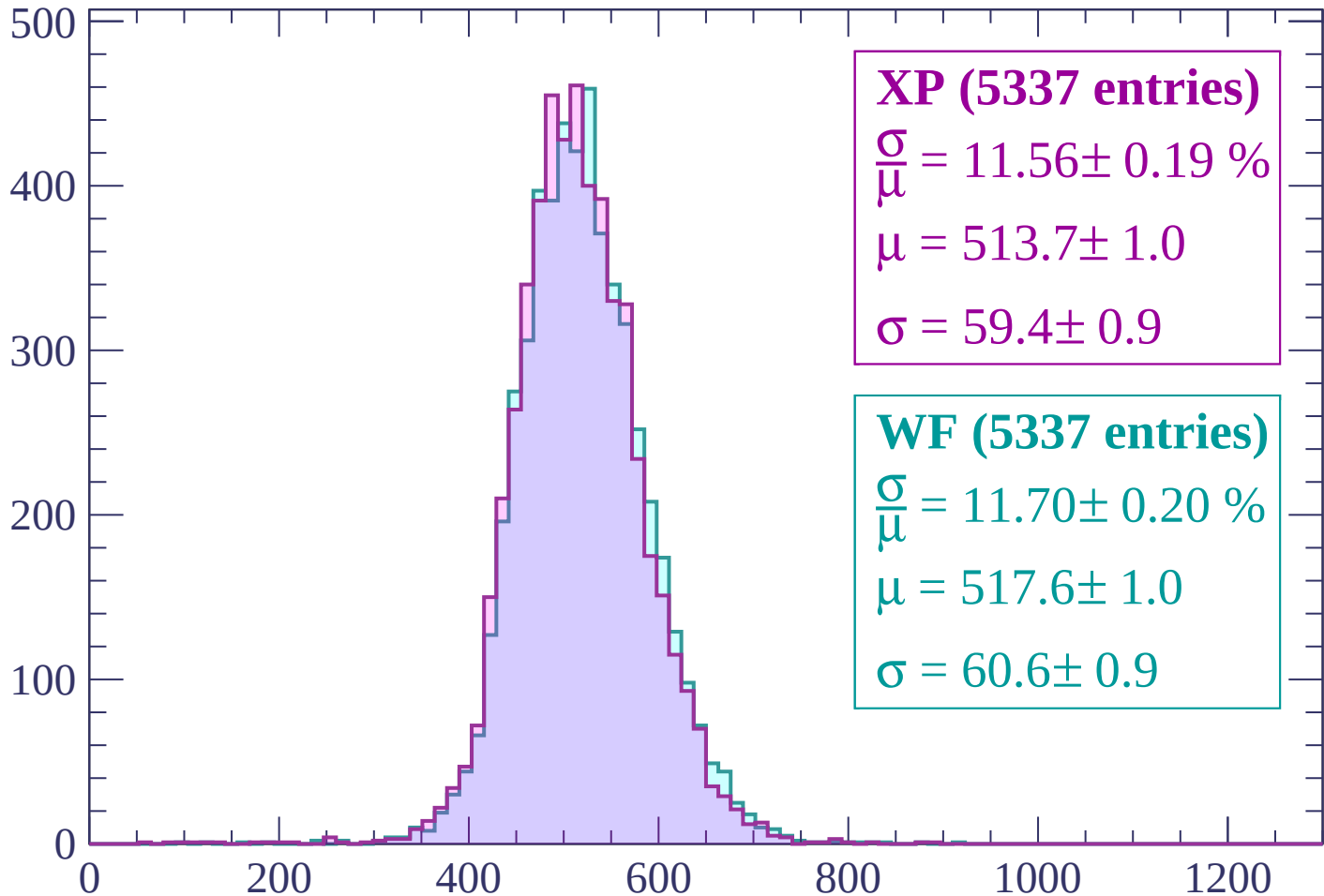
$\mu = 517.8 \pm 1.0$

$\sigma = 60.4 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-2460 < p < -2400$

Count



XP (5337 entries)

$$\frac{\sigma}{\mu} = 11.56 \pm 0.19 \%$$

$$\mu = 513.7 \pm 1.0$$

$$\sigma = 59.4 \pm 0.9$$

WF (5337 entries)

$$\frac{\sigma}{\mu} = 11.70 \pm 0.20 \%$$

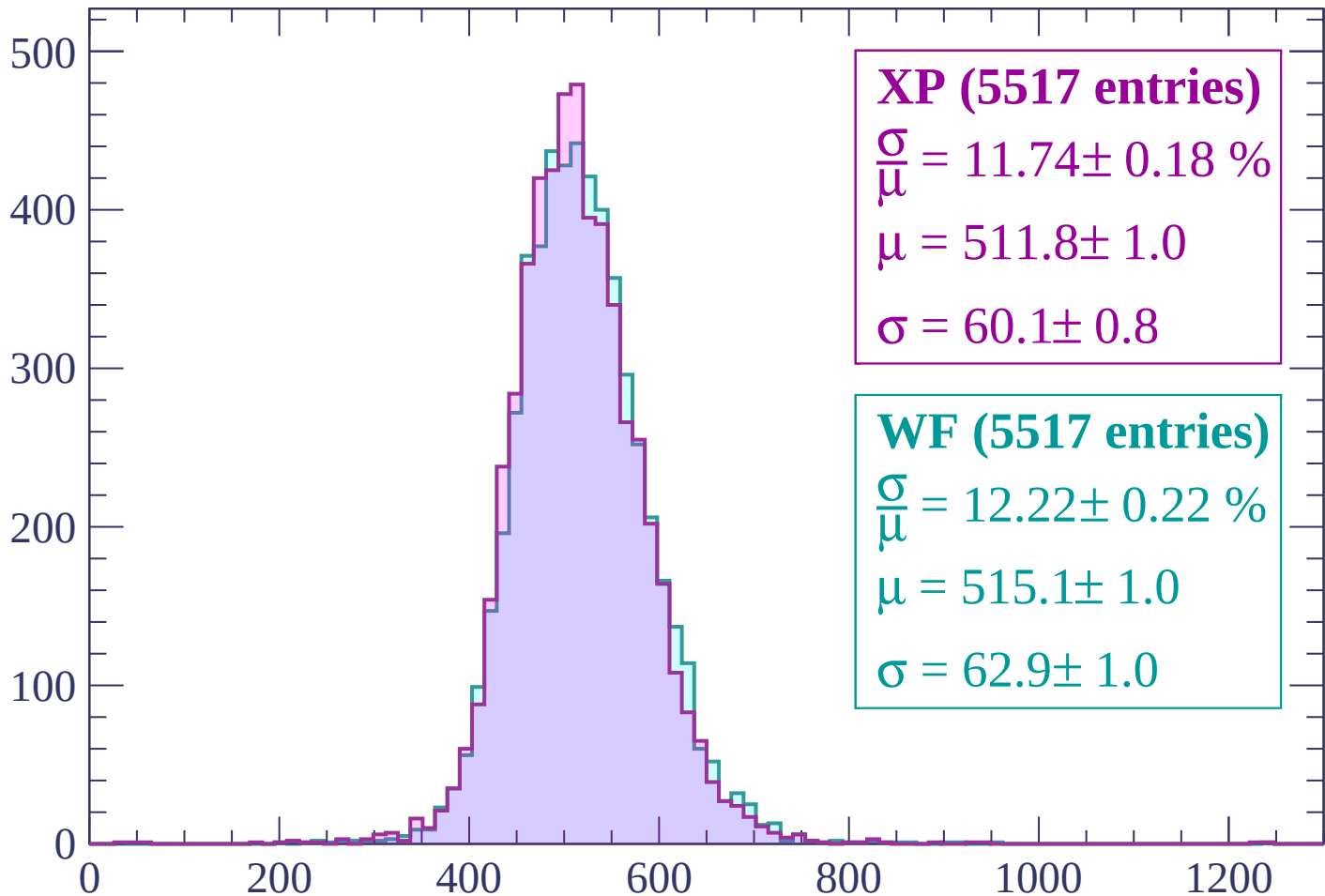
$$\mu = 517.6 \pm 1.0$$

$$\sigma = 60.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -2400 < p < -2340

Count



XP (5517 entries)

$$\frac{\sigma}{\mu} = 11.74 \pm 0.18 \%$$

$$\mu = 511.8 \pm 1.0$$

$$\sigma = 60.1 \pm 0.8$$

WF (5517 entries)

$$\frac{\sigma}{\mu} = 12.22 \pm 0.22 \%$$

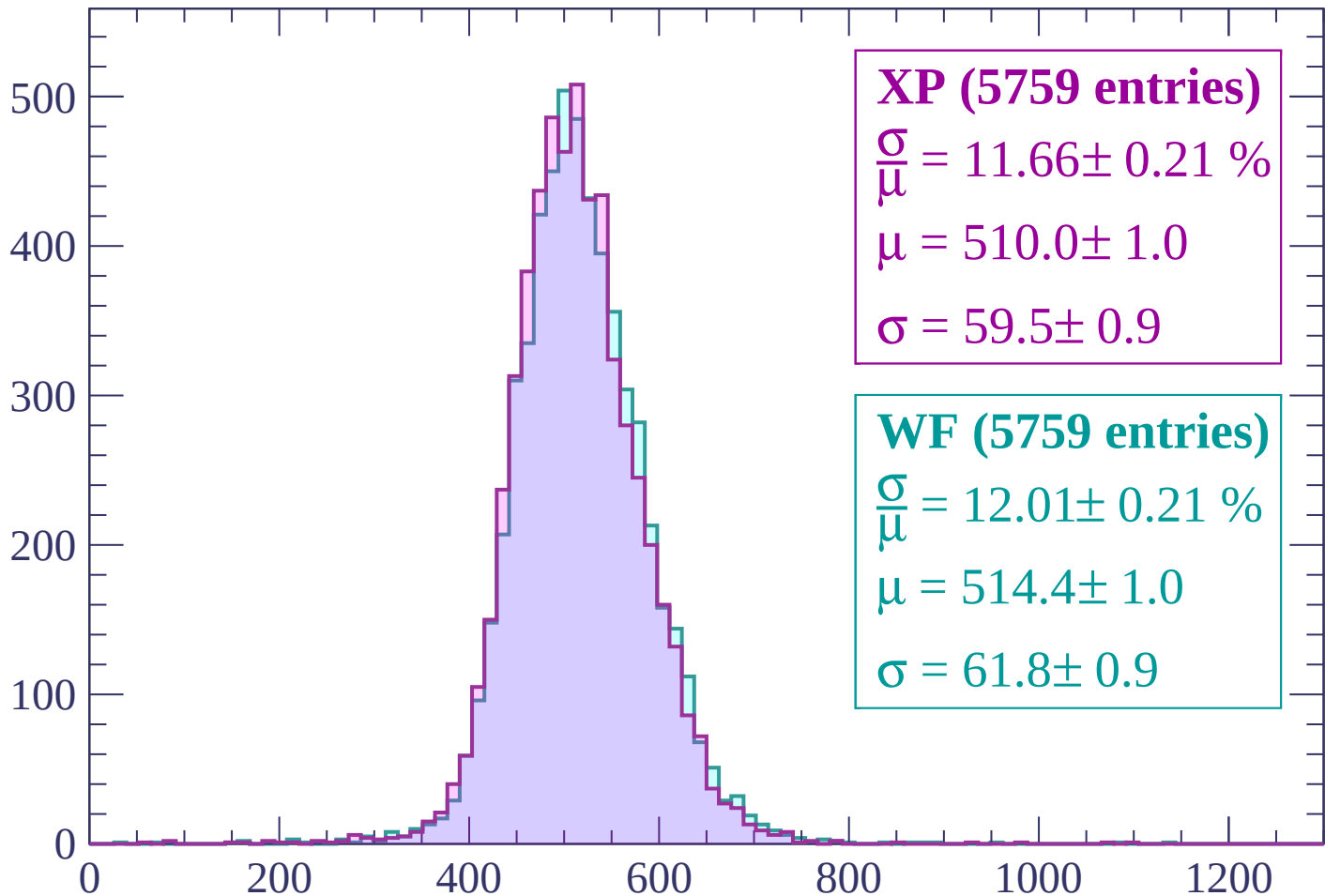
$$\mu = 515.1 \pm 1.0$$

$$\sigma = 62.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-2340 < p < -2280$

Count



XP (5759 entries)

$\frac{\sigma}{\mu} = 11.66 \pm 0.21 \%$

$\mu = 510.0 \pm 1.0$

$\sigma = 59.5 \pm 0.9$

WF (5759 entries)

$\frac{\sigma}{\mu} = 12.01 \pm 0.21 \%$

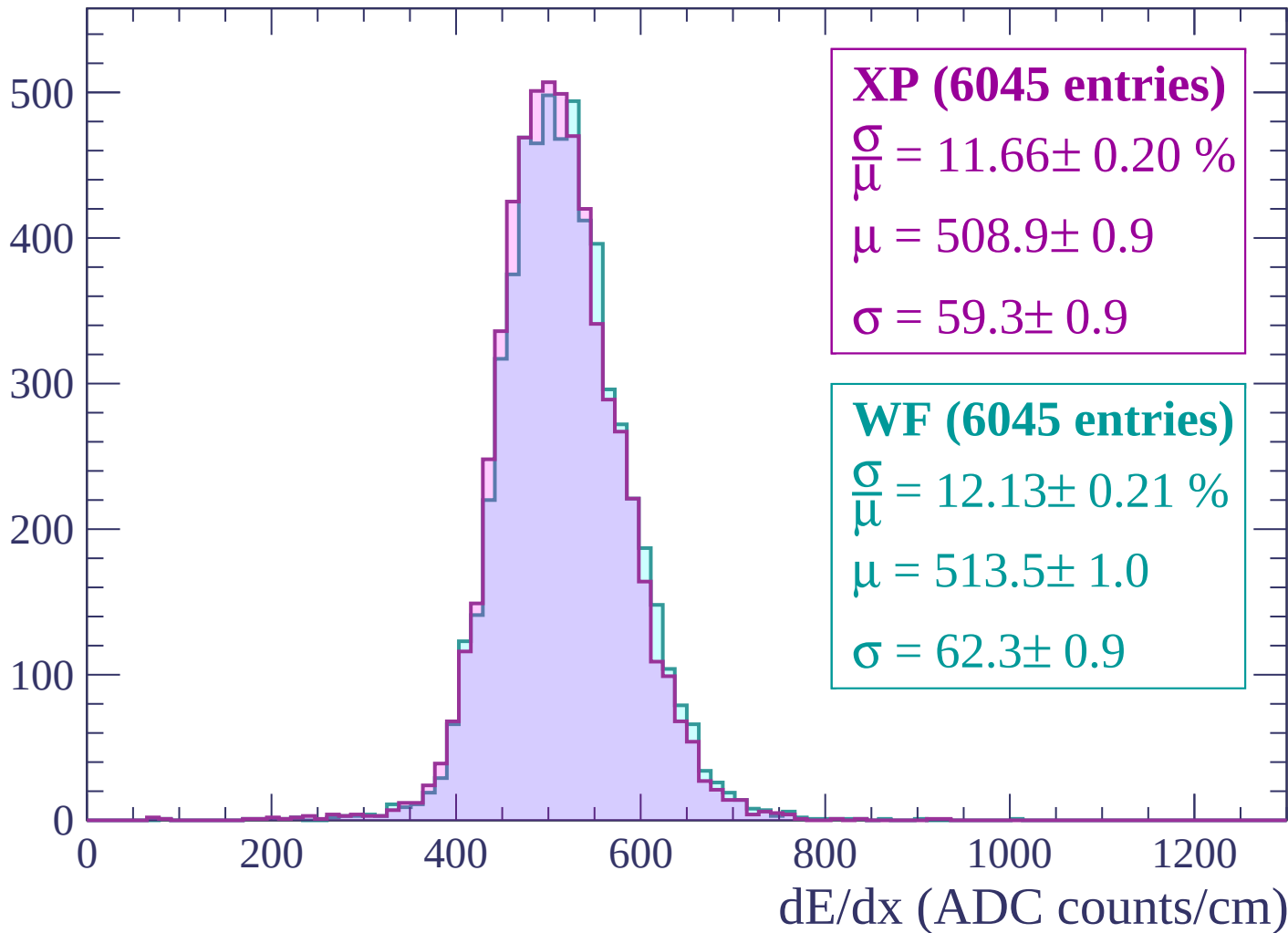
$\mu = 514.4 \pm 1.0$

$\sigma = 61.8 \pm 0.9$

dE/dx (ADC counts/cm)

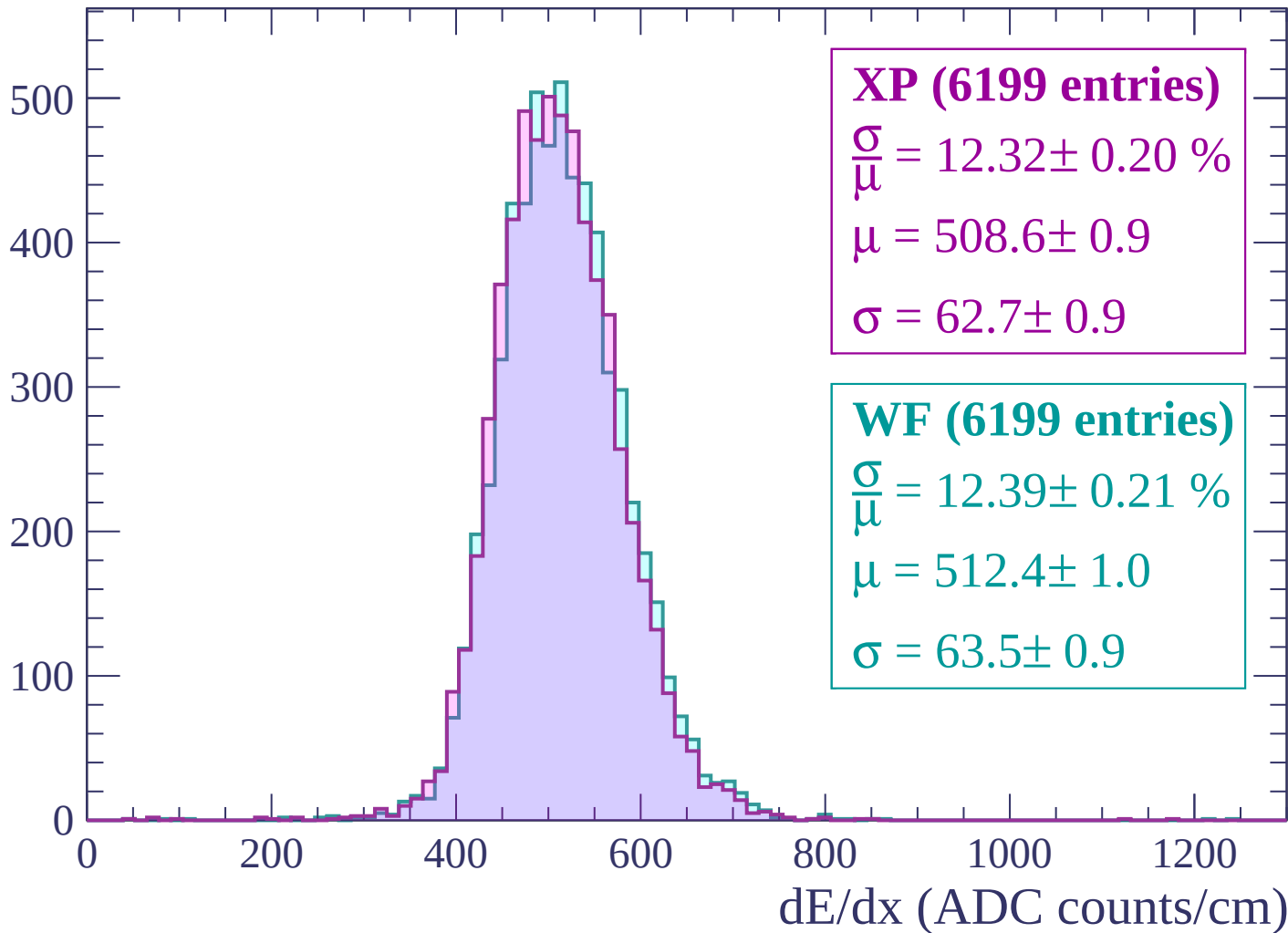
Energy loss | $-2280 < p < -2220$

Count



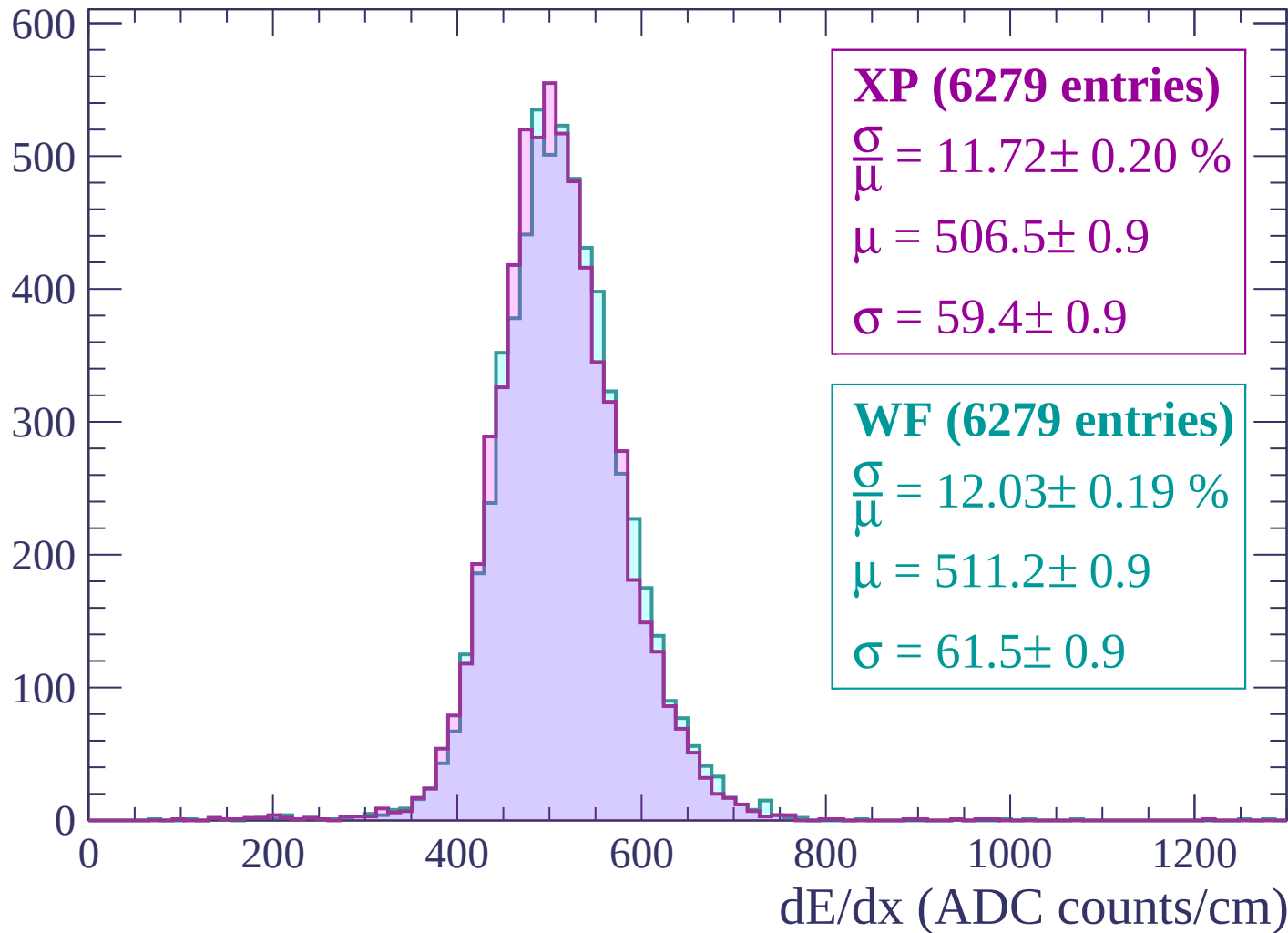
Energy loss | -2220 < p < -2160

Count



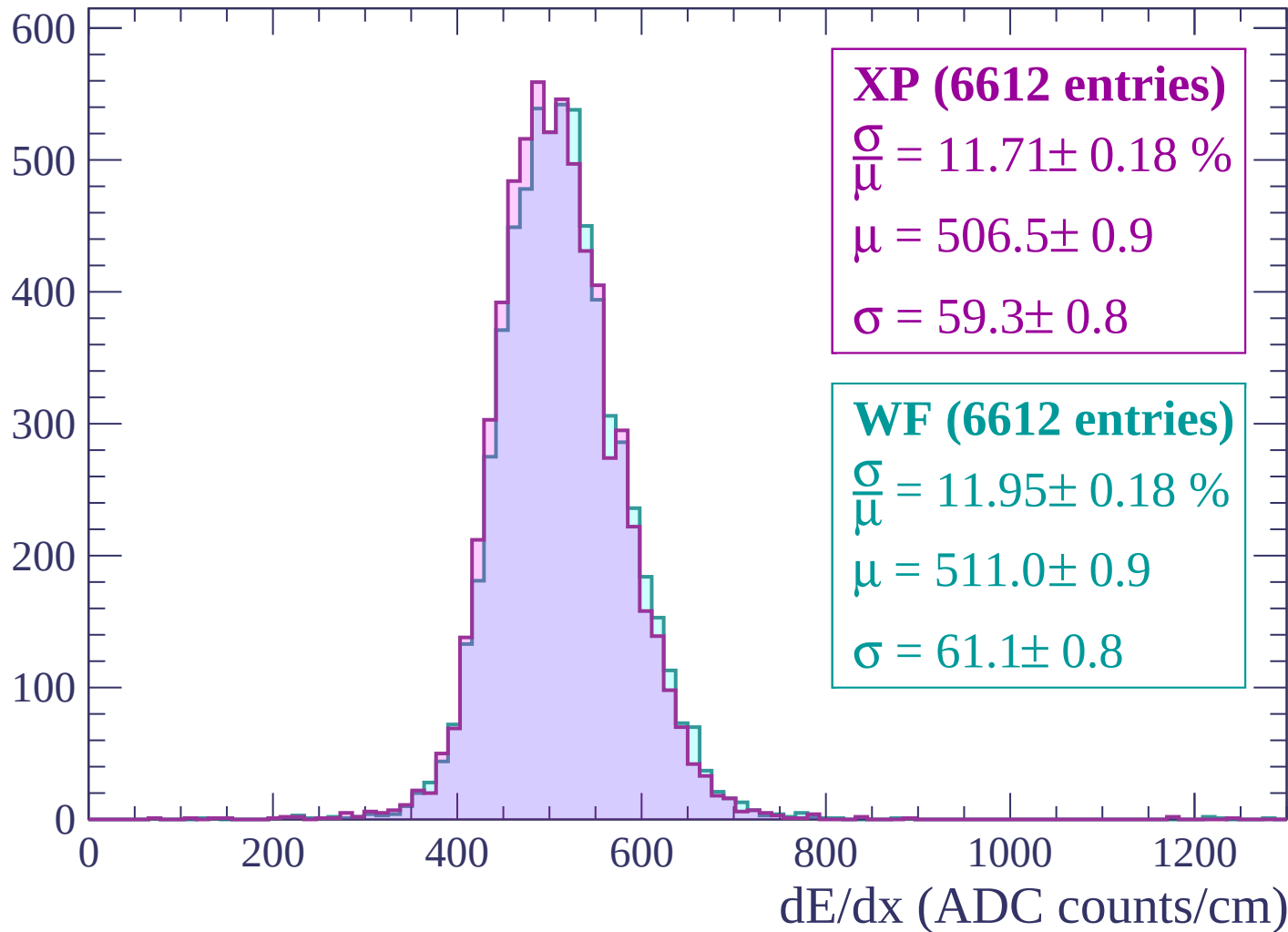
Energy loss | -2160 < p < -2100

Count



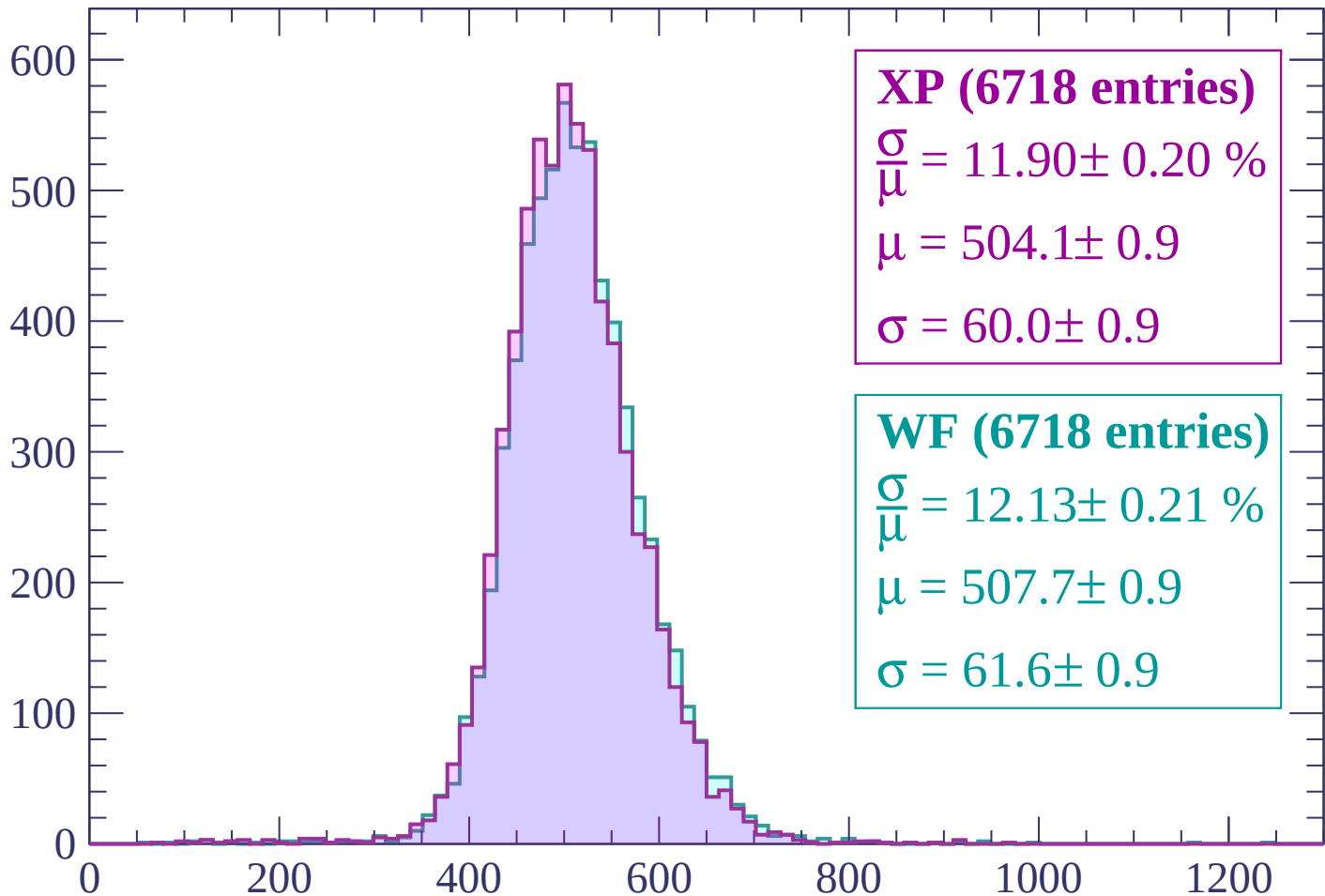
Energy loss | -2100 < p < -2040

Count



Energy loss | -2040 < p < -1980

Count



XP (6718 entries)

$$\frac{\sigma}{\mu} = 11.90 \pm 0.20 \%$$

$$\mu = 504.1 \pm 0.9$$

$$\sigma = 60.0 \pm 0.9$$

WF (6718 entries)

$$\frac{\sigma}{\mu} = 12.13 \pm 0.21 \%$$

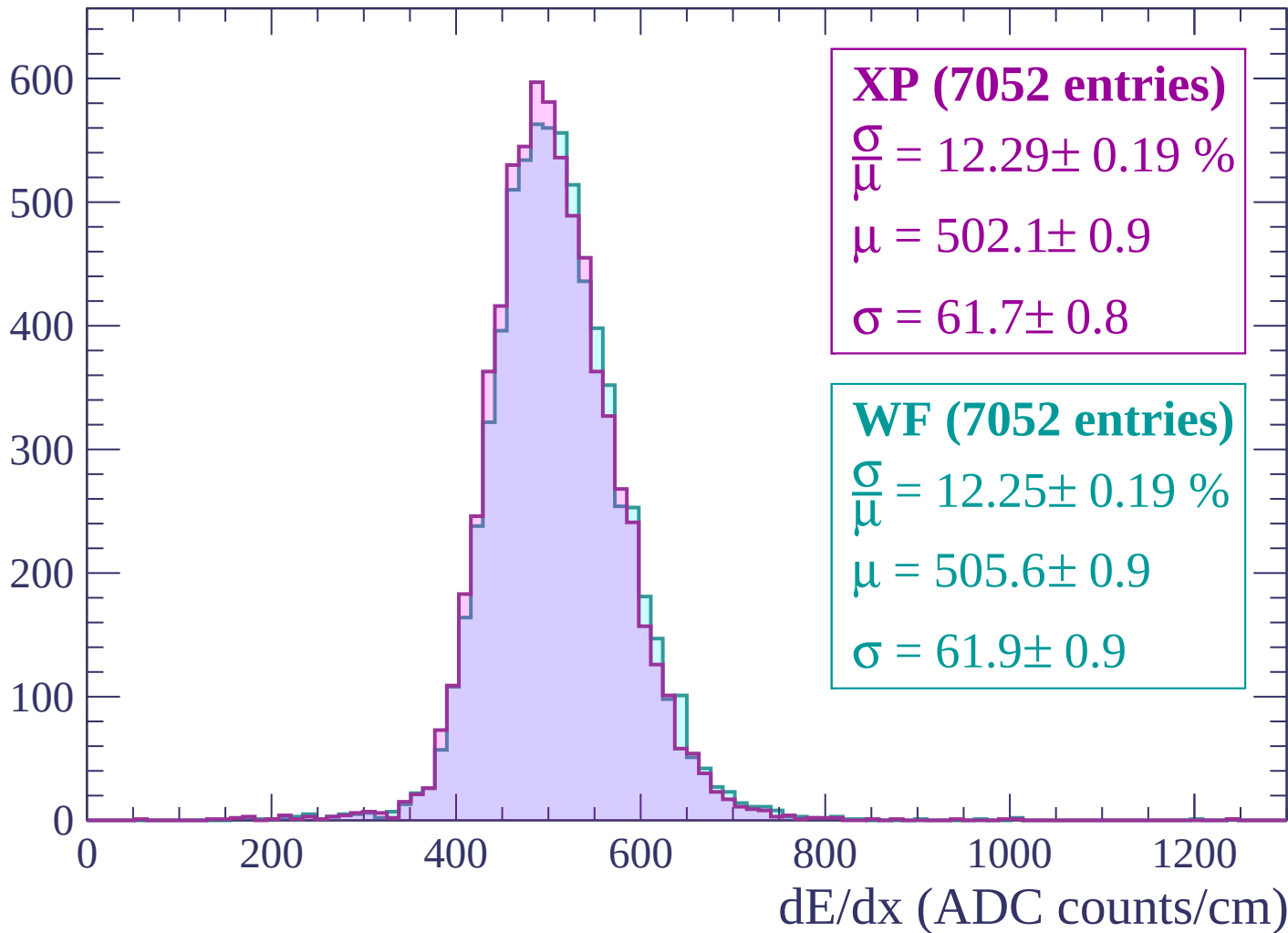
$$\mu = 507.7 \pm 0.9$$

$$\sigma = 61.6 \pm 0.9$$

dE/dx (ADC counts/cm)

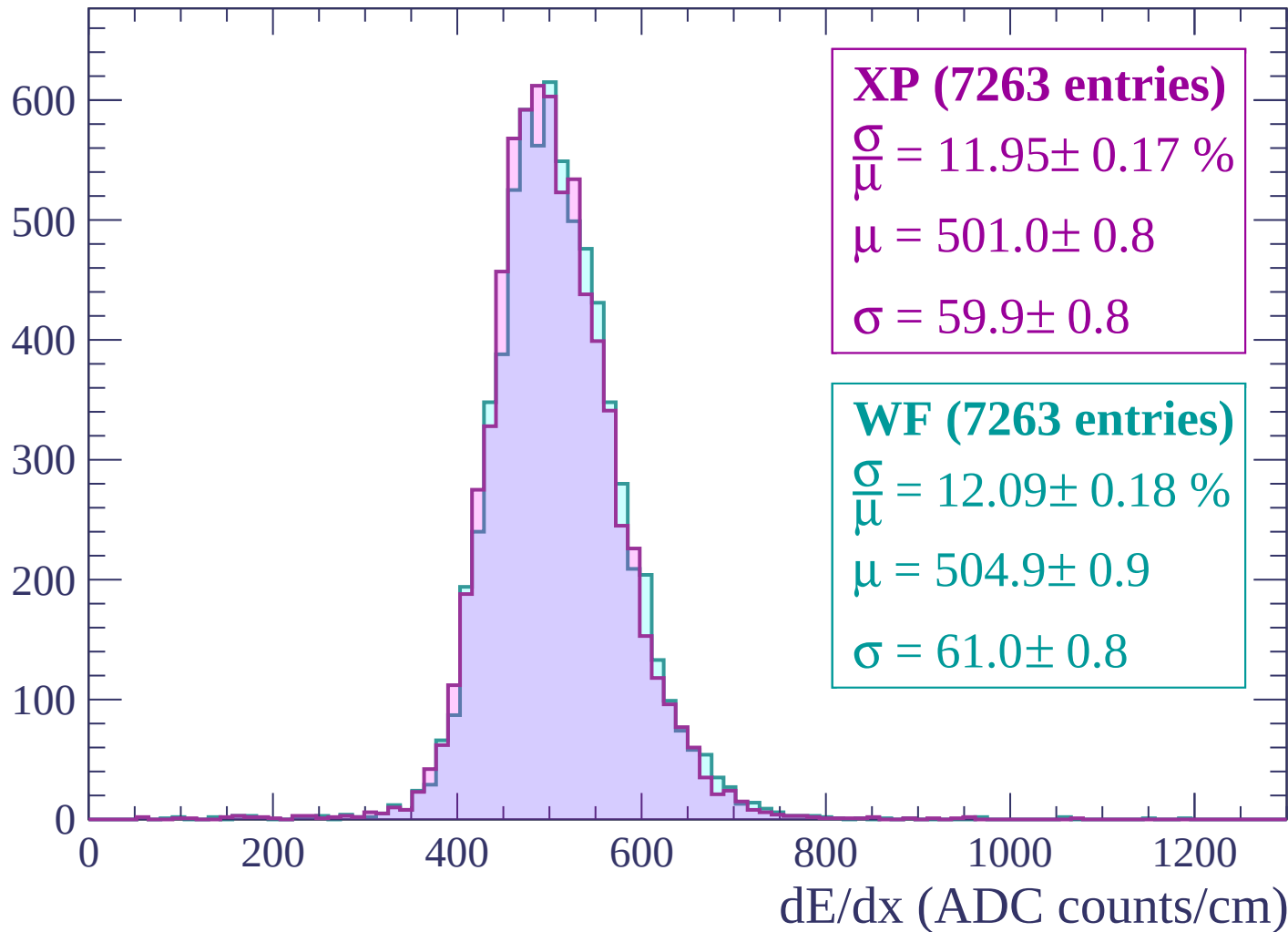
Energy loss | $-1980 < p < -1920$

Count



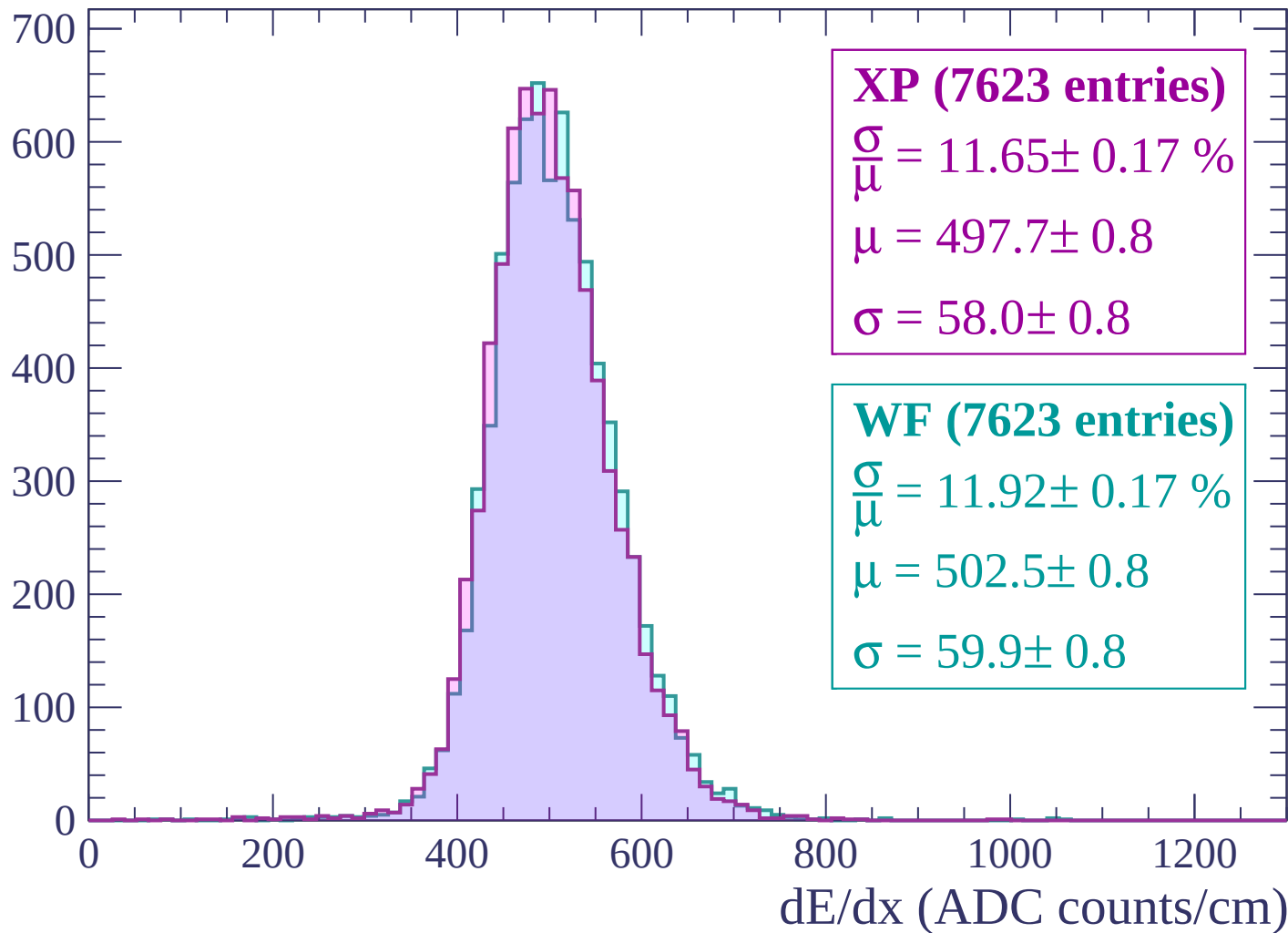
Energy loss | -1920 < p < -1860

Count



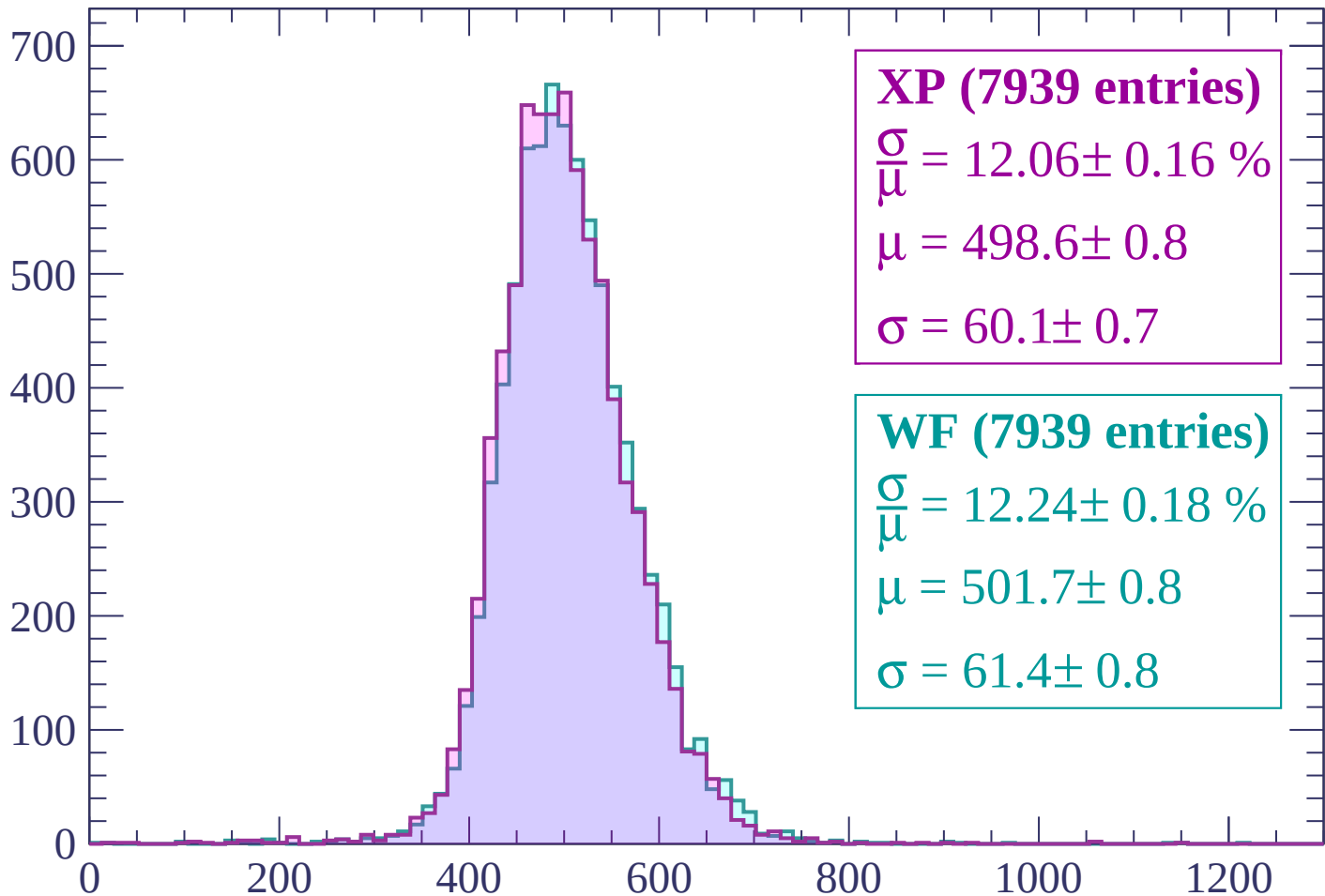
Energy loss | -1860 < p < -1800

Count



Energy loss | -1800 < p < -1740

Count



XP (7939 entries)

$$\frac{\sigma}{\mu} = 12.06 \pm 0.16 \%$$

$$\mu = 498.6 \pm 0.8$$

$$\sigma = 60.1 \pm 0.7$$

WF (7939 entries)

$$\frac{\sigma}{\mu} = 12.24 \pm 0.18 \%$$

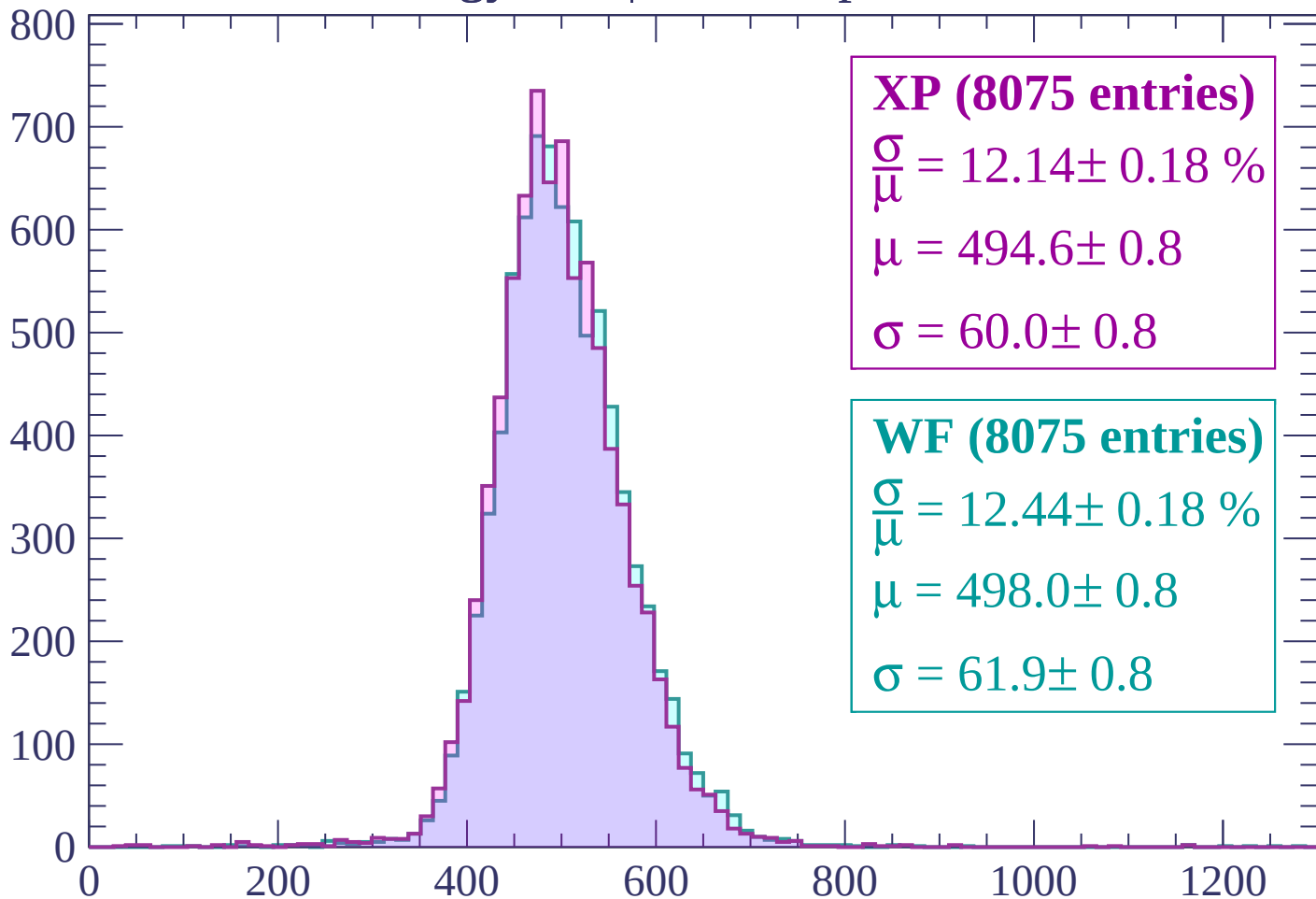
$$\mu = 501.7 \pm 0.8$$

$$\sigma = 61.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (8075 entries)

$$\frac{\sigma}{\mu} = 12.14 \pm 0.18 \%$$

$$\mu = 494.6 \pm 0.8$$

$$\sigma = 60.0 \pm 0.8$$

WF (8075 entries)

$$\frac{\sigma}{\mu} = 12.44 \pm 0.18 \%$$

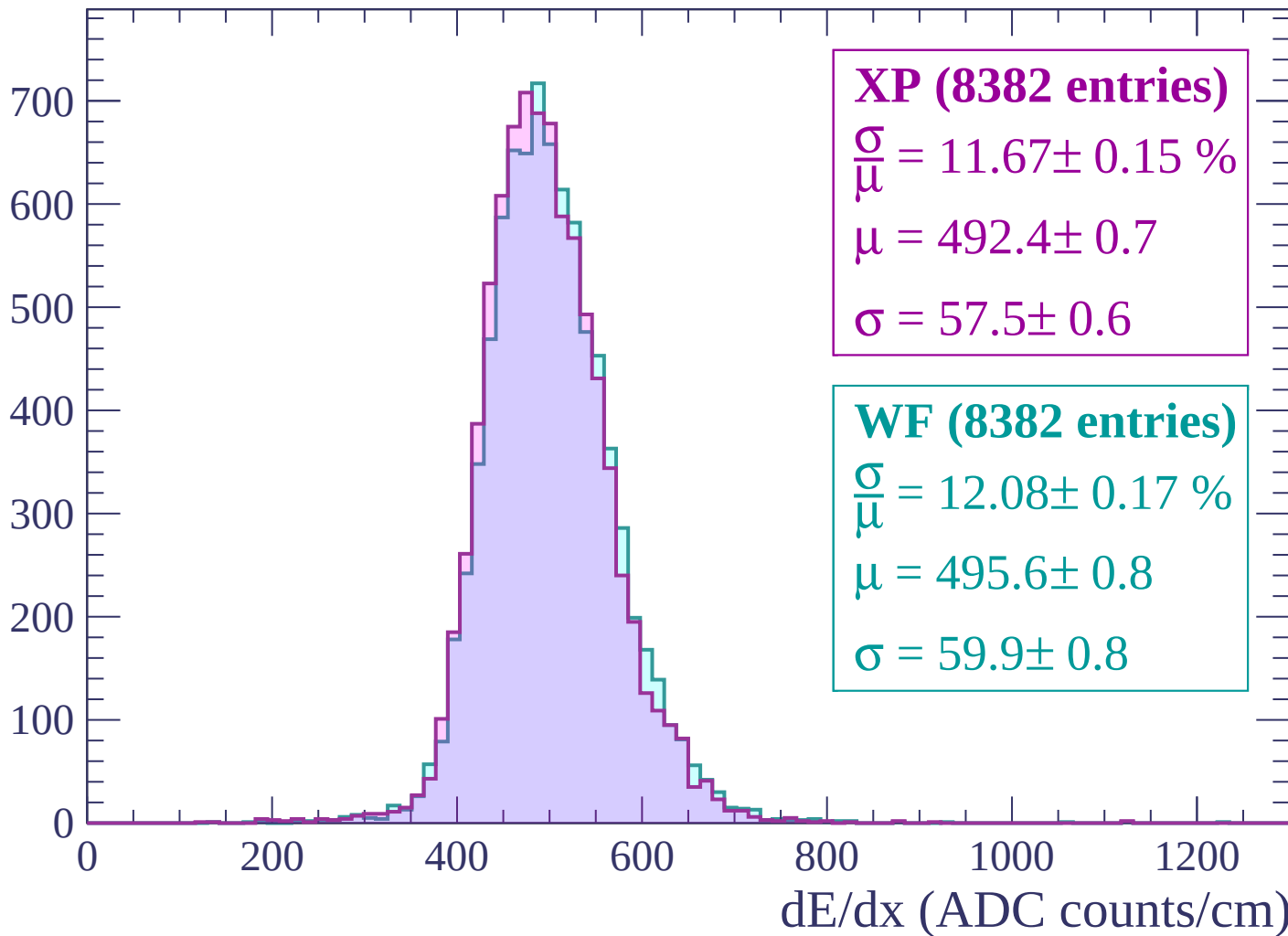
$$\mu = 498.0 \pm 0.8$$

$$\sigma = 61.9 \pm 0.8$$

dE/dx (ADC counts/cm)

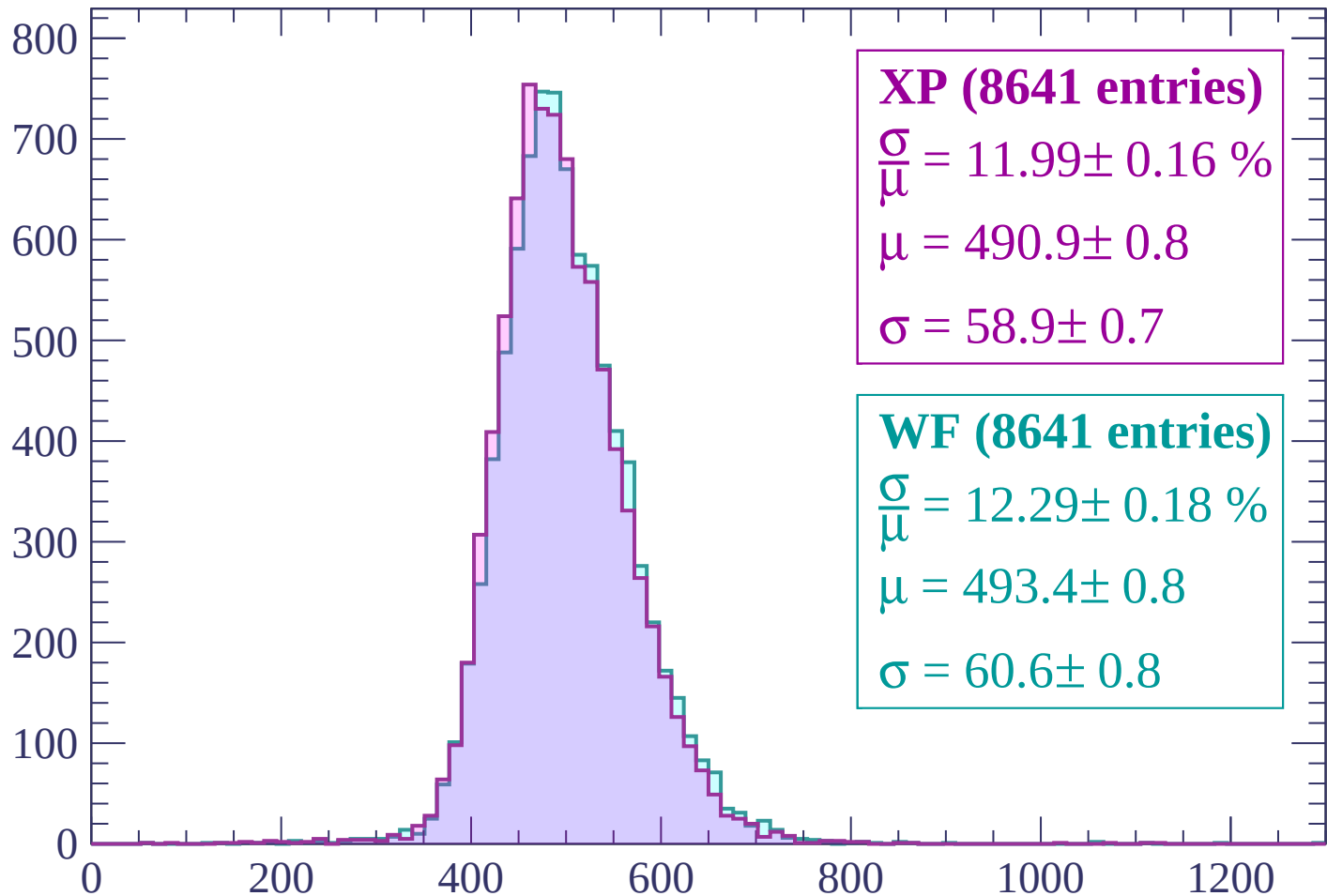
Energy loss | $-1680 < p < -1620$

Count



Energy loss | -1620 < p < -1560

Count



XP (8641 entries)

$$\frac{\sigma}{\mu} = 11.99 \pm 0.16 \%$$

$$\mu = 490.9 \pm 0.8$$

$$\sigma = 58.9 \pm 0.7$$

WF (8641 entries)

$$\frac{\sigma}{\mu} = 12.29 \pm 0.18 \%$$

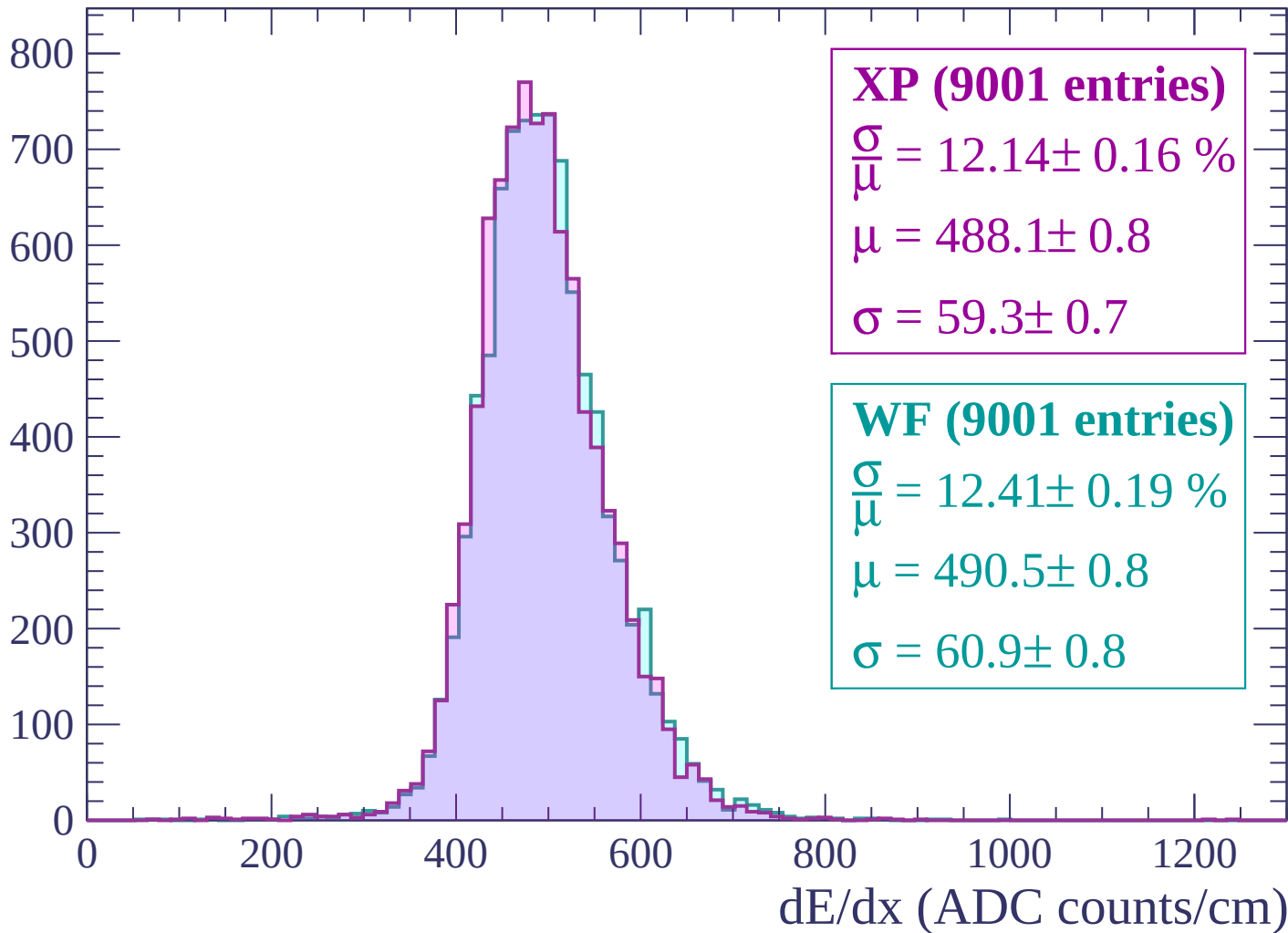
$$\mu = 493.4 \pm 0.8$$

$$\sigma = 60.6 \pm 0.8$$

dE/dx (ADC counts/cm)

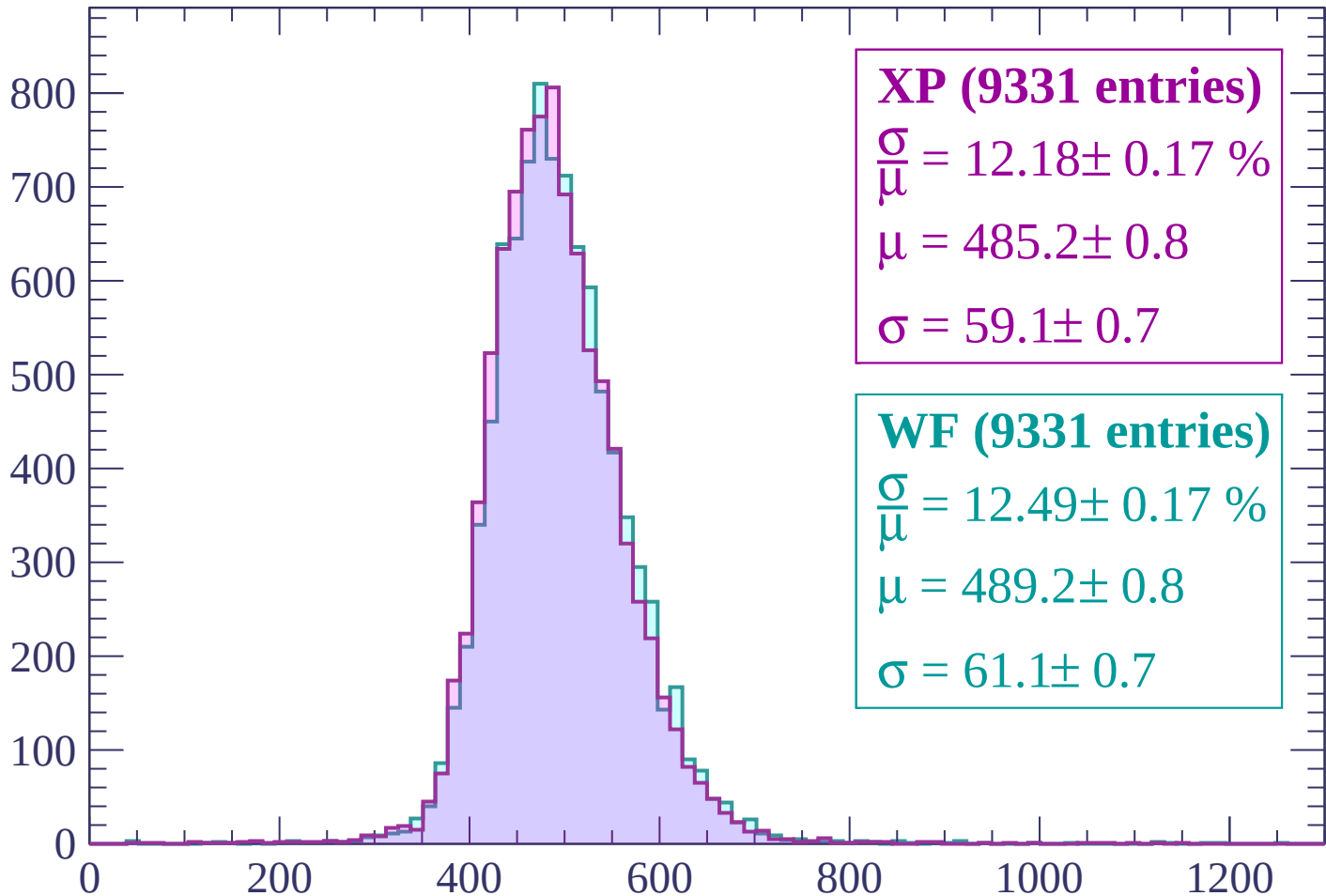
Energy loss | -1560 < p < -1500

Count



Energy loss | -1500 < p < -1440

Count



XP (9331 entries)

$$\frac{\sigma}{\mu} = 12.18 \pm 0.17 \%$$

$$\mu = 485.2 \pm 0.8$$

$$\sigma = 59.1 \pm 0.7$$

WF (9331 entries)

$$\frac{\sigma}{\mu} = 12.49 \pm 0.17 \%$$

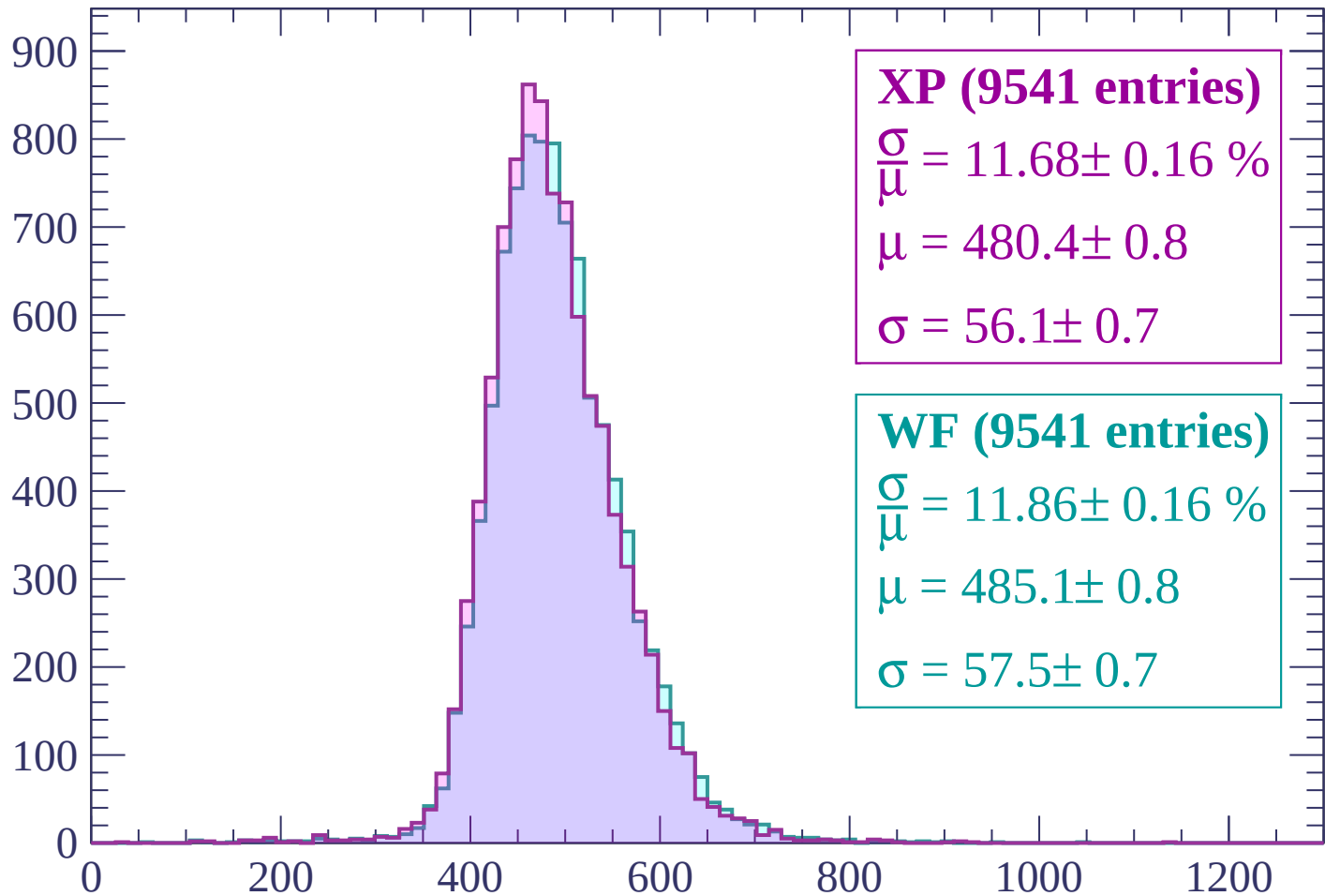
$$\mu = 489.2 \pm 0.8$$

$$\sigma = 61.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1380$

Count



XP (9541 entries)

$$\frac{\sigma}{\mu} = 11.68 \pm 0.16 \%$$

$$\mu = 480.4 \pm 0.8$$

$$\sigma = 56.1 \pm 0.7$$

WF (9541 entries)

$$\frac{\sigma}{\mu} = 11.86 \pm 0.16 \%$$

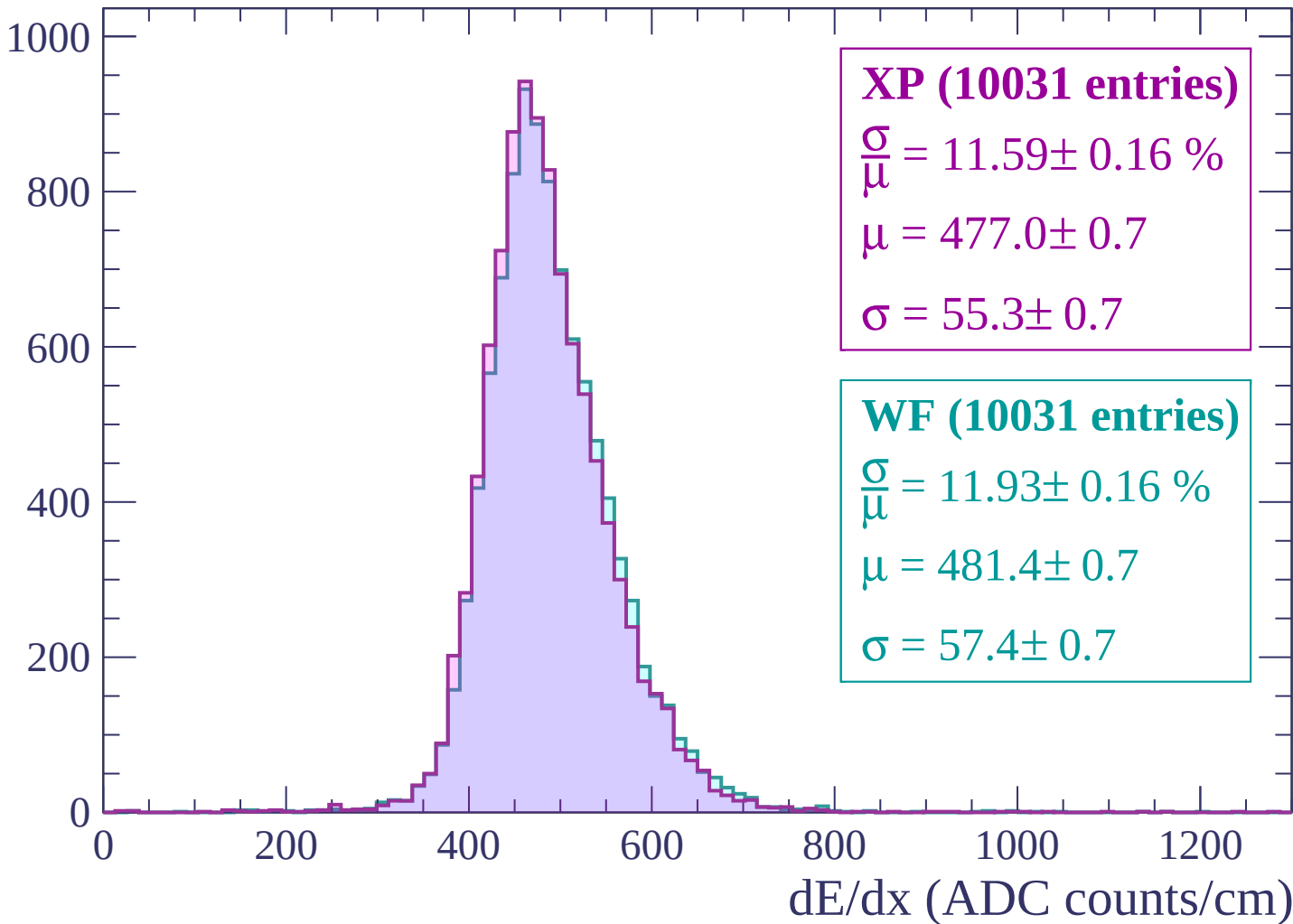
$$\mu = 485.1 \pm 0.8$$

$$\sigma = 57.5 \pm 0.7$$

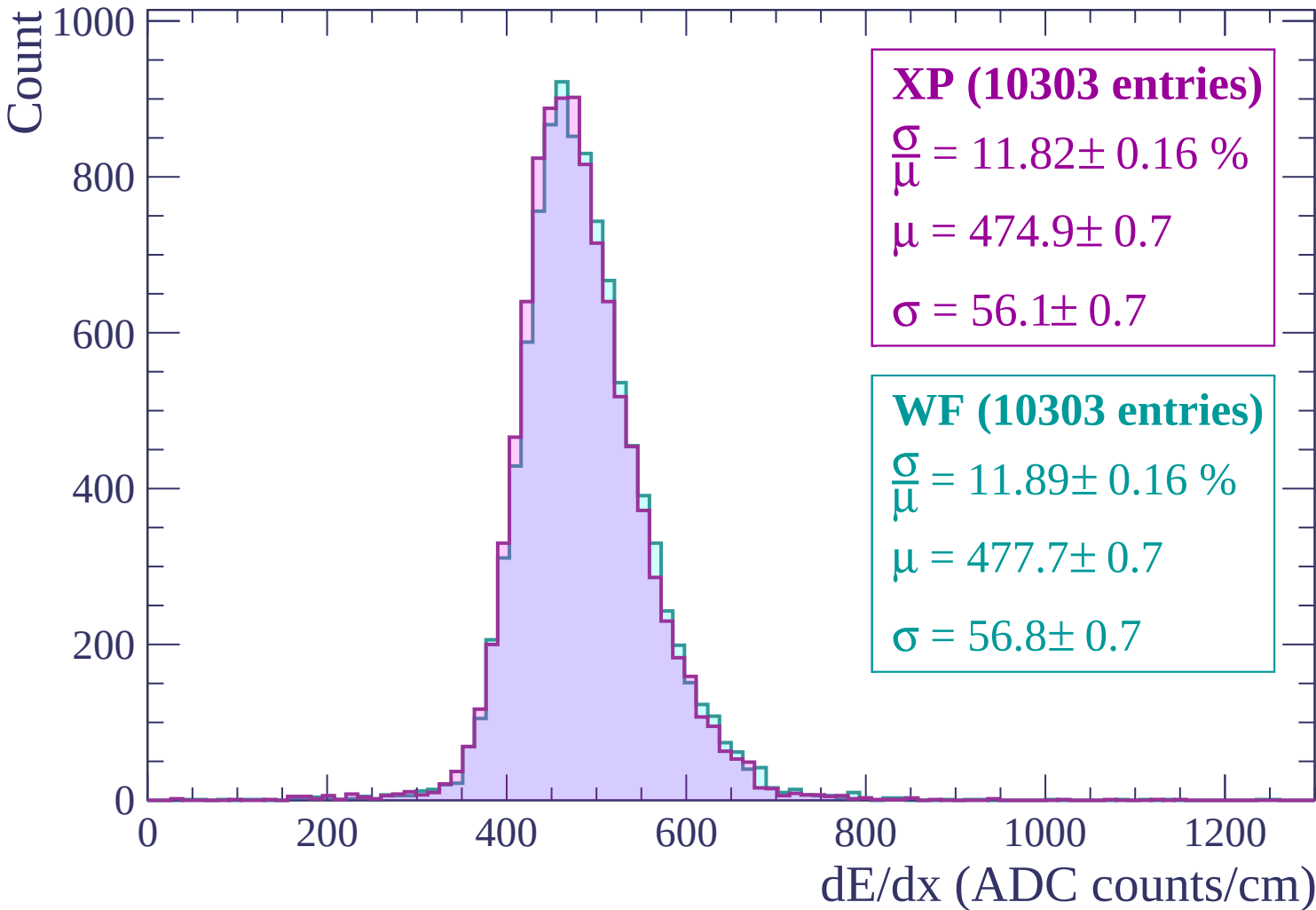
dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



Energy loss | -1320 < p < -1260



Energy loss | $-1260 < p < -1200$

Count

1000

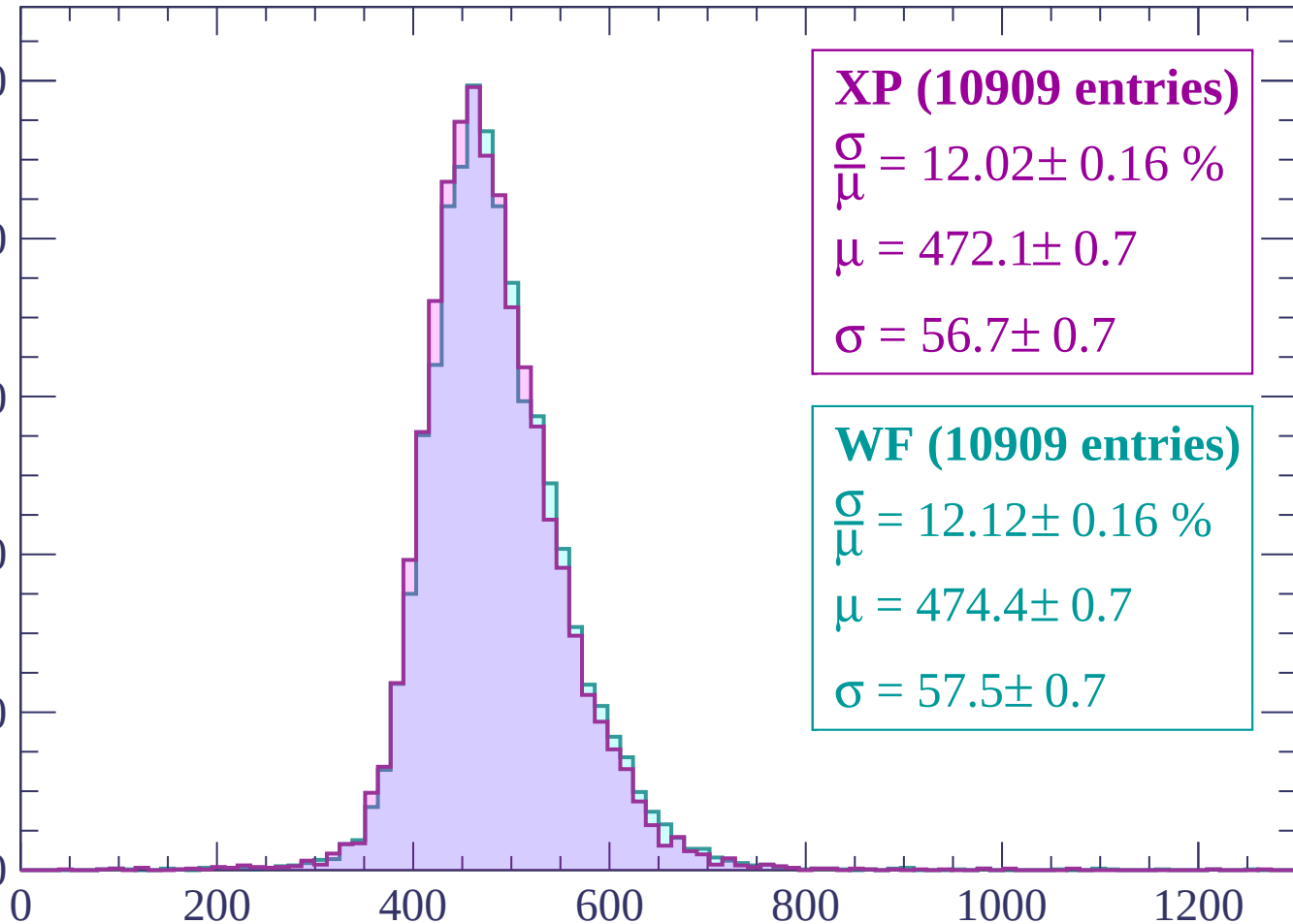
800

600

400

200

0



XP (10909 entries)

$\frac{\sigma}{\mu} = 12.02 \pm 0.16 \%$

$\mu = 472.1 \pm 0.7$

$\sigma = 56.7 \pm 0.7$

WF (10909 entries)

$\frac{\sigma}{\mu} = 12.12 \pm 0.16 \%$

$\mu = 474.4 \pm 0.7$

$\sigma = 57.5 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1140$

Count

1000

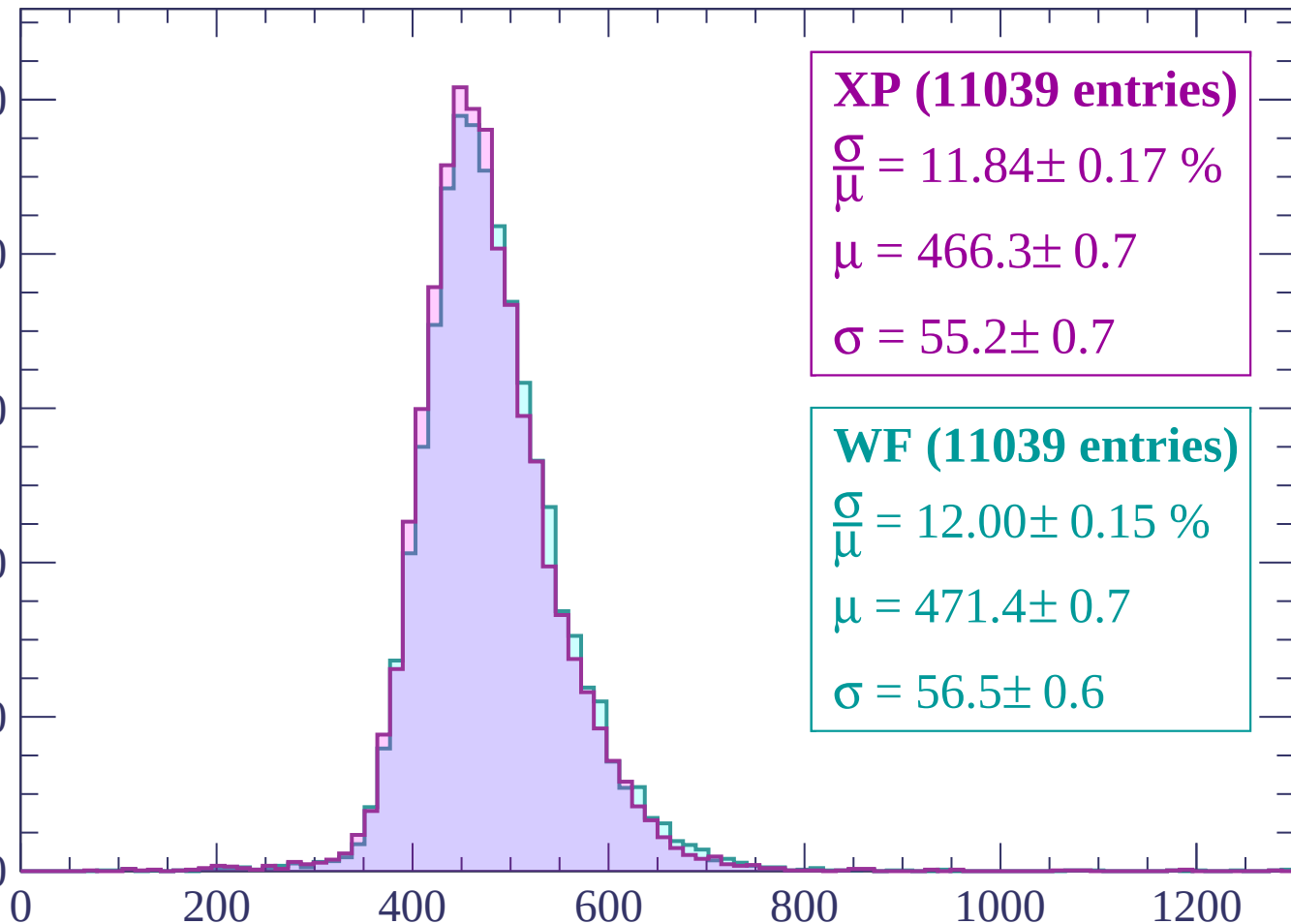
800

600

400

200

0



XP (11039 entries)

$\frac{\sigma}{\mu} = 11.84 \pm 0.17 \%$

$\mu = 466.3 \pm 0.7$

$\sigma = 55.2 \pm 0.7$

WF (11039 entries)

$\frac{\sigma}{\mu} = 12.00 \pm 0.15 \%$

$\mu = 471.4 \pm 0.7$

$\sigma = 56.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | -1140 < p < -1080

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (11464 entries)

$$\frac{\sigma}{\mu} = 11.26 \pm 0.14 \%$$

$$\mu = 462.9 \pm 0.6$$

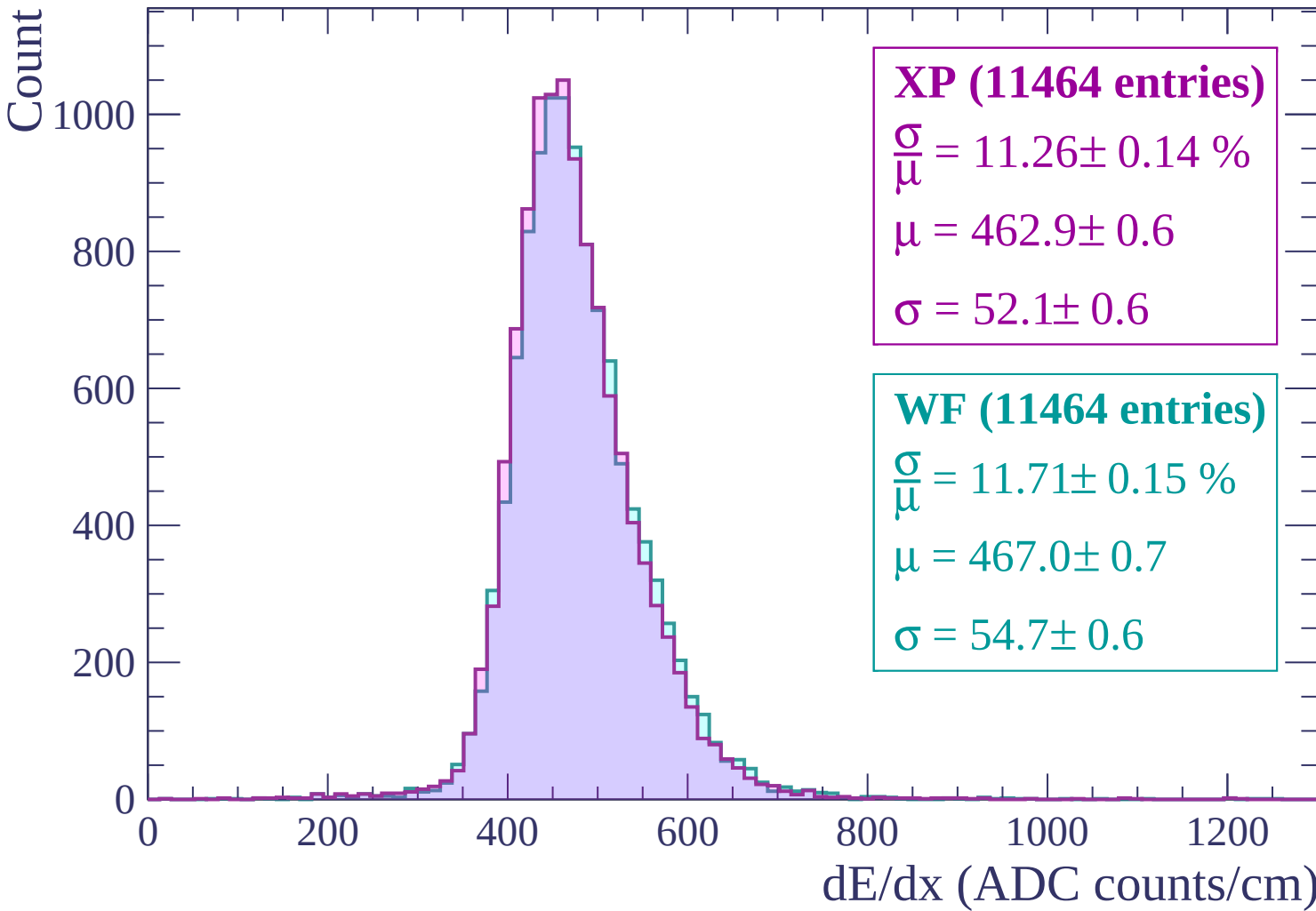
$$\sigma = 52.1 \pm 0.6$$

WF (11464 entries)

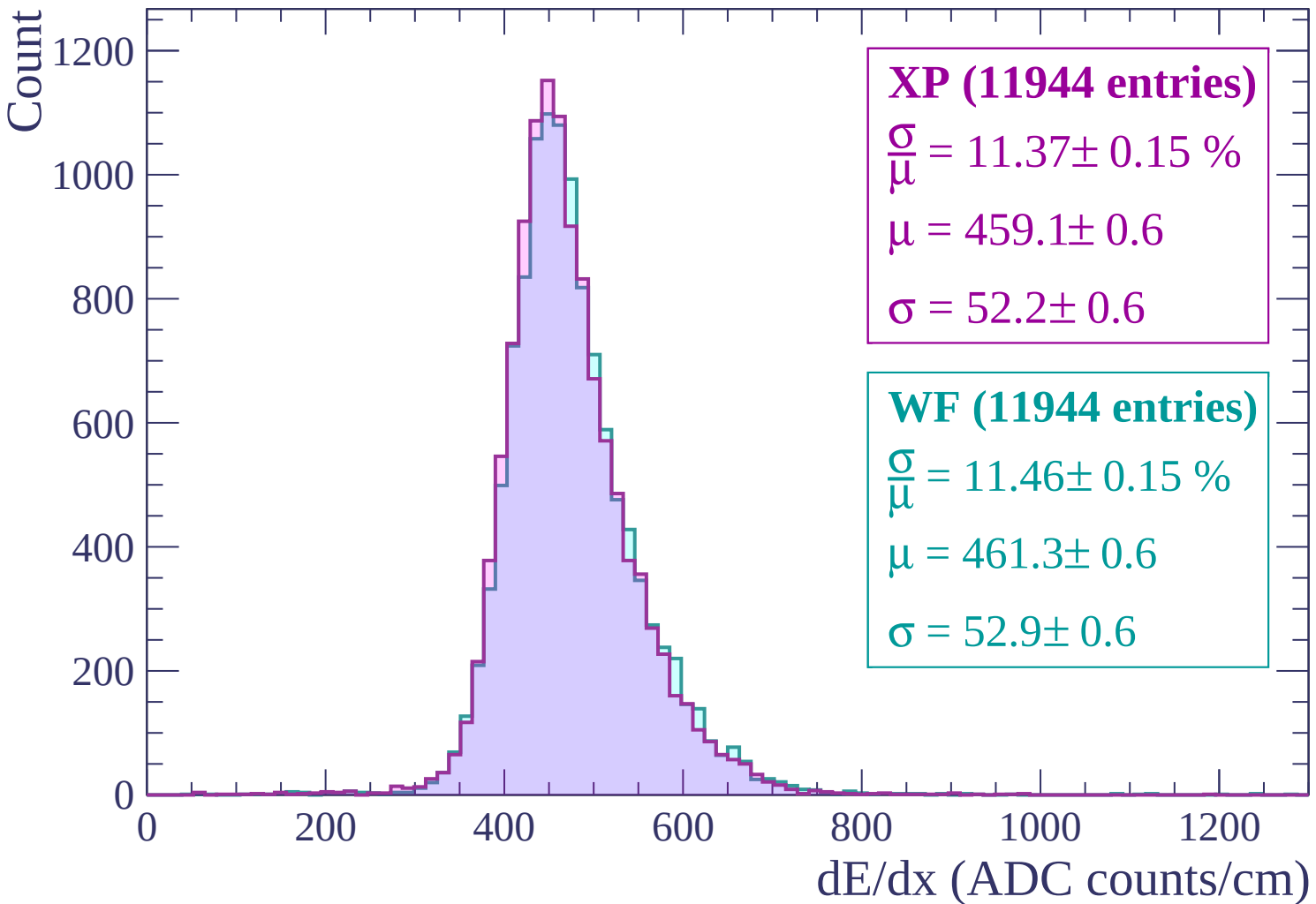
$$\frac{\sigma}{\mu} = 11.71 \pm 0.15 \%$$

$$\mu = 467.0 \pm 0.7$$

$$\sigma = 54.7 \pm 0.6$$



Energy loss | -1080 < p < -1020



Energy loss | $-1020 < p < -960$

Count

1200

1000

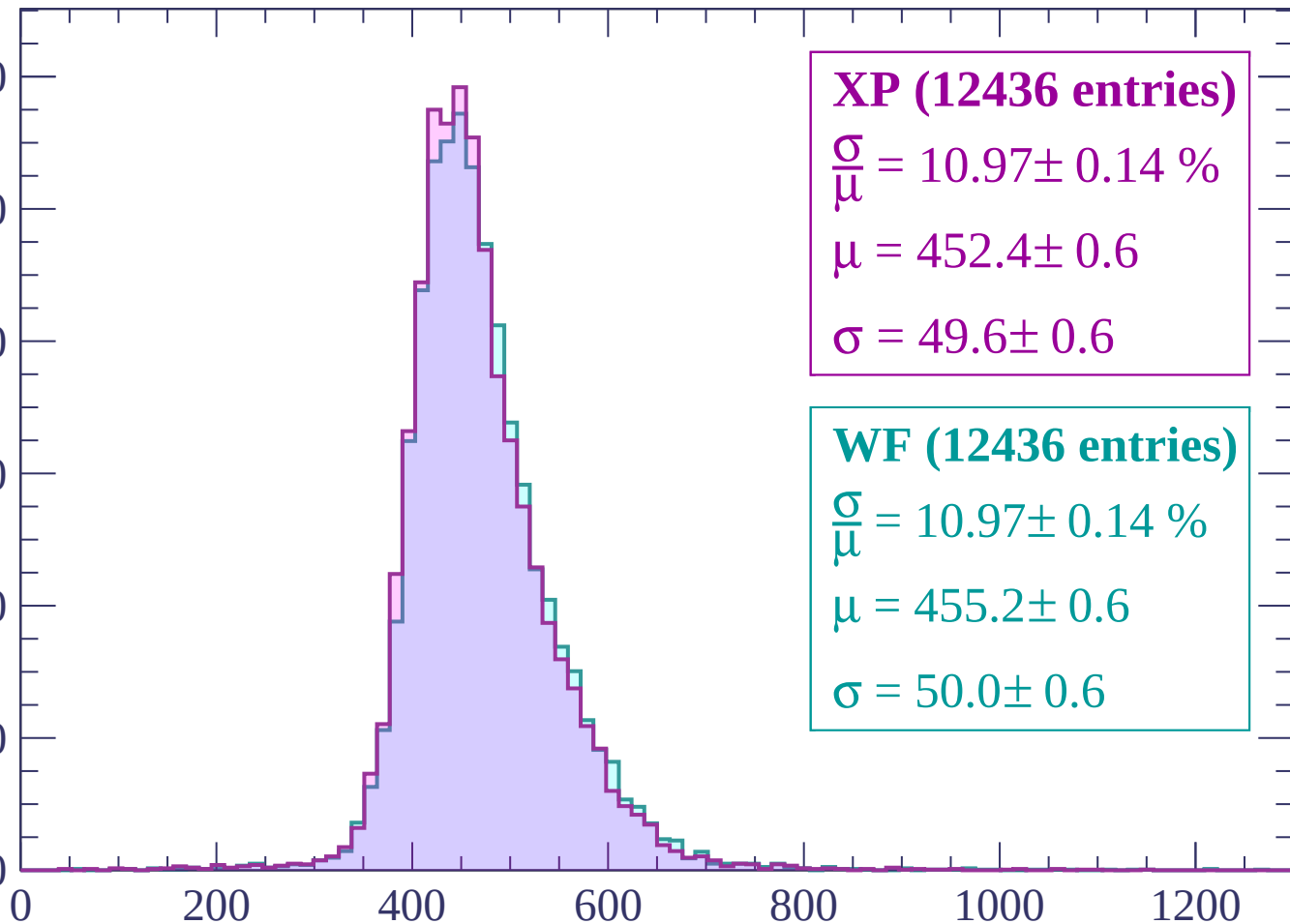
800

600

400

200

0



XP (12436 entries)

$\frac{\sigma}{\mu} = 10.97 \pm 0.14 \%$

$\mu = 452.4 \pm 0.6$

$\sigma = 49.6 \pm 0.6$

WF (12436 entries)

$\frac{\sigma}{\mu} = 10.97 \pm 0.14 \%$

$\mu = 455.2 \pm 0.6$

$\sigma = 50.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (12619 entries)

$$\frac{\sigma}{\mu} = 11.37 \pm 0.15 \%$$

$$\mu = 450.0 \pm 0.6$$

$$\sigma = 51.2 \pm 0.6$$

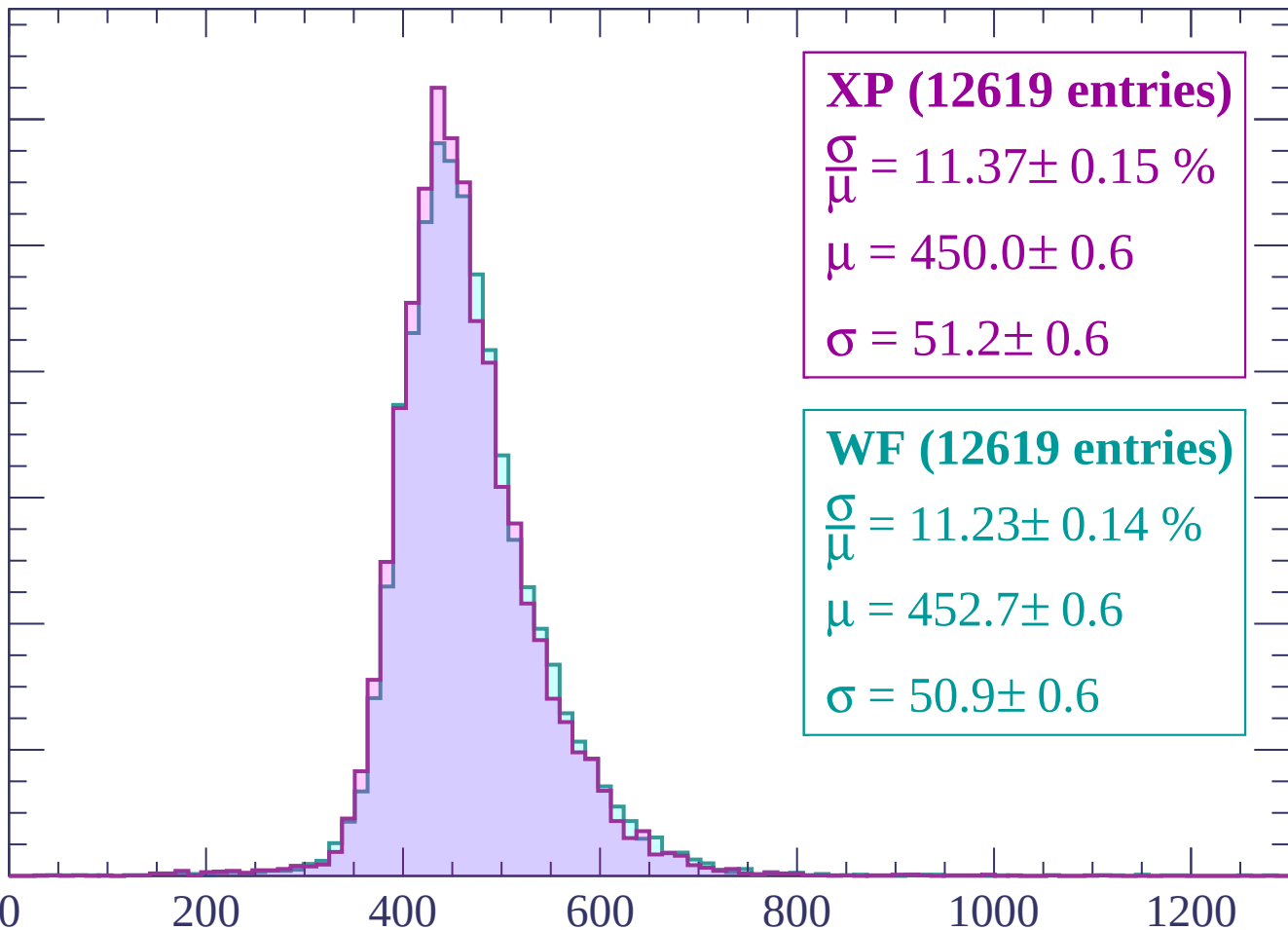
WF (12619 entries)

$$\frac{\sigma}{\mu} = 11.23 \pm 0.14 \%$$

$$\mu = 452.7 \pm 0.6$$

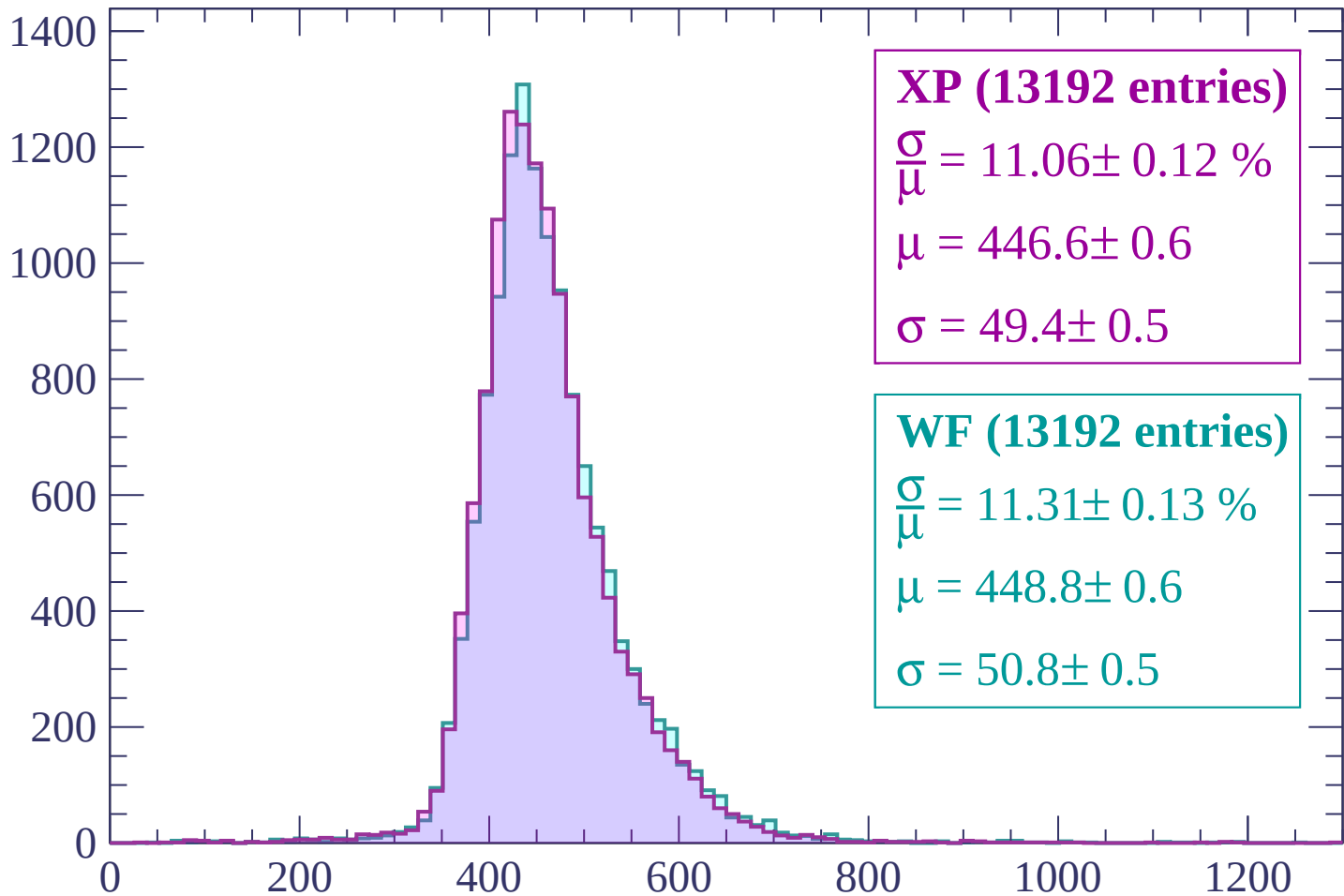
$$\sigma = 50.9 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $-900 < p < -840$

Count



XP (13192 entries)

$$\frac{\sigma}{\mu} = 11.06 \pm 0.12 \%$$

$$\mu = 446.6 \pm 0.6$$

$$\sigma = 49.4 \pm 0.5$$

WF (13192 entries)

$$\frac{\sigma}{\mu} = 11.31 \pm 0.13 \%$$

$$\mu = 448.8 \pm 0.6$$

$$\sigma = 50.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count

1400
1200
1000
800
600
400
200
0

XP (13779 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.13 \%$$

$$\mu = 440.5 \pm 0.5$$

$$\sigma = 47.8 \pm 0.5$$

WF (13779 entries)

$$\frac{\sigma}{\mu} = 11.00 \pm 0.13 \%$$

$$\mu = 442.9 \pm 0.6$$

$$\sigma = 48.7 \pm 0.5$$

0

200

400

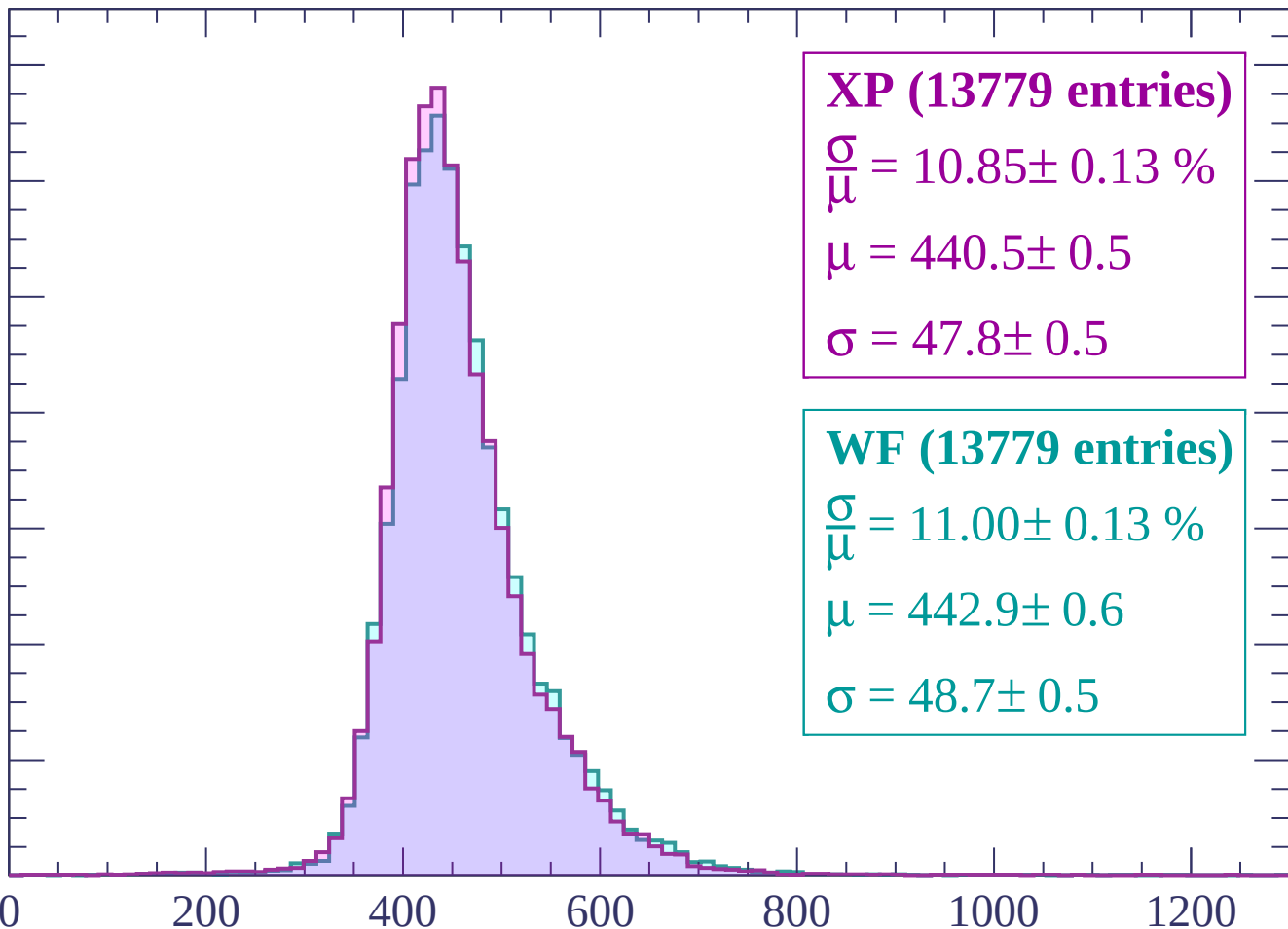
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $-780 < p < -720$

Count

1400

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (13966 entries)

$\frac{\sigma}{\mu} = 10.66 \pm 0.13 \%$

$\mu = 434.6 \pm 0.5$

$\sigma = 46.3 \pm 0.5$

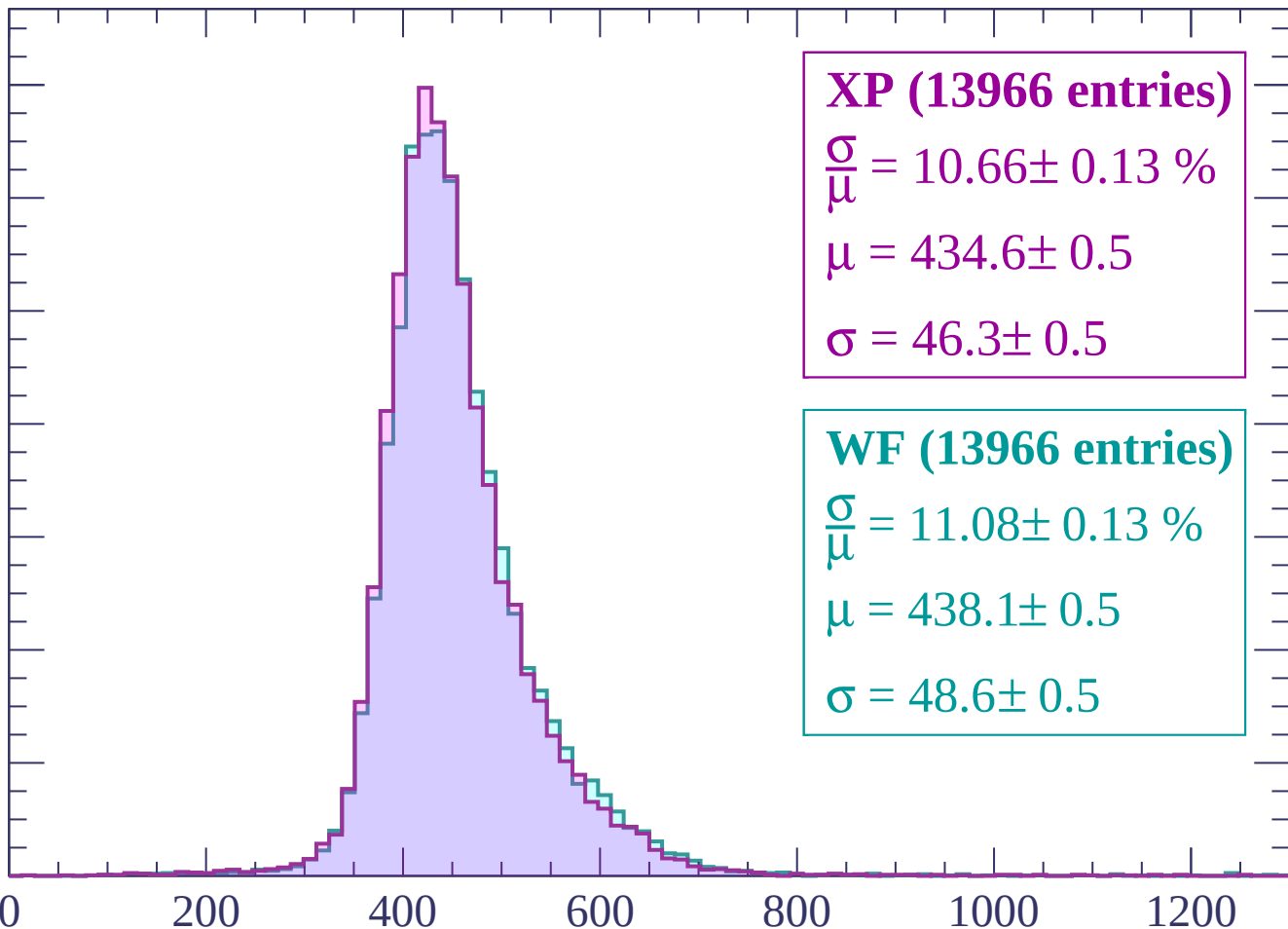
WF (13966 entries)

$\frac{\sigma}{\mu} = 11.08 \pm 0.13 \%$

$\mu = 438.1 \pm 0.5$

$\sigma = 48.6 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $-720 < p < -660$

Count

1400
1200
1000
800
600
400
200
0

XP (14242 entries)

$$\frac{\sigma}{\mu} = 10.66 \pm 0.13 \%$$

$$\mu = 430.0 \pm 0.5$$

$$\sigma = 45.8 \pm 0.5$$

WF (14242 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.13 \%$$

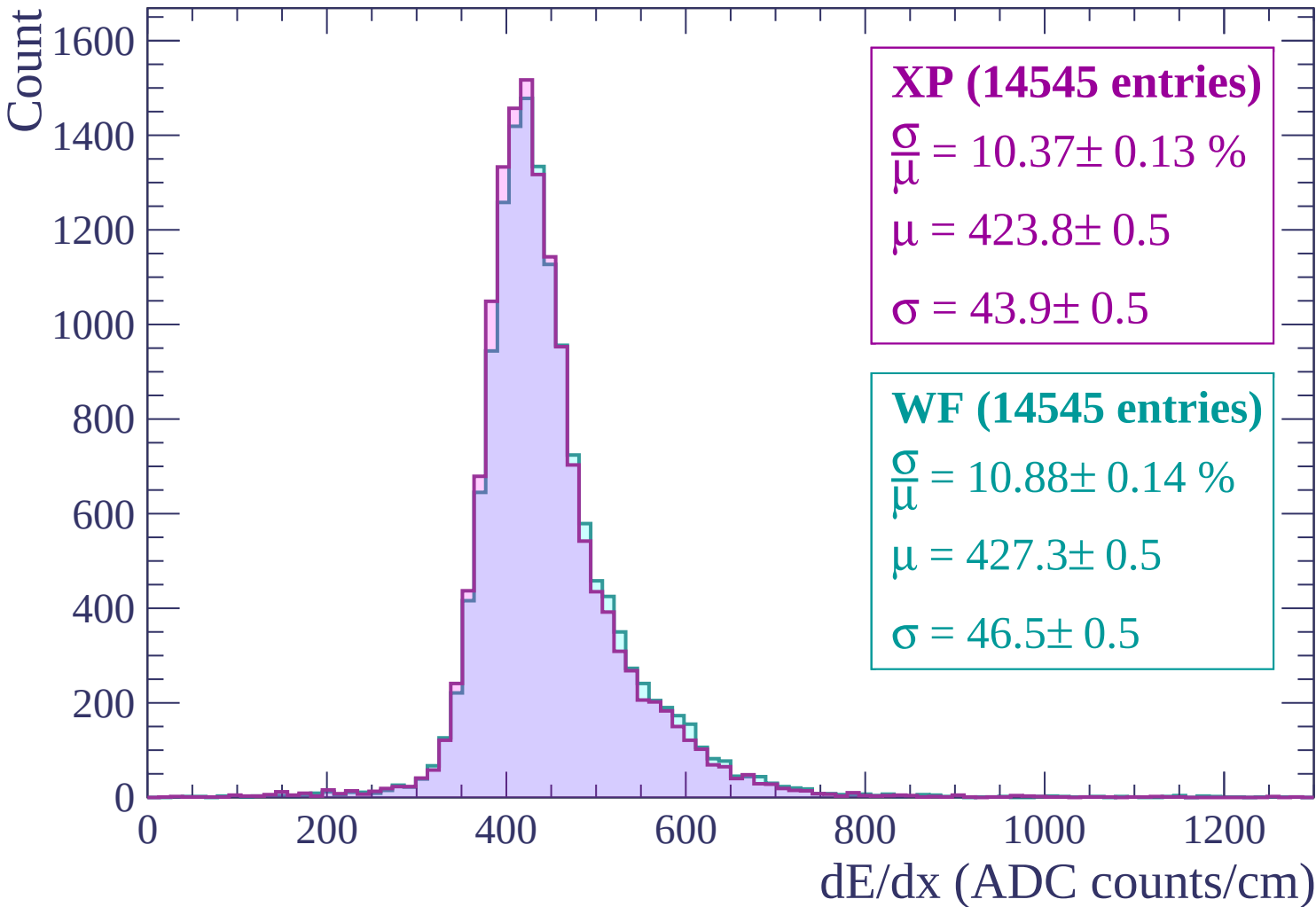
$$\mu = 432.0 \pm 0.5$$

$$\sigma = 46.9 \pm 0.5$$

0 200 400 600 800 1000 1200

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$



Energy loss | $-600 < p < -540$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14820 entries)

$$\frac{\sigma}{\mu} = 10.36 \pm 0.13 \%$$

$$\mu = 419.9 \pm 0.5$$

$$\sigma = 43.5 \pm 0.5$$

WF (14820 entries)

$$\frac{\sigma}{\mu} = 10.47 \pm 0.13 \%$$

$$\mu = 421.6 \pm 0.5$$

$$\sigma = 44.1 \pm 0.5$$

0

200

400

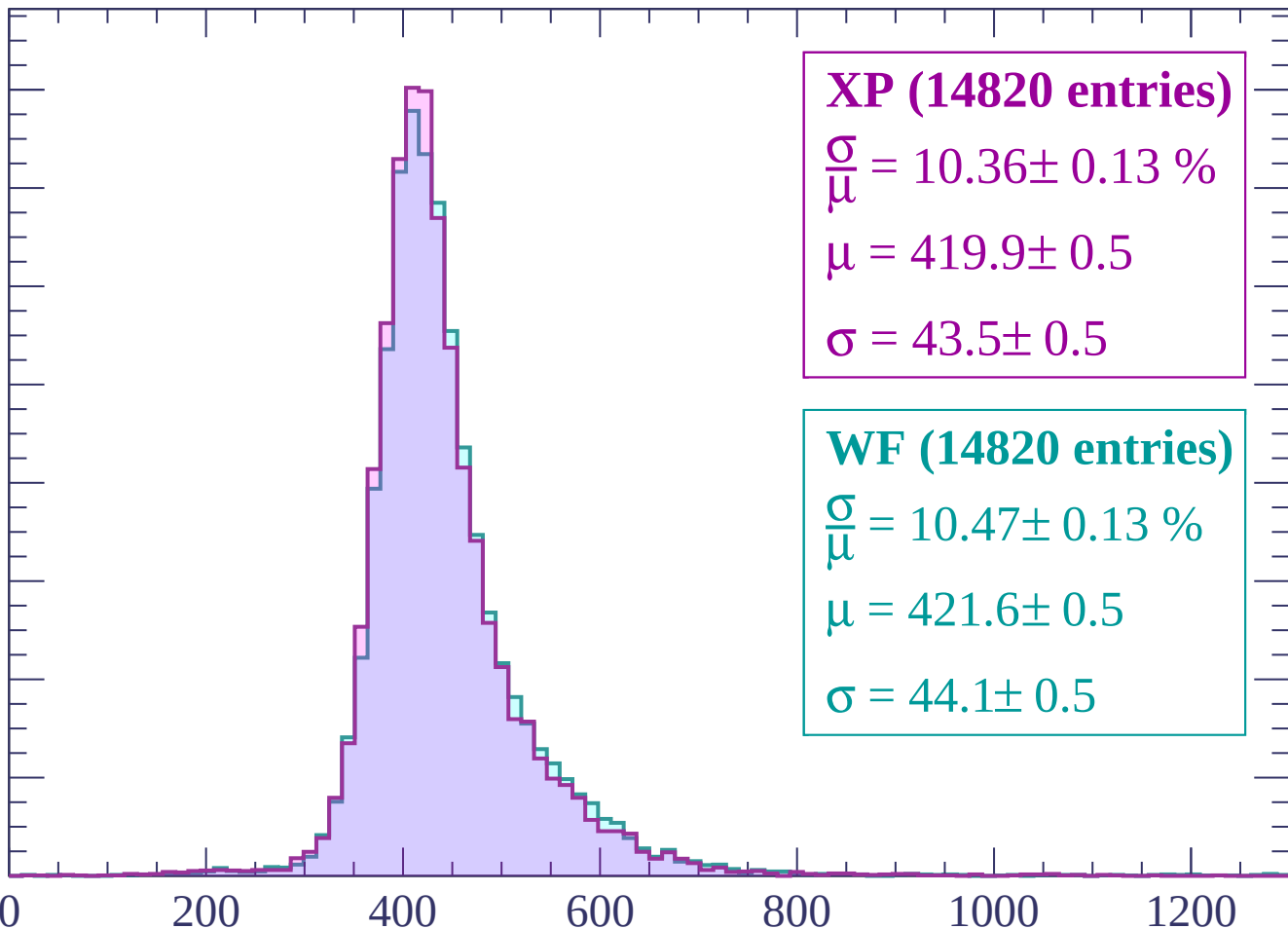
600

800

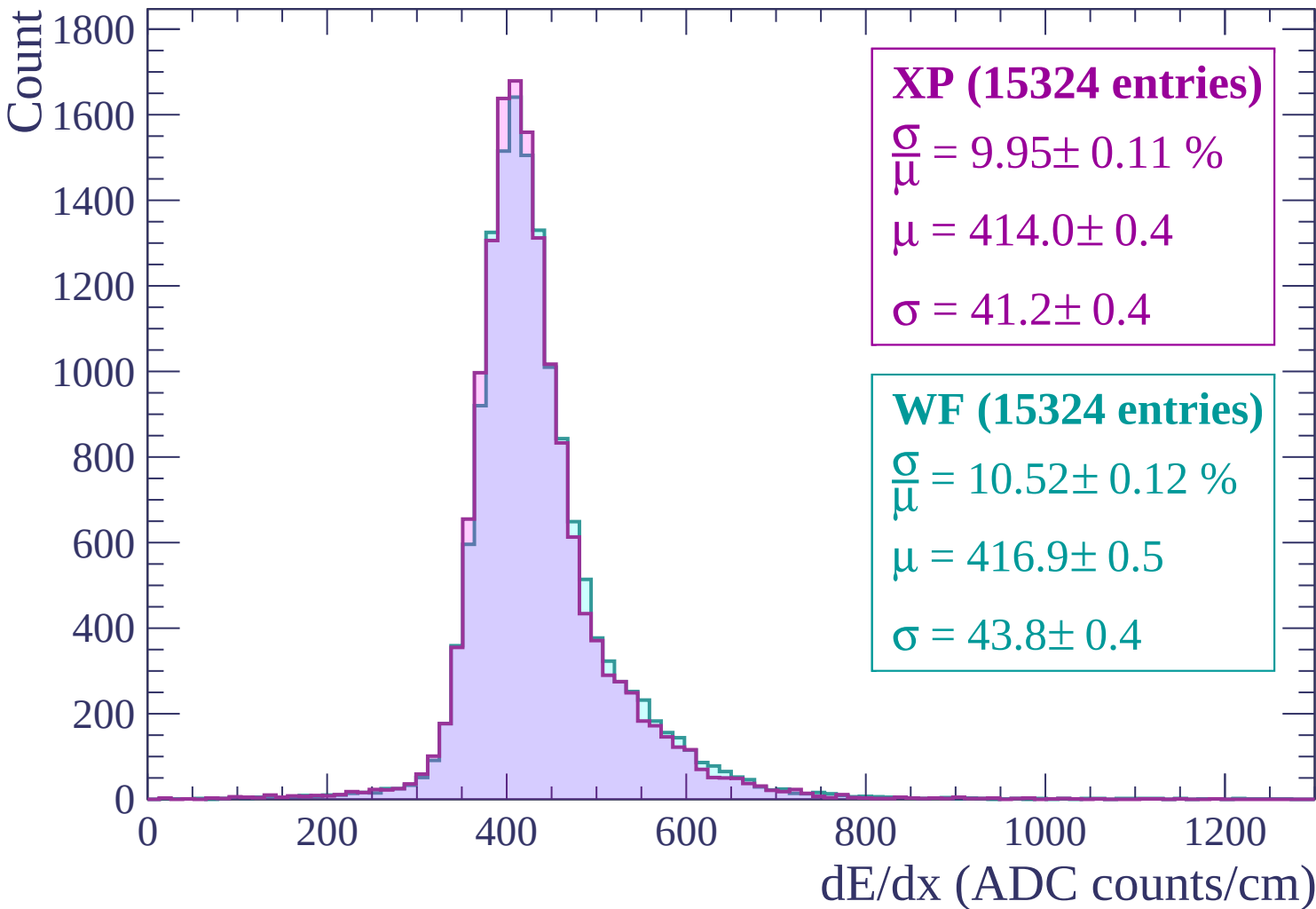
1000

1200

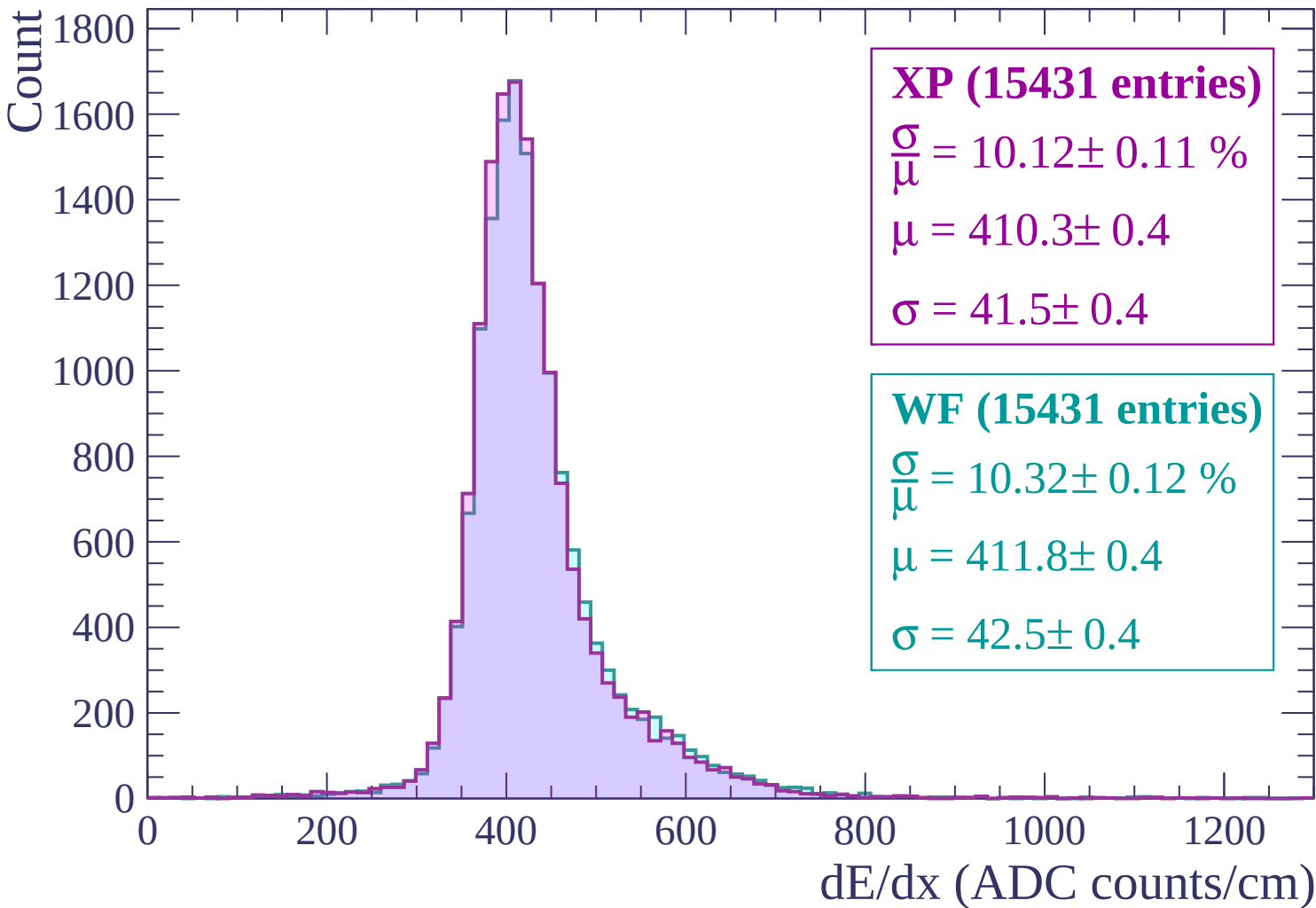
dE/dx (ADC counts/cm)



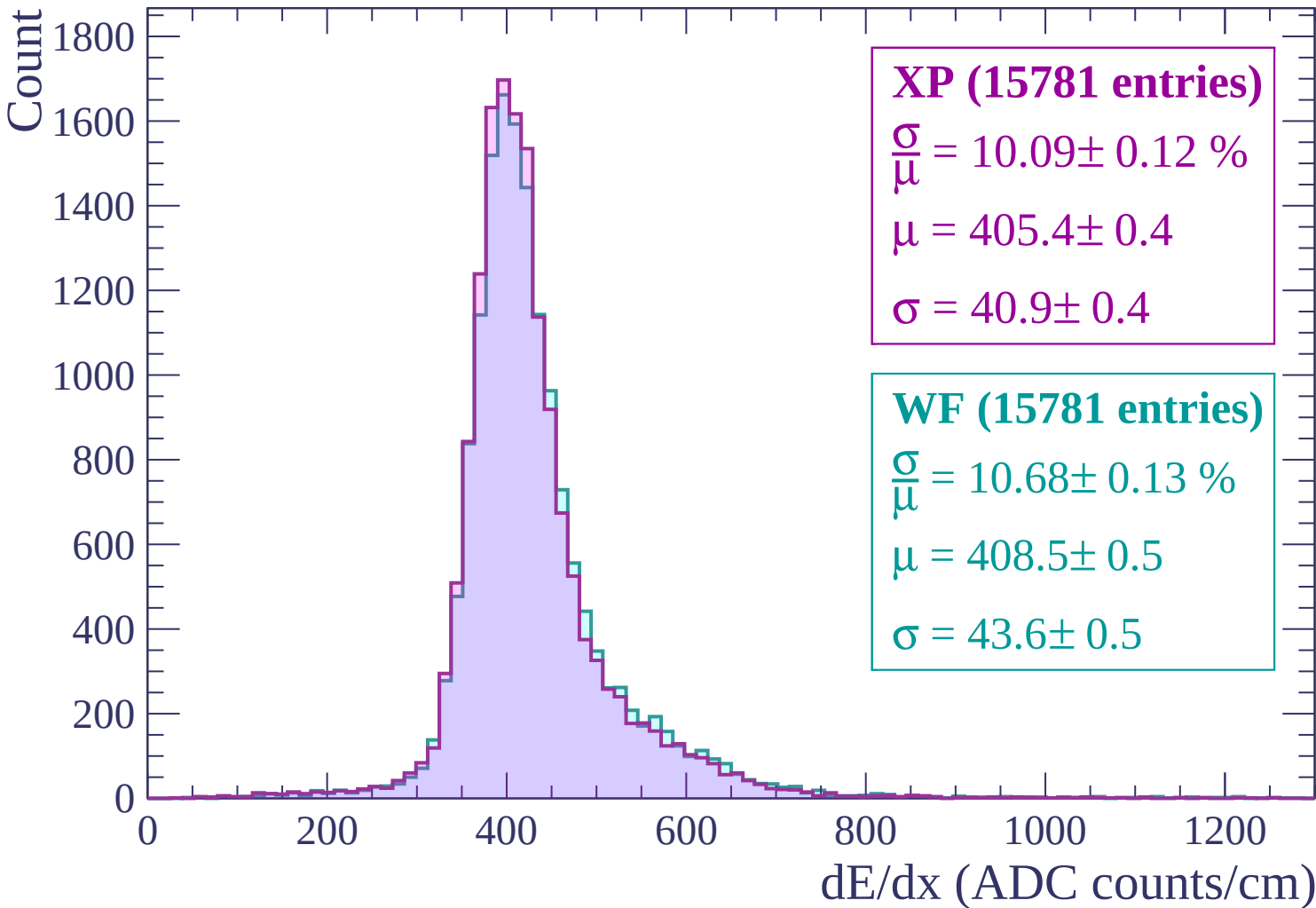
Energy loss | $-540 < p < -480$



Energy loss | $-480 < p < -420$



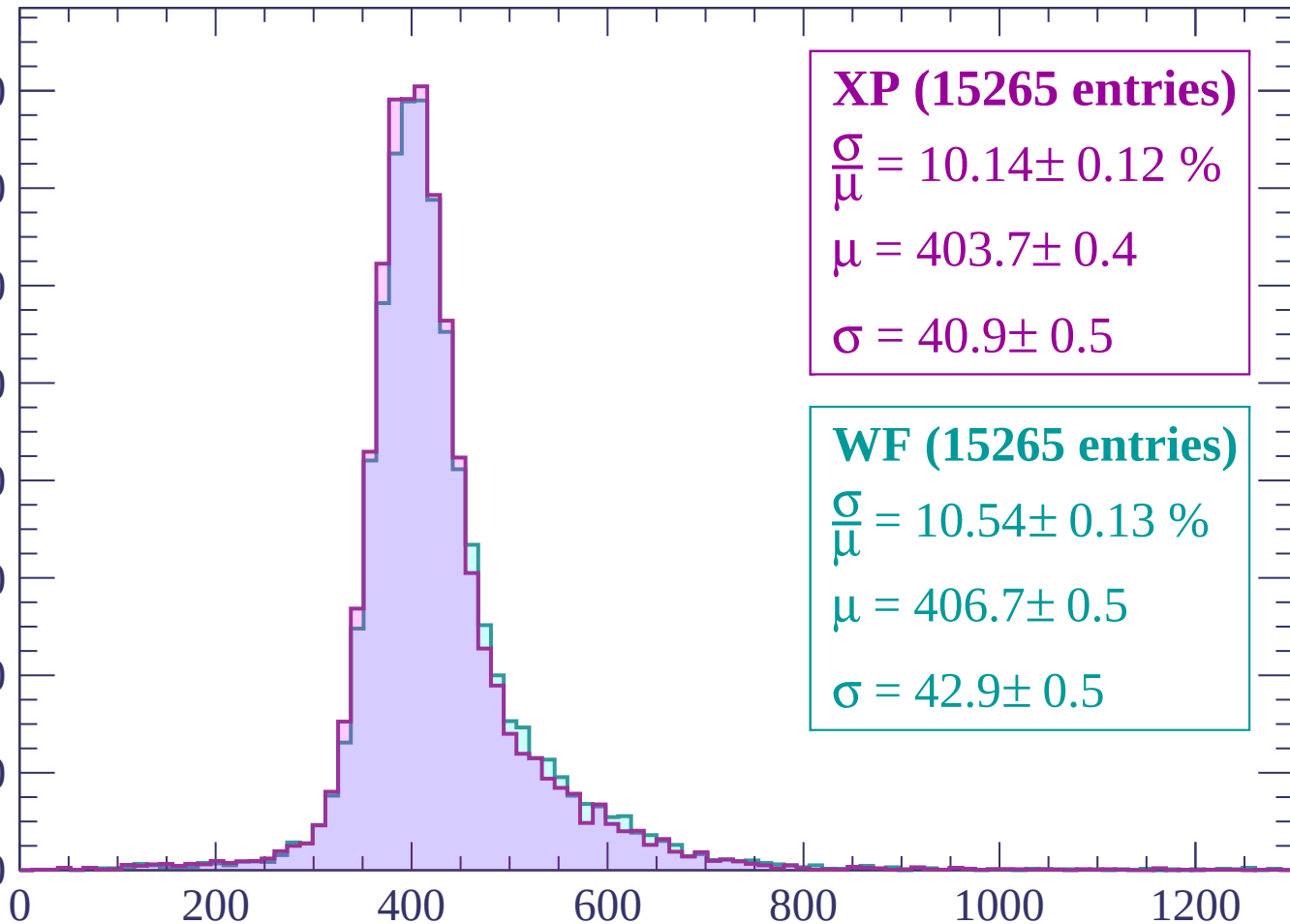
Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (15265 entries)

$\frac{\sigma}{\mu} = 10.14 \pm 0.12 \%$

$\mu = 403.7 \pm 0.4$

$\sigma = 40.9 \pm 0.5$

WF (15265 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.13 \%$

$\mu = 406.7 \pm 0.5$

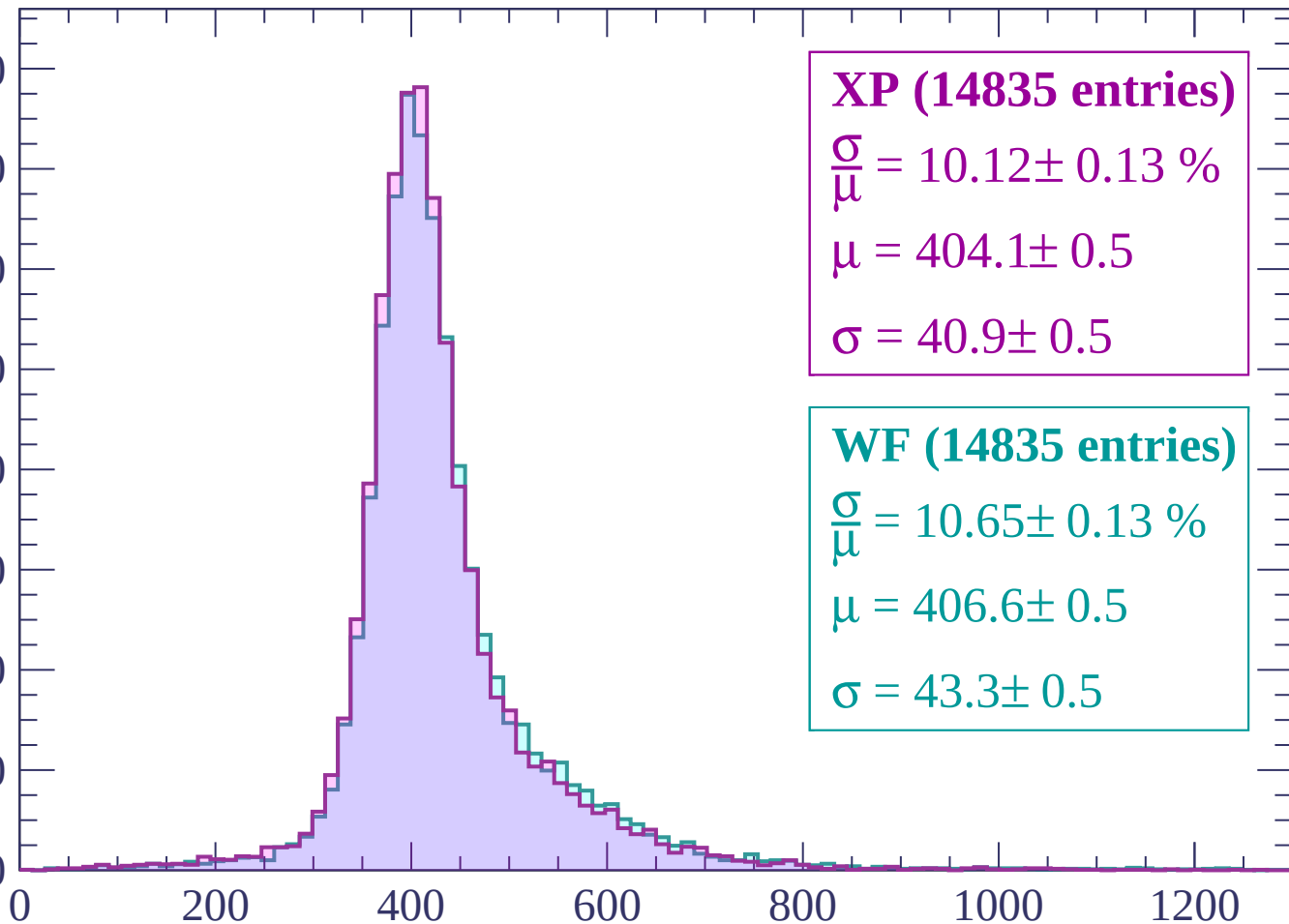
$\sigma = 42.9 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (14835 entries)

$\frac{\sigma}{\mu} = 10.12 \pm 0.13 \%$

$\mu = 404.1 \pm 0.5$

$\sigma = 40.9 \pm 0.5$

WF (14835 entries)

$\frac{\sigma}{\mu} = 10.65 \pm 0.13 \%$

$\mu = 406.6 \pm 0.5$

$\sigma = 43.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (13131 entries)

$$\frac{\sigma}{\mu} = 11.53 \pm 0.16 \%$$

$$\mu = 416.1 \pm 0.6$$

$$\sigma = 48.0 \pm 0.6$$

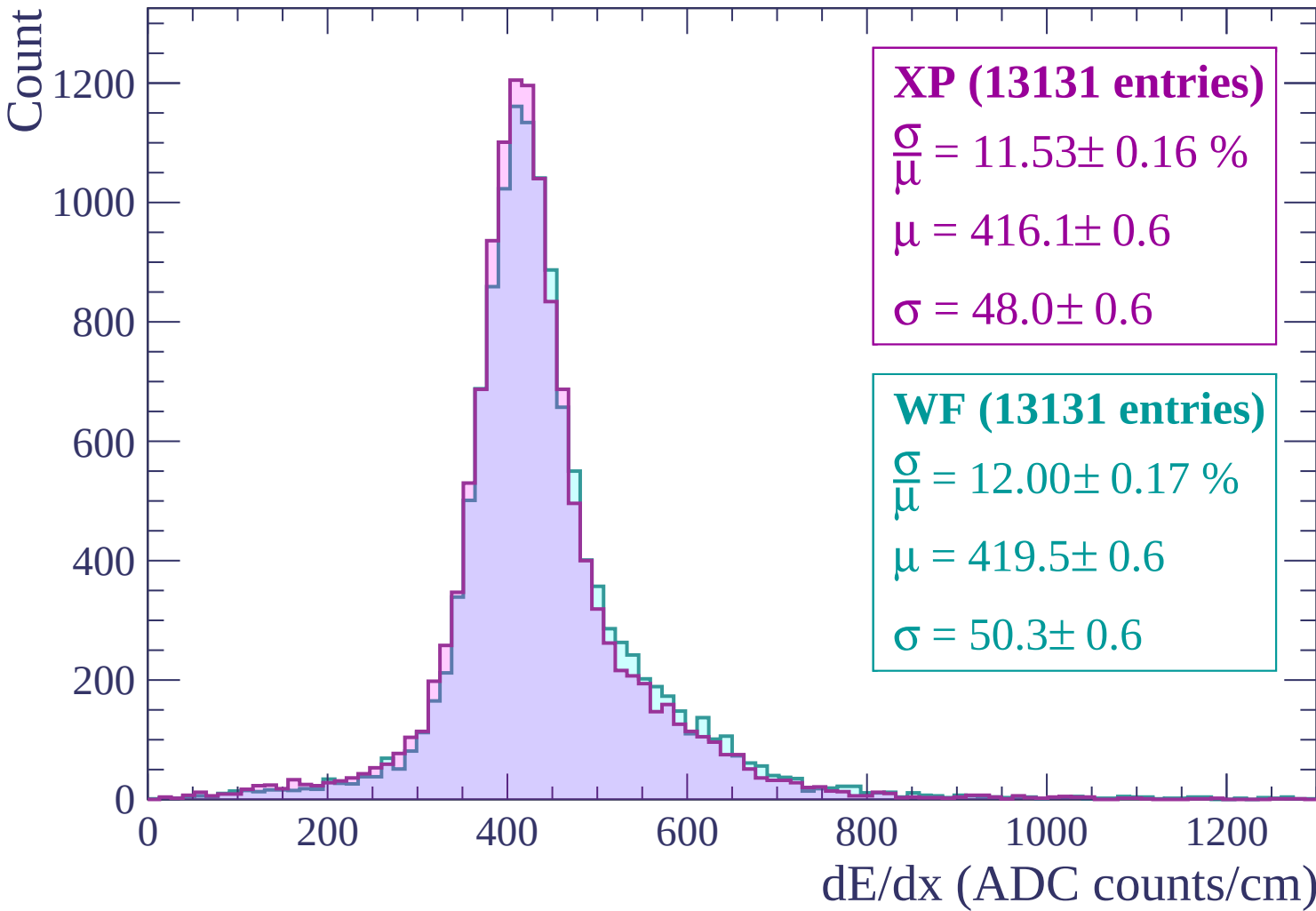
WF (13131 entries)

$$\frac{\sigma}{\mu} = 12.00 \pm 0.17 \%$$

$$\mu = 419.5 \pm 0.6$$

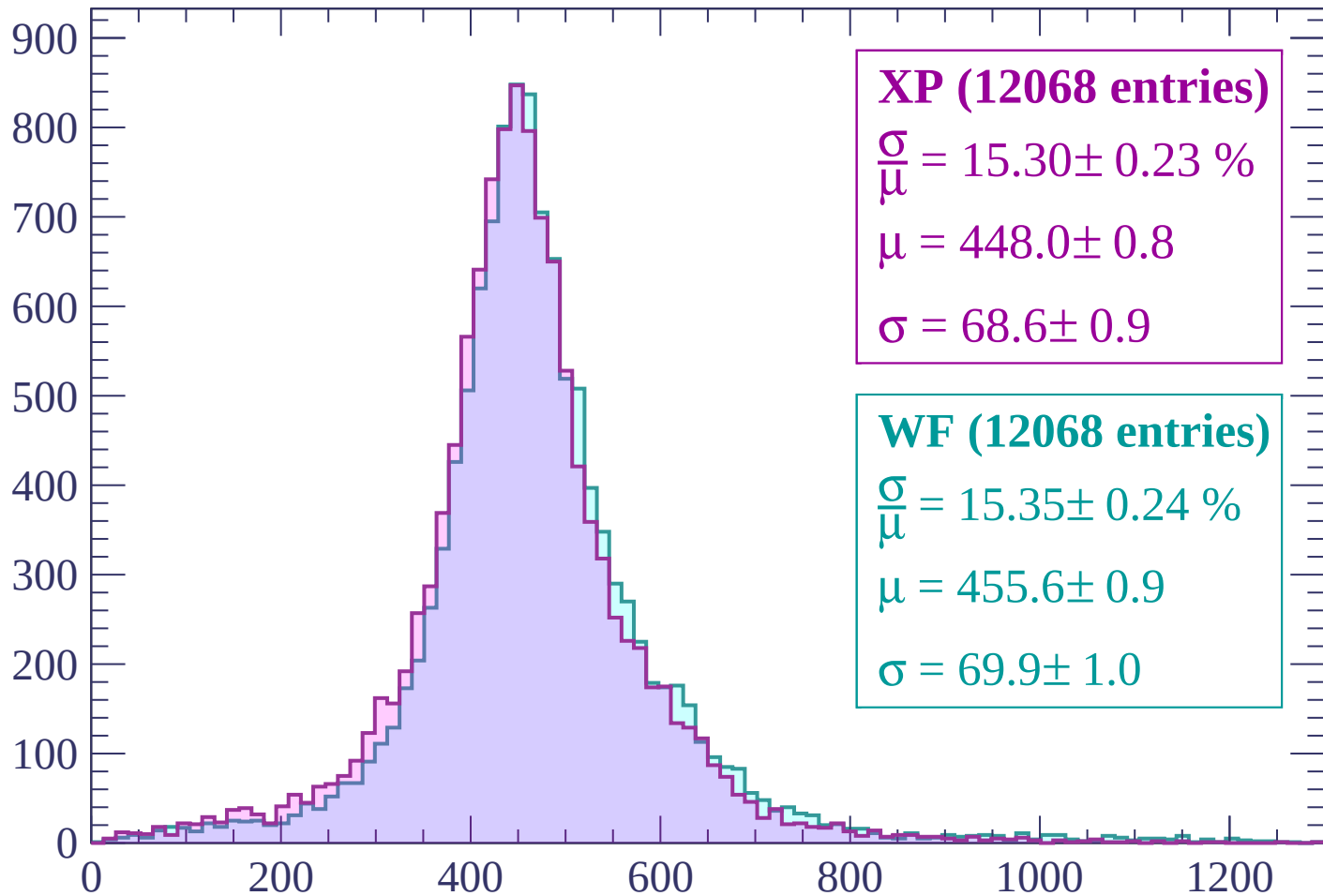
$$\sigma = 50.3 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | -180 < p < -120

Count



XP (12068 entries)

$$\frac{\sigma}{\mu} = 15.30 \pm 0.23 \%$$

$$\mu = 448.0 \pm 0.8$$

$$\sigma = 68.6 \pm 0.9$$

WF (12068 entries)

$$\frac{\sigma}{\mu} = 15.35 \pm 0.24 \%$$

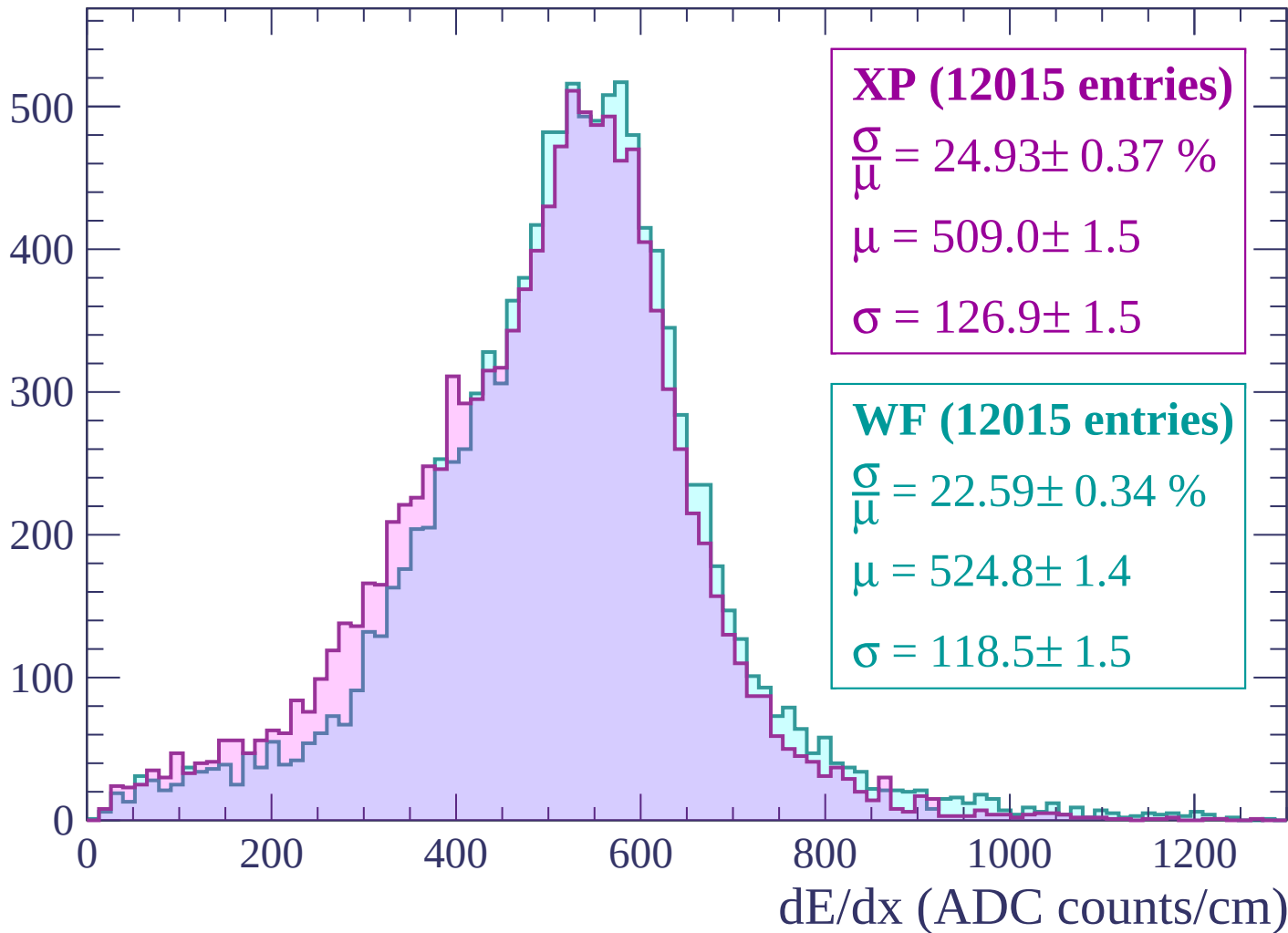
$$\mu = 455.6 \pm 0.9$$

$$\sigma = 69.9 \pm 1.0$$

dE/dx (ADC counts/cm)

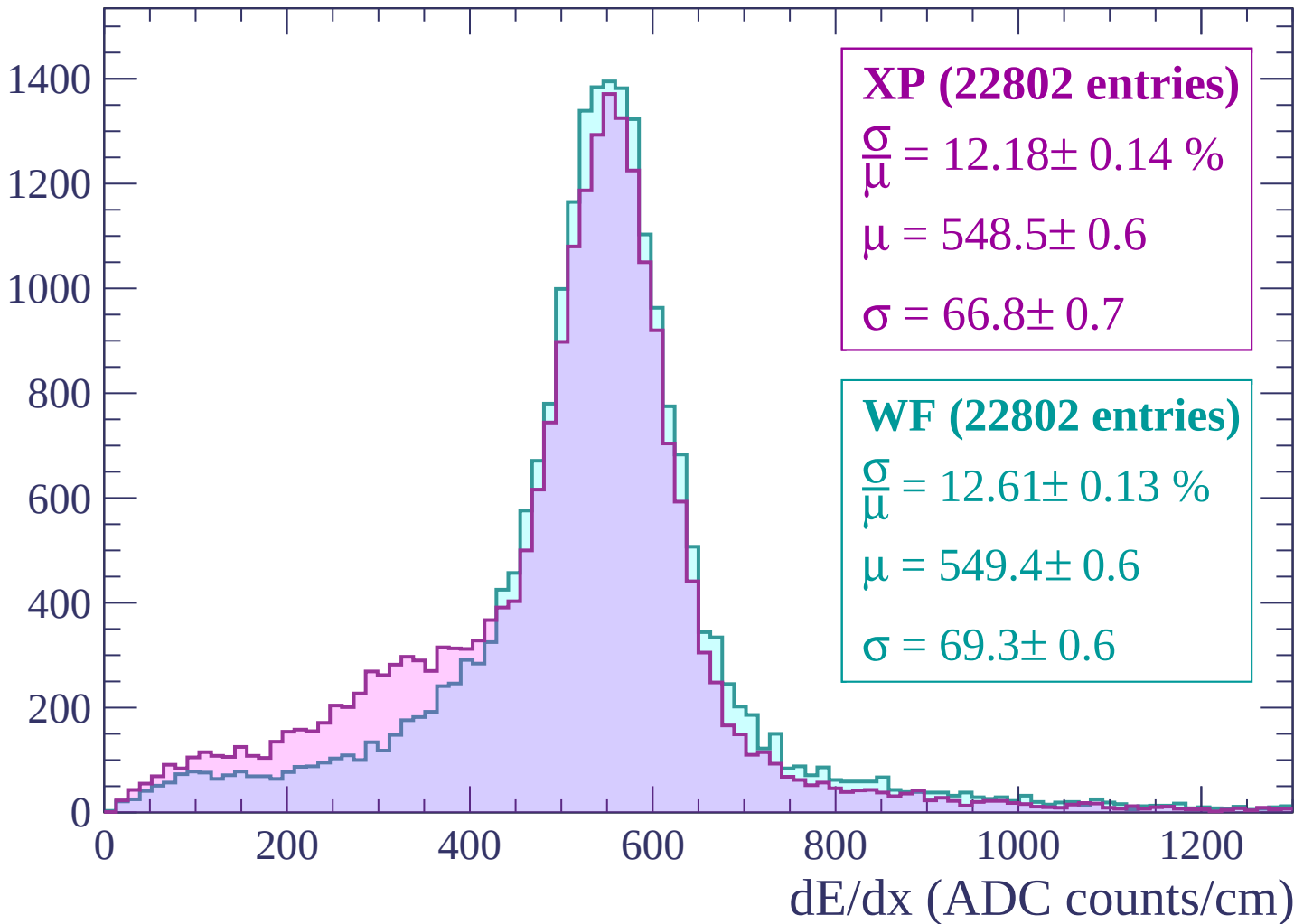
Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count

3500
3000
2500
2000
1500
1000
500
0

XP (54448 entries)

$\frac{\sigma}{\mu} = 15.38 \pm 0.11 \%$

$\mu = 496.0 \pm 0.4$

$\sigma = 76.3 \pm 0.5$

WF (54448 entries)

$\frac{\sigma}{\mu} = 14.78 \pm 0.10 \%$

$\mu = 501.4 \pm 0.4$

$\sigma = 74.1 \pm 0.4$

0

200

400

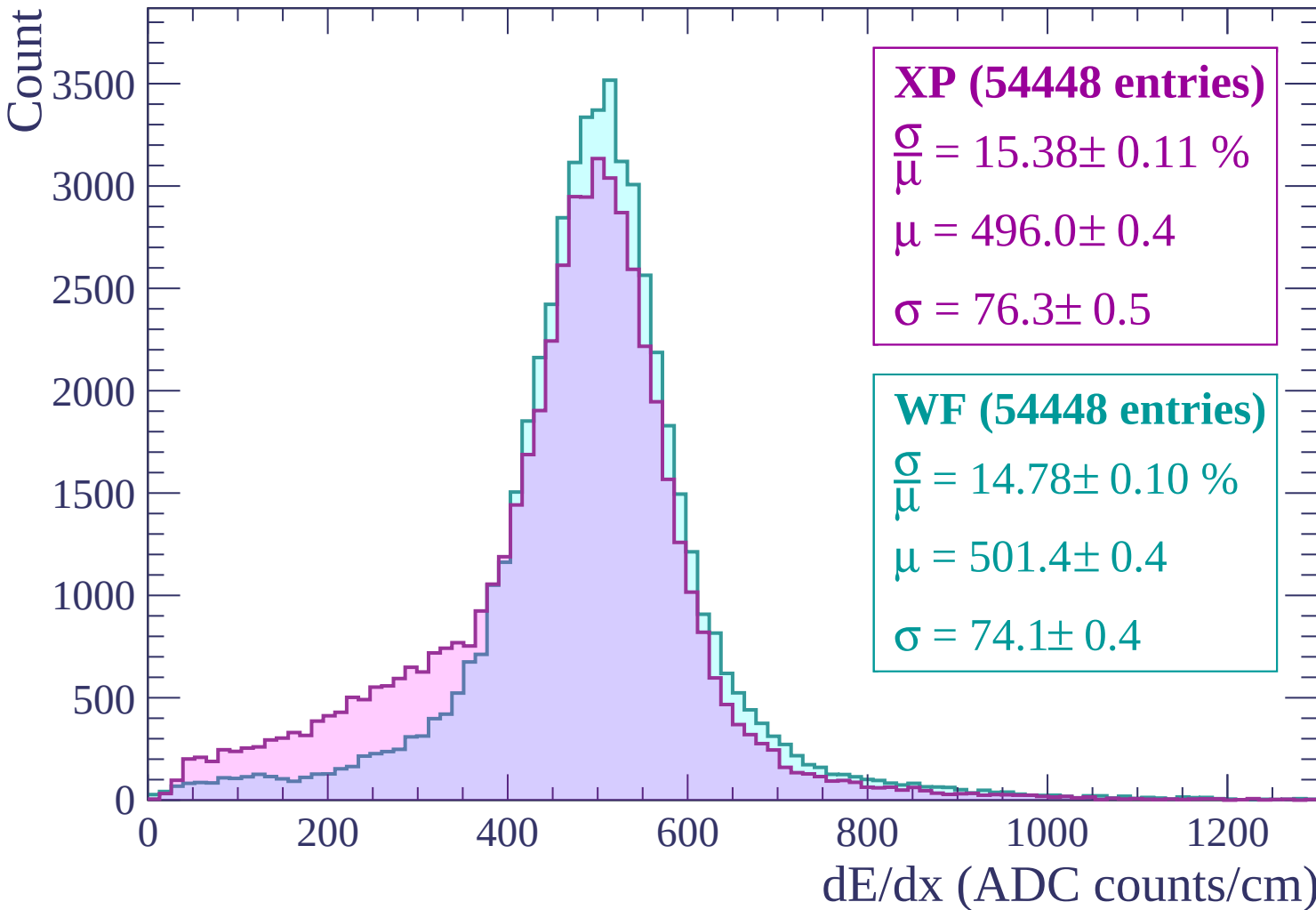
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $60 < p < 120$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (18423 entries)

$\frac{\sigma}{\mu} = 11.67 \pm 0.14 \%$

$\mu = 542.6 \pm 0.7$

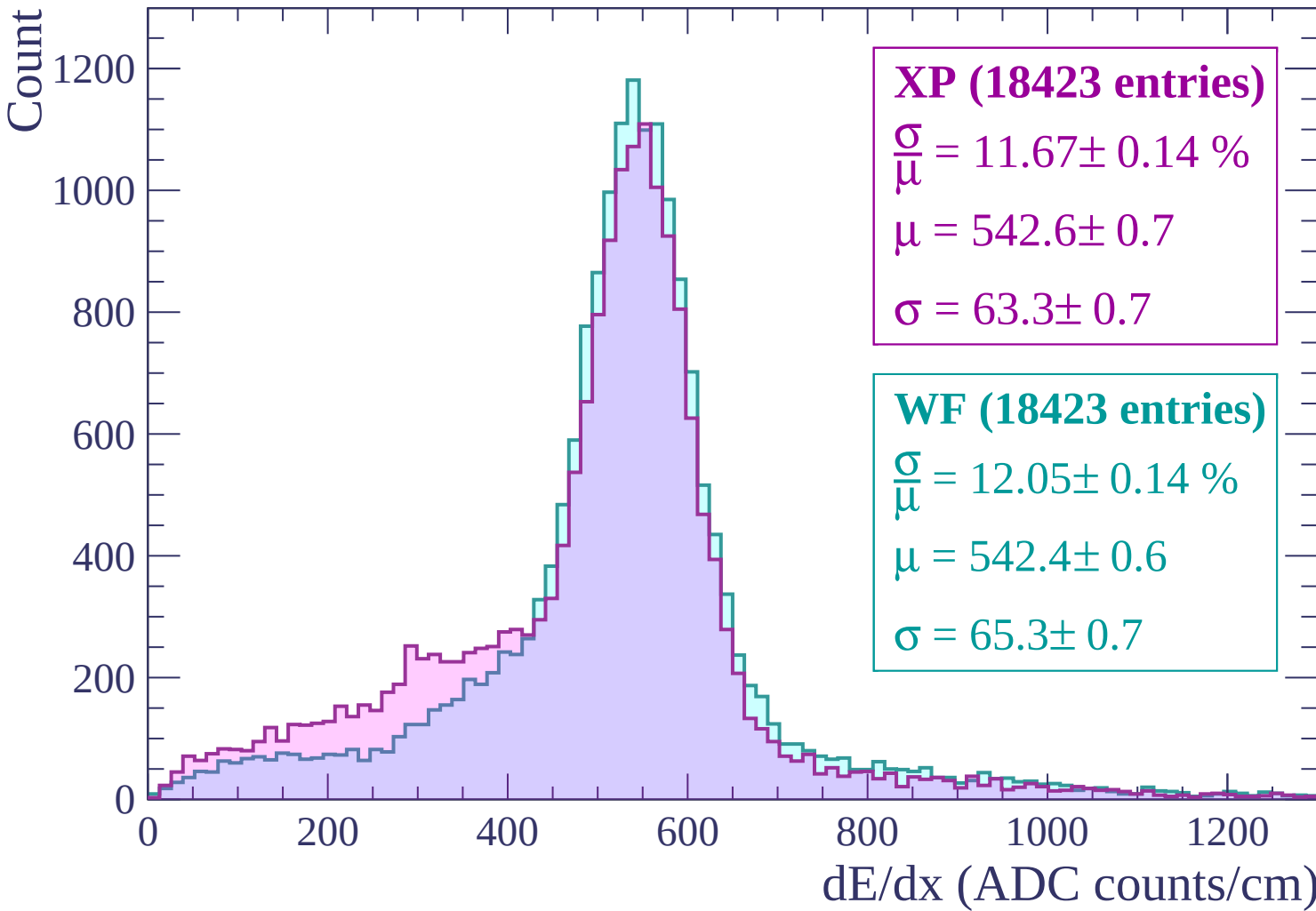
$\sigma = 63.3 \pm 0.7$

WF (18423 entries)

$\frac{\sigma}{\mu} = 12.05 \pm 0.14 \%$

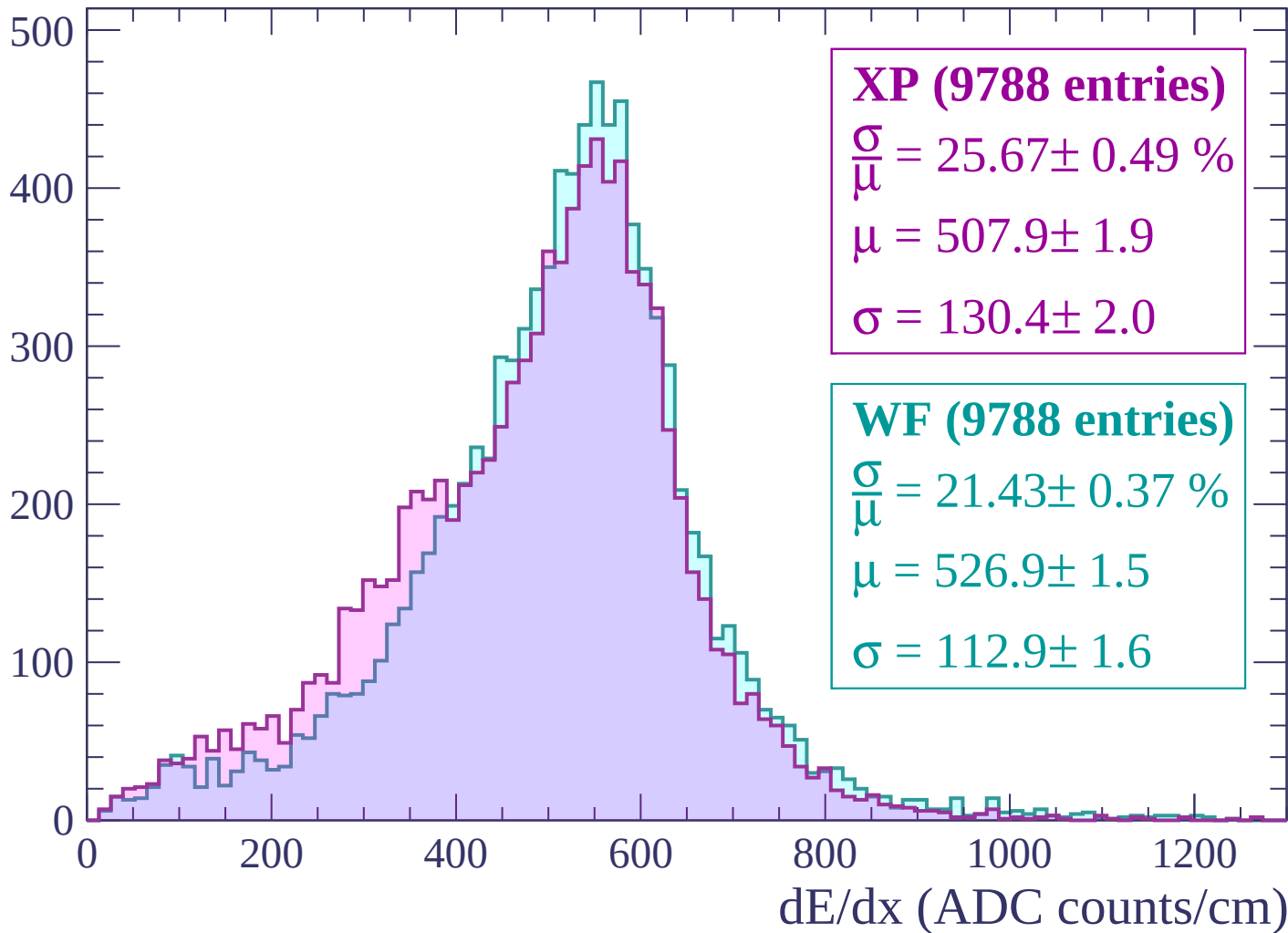
$\mu = 542.4 \pm 0.6$

$\sigma = 65.3 \pm 0.7$



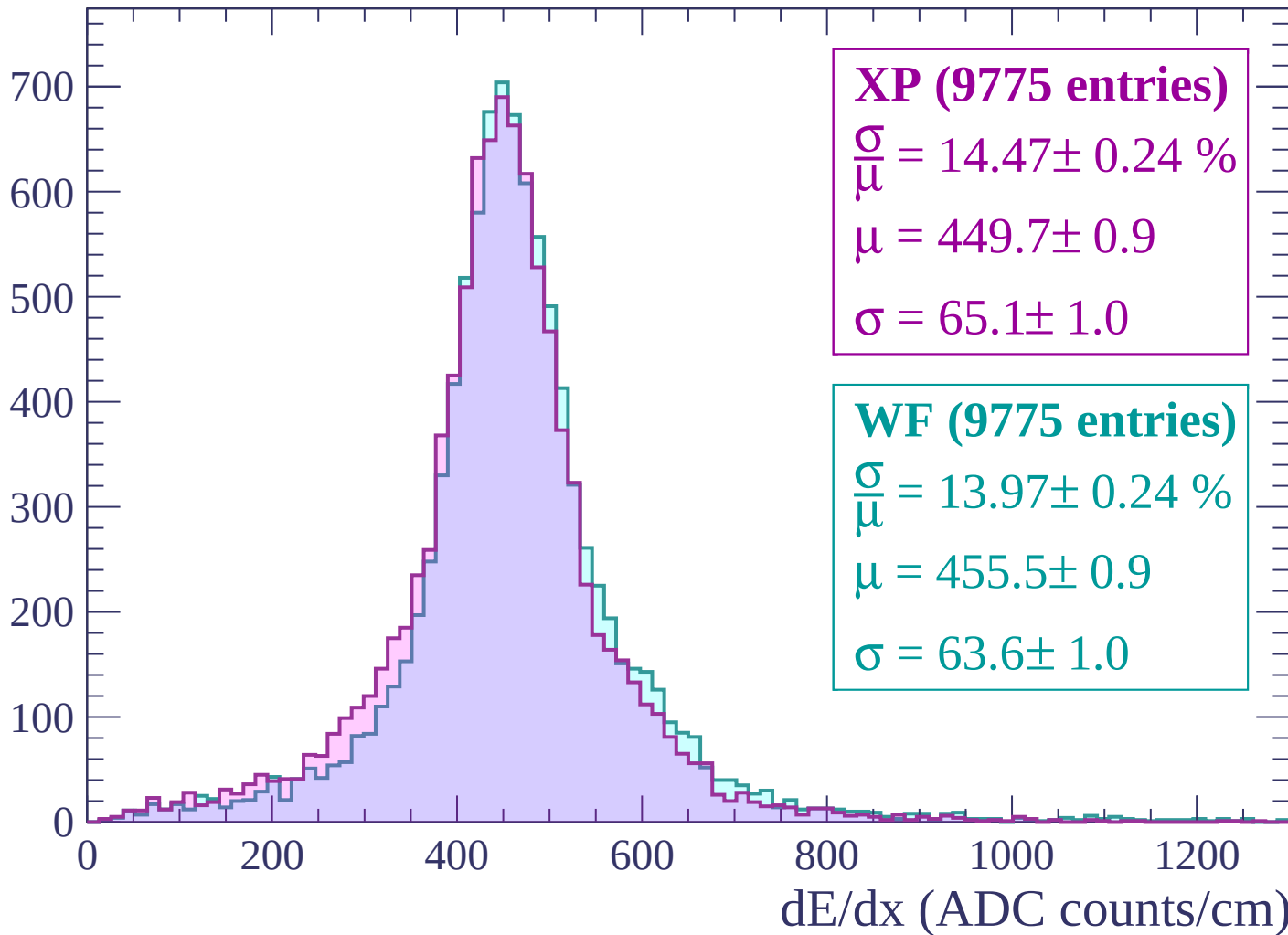
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count

1000

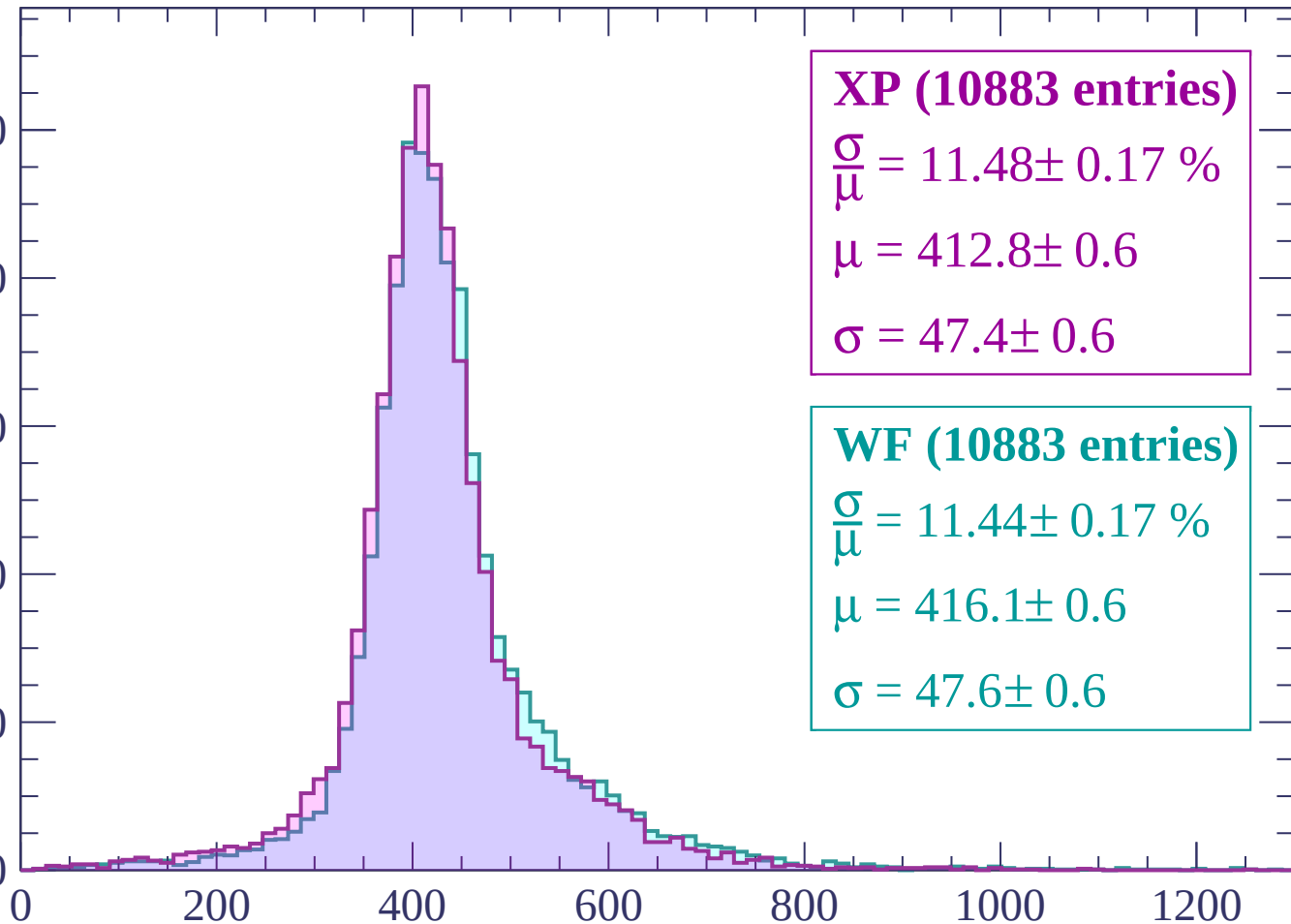
800

600

400

200

0



XP (10883 entries)

$\frac{\sigma}{\mu} = 11.48 \pm 0.17 \%$

$\mu = 412.8 \pm 0.6$

$\sigma = 47.4 \pm 0.6$

WF (10883 entries)

$\frac{\sigma}{\mu} = 11.44 \pm 0.17 \%$

$\mu = 416.1 \pm 0.6$

$\sigma = 47.6 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count

1400
1200
1000
800
600
400
200
0

XP (12927 entries)

$$\frac{\sigma}{\mu} = 10.02 \pm 0.13 \%$$

$$\mu = 400.9 \pm 0.5$$

$$\sigma = 40.2 \pm 0.5$$

WF (12927 entries)

$$\frac{\sigma}{\mu} = 10.01 \pm 0.13 \%$$

$$\mu = 403.7 \pm 0.5$$

$$\sigma = 40.4 \pm 0.5$$

0

200

400

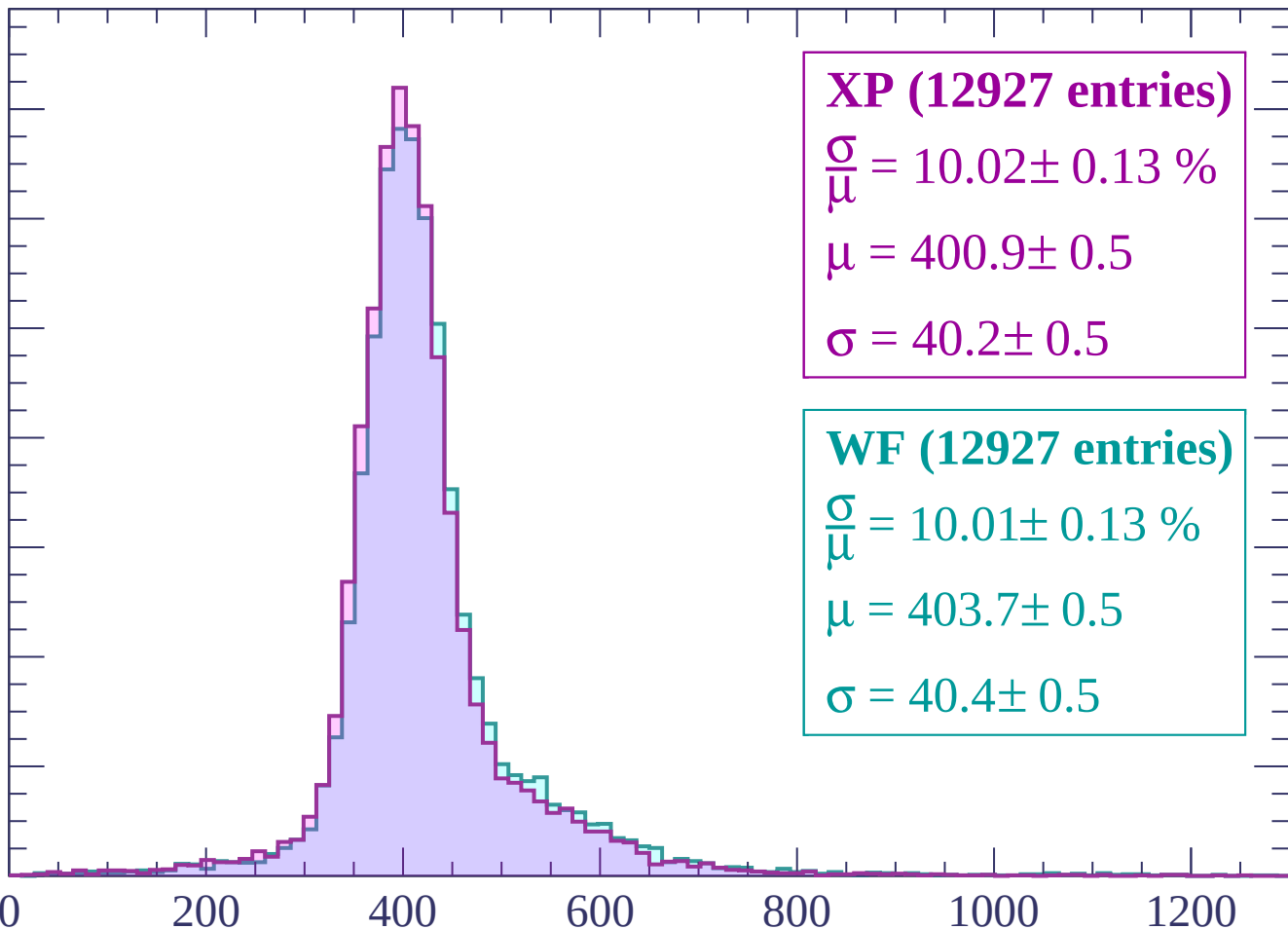
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $360 < p < 420$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13426 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.13 \%$$

$$\mu = 398.4 \pm 0.4$$

$$\sigma = 39.8 \pm 0.5$$

WF (13426 entries)

$$\frac{\sigma}{\mu} = 10.20 \pm 0.13 \%$$

$$\mu = 401.0 \pm 0.5$$

$$\sigma = 40.9 \pm 0.5$$

0

200

400

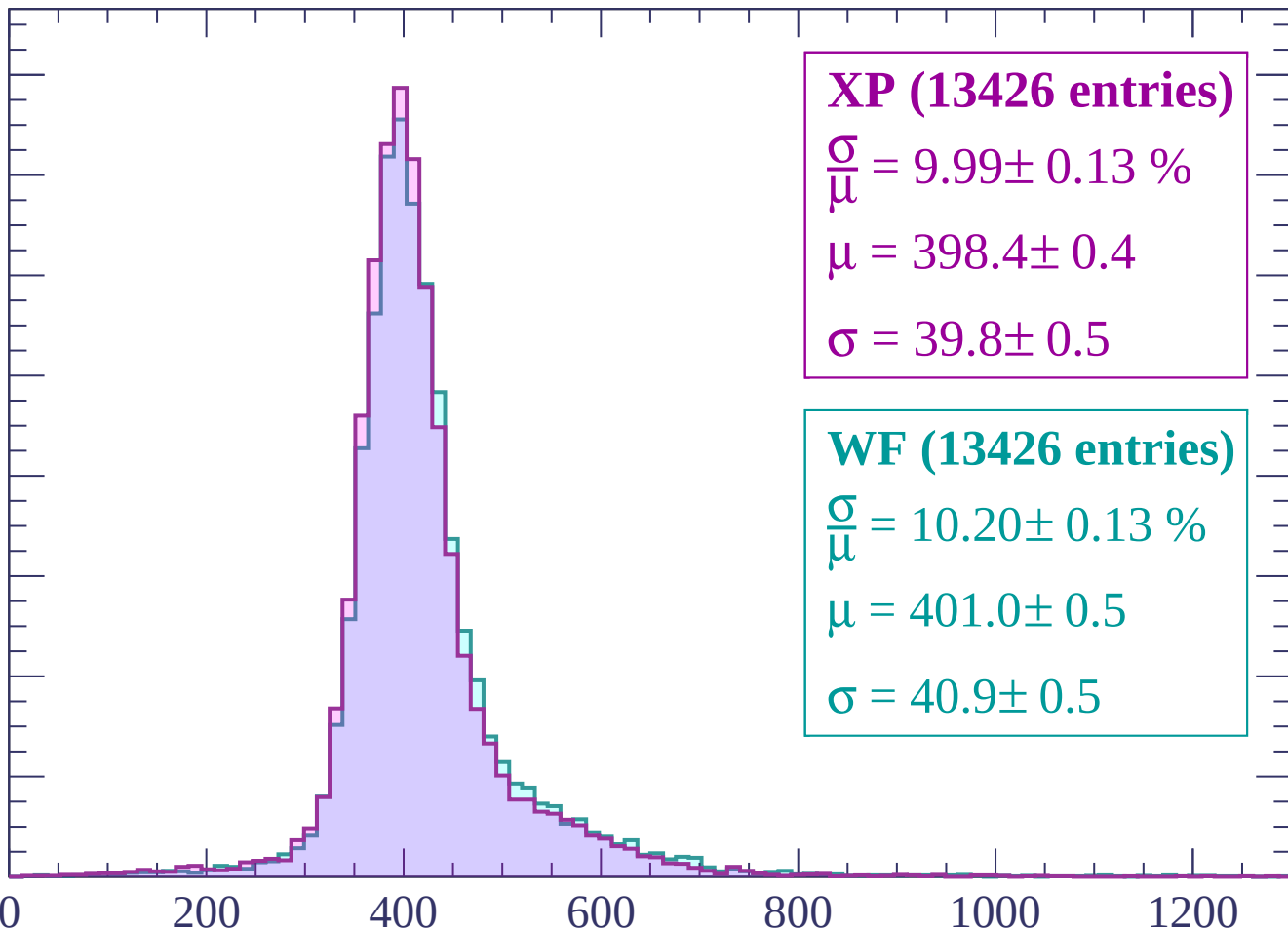
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $420 < p < 480$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13038 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.12 \%$$

$$\mu = 400.5 \pm 0.4$$

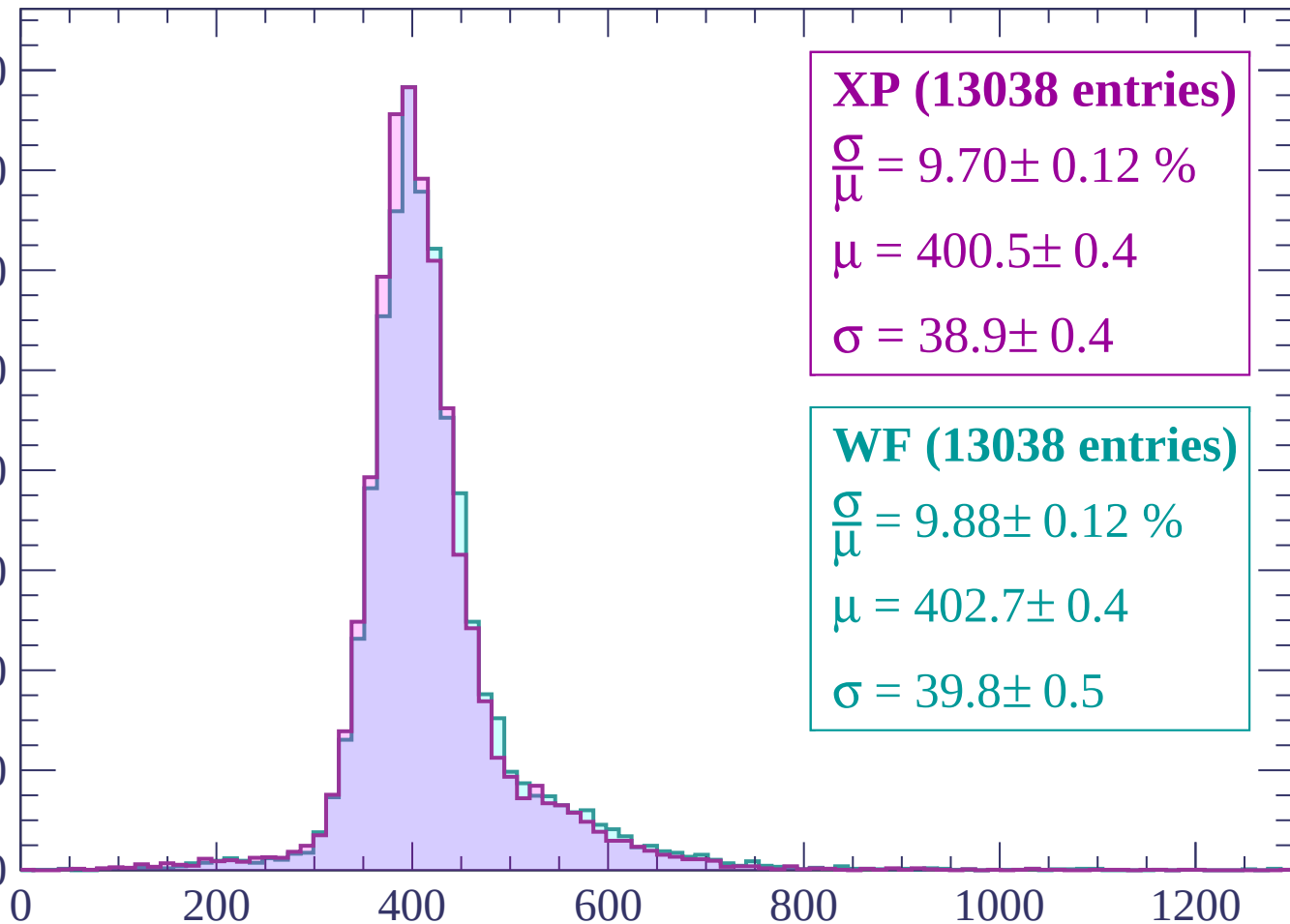
$$\sigma = 38.9 \pm 0.4$$

WF (13038 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.12 \%$$

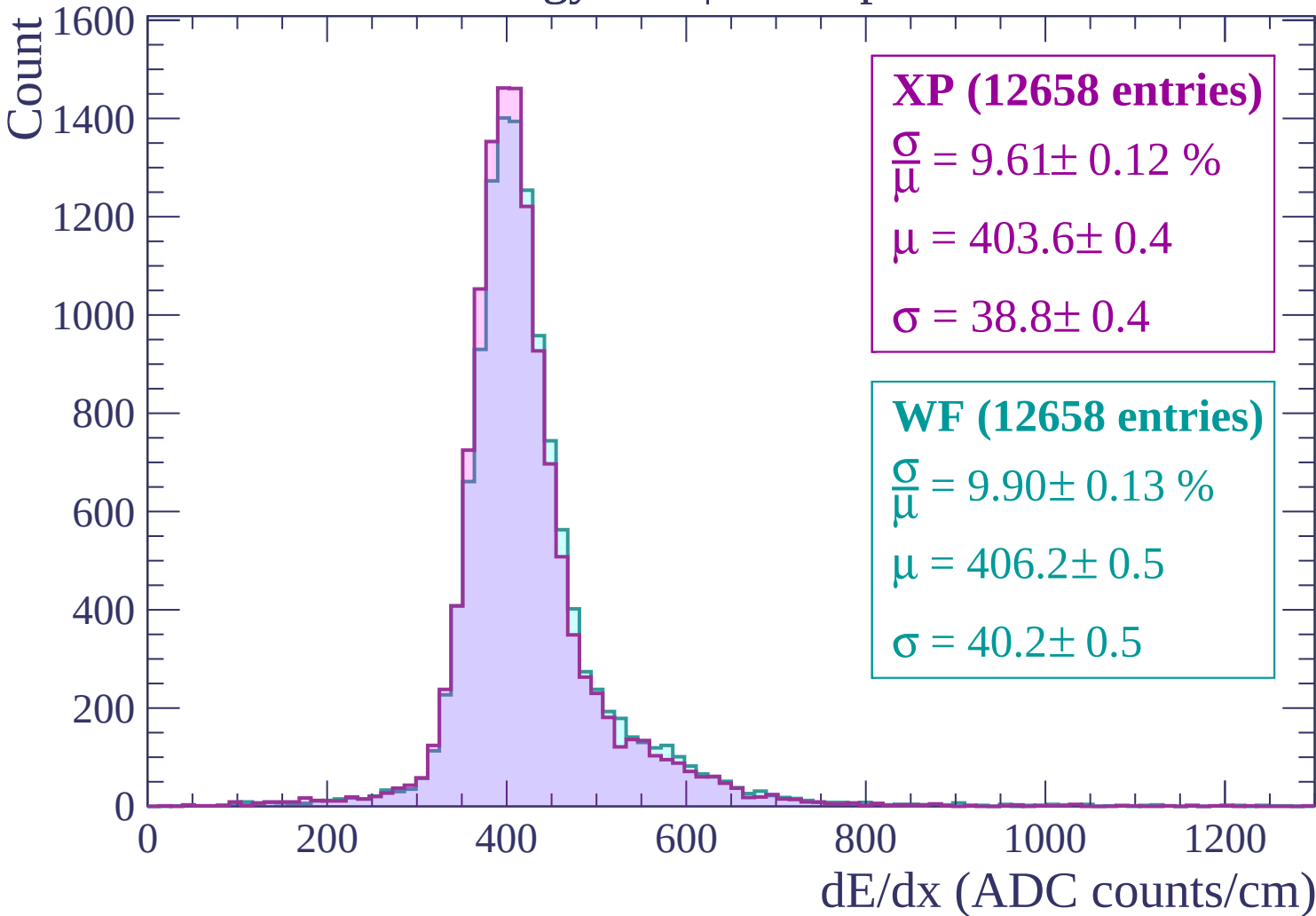
$$\mu = 402.7 \pm 0.4$$

$$\sigma = 39.8 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$

Count

1400
1200
1000
800
600
400
200
0

XP (12149 entries)

$$\frac{\sigma}{\mu} = 9.56 \pm 0.12 \%$$

$$\mu = 406.6 \pm 0.5$$

$$\sigma = 38.9 \pm 0.4$$

WF (12149 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.12 \%$$

$$\mu = 410.0 \pm 0.5$$

$$\sigma = 41.0 \pm 0.4$$

0

200

400

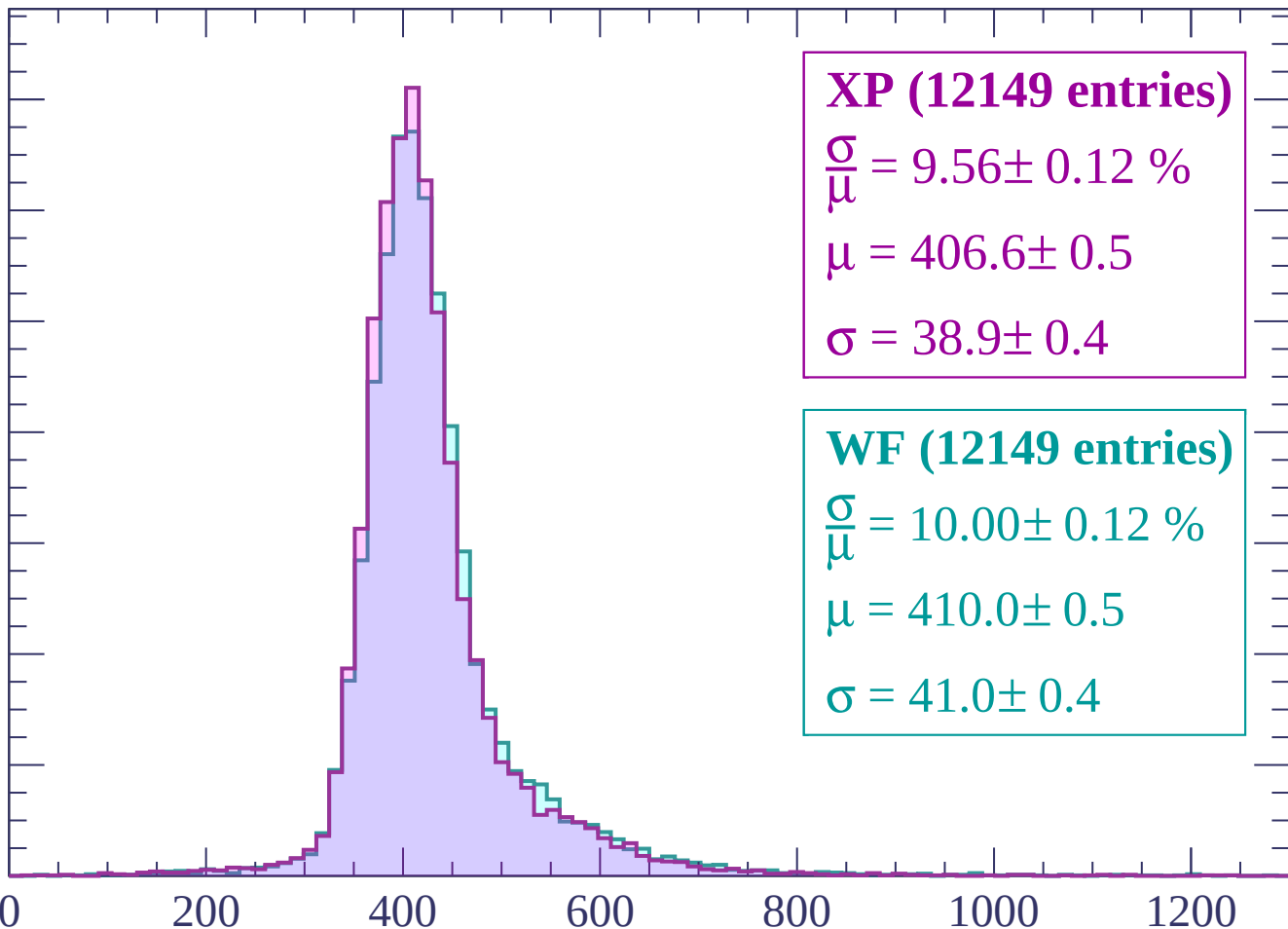
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $600 < p < 660$

Count

1400
1200
1000
800
600
400
200
0

XP (11583 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.12 \%$$

$$\mu = 409.7 \pm 0.5$$

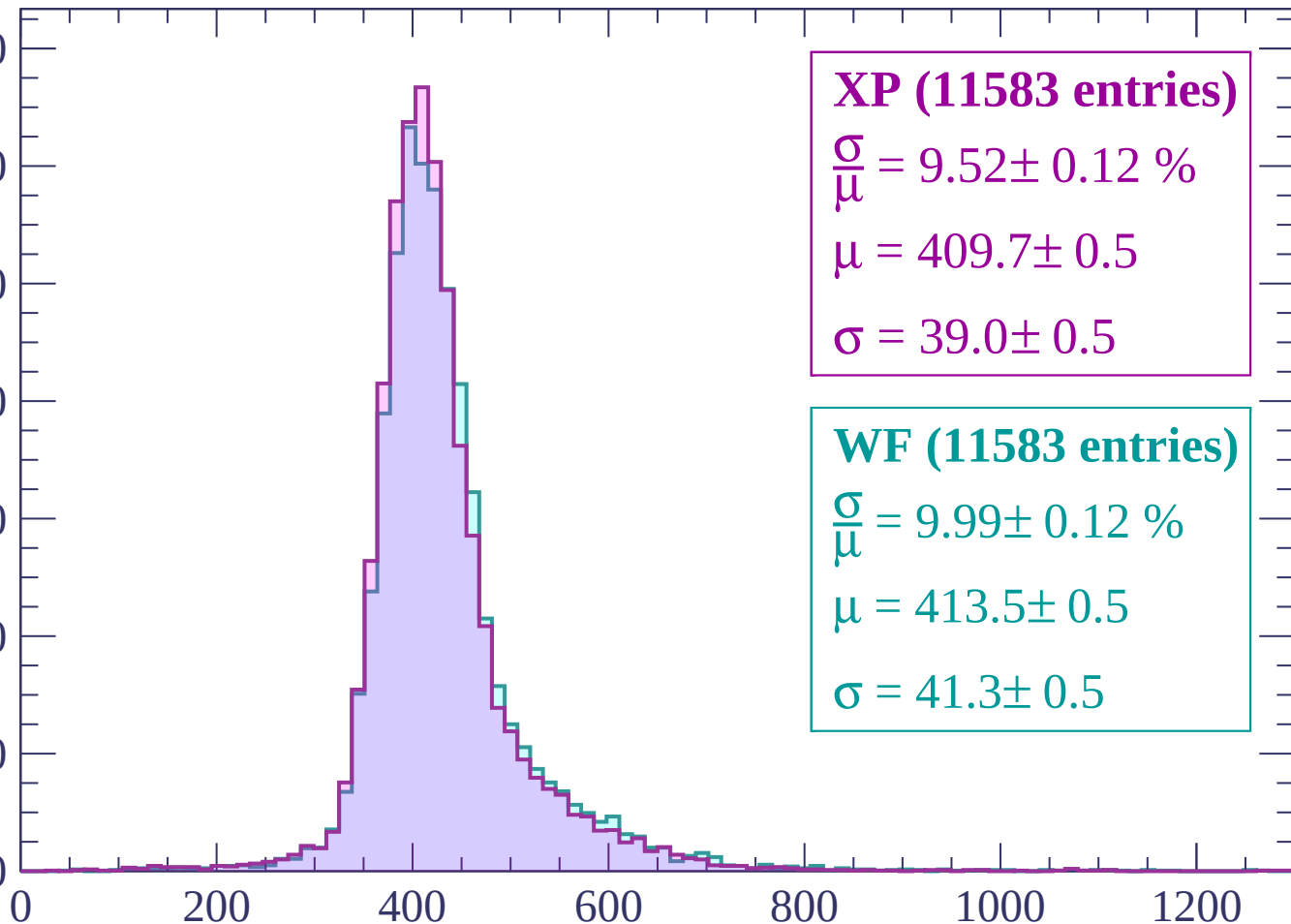
$$\sigma = 39.0 \pm 0.5$$

WF (11583 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.12 \%$$

$$\mu = 413.5 \pm 0.5$$

$$\sigma = 41.3 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (11178 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.14 \%$$

$$\mu = 413.0 \pm 0.5$$

$$\sigma = 40.6 \pm 0.5$$

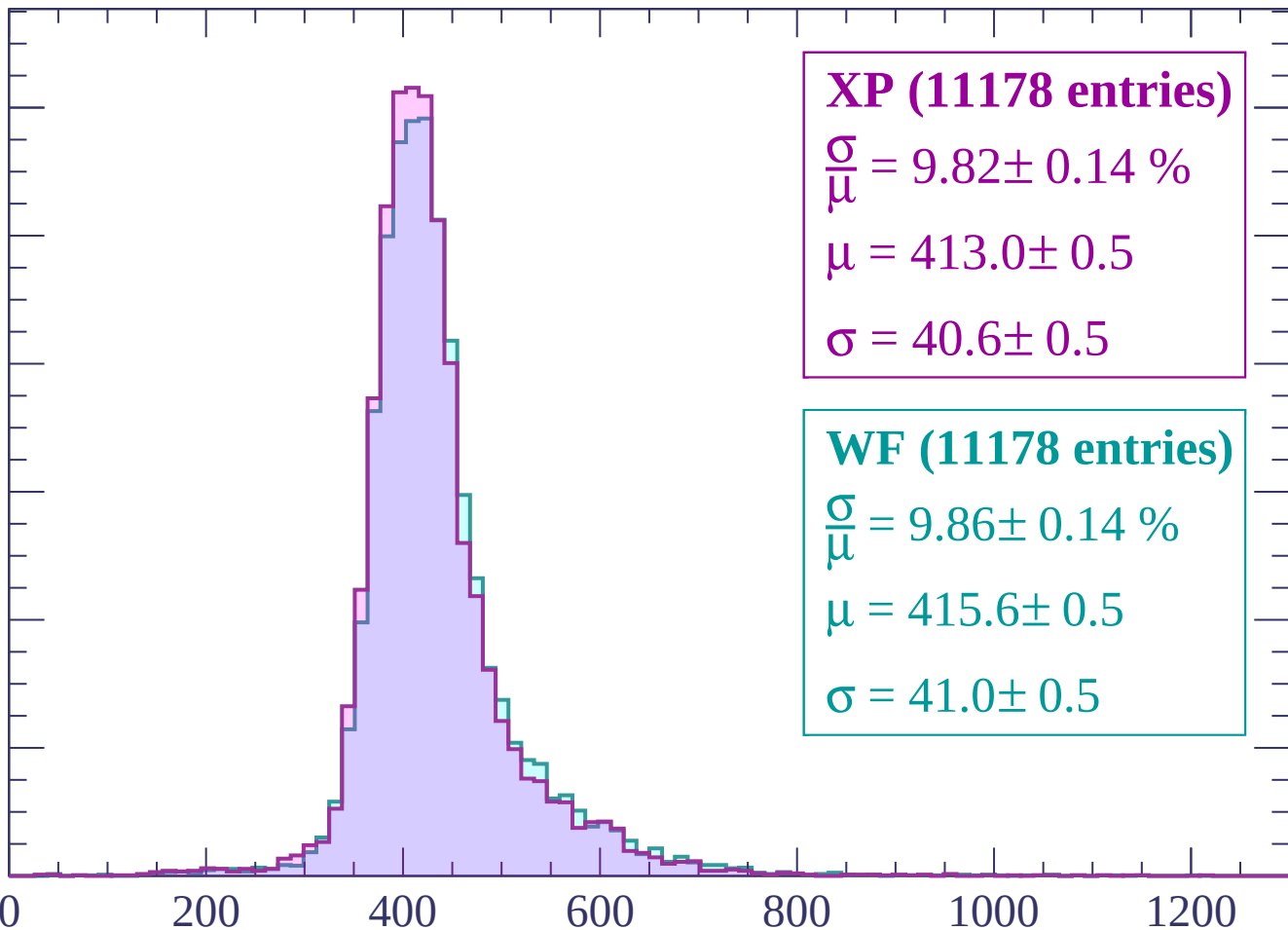
WF (11178 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.14 \%$$

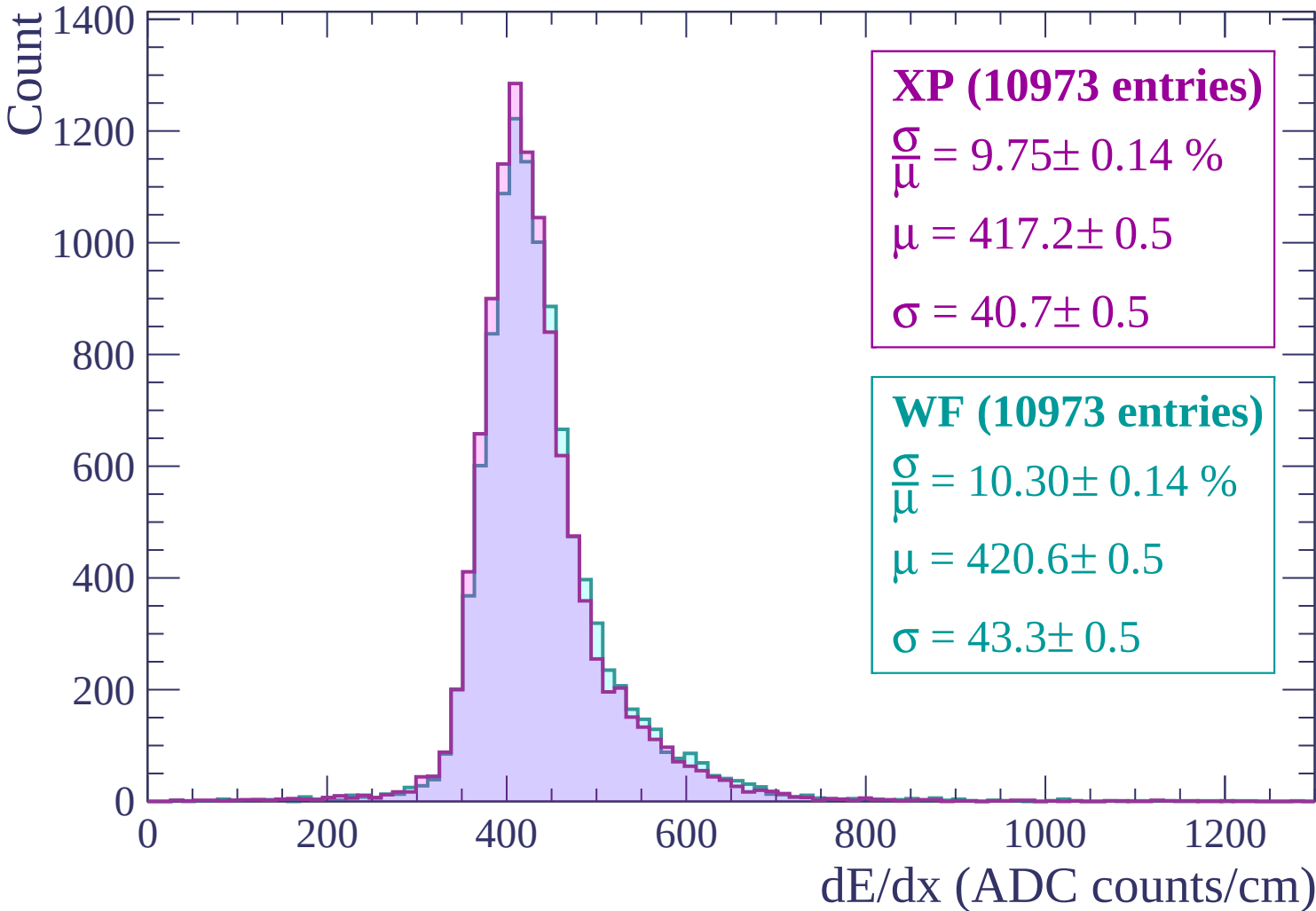
$$\mu = 415.6 \pm 0.5$$

$$\sigma = 41.0 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $720 < p < 780$



Energy loss | $780 < p < 840$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (10256 entries)

$\frac{\sigma}{\mu} = 9.51 \pm 0.13 \%$

$\mu = 420.5 \pm 0.5$

$\sigma = 40.0 \pm 0.5$

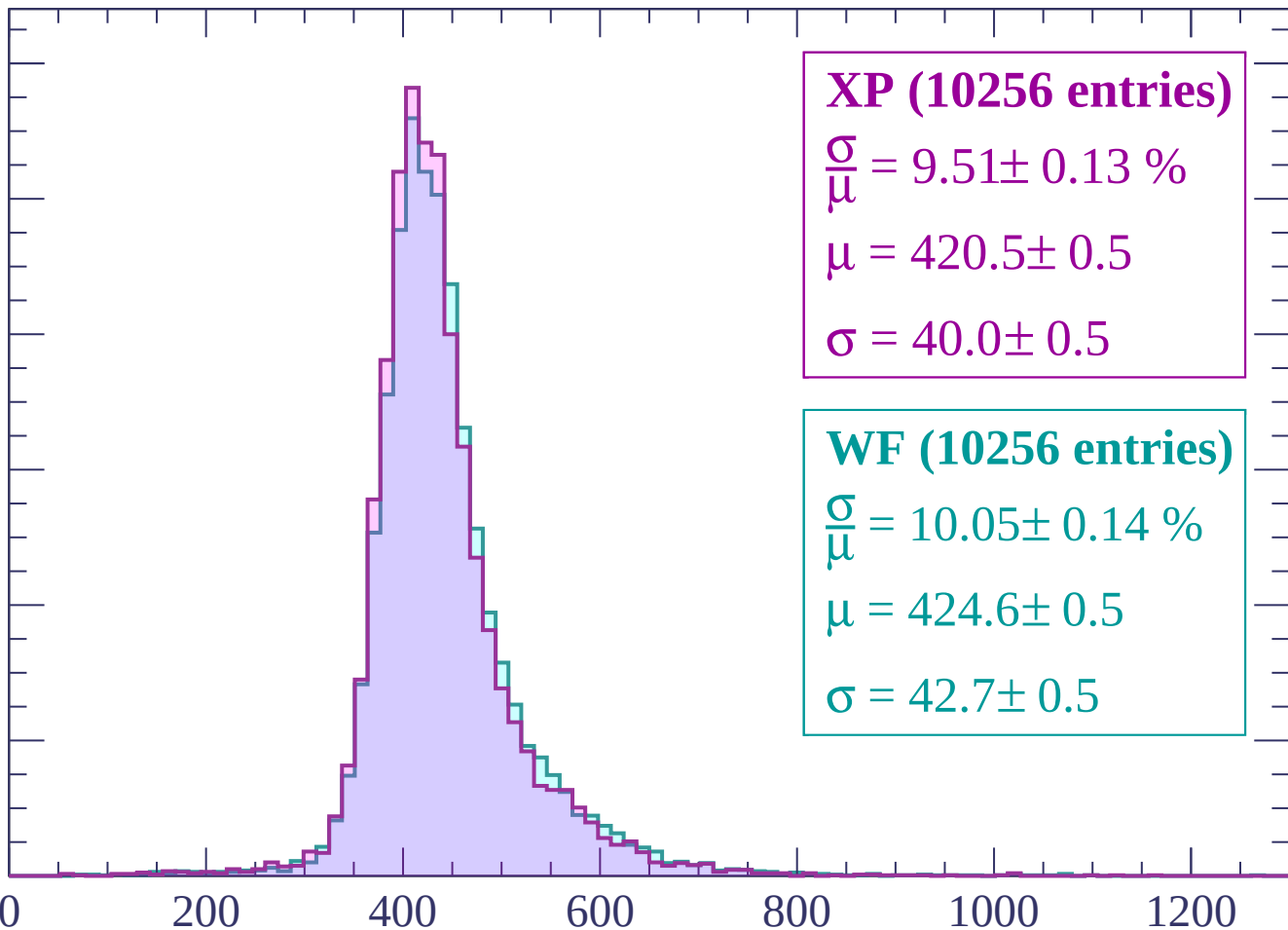
WF (10256 entries)

$\frac{\sigma}{\mu} = 10.05 \pm 0.14 \%$

$\mu = 424.6 \pm 0.5$

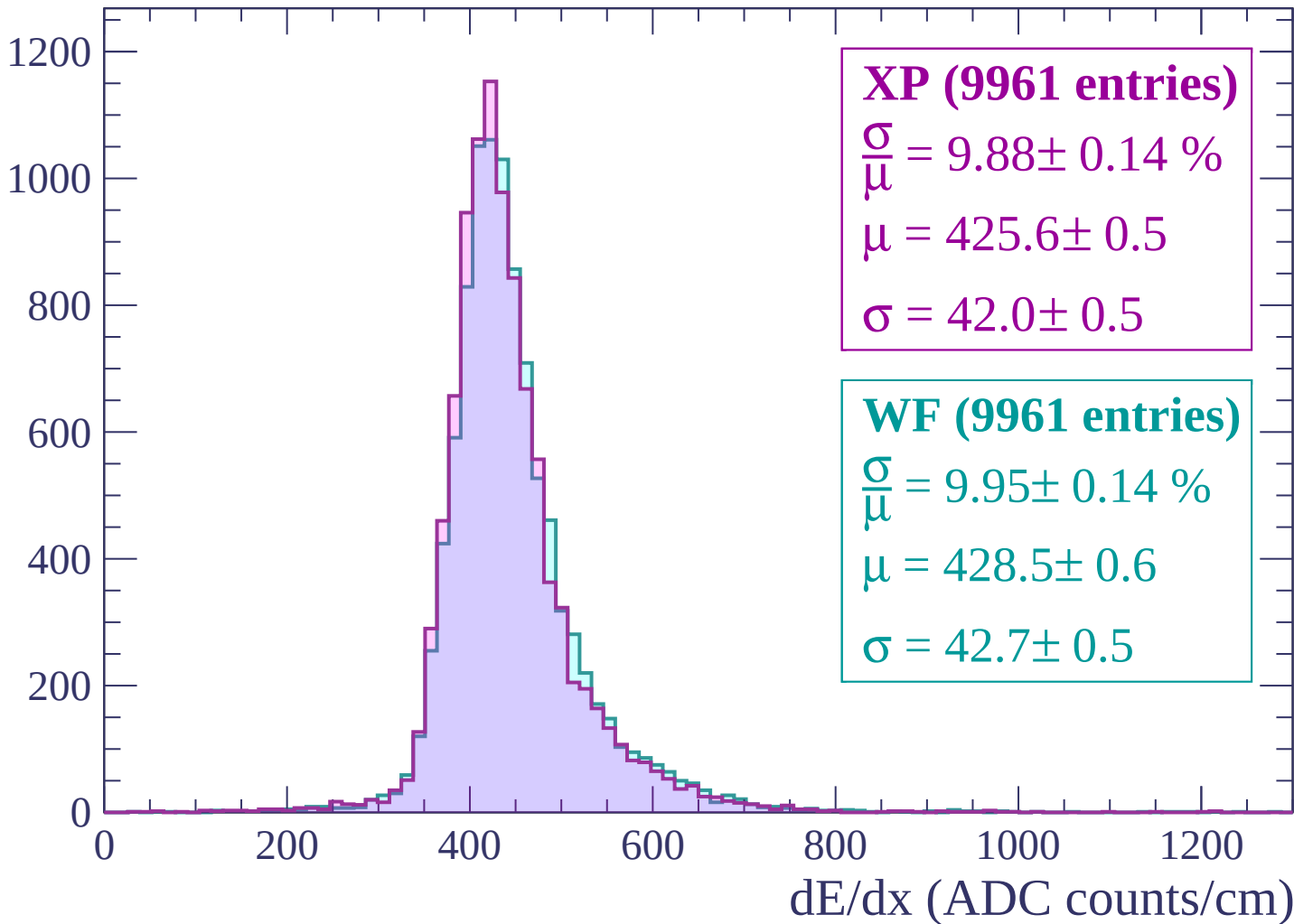
$\sigma = 42.7 \pm 0.5$

dE/dx (ADC counts/cm)

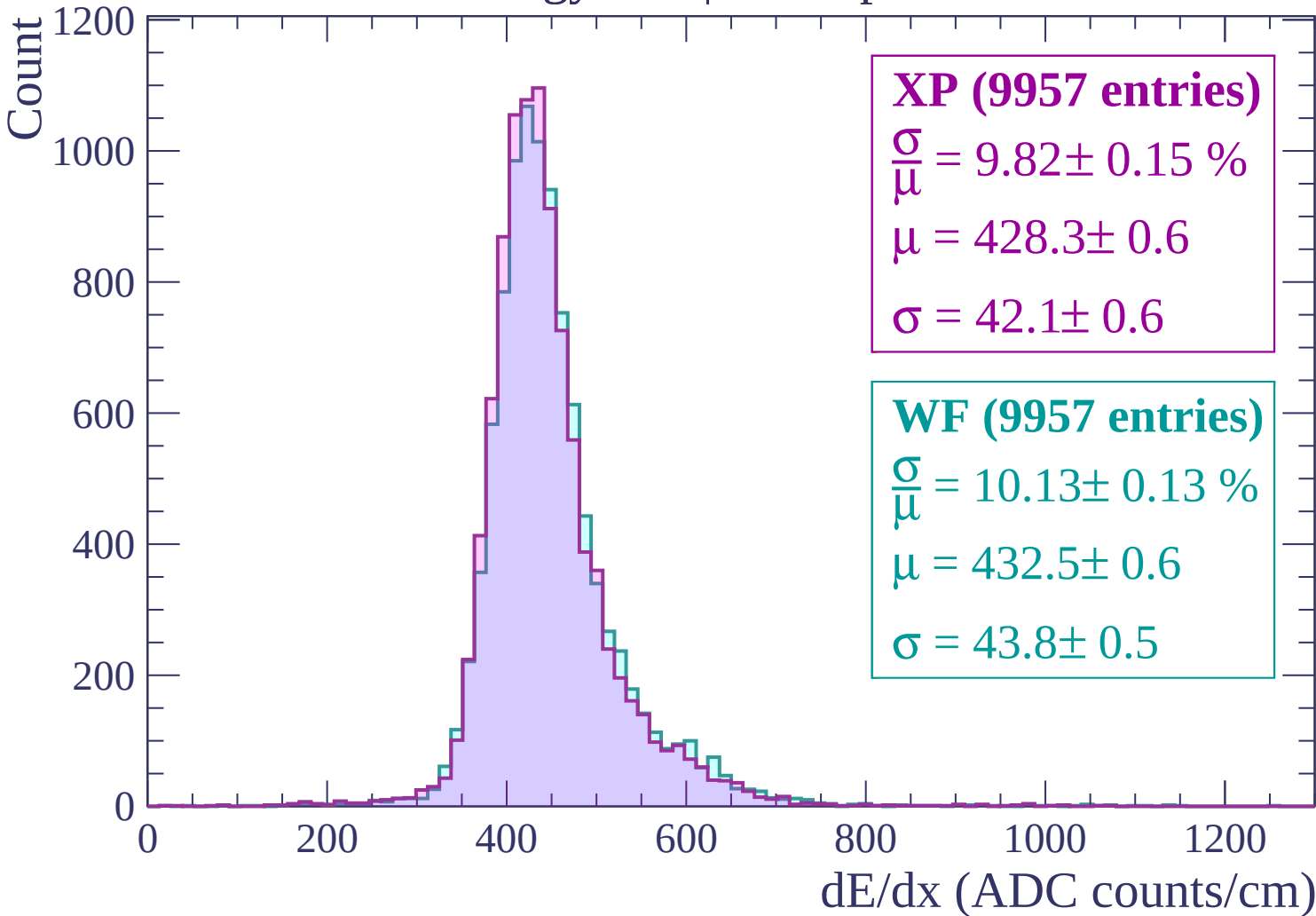


Energy loss | $840 < p < 900$

Count



Energy loss | $900 < p < 960$



Energy loss | $960 < p < 1020$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (9353 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.15 \%$$

$$\mu = 431.3 \pm 0.6$$

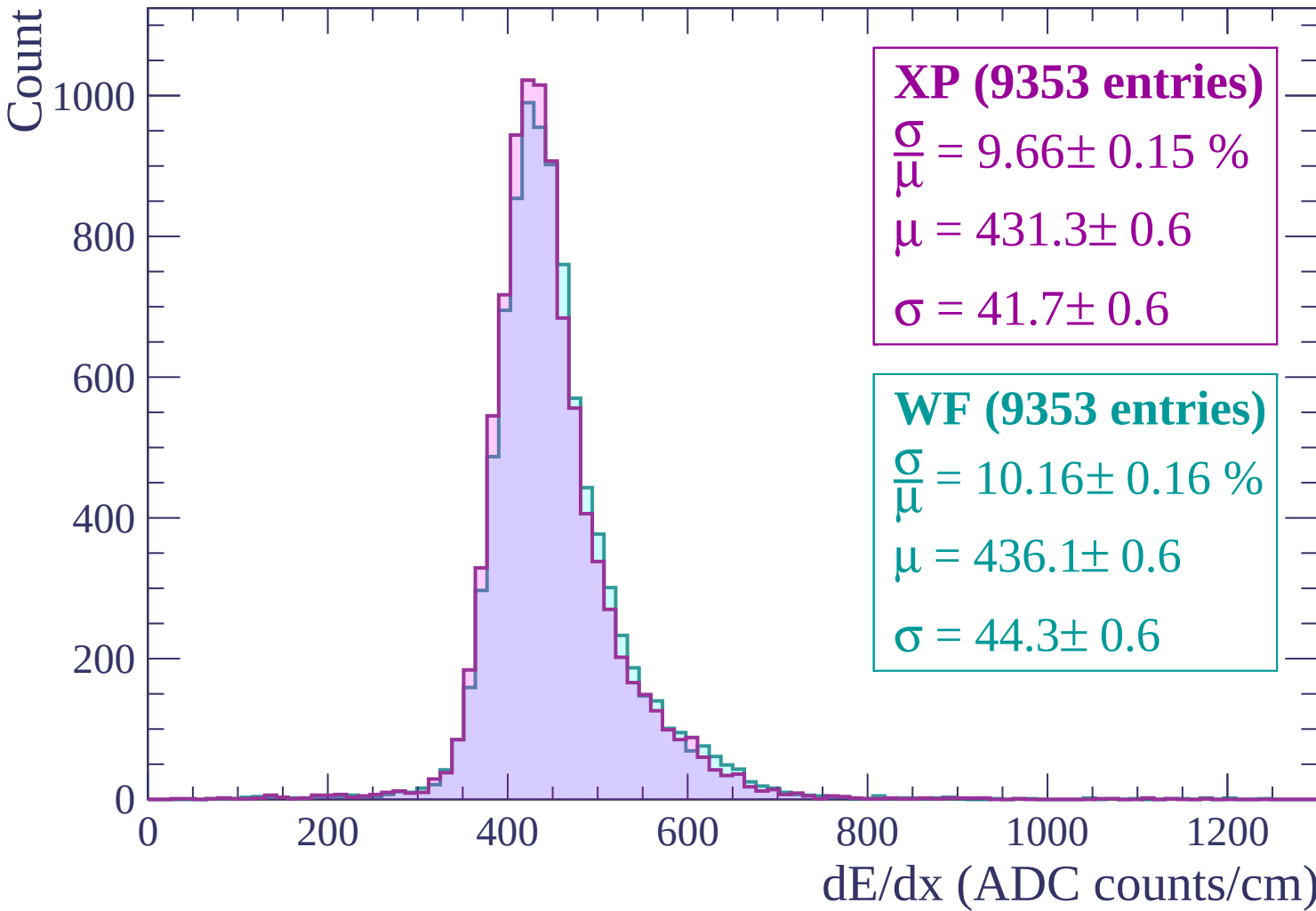
$$\sigma = 41.7 \pm 0.6$$

WF (9353 entries)

$$\frac{\sigma}{\mu} = 10.16 \pm 0.16 \%$$

$$\mu = 436.1 \pm 0.6$$

$$\sigma = 44.3 \pm 0.6$$



Energy loss | $1020 < p < 1080$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9094 entries)

$$\frac{\sigma}{\mu} = 9.83 \pm 0.14 \%$$

$$\mu = 436.7 \pm 0.6$$

$$\sigma = 43.0 \pm 0.6$$

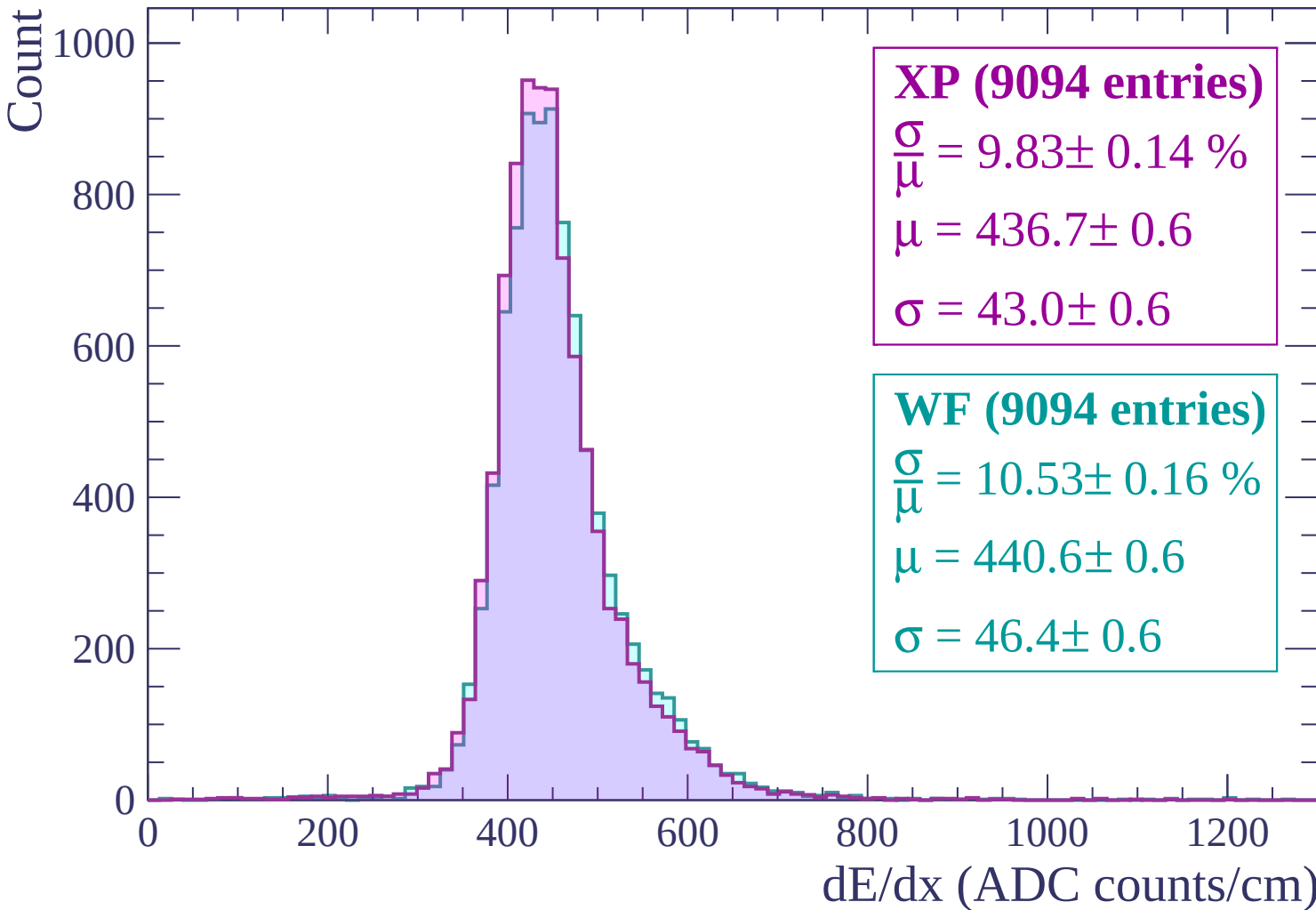
WF (9094 entries)

$$\frac{\sigma}{\mu} = 10.53 \pm 0.16 \%$$

$$\mu = 440.6 \pm 0.6$$

$$\sigma = 46.4 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $1080 < p < 1140$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8770 entries)

$\frac{\sigma}{\mu} = 9.94 \pm 0.15 \%$

$\mu = 439.1 \pm 0.6$

$\sigma = 43.6 \pm 0.6$

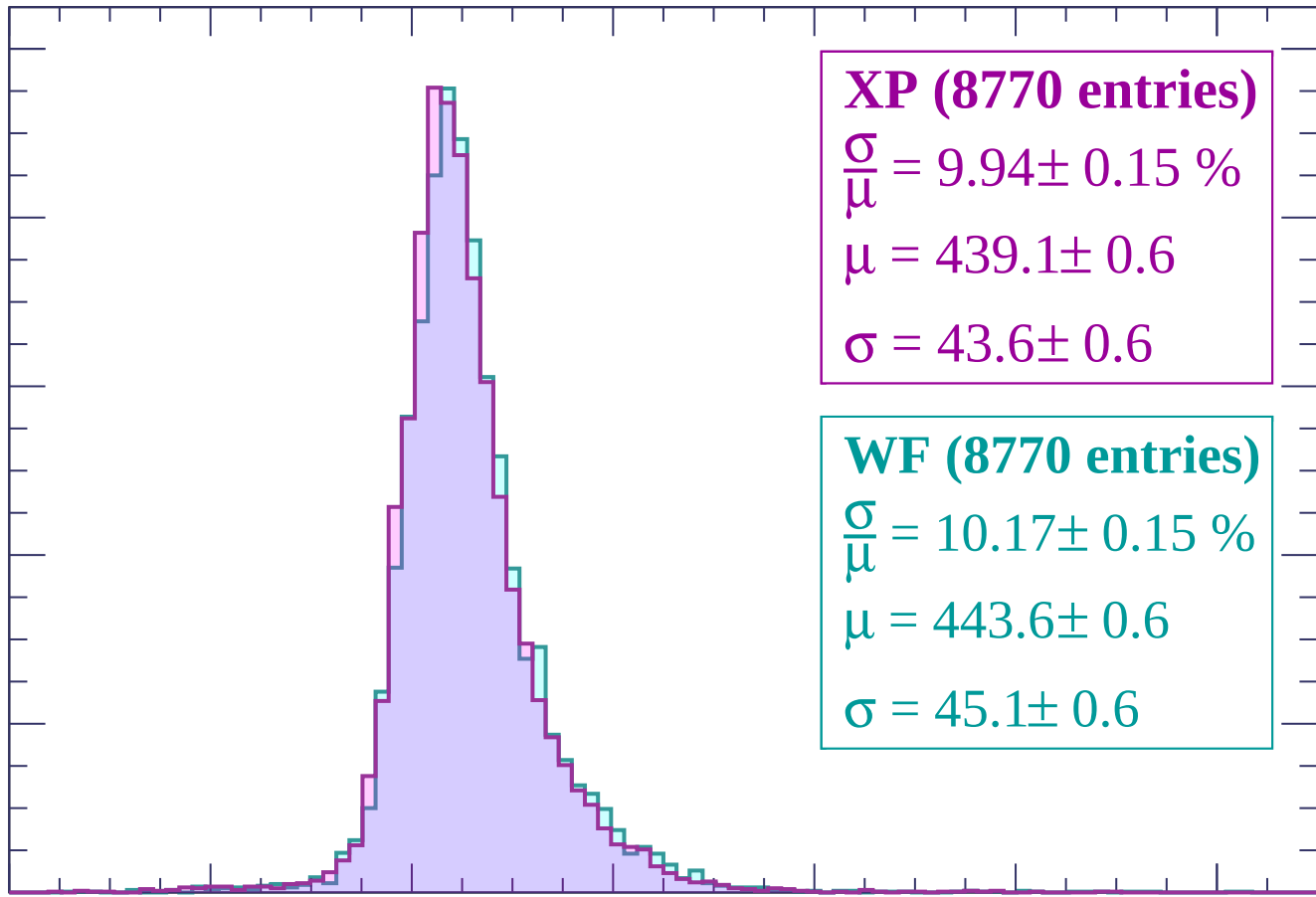
WF (8770 entries)

$\frac{\sigma}{\mu} = 10.17 \pm 0.15 \%$

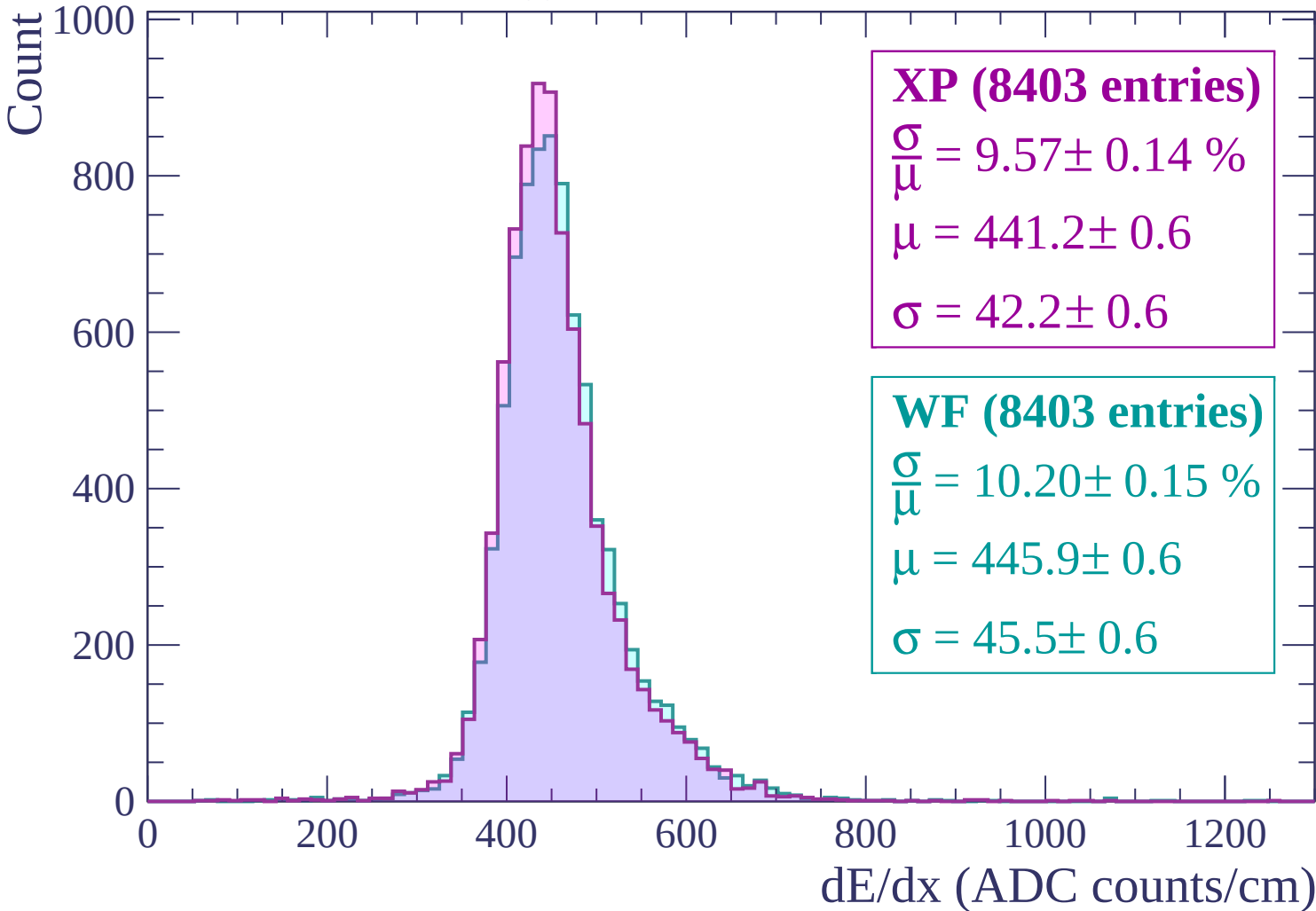
$\mu = 443.6 \pm 0.6$

$\sigma = 45.1 \pm 0.6$

dE/dx (ADC counts/cm)

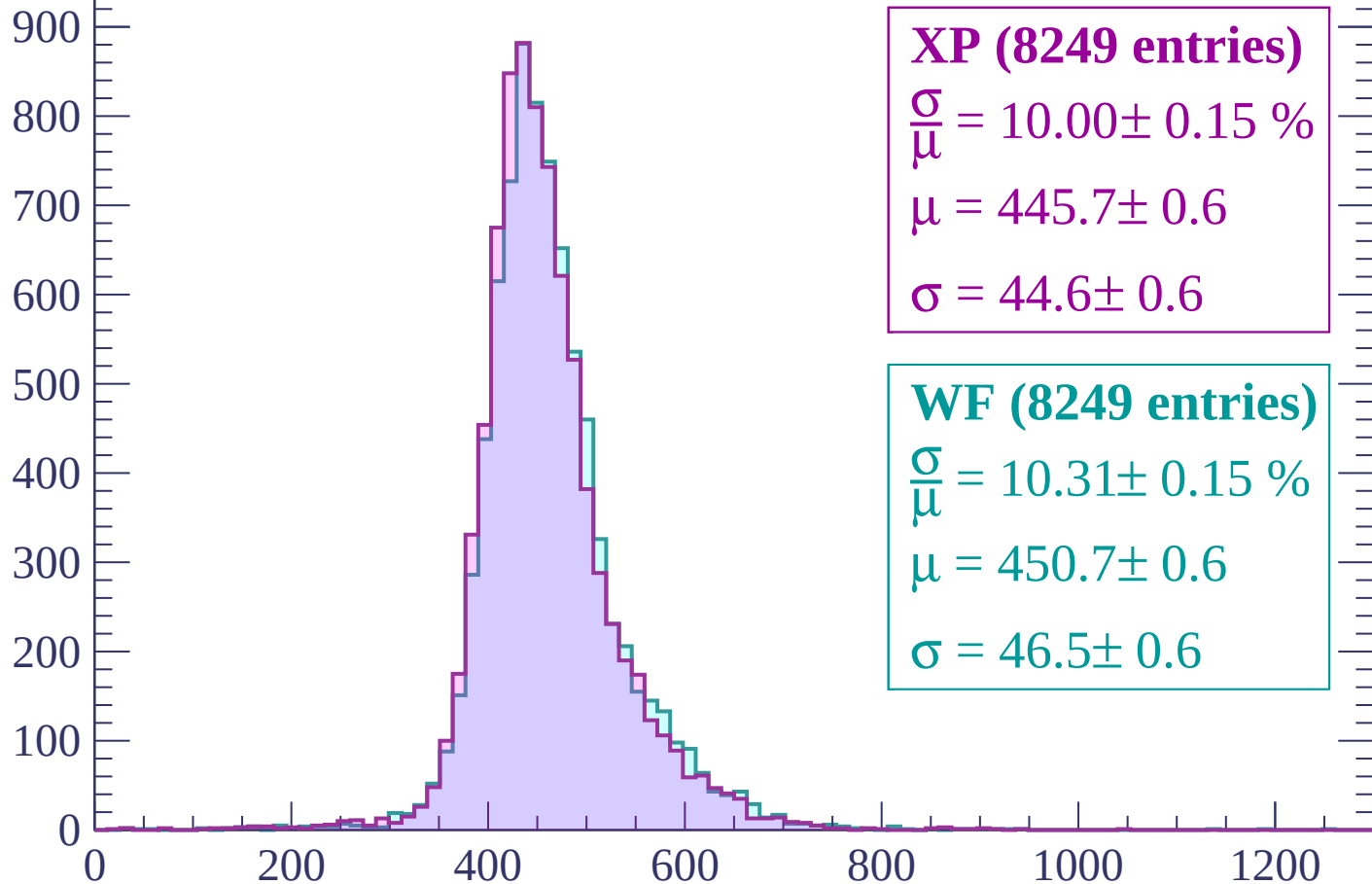


Energy loss | $1140 < p < 1200$



Energy loss | $1200 < p < 1260$

Count



XP (8249 entries)

$\frac{\sigma}{\mu} = 10.00 \pm 0.15 \%$

$\mu = 445.7 \pm 0.6$

$\sigma = 44.6 \pm 0.6$

WF (8249 entries)

$\frac{\sigma}{\mu} = 10.31 \pm 0.15 \%$

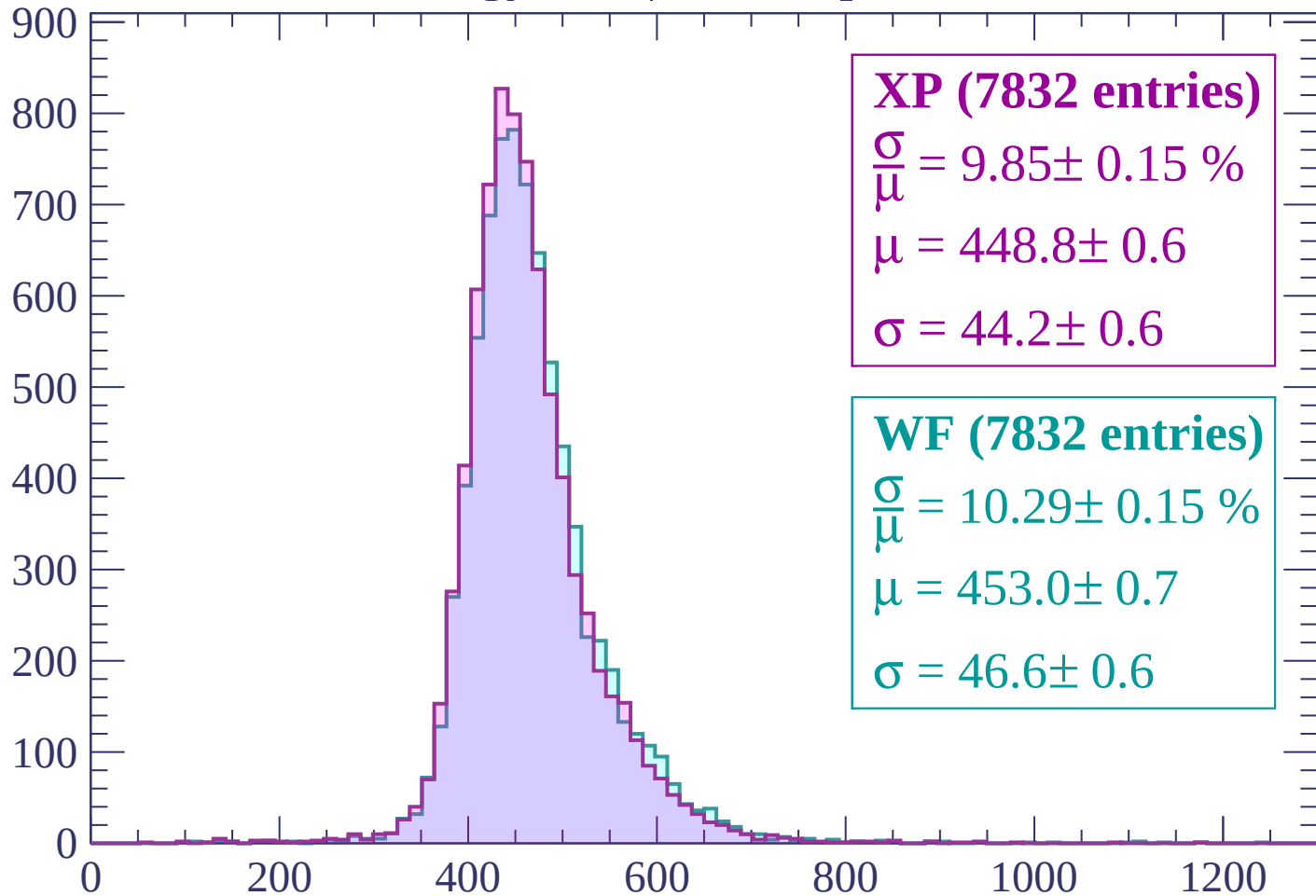
$\mu = 450.7 \pm 0.6$

$\sigma = 46.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (7832 entries)

$$\frac{\sigma}{\mu} = 9.85 \pm 0.15 \%$$

$$\mu = 448.8 \pm 0.6$$

$$\sigma = 44.2 \pm 0.6$$

WF (7832 entries)

$$\frac{\sigma}{\mu} = 10.29 \pm 0.15 \%$$

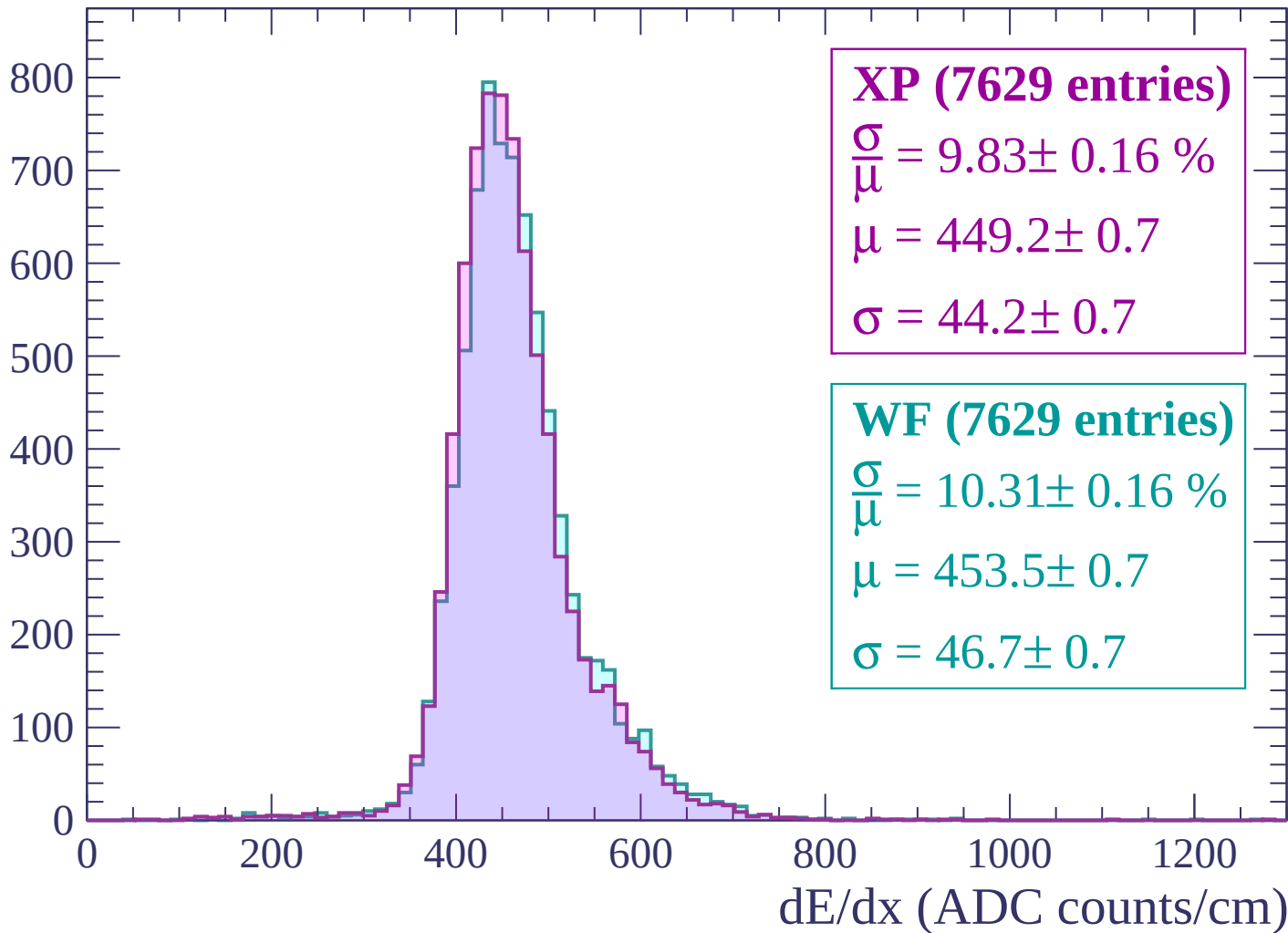
$$\mu = 453.0 \pm 0.7$$

$$\sigma = 46.6 \pm 0.6$$

dE/dx (ADC counts/cm)

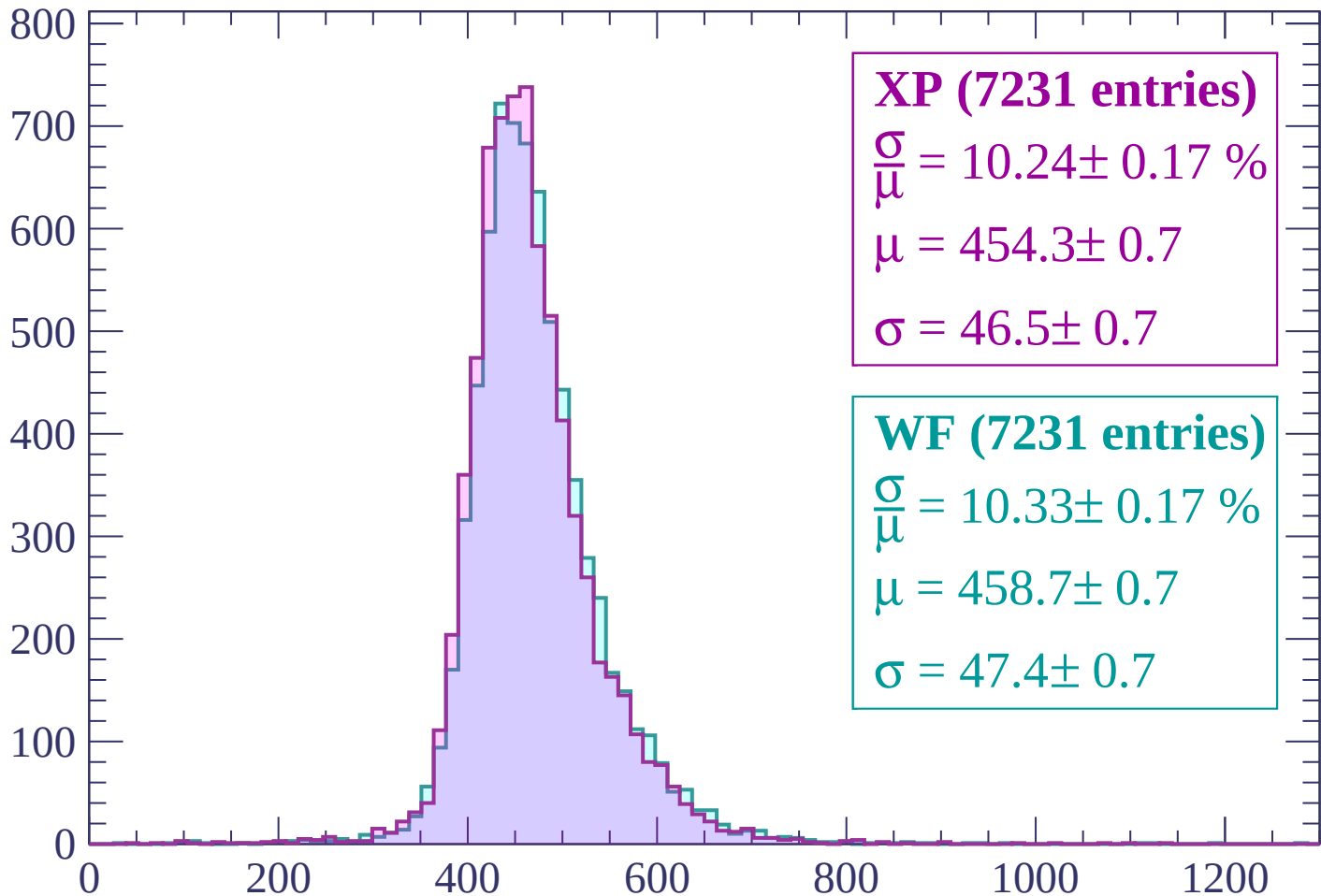
Energy loss | $1320 < p < 1380$

Count



Energy loss | $1380 < p < 1440$

Count



XP (7231 entries)

$$\frac{\sigma}{\mu} = 10.24 \pm 0.17 \%$$

$$\mu = 454.3 \pm 0.7$$

$$\sigma = 46.5 \pm 0.7$$

WF (7231 entries)

$$\frac{\sigma}{\mu} = 10.33 \pm 0.17 \%$$

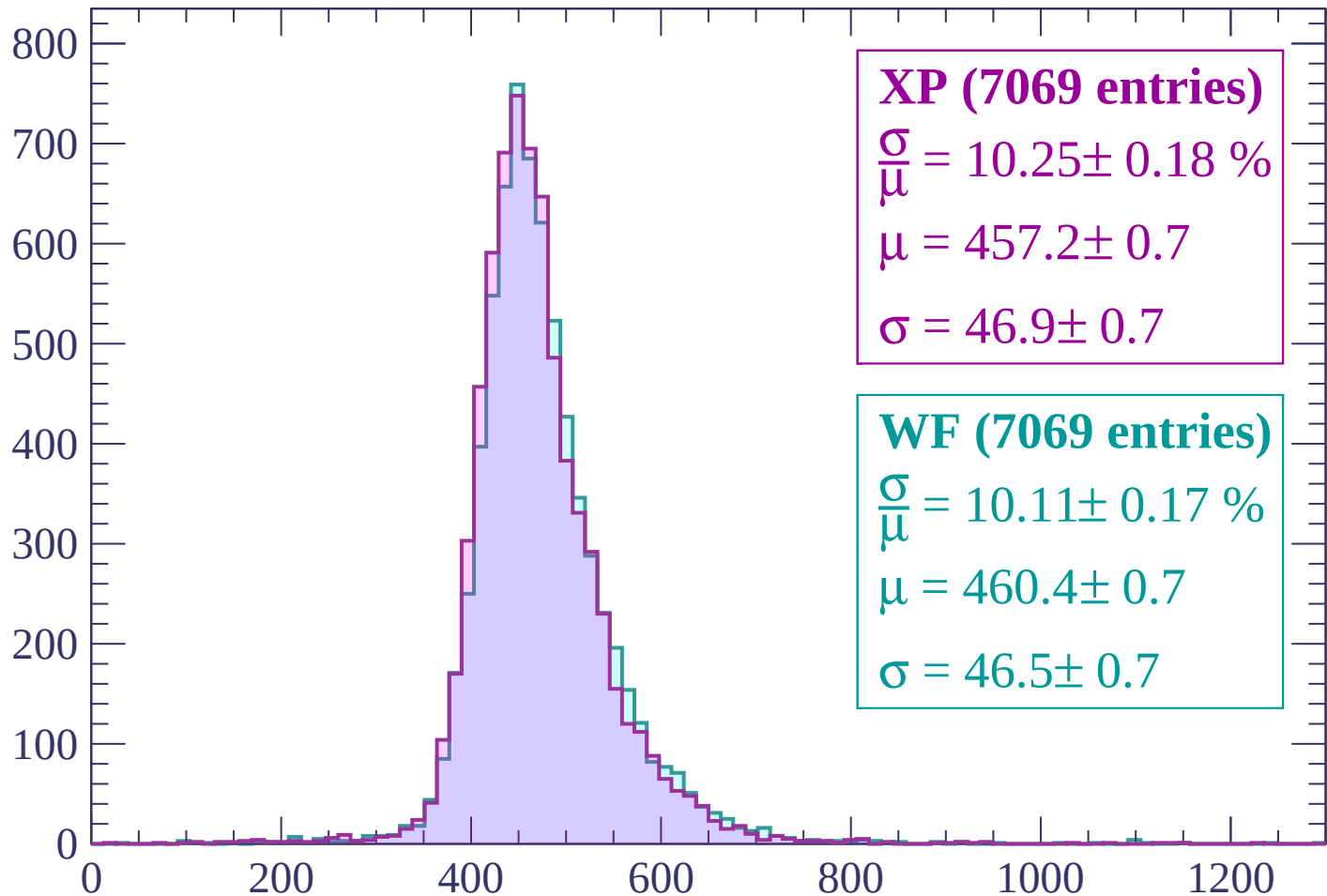
$$\mu = 458.7 \pm 0.7$$

$$\sigma = 47.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (7069 entries)

$\frac{\sigma}{\mu} = 10.25 \pm 0.18 \%$

$\mu = 457.2 \pm 0.7$

$\sigma = 46.9 \pm 0.7$

WF (7069 entries)

$\frac{\sigma}{\mu} = 10.11 \pm 0.17 \%$

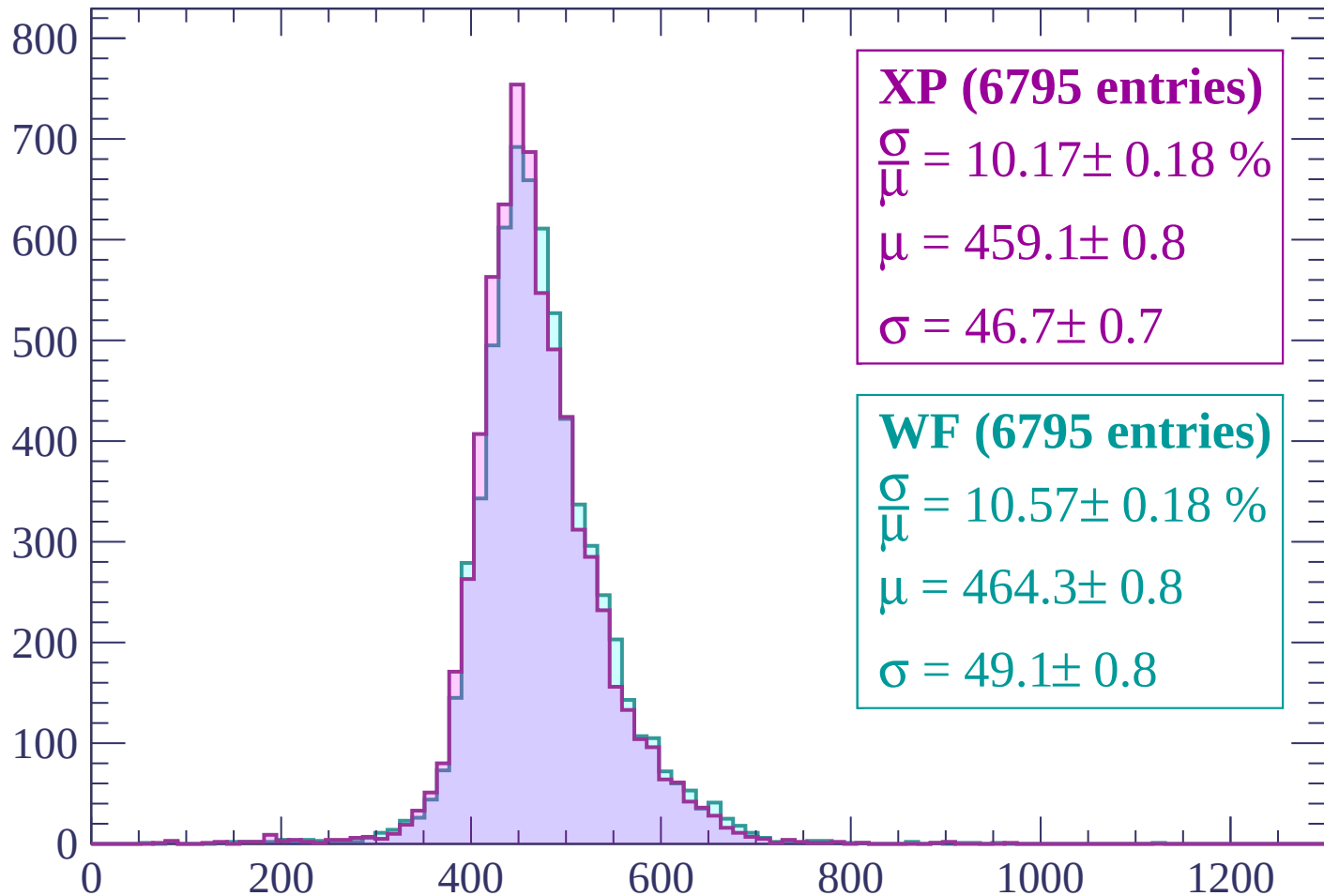
$\mu = 460.4 \pm 0.7$

$\sigma = 46.5 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (6795 entries)

$$\frac{\sigma}{\mu} = 10.17 \pm 0.18 \%$$

$$\mu = 459.1 \pm 0.8$$

$$\sigma = 46.7 \pm 0.7$$

WF (6795 entries)

$$\frac{\sigma}{\mu} = 10.57 \pm 0.18 \%$$

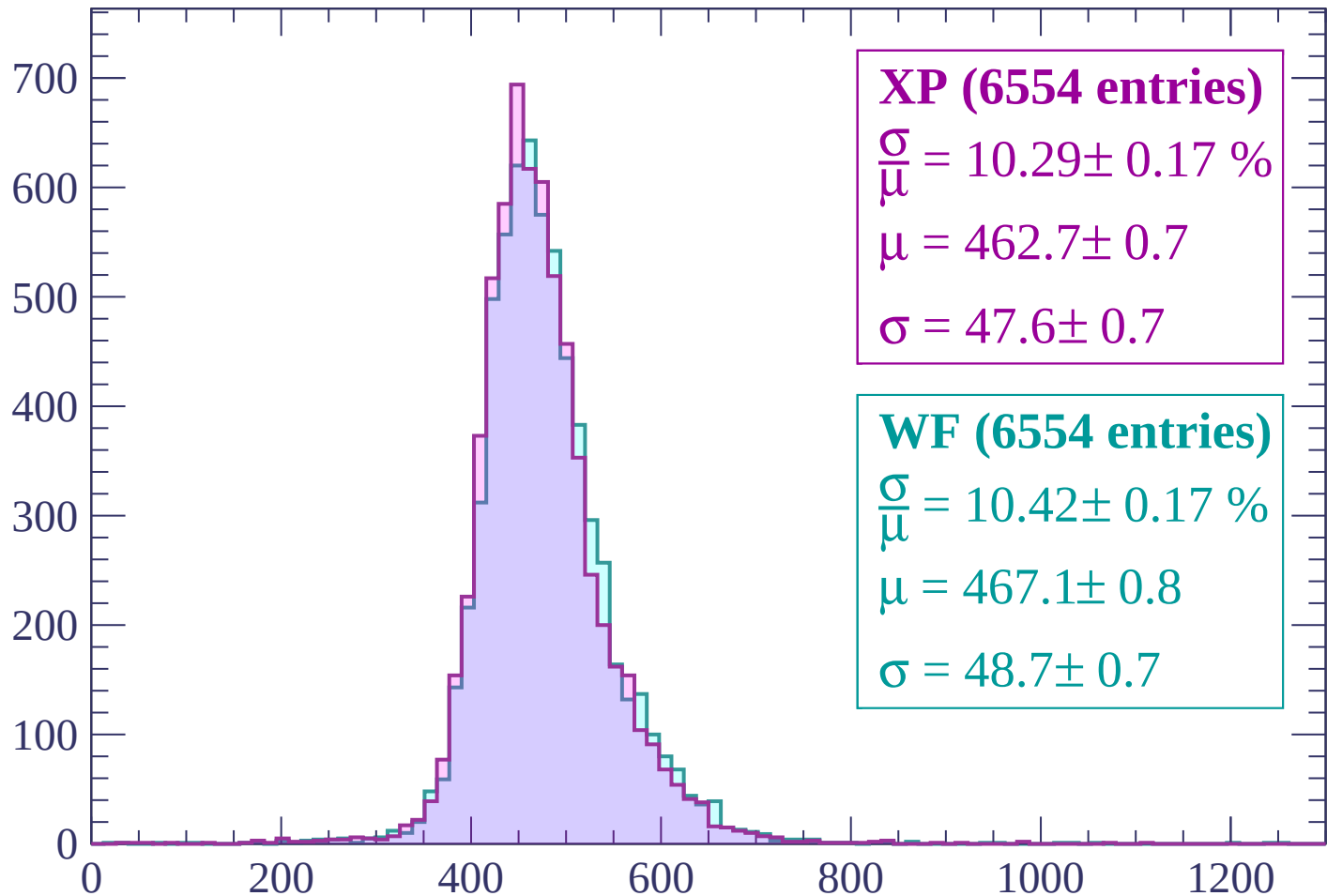
$$\mu = 464.3 \pm 0.8$$

$$\sigma = 49.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (6554 entries)

$\frac{\sigma}{\mu} = 10.29 \pm 0.17 \%$

$\mu = 462.7 \pm 0.7$

$\sigma = 47.6 \pm 0.7$

WF (6554 entries)

$\frac{\sigma}{\mu} = 10.42 \pm 0.17 \%$

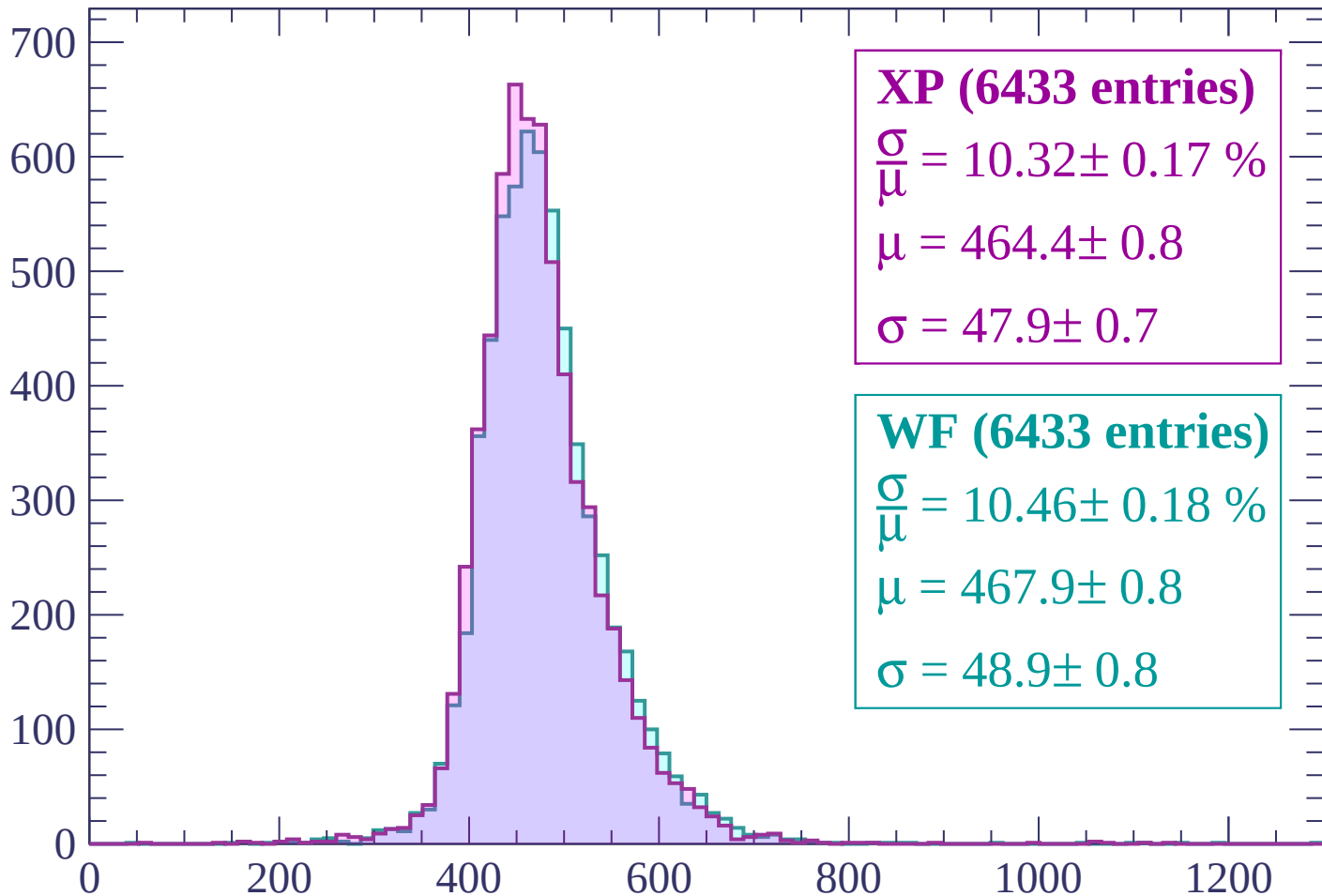
$\mu = 467.1 \pm 0.8$

$\sigma = 48.7 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (6433 entries)

$$\frac{\sigma}{\mu} = 10.32 \pm 0.17 \%$$

$$\mu = 464.4 \pm 0.8$$

$$\sigma = 47.9 \pm 0.7$$

WF (6433 entries)

$$\frac{\sigma}{\mu} = 10.46 \pm 0.18 \%$$

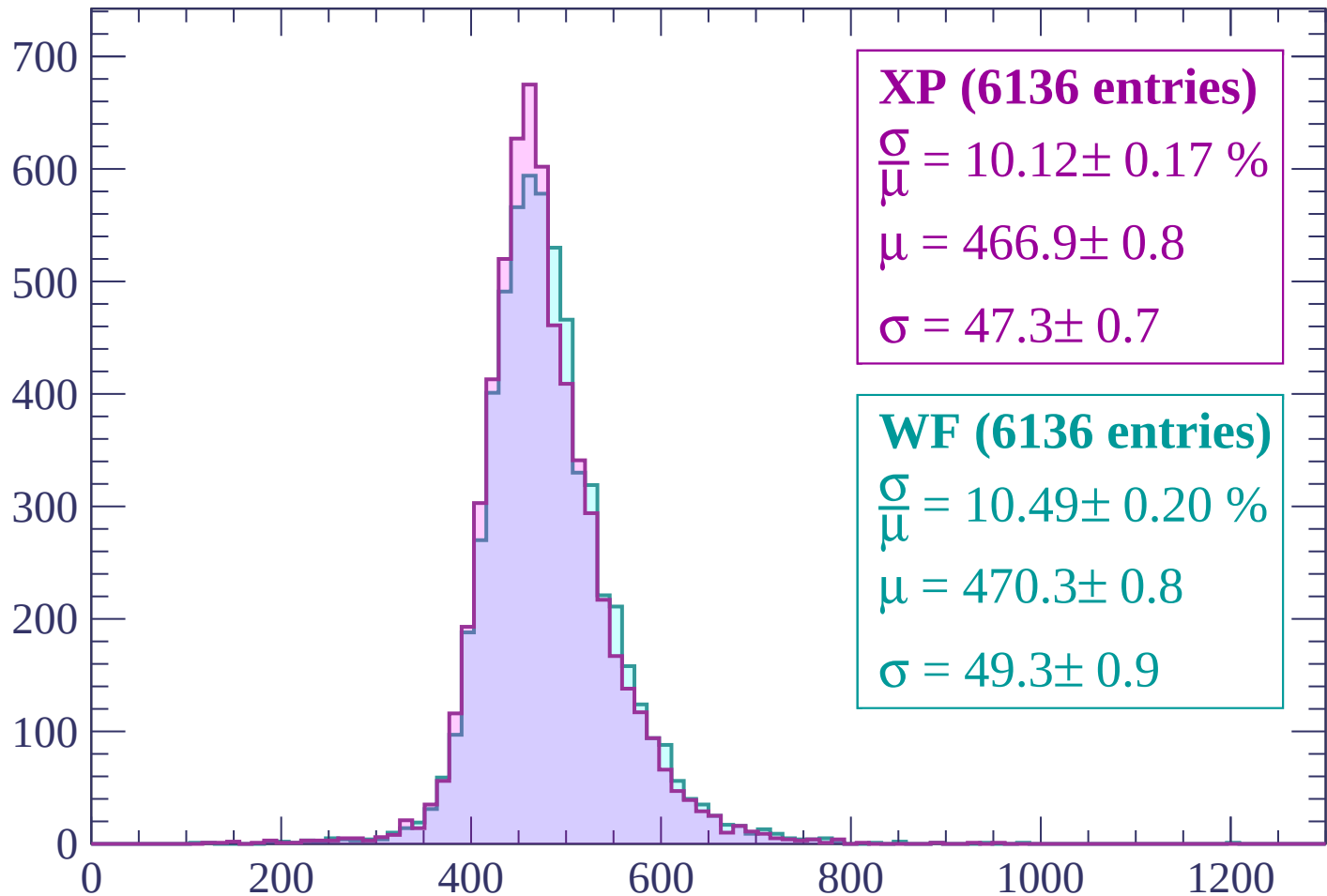
$$\mu = 467.9 \pm 0.8$$

$$\sigma = 48.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (6136 entries)

$$\frac{\sigma}{\mu} = 10.12 \pm 0.17 \%$$

$$\mu = 466.9 \pm 0.8$$

$$\sigma = 47.3 \pm 0.7$$

WF (6136 entries)

$$\frac{\sigma}{\mu} = 10.49 \pm 0.20 \%$$

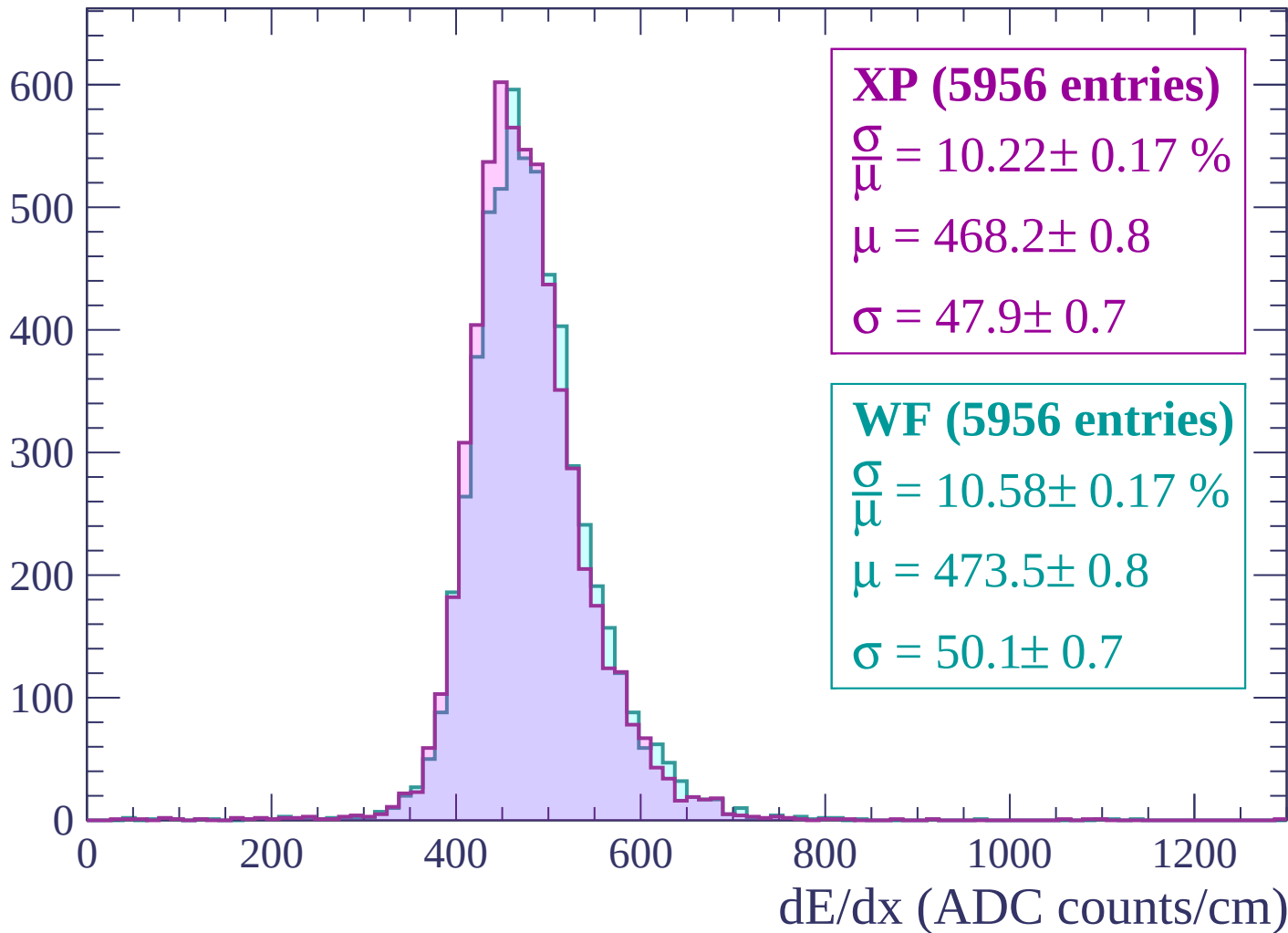
$$\mu = 470.3 \pm 0.8$$

$$\sigma = 49.3 \pm 0.9$$

dE/dx (ADC counts/cm)

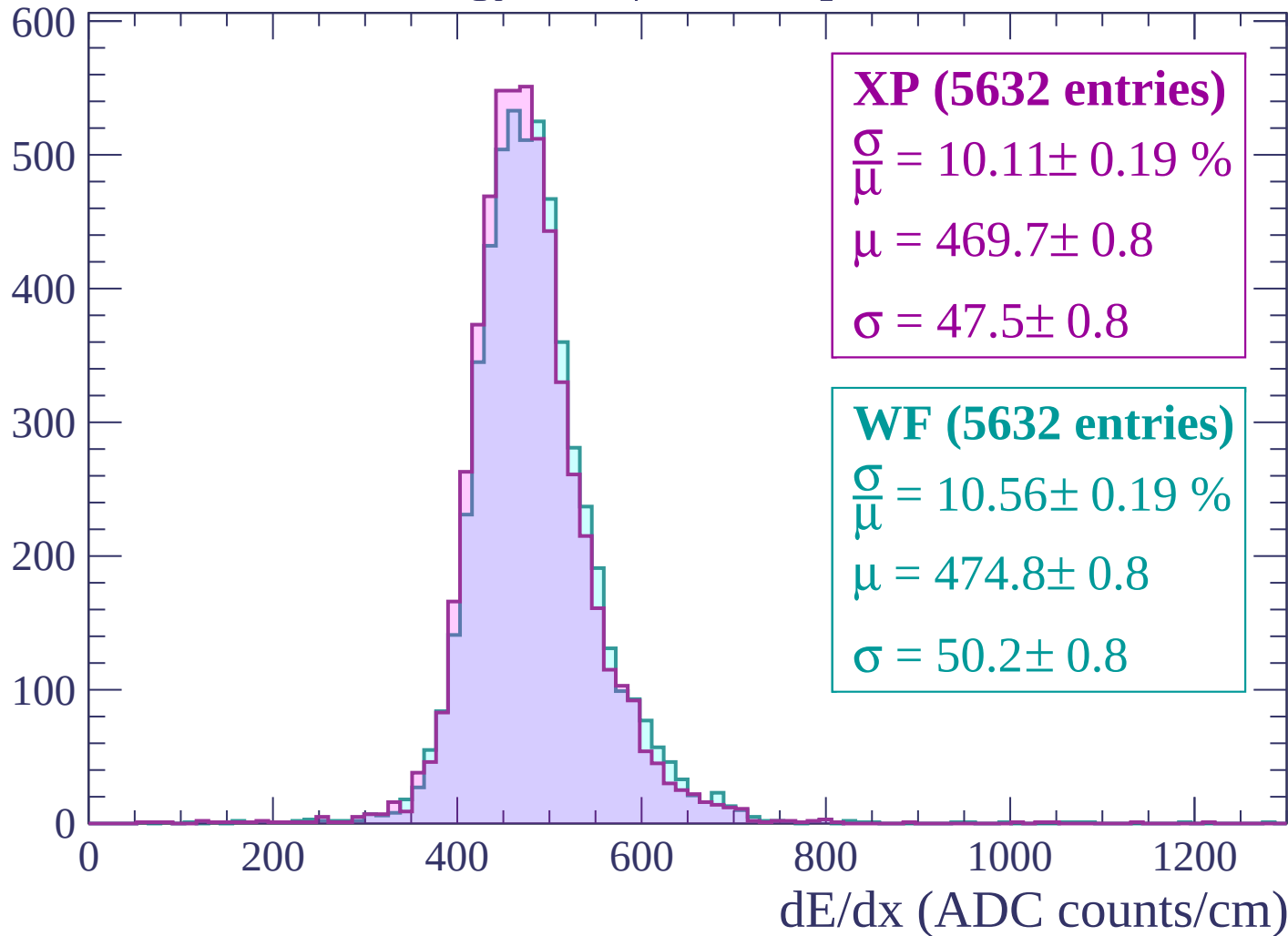
Energy loss | $1740 < p < 1800$

Count



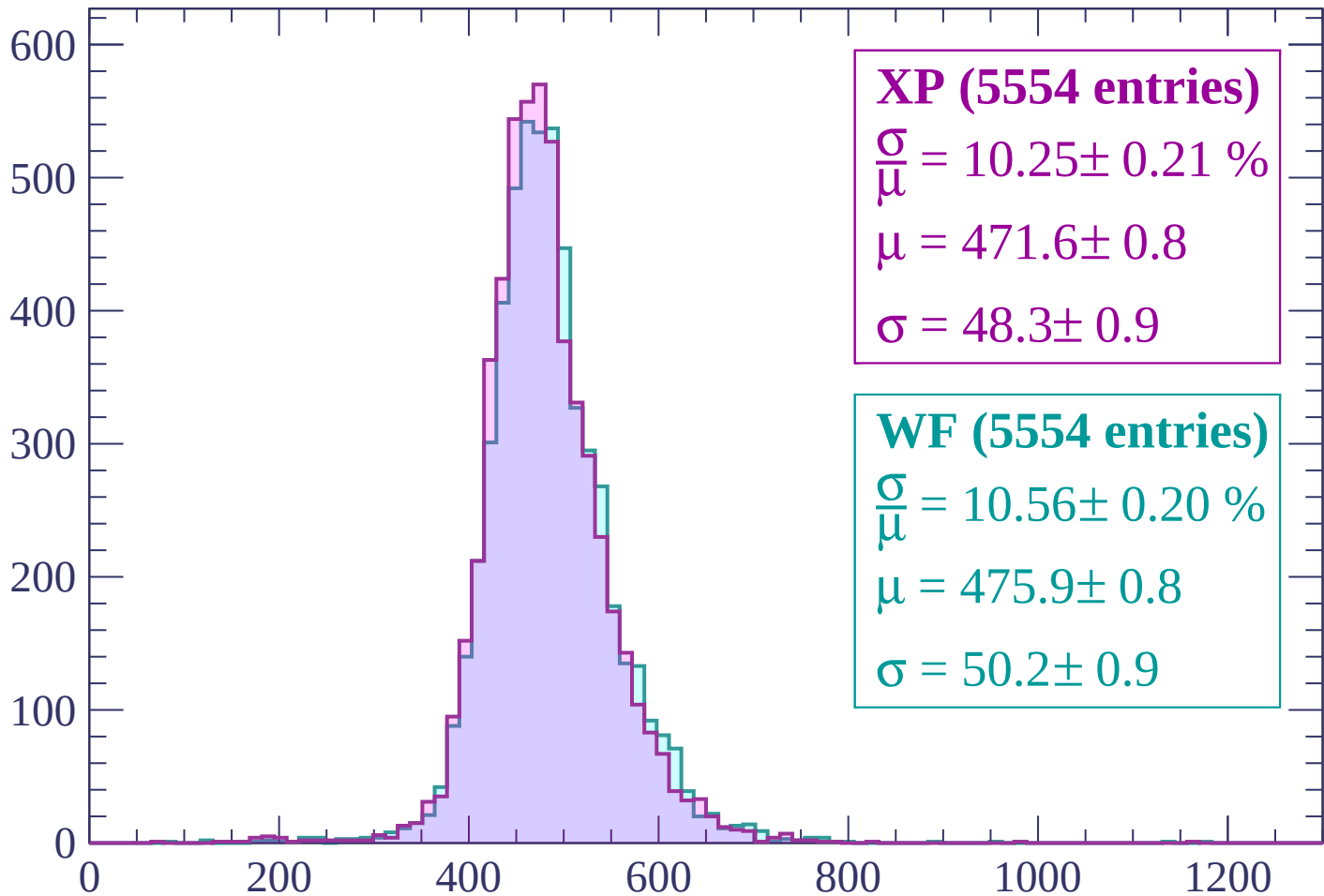
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (5554 entries)

$$\frac{\sigma}{\mu} = 10.25 \pm 0.21 \%$$

$$\mu = 471.6 \pm 0.8$$

$$\sigma = 48.3 \pm 0.9$$

WF (5554 entries)

$$\frac{\sigma}{\mu} = 10.56 \pm 0.20 \%$$

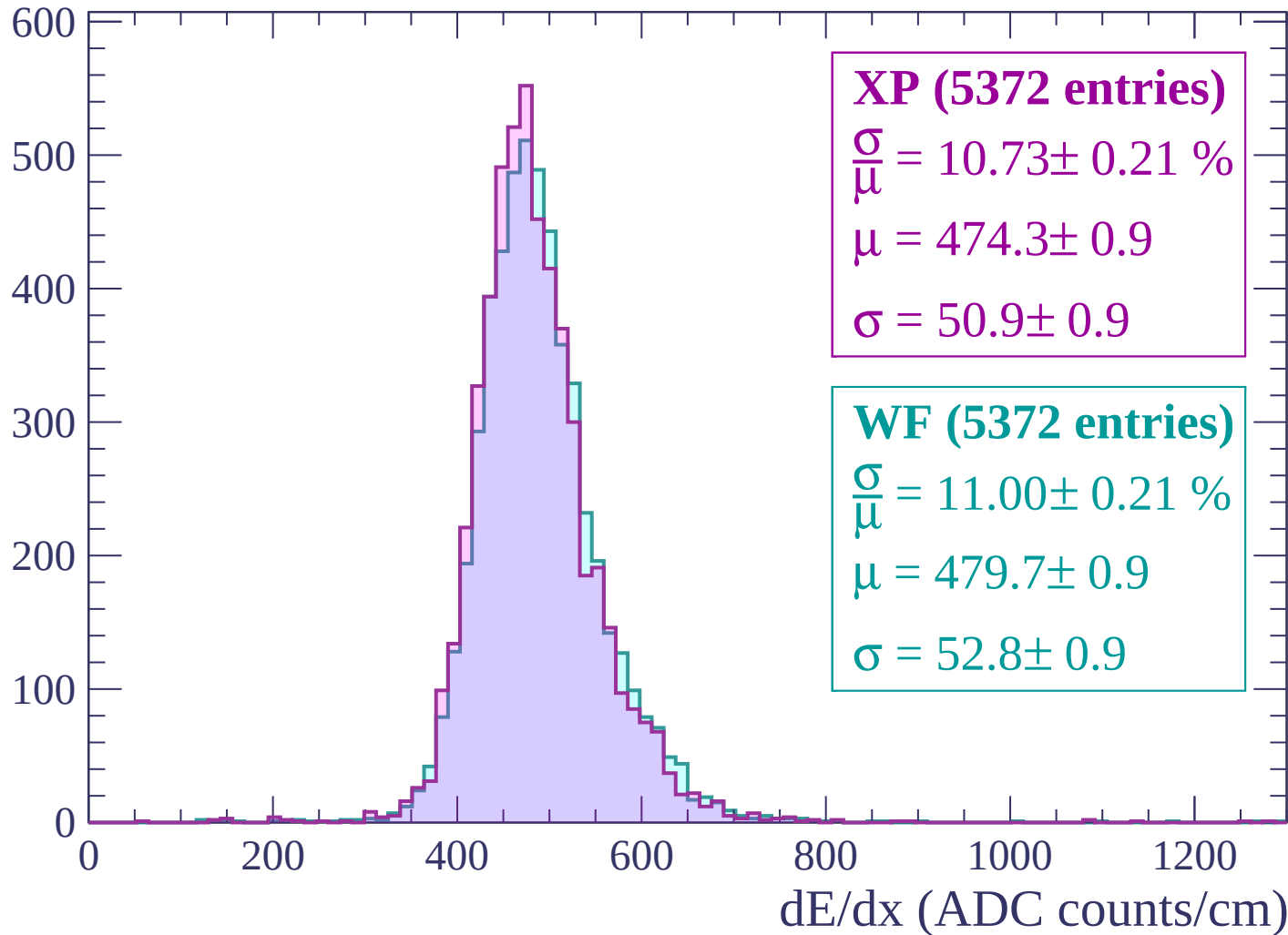
$$\mu = 475.9 \pm 0.8$$

$$\sigma = 50.2 \pm 0.9$$

dE/dx (ADC counts/cm)

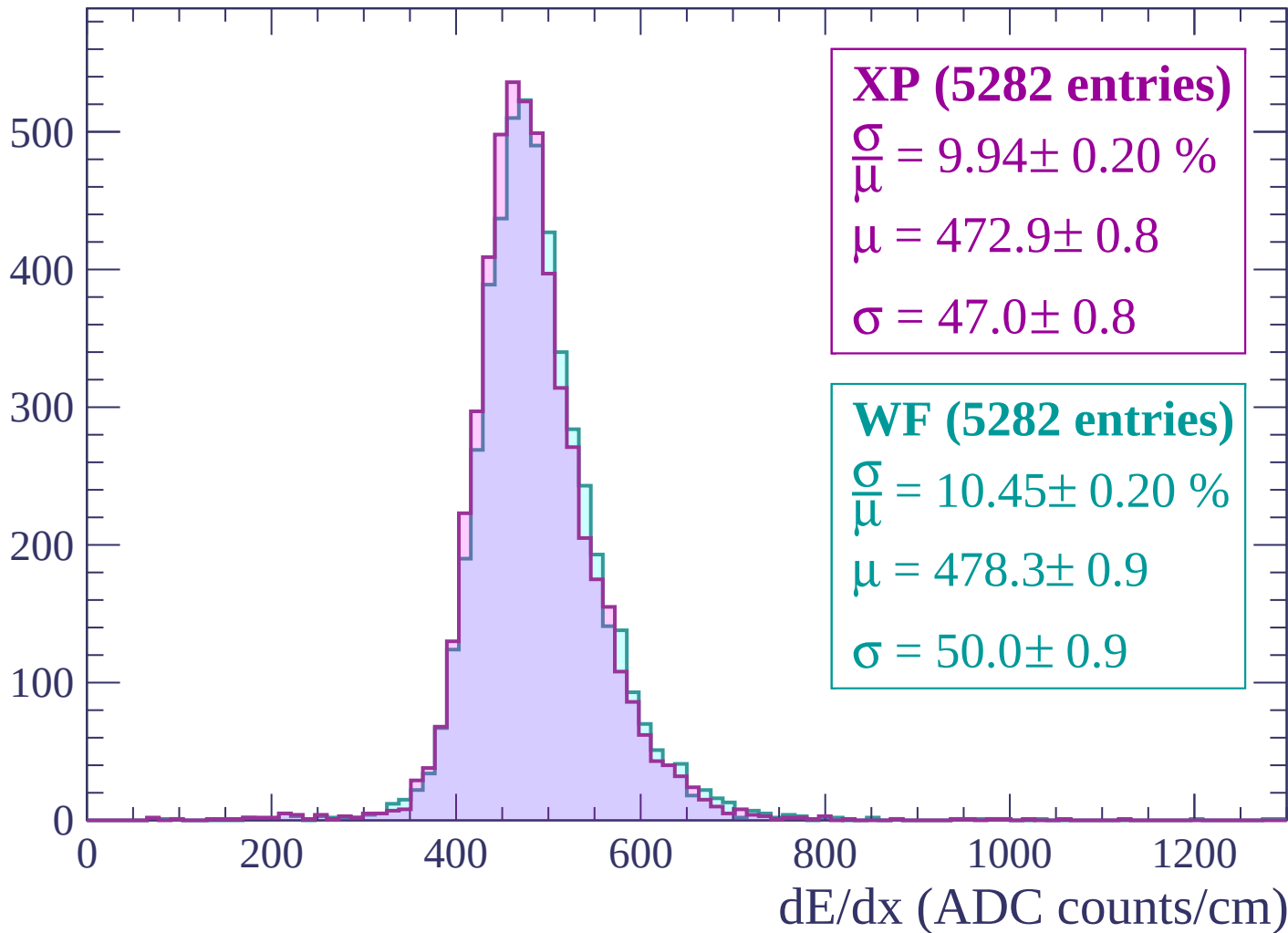
Energy loss | $1920 < p < 1980$

Count



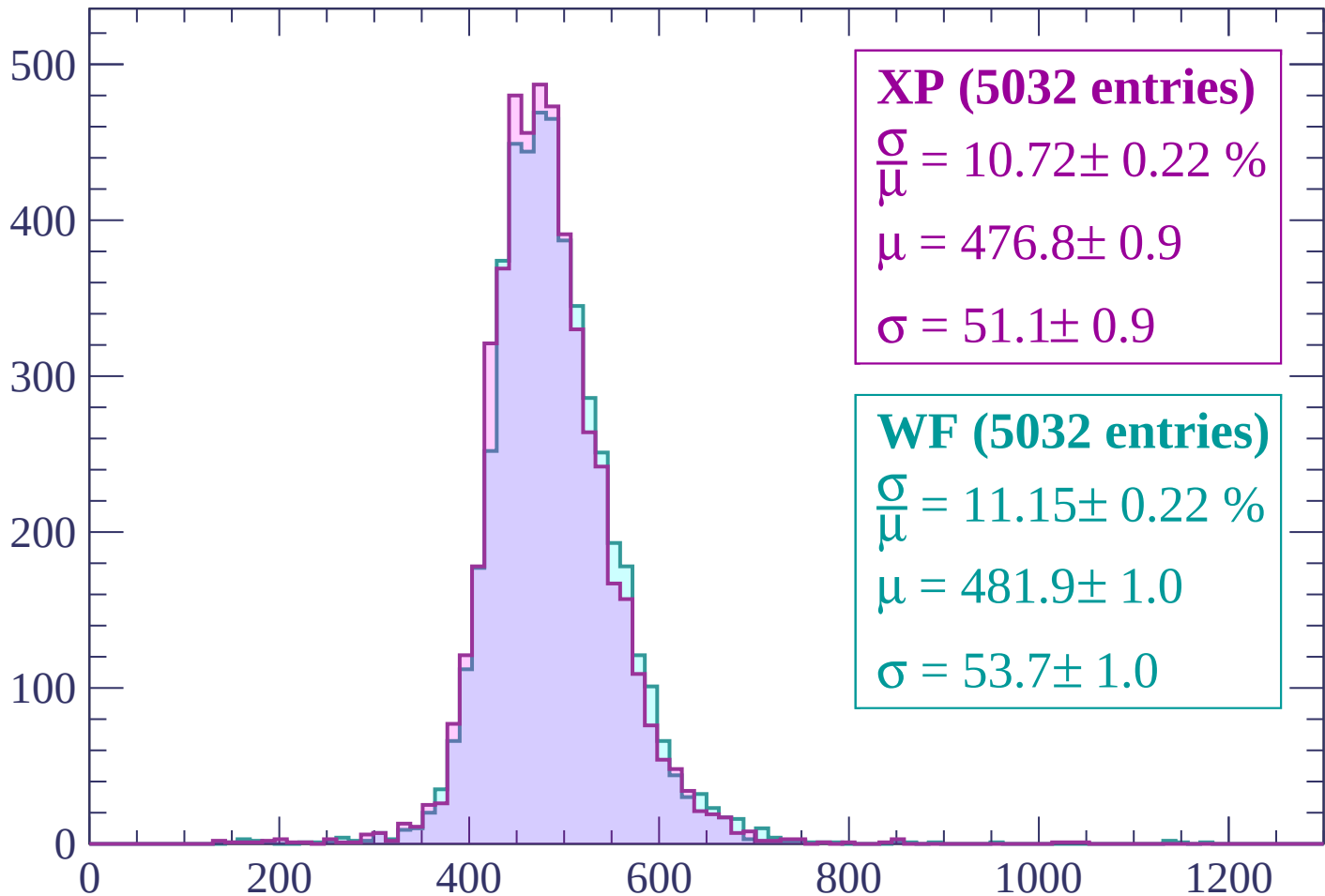
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (5032 entries)

$$\frac{\sigma}{\mu} = 10.72 \pm 0.22 \%$$

$$\mu = 476.8 \pm 0.9$$

$$\sigma = 51.1 \pm 0.9$$

WF (5032 entries)

$$\frac{\sigma}{\mu} = 11.15 \pm 0.22 \%$$

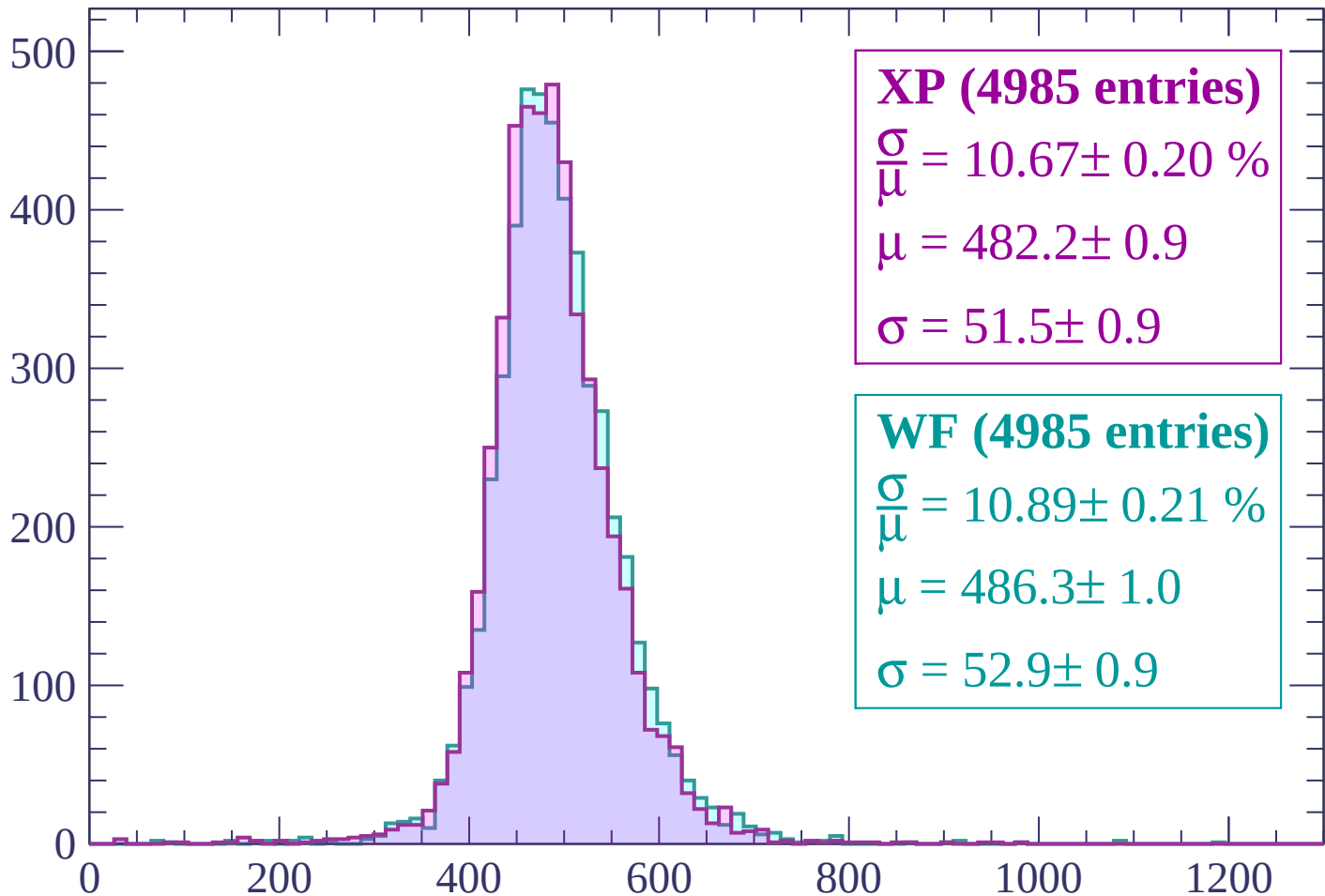
$$\mu = 481.9 \pm 1.0$$

$$\sigma = 53.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP (4985 entries)

$\frac{\sigma}{\mu} = 10.67 \pm 0.20 \%$

$\mu = 482.2 \pm 0.9$

$\sigma = 51.5 \pm 0.9$

WF (4985 entries)

$\frac{\sigma}{\mu} = 10.89 \pm 0.21 \%$

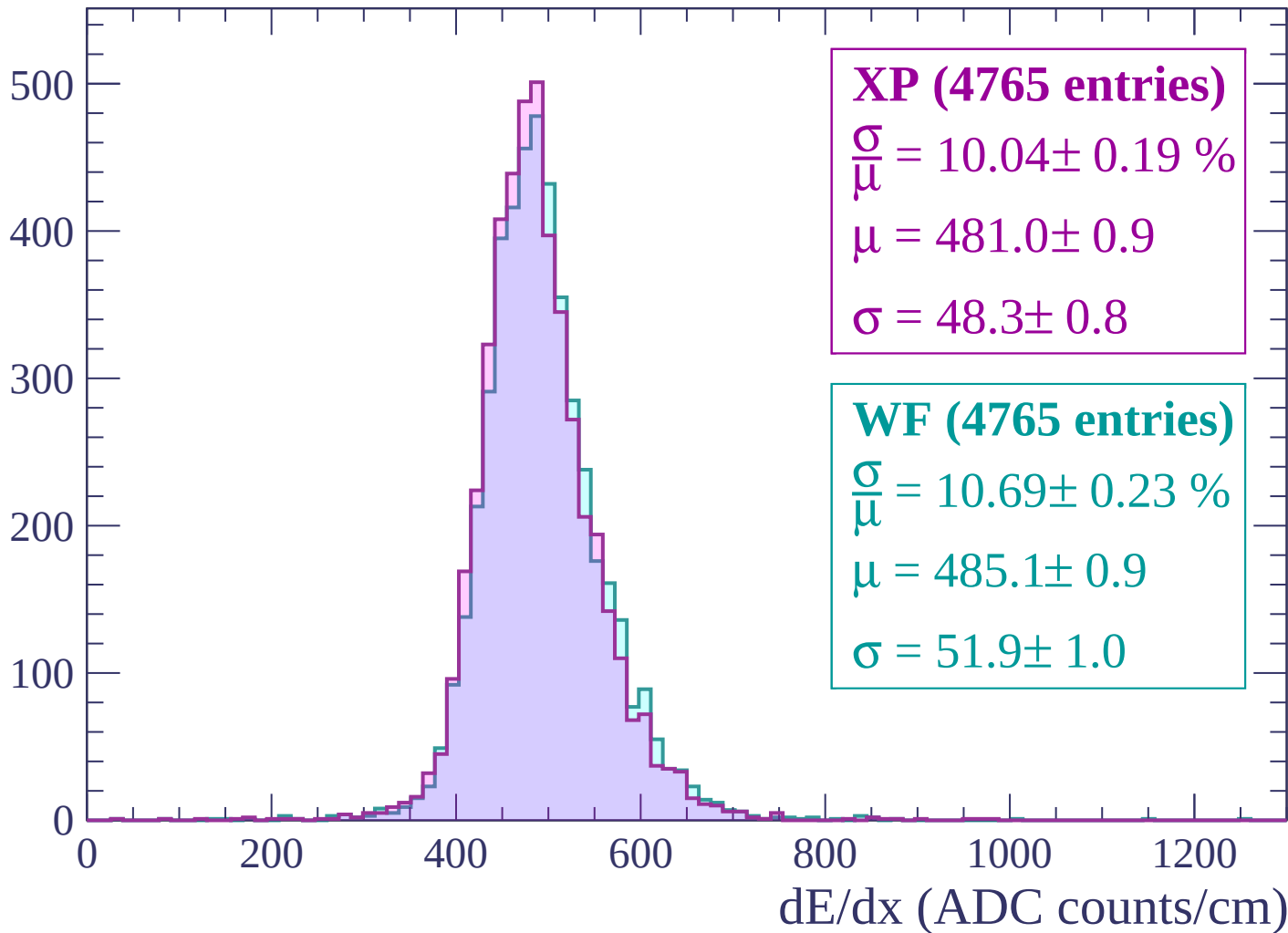
$\mu = 486.3 \pm 1.0$

$\sigma = 52.9 \pm 0.9$

dE/dx (ADC counts/cm)

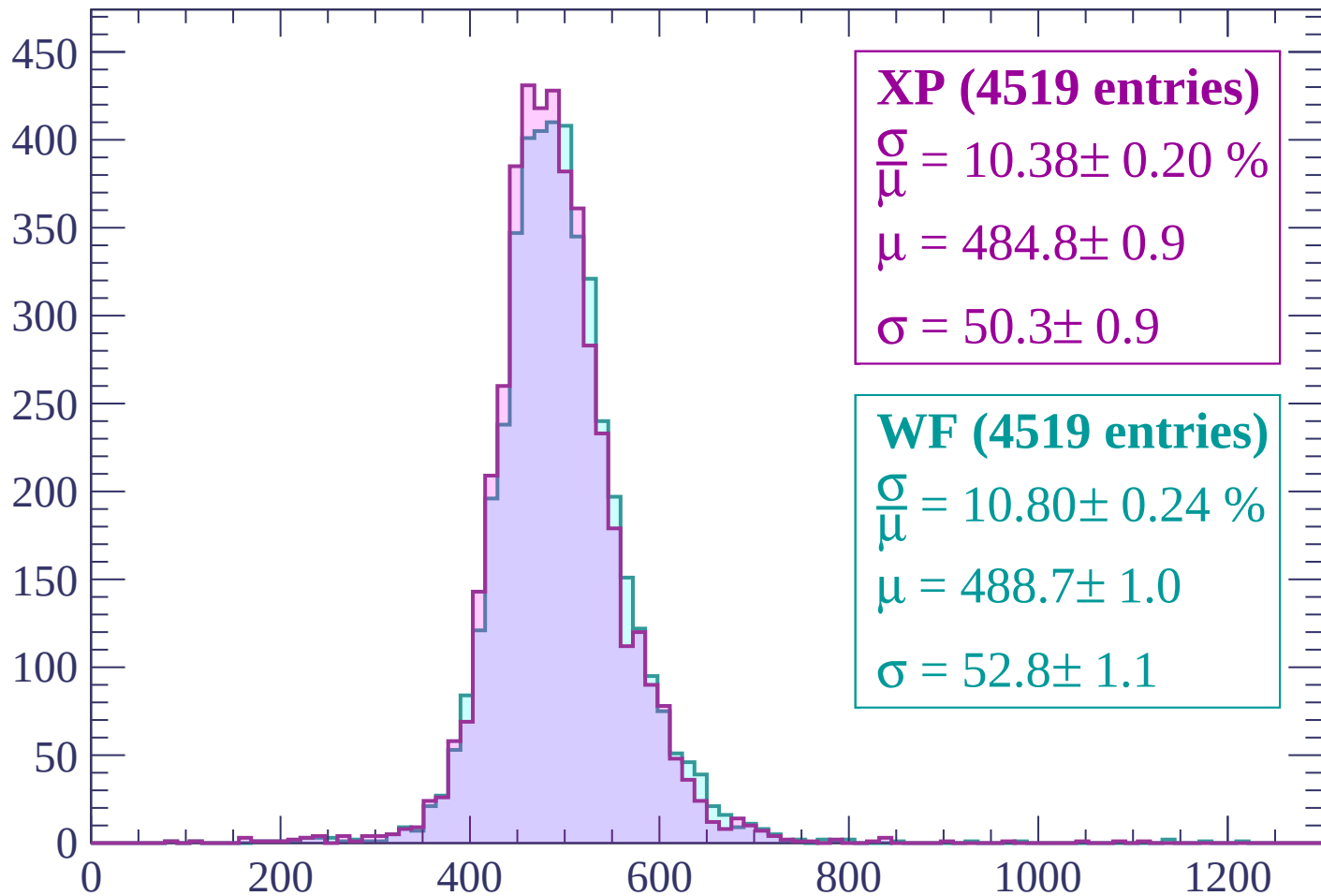
Energy loss | $2160 < p < 2220$

Count



Energy loss | $2220 < p < 2280$

Count



XP (4519 entries)

$\frac{\sigma}{\mu} = 10.38 \pm 0.20 \%$

$\mu = 484.8 \pm 0.9$

$\sigma = 50.3 \pm 0.9$

WF (4519 entries)

$\frac{\sigma}{\mu} = 10.80 \pm 0.24 \%$

$\mu = 488.7 \pm 1.0$

$\sigma = 52.8 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (4412 entries)

$\frac{\sigma}{\mu} = 10.25 \pm 0.24 \%$

$\mu = 484.2 \pm 1.0$

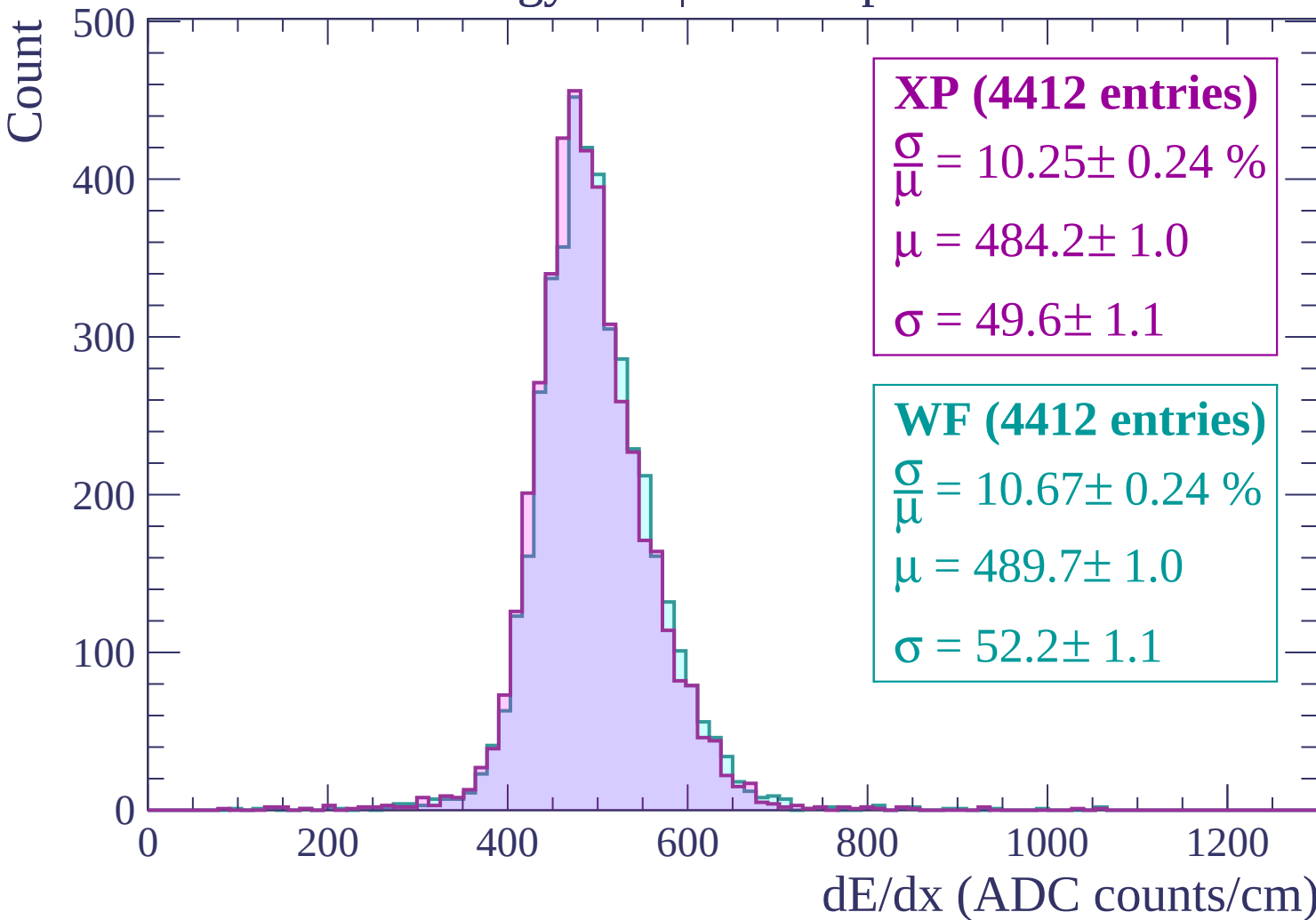
$\sigma = 49.6 \pm 1.1$

WF (4412 entries)

$\frac{\sigma}{\mu} = 10.67 \pm 0.24 \%$

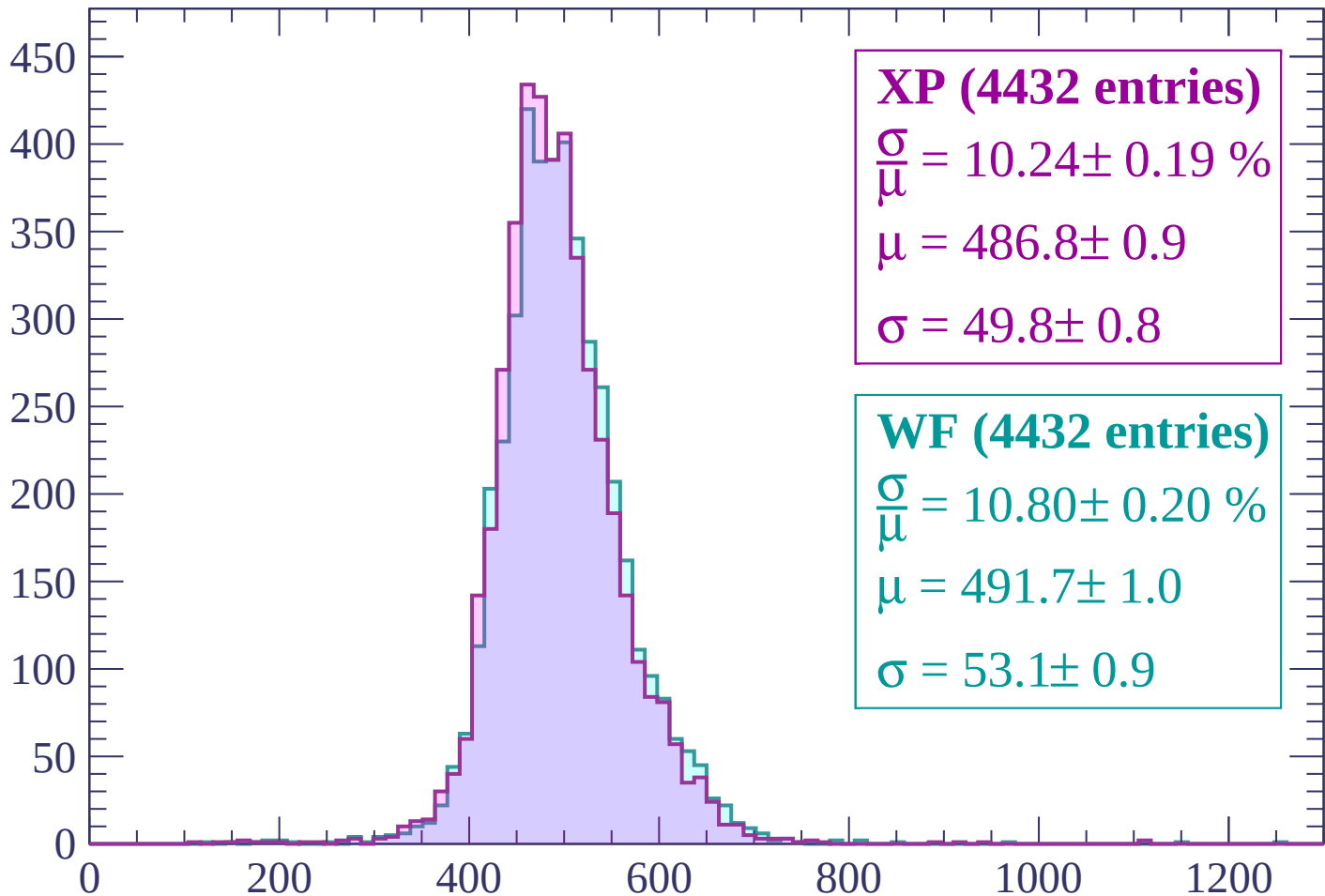
$\mu = 489.7 \pm 1.0$

$\sigma = 52.2 \pm 1.1$



Energy loss | $2340 < p < 2400$

Count



XP (4432 entries)

$$\frac{\sigma}{\mu} = 10.24 \pm 0.19 \%$$

$$\mu = 486.8 \pm 0.9$$

$$\sigma = 49.8 \pm 0.8$$

WF (4432 entries)

$$\frac{\sigma}{\mu} = 10.80 \pm 0.20 \%$$

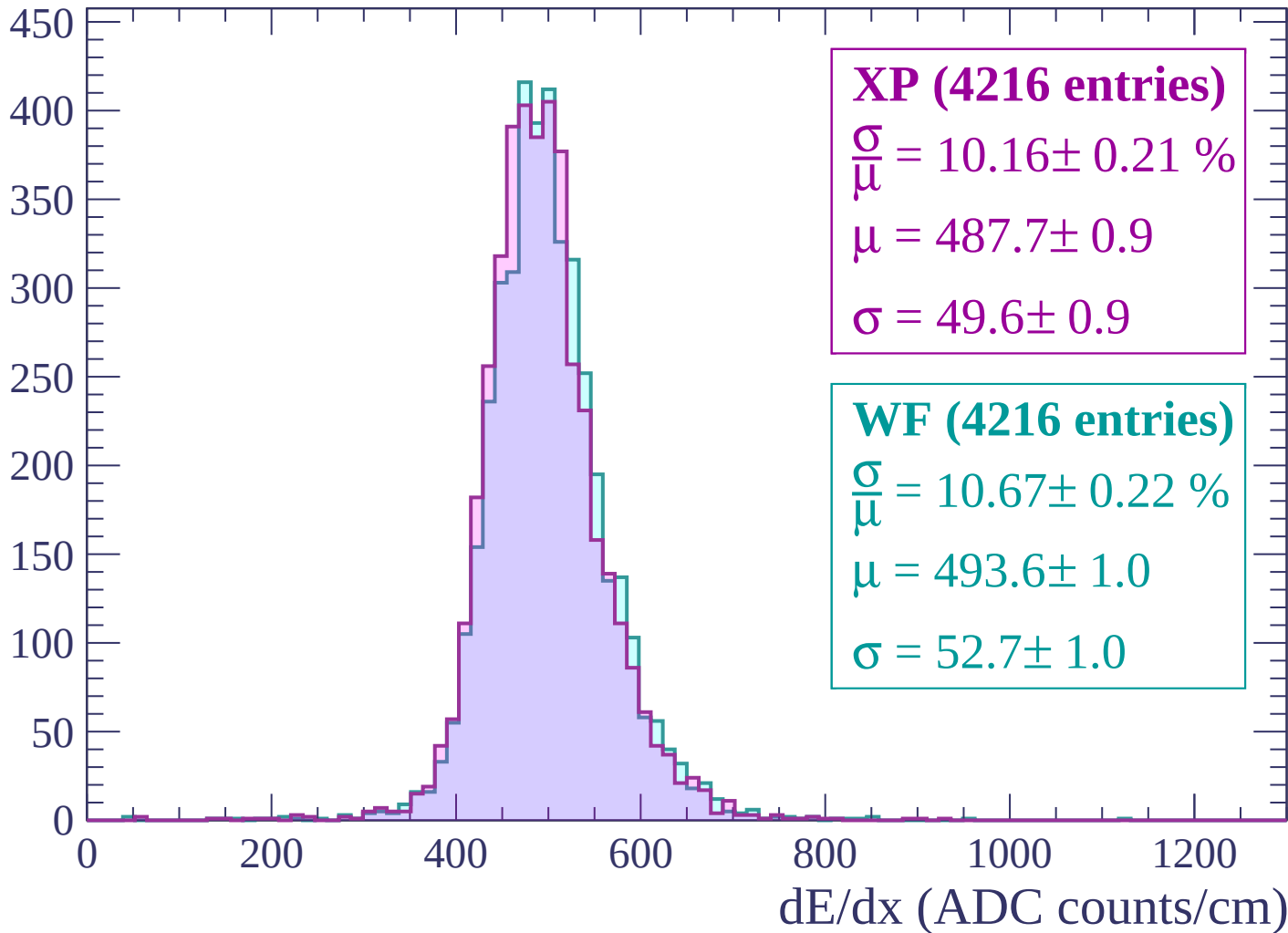
$$\mu = 491.7 \pm 1.0$$

$$\sigma = 53.1 \pm 0.9$$

dE/dx (ADC counts/cm)

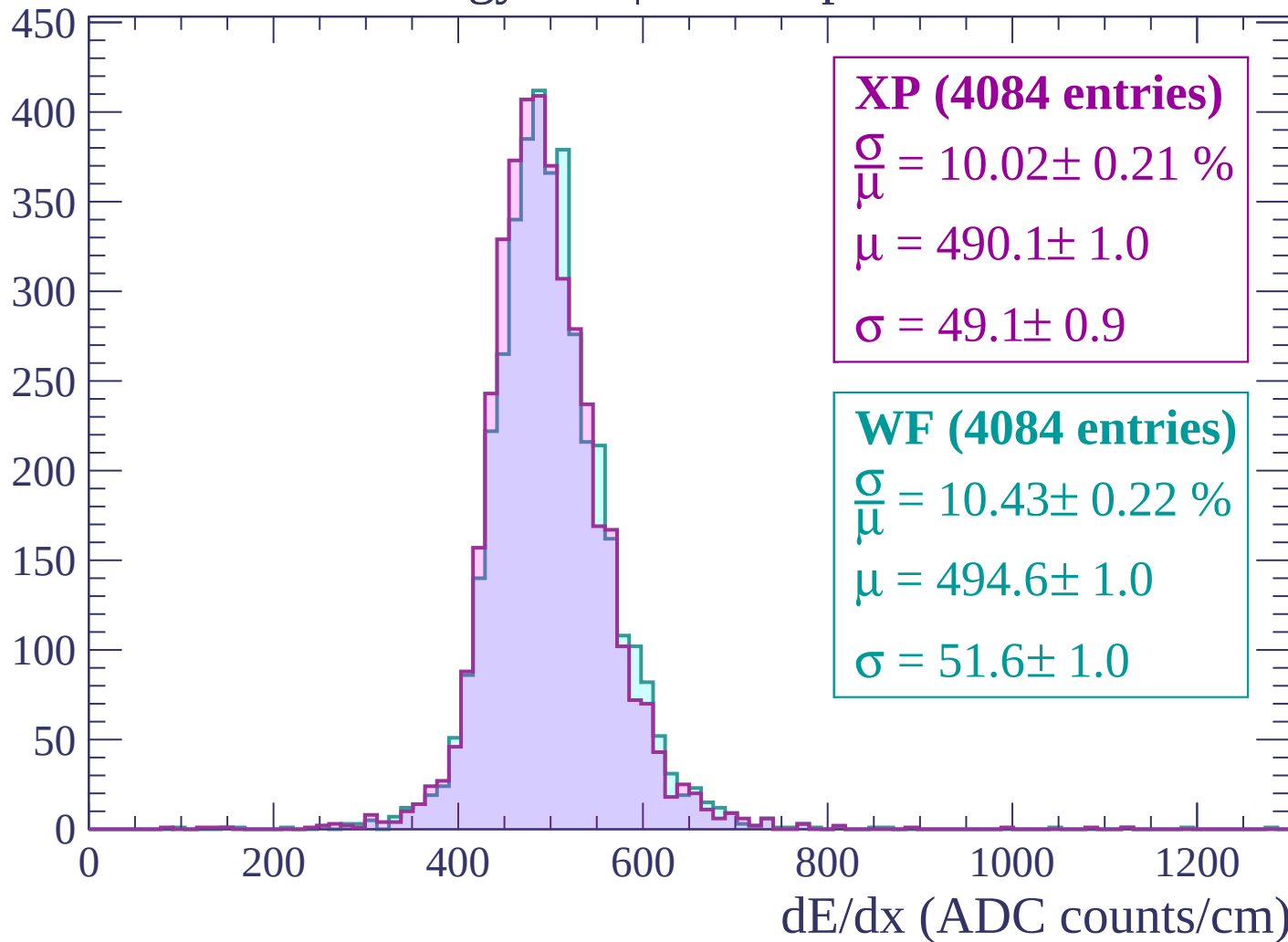
Energy loss | $2400 < p < 2460$

Count



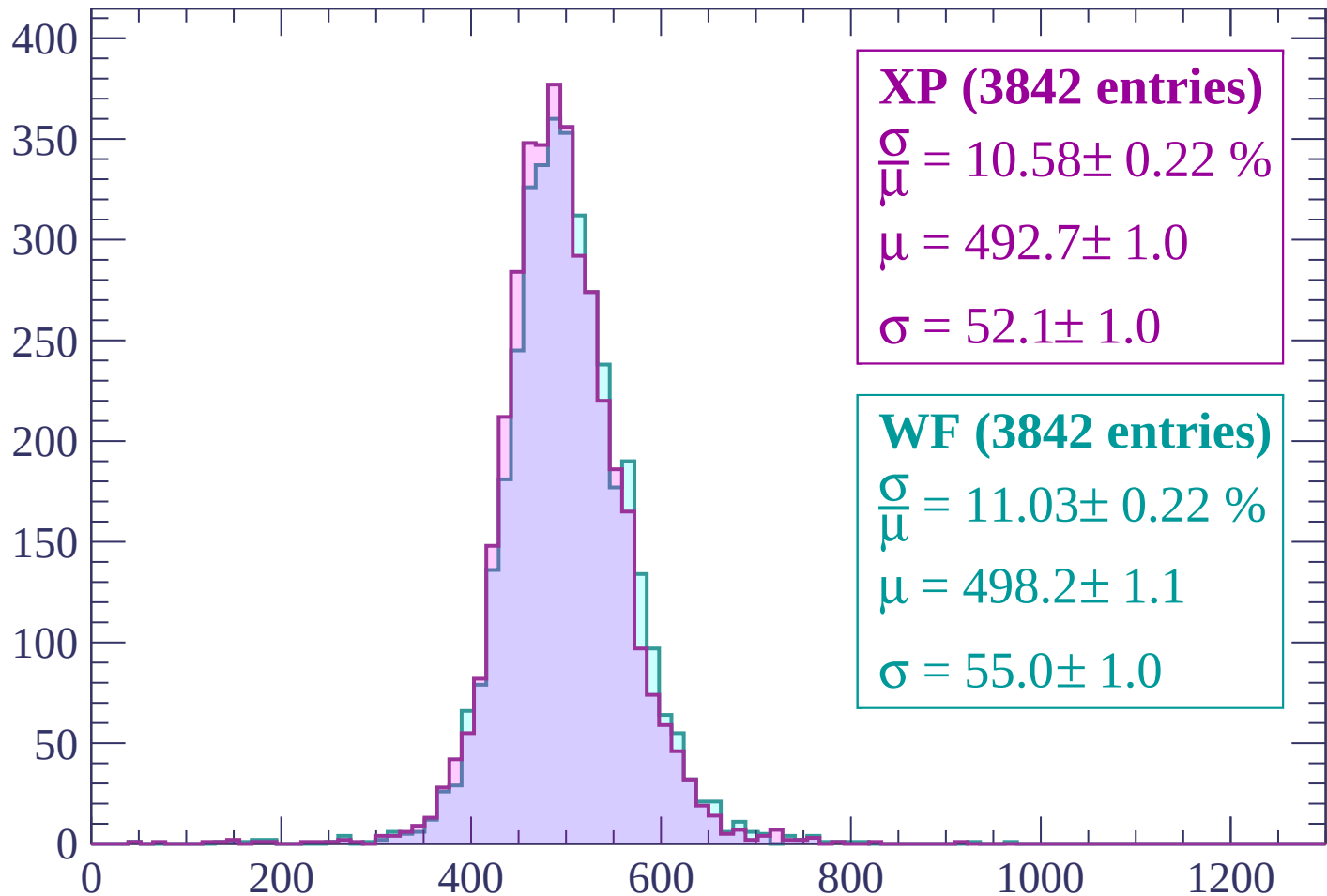
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (3842 entries)

$$\frac{\sigma}{\mu} = 10.58 \pm 0.22 \%$$

$$\mu = 492.7 \pm 1.0$$

$$\sigma = 52.1 \pm 1.0$$

WF (3842 entries)

$$\frac{\sigma}{\mu} = 11.03 \pm 0.22 \%$$

$$\mu = 498.2 \pm 1.1$$

$$\sigma = 55.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count

400
350
300
250
200
150
100
50
0

XP (3804 entries)

$\frac{\sigma}{\mu} = 10.21 \pm 0.23 \%$

$\mu = 490.9 \pm 1.0$

$\sigma = 50.1 \pm 1.0$

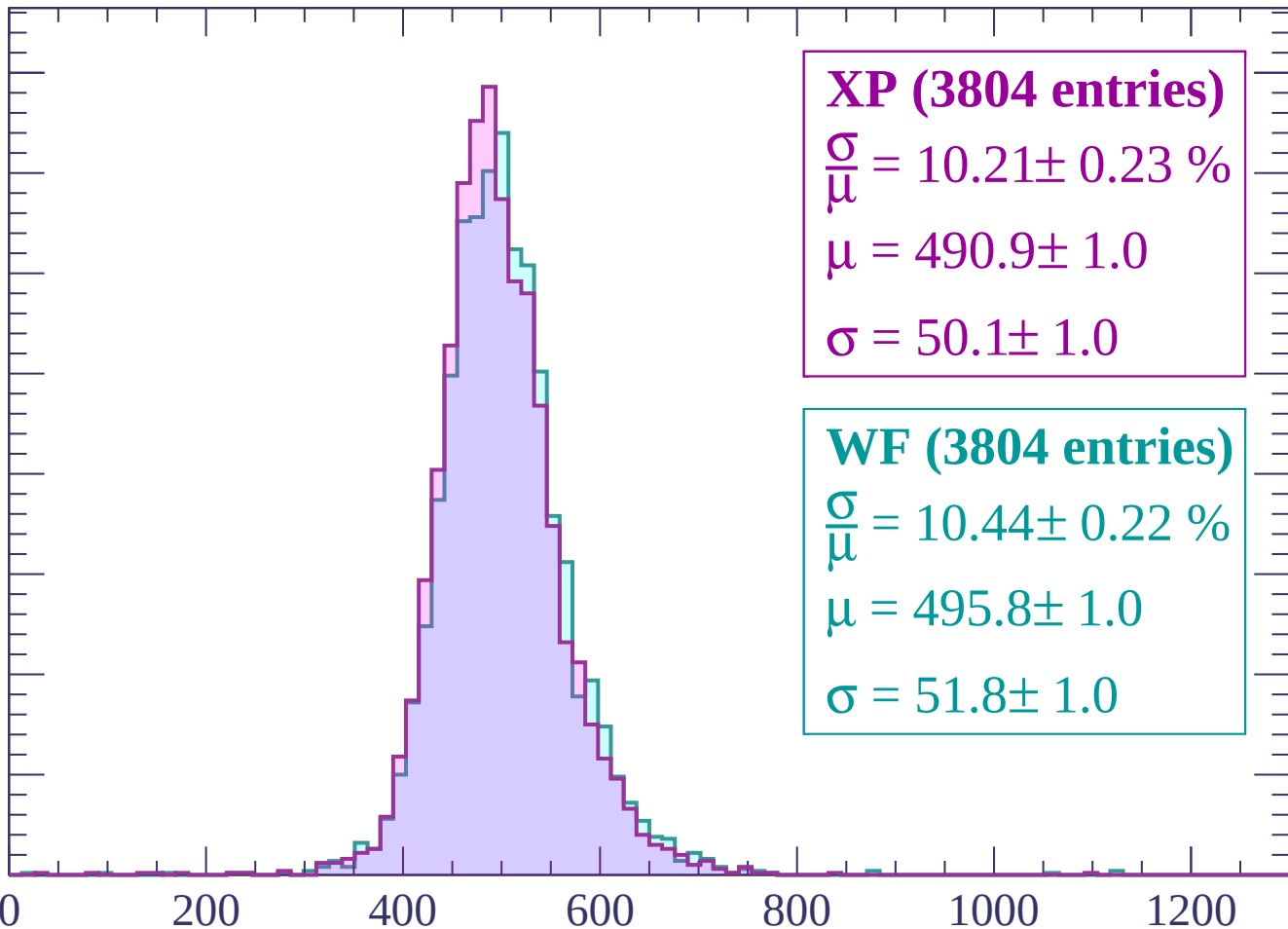
WF (3804 entries)

$\frac{\sigma}{\mu} = 10.44 \pm 0.22 \%$

$\mu = 495.8 \pm 1.0$

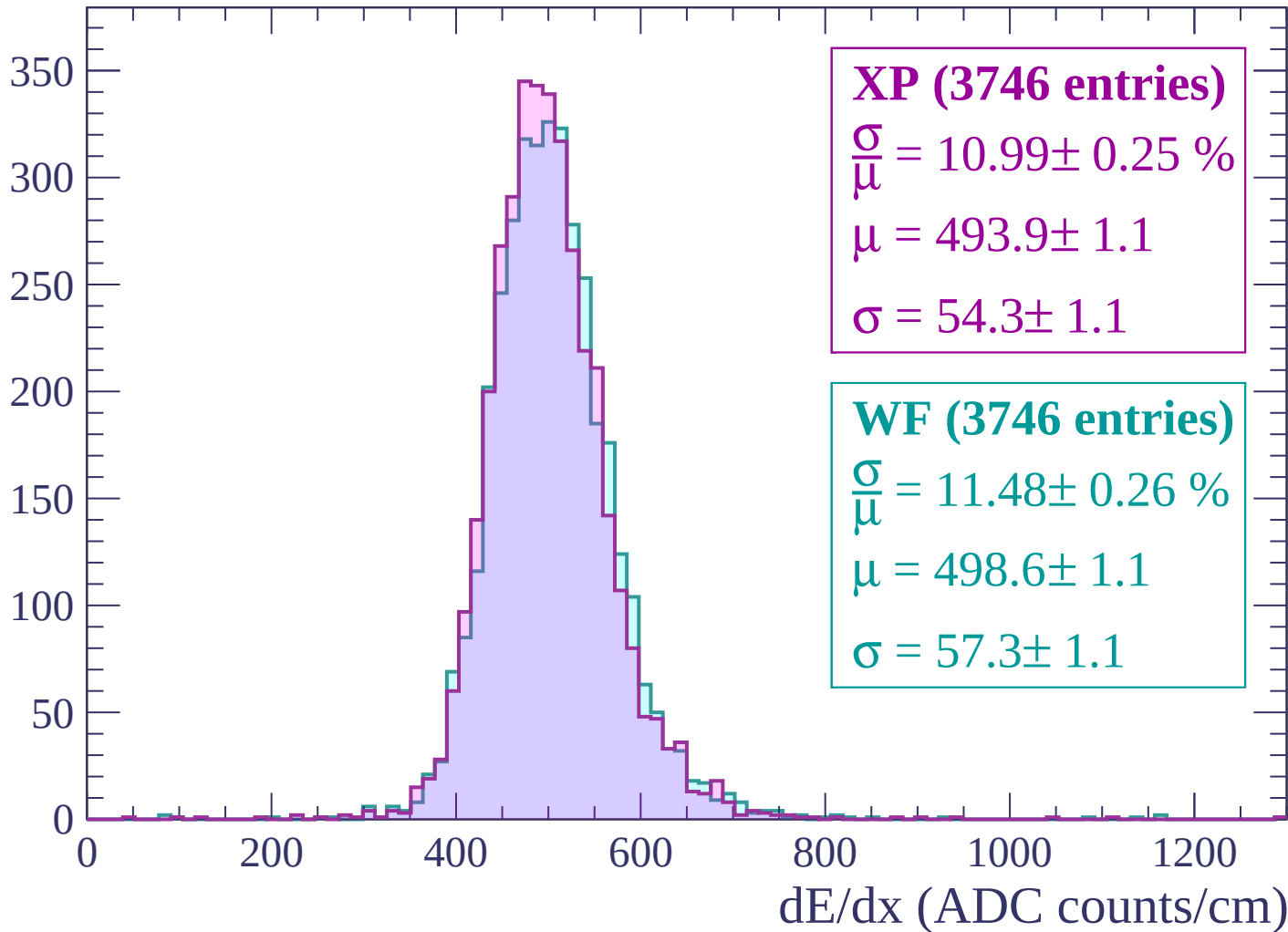
$\sigma = 51.8 \pm 1.0$

0 200 400 600 800 1000 1200
dE/dx (ADC counts/cm)



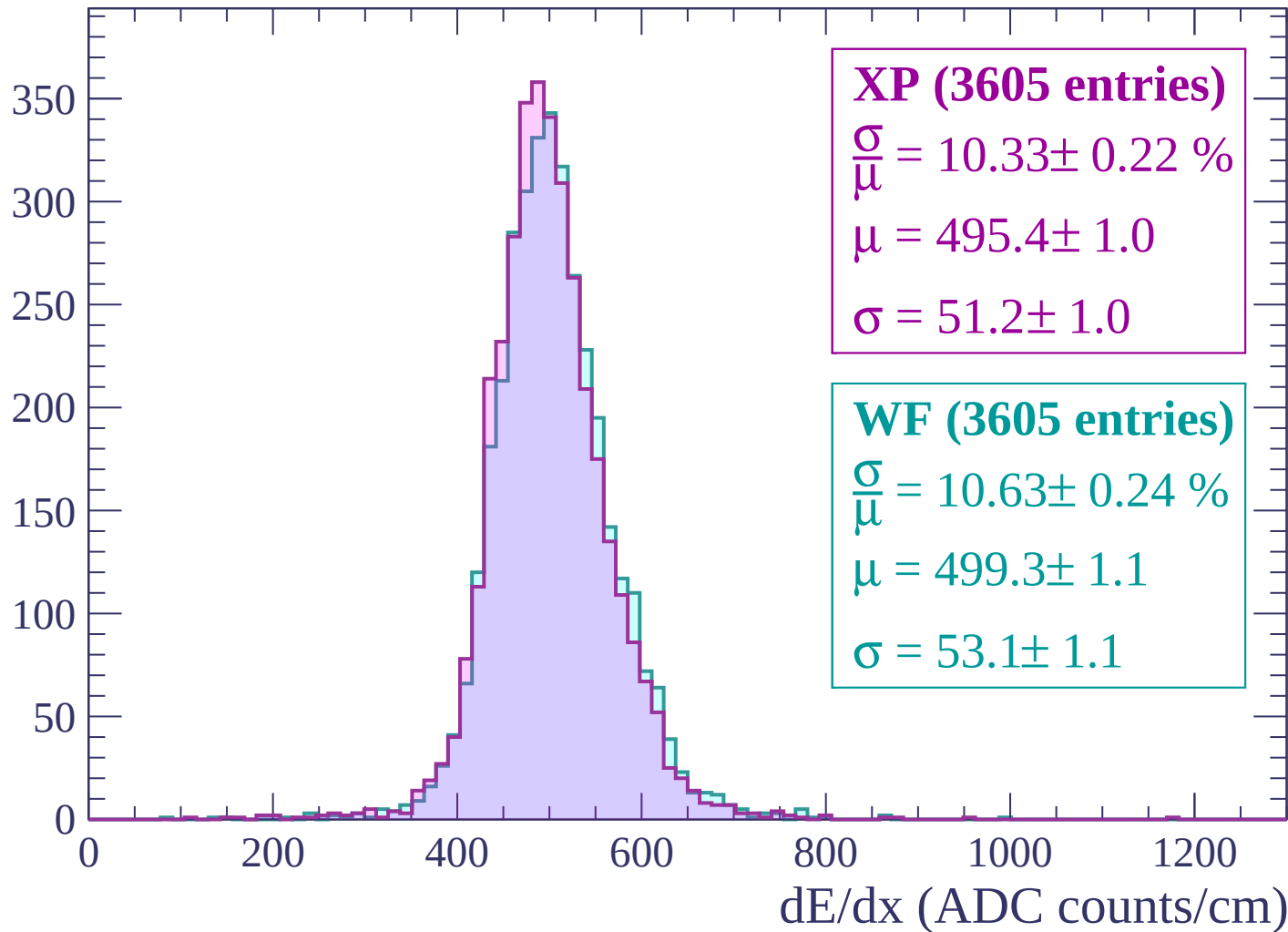
Energy loss | $2640 < p < 2700$

Count



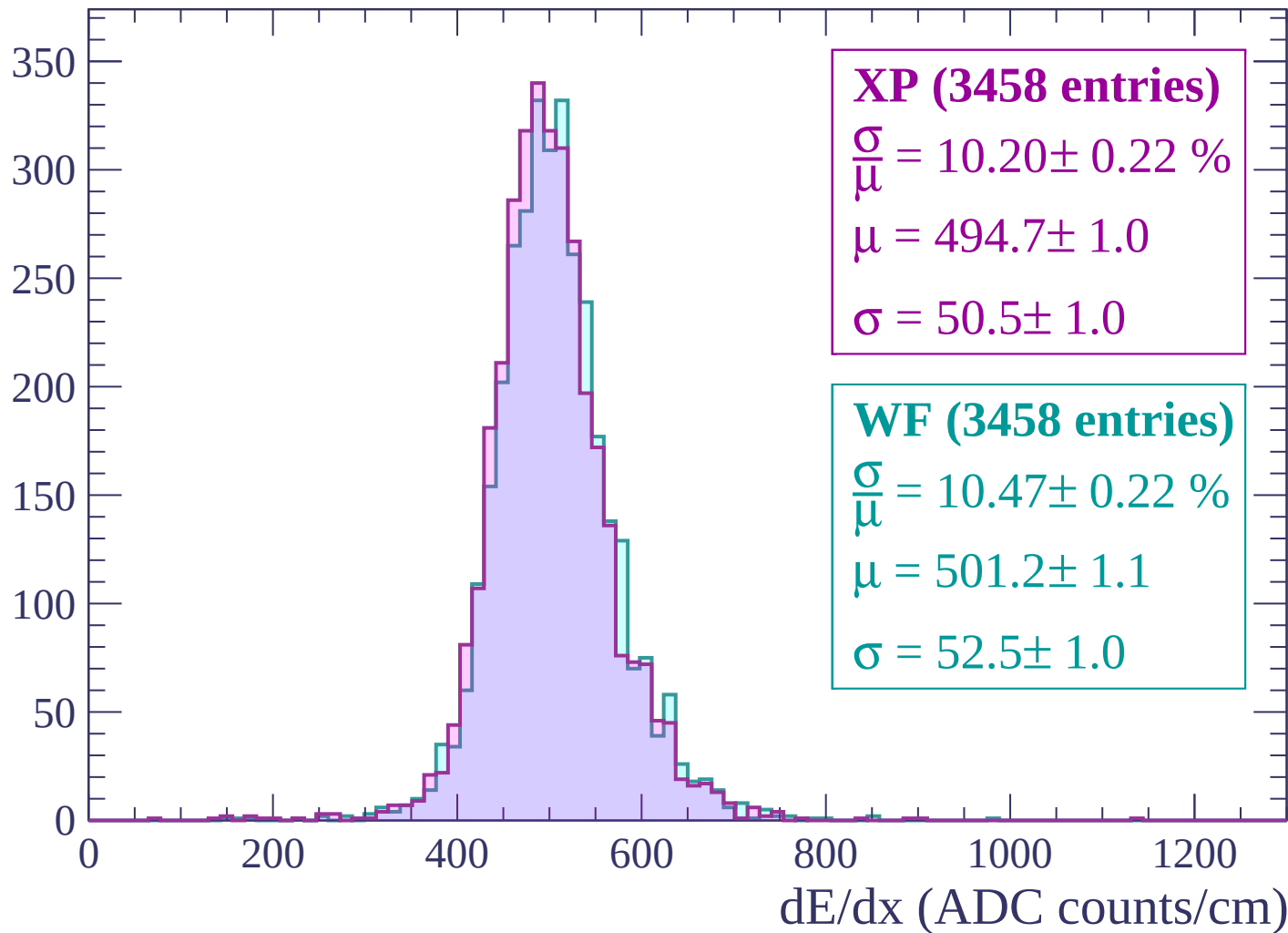
Energy loss | $2700 < p < 2760$

Count



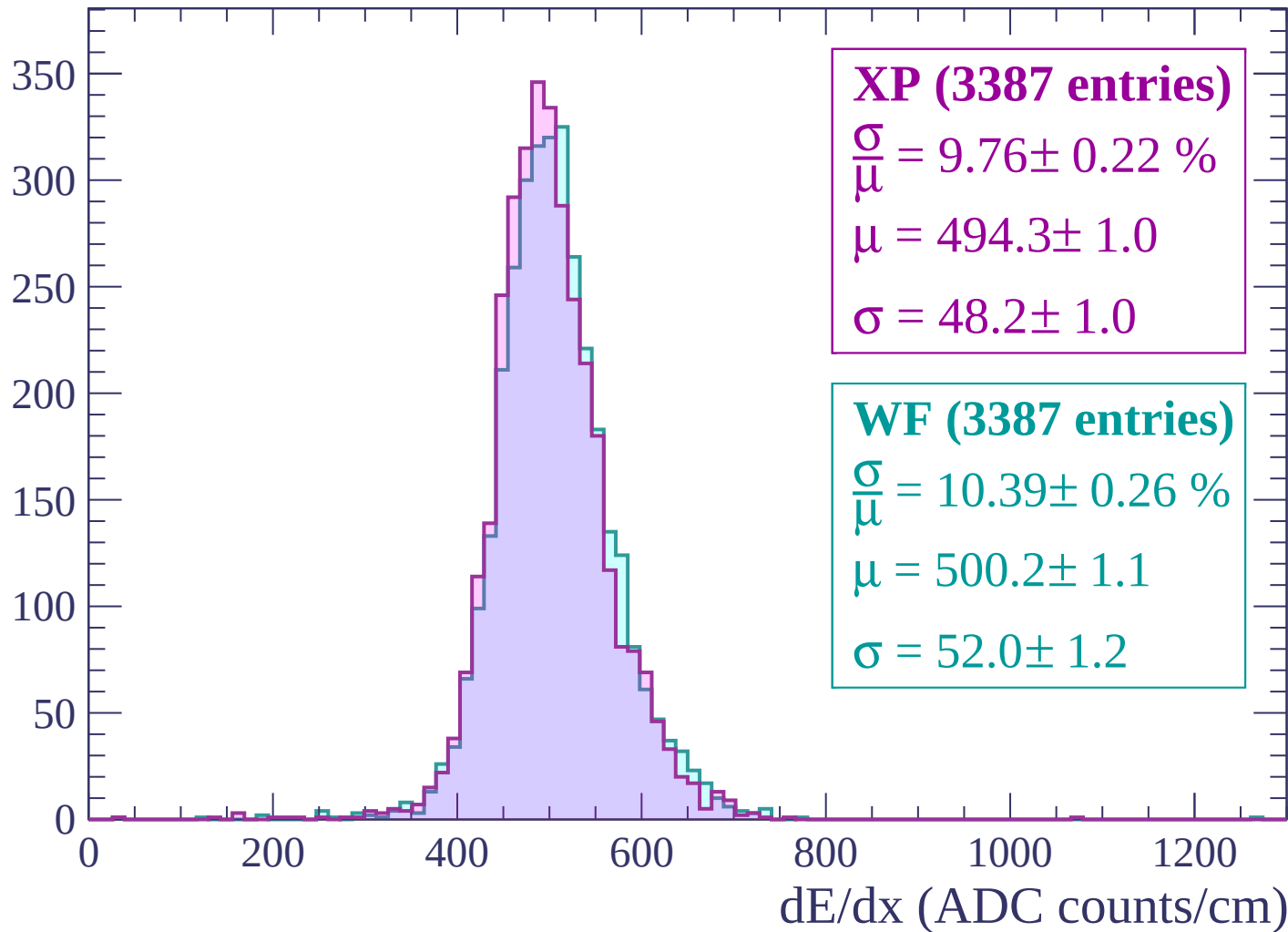
Energy loss | $2760 < p < 2820$

Count



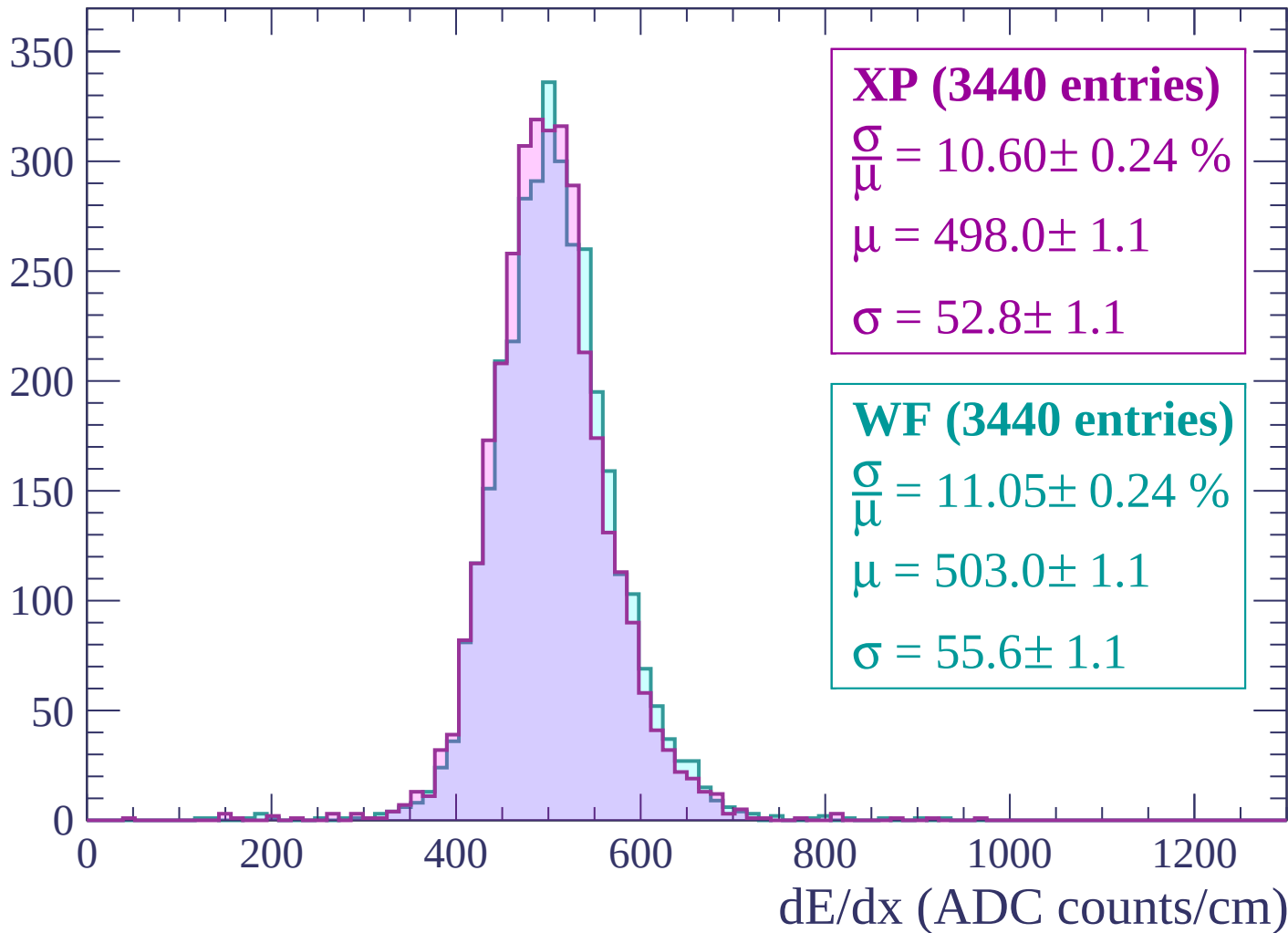
Energy loss | $2820 < p < 2880$

Count



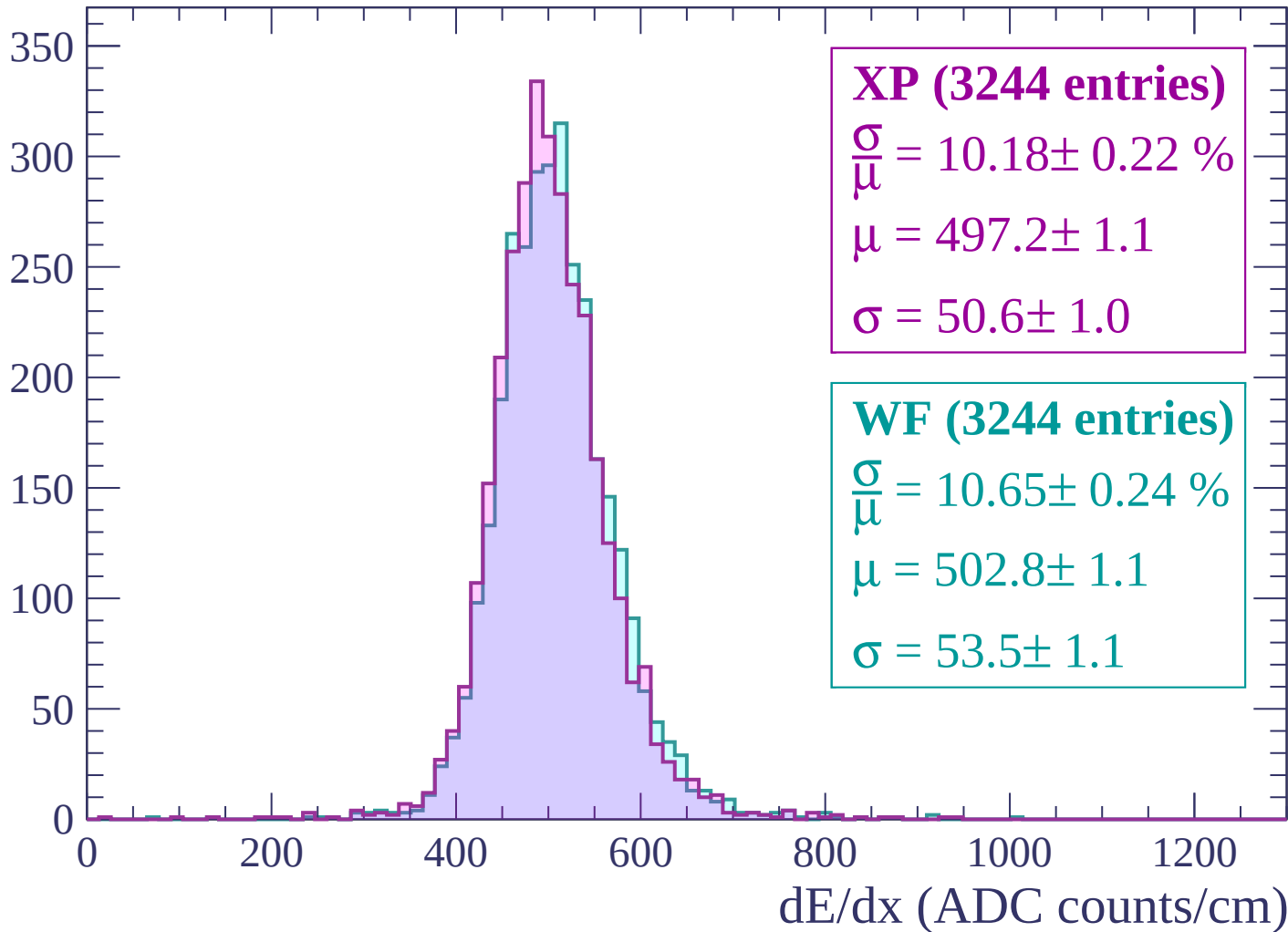
Energy loss | $2880 < p < 2940$

Count



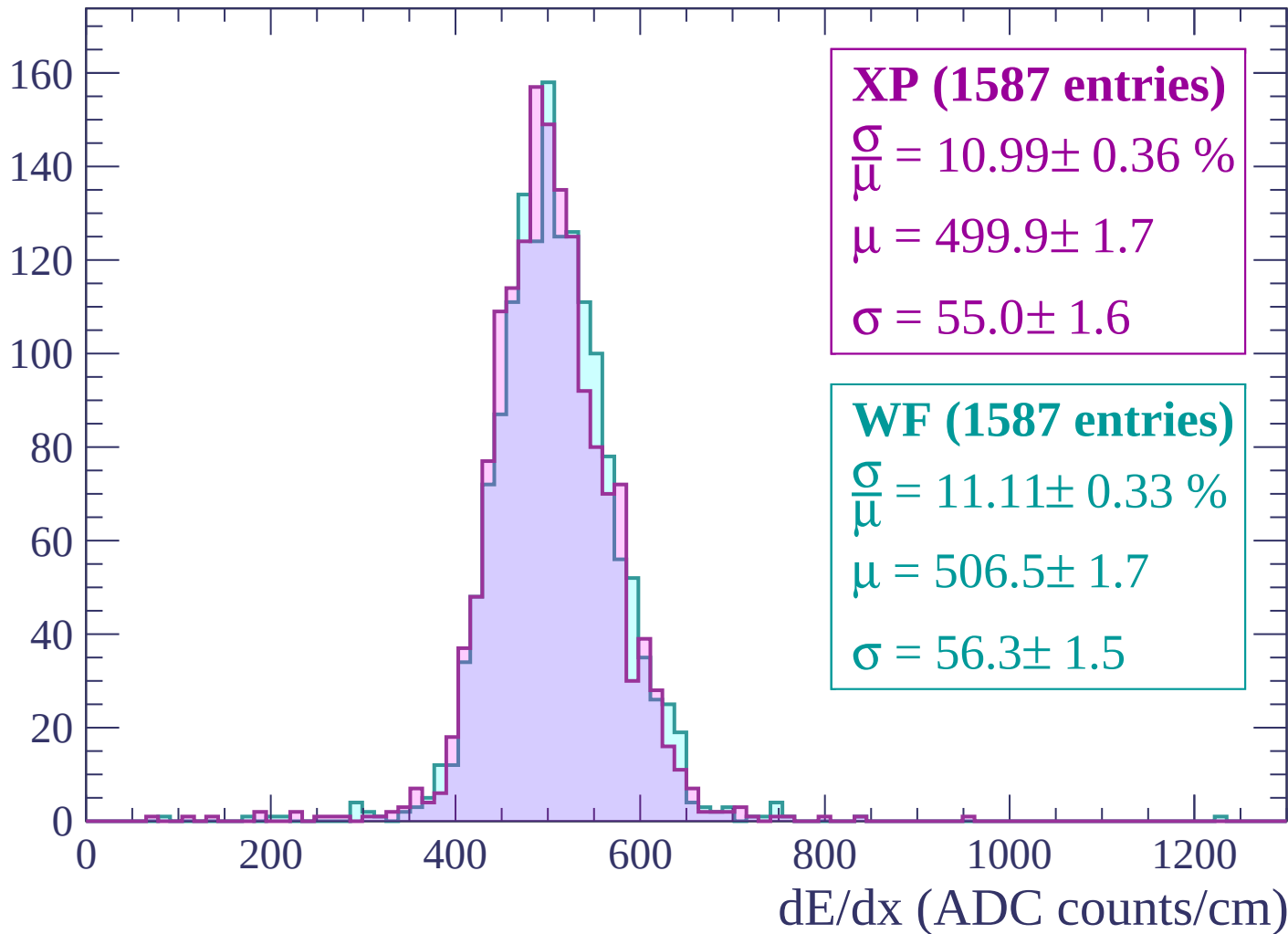
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

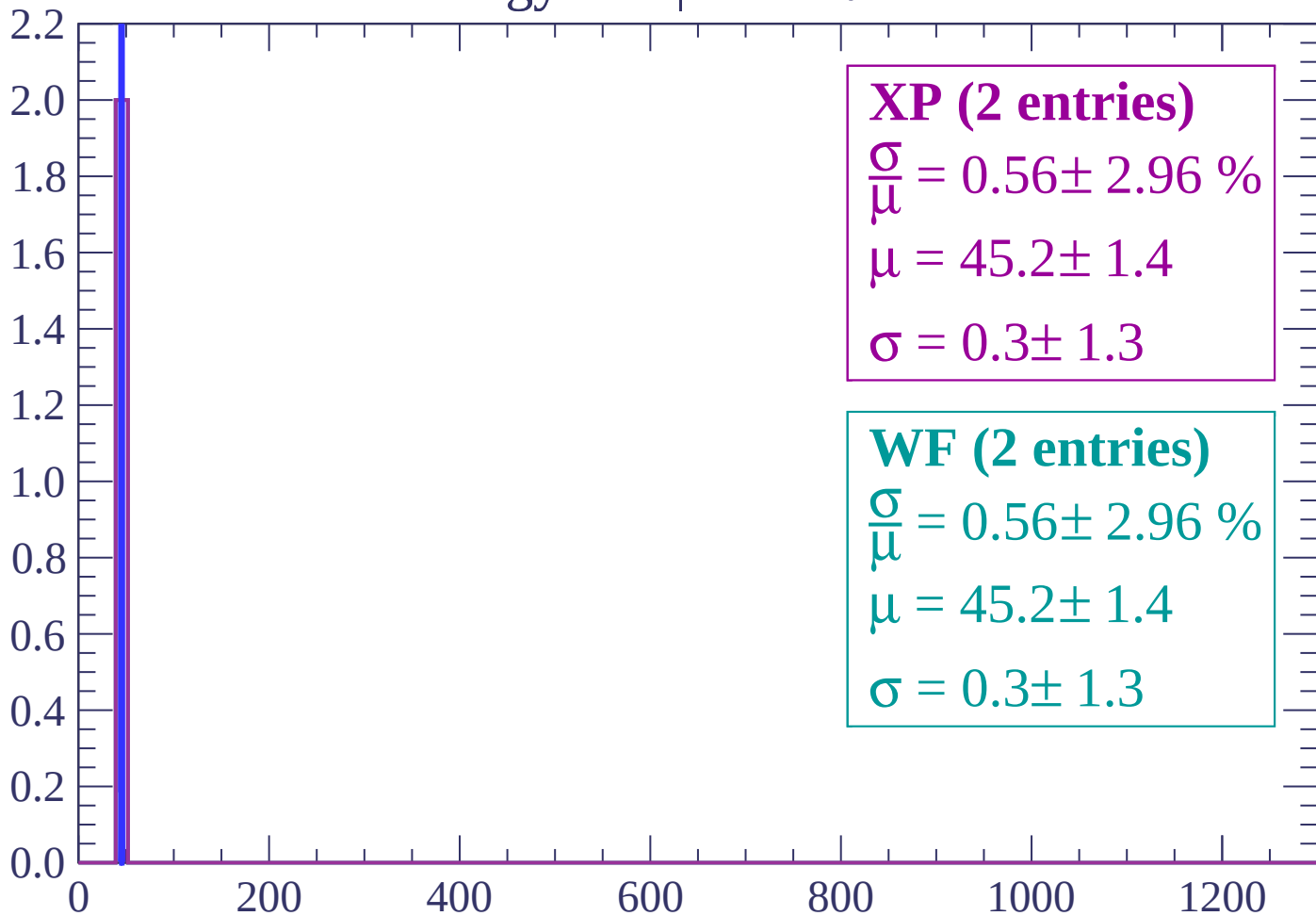
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count



XP (2 entries)

$\frac{\sigma}{\mu} = 0.56 \pm 2.96 \%$

$\mu = 45.2 \pm 1.4$

$\sigma = 0.3 \pm 1.3$

WF (2 entries)

$\frac{\sigma}{\mu} = 0.56 \pm 2.96 \%$

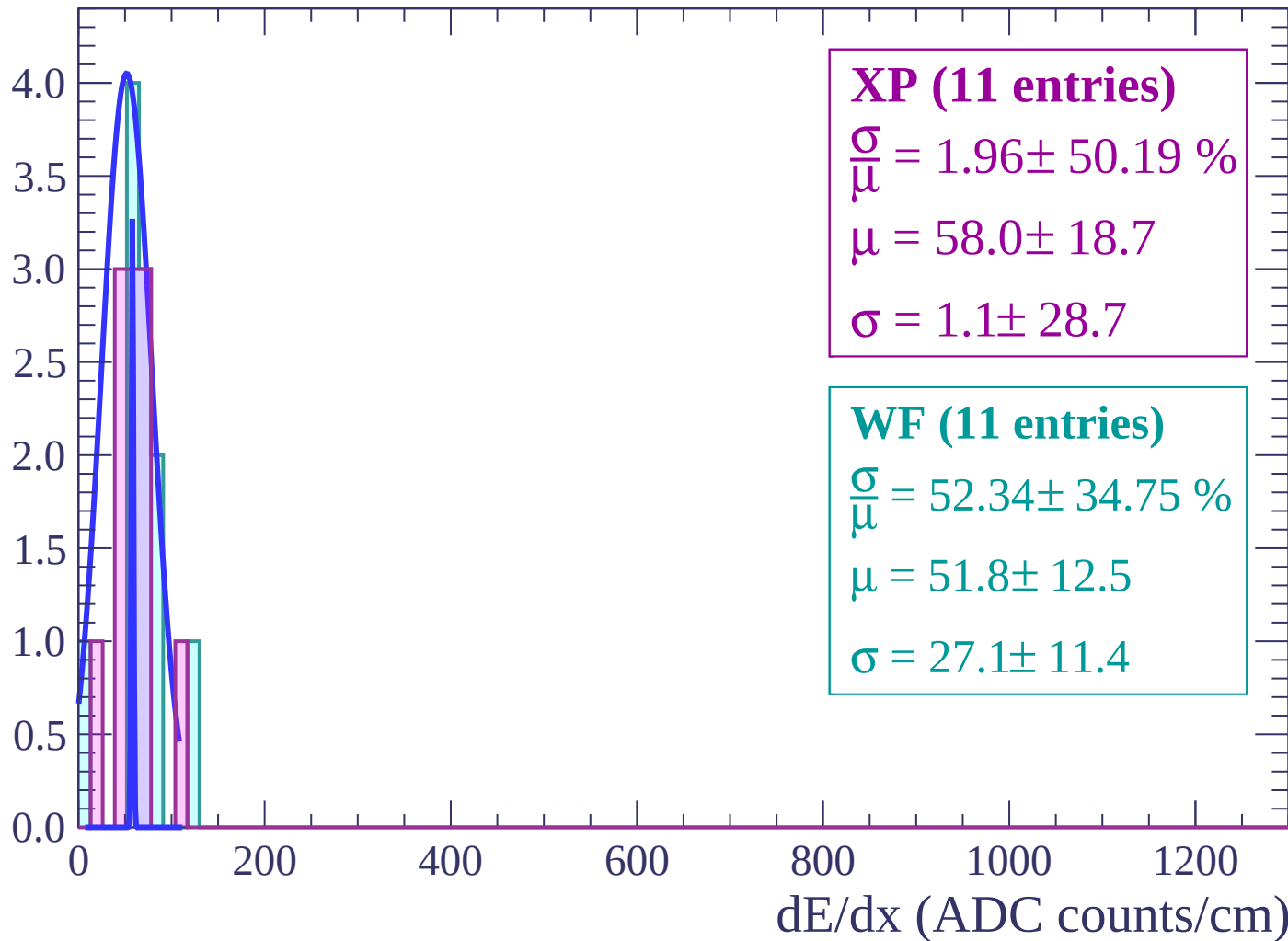
$\mu = 45.2 \pm 1.4$

$\sigma = 0.3 \pm 1.3$

dE/dx (ADC counts/cm)

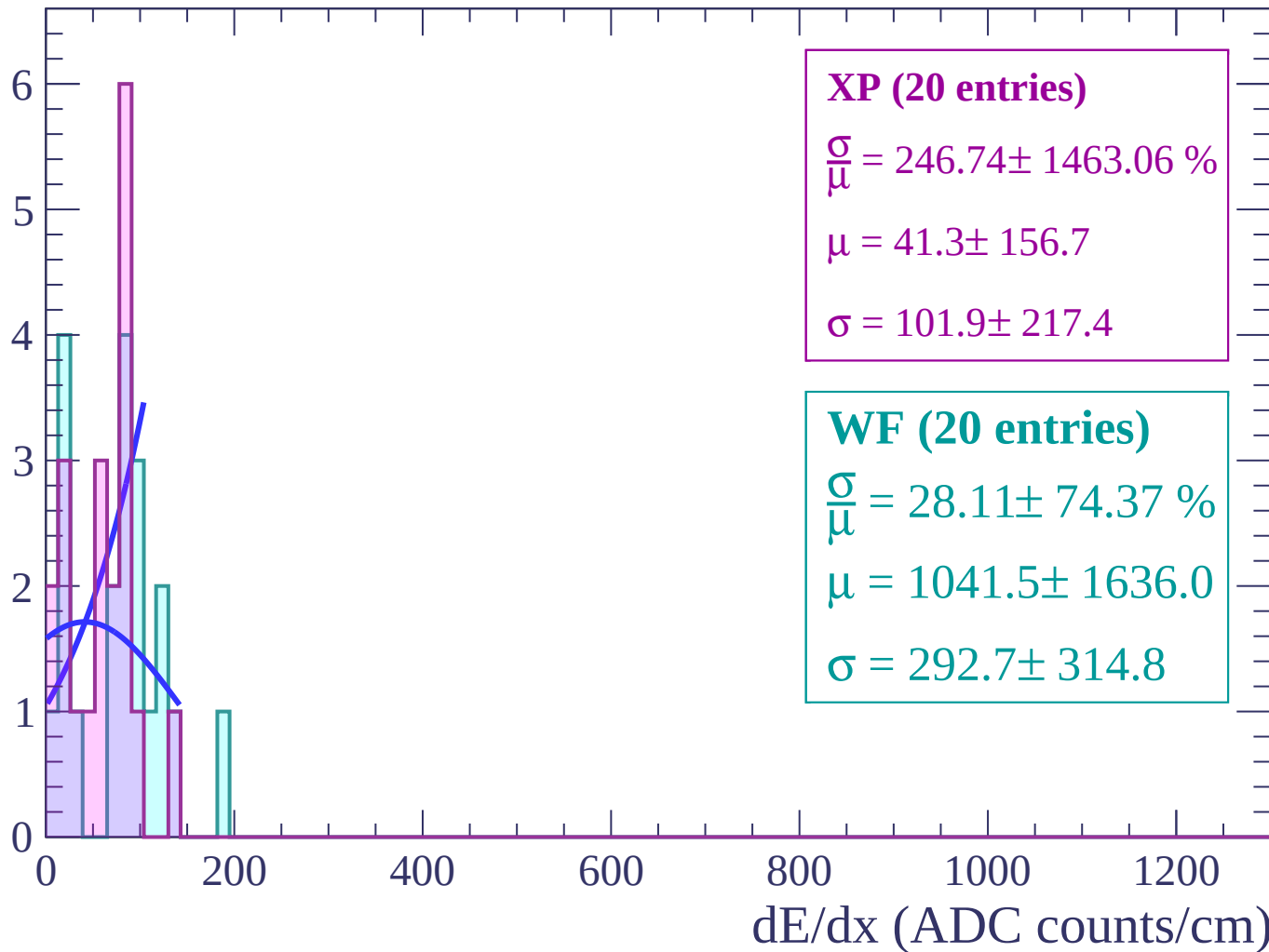
Energy loss | $-82 < \theta < -80$

Count



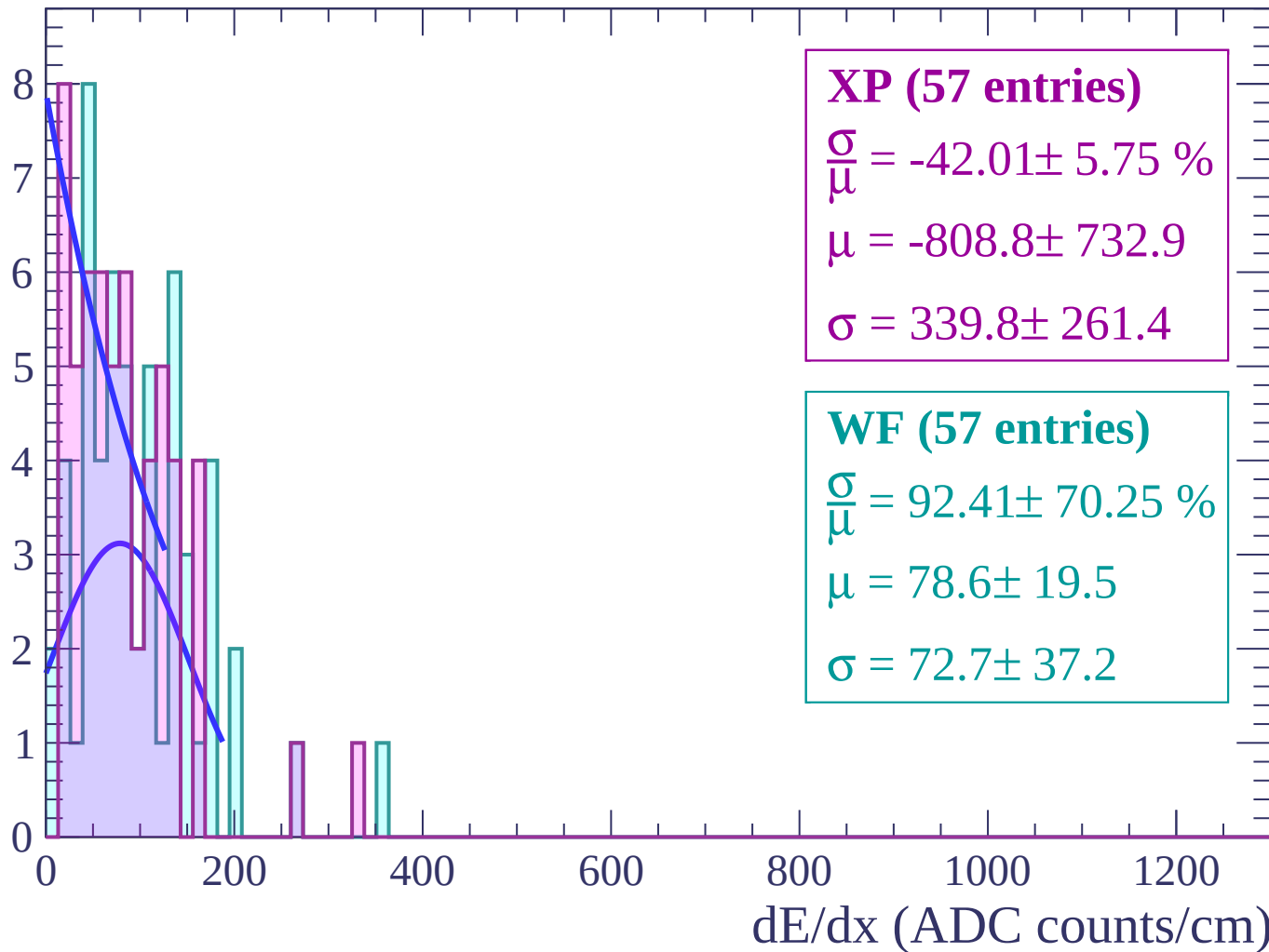
Energy loss | $-80 < \theta < -78$

Count



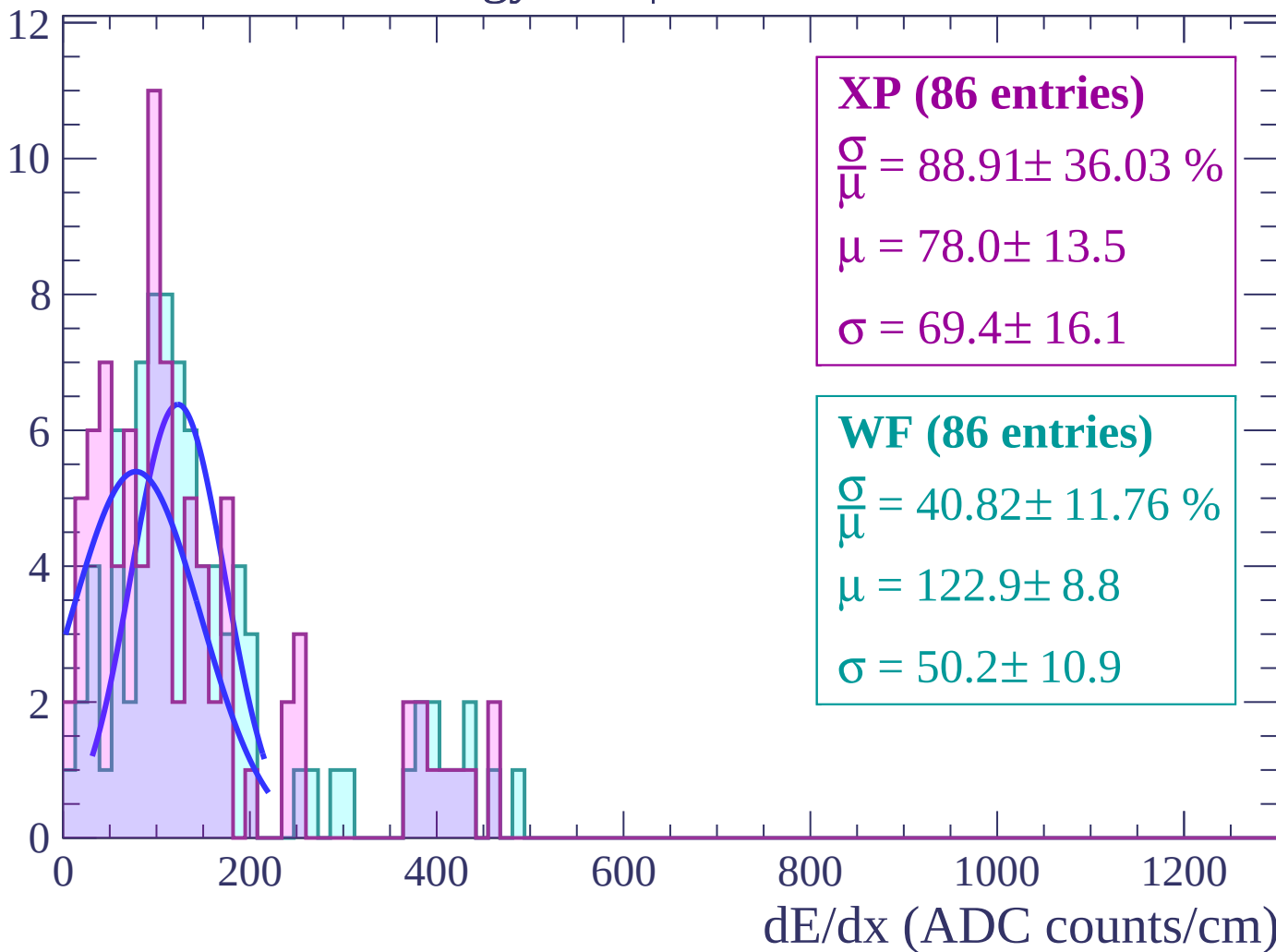
Energy loss | $-78 < \theta < -76$

Count



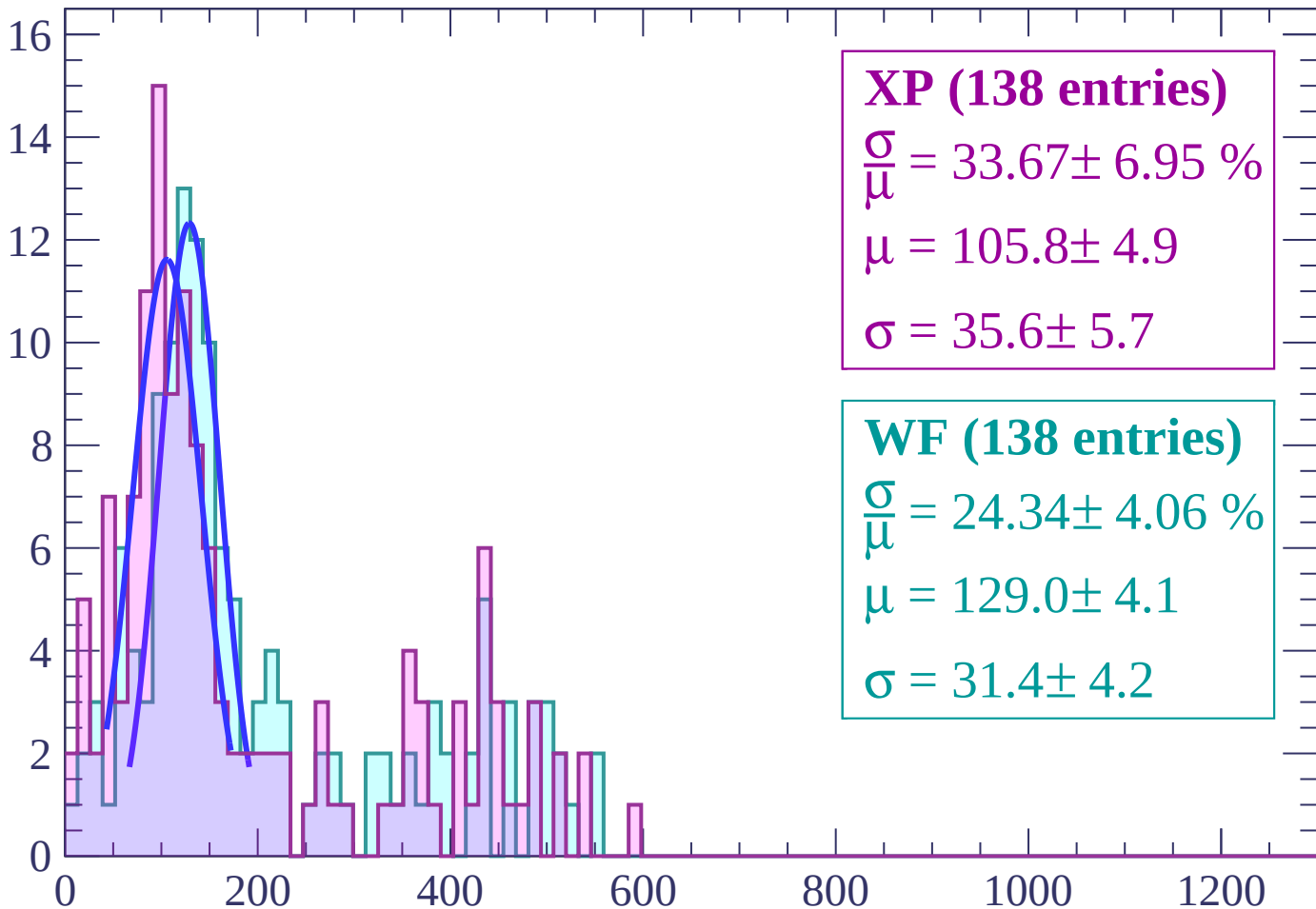
Energy loss | $-76 < \theta < -74$

Count



Energy loss | $-74 < \theta < -72$

Count



XP (138 entries)

$$\frac{\sigma}{\mu} = 33.67 \pm 6.95 \%$$

$$\mu = 105.8 \pm 4.9$$

$$\sigma = 35.6 \pm 5.7$$

WF (138 entries)

$$\frac{\sigma}{\mu} = 24.34 \pm 4.06 \%$$

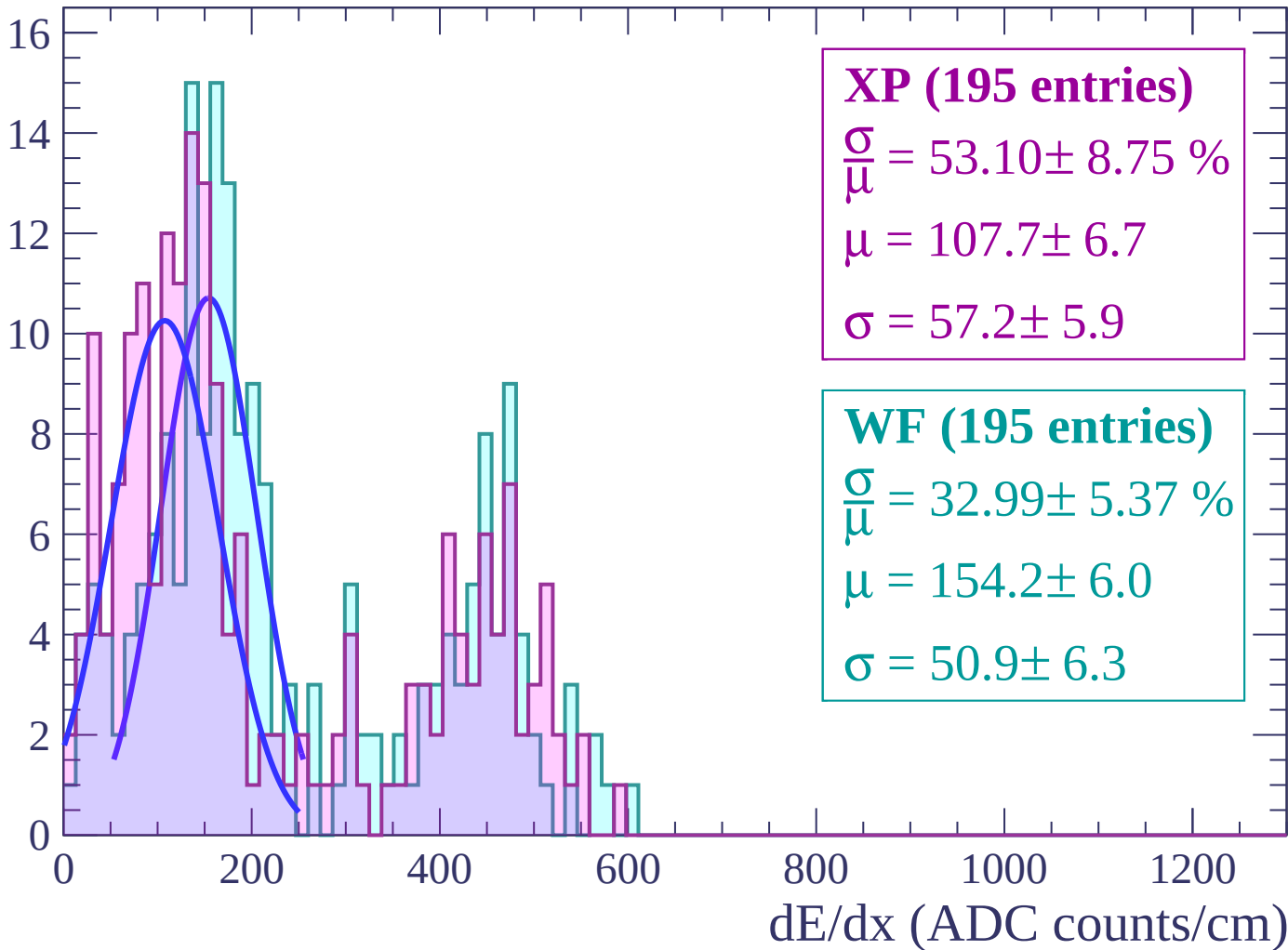
$$\mu = 129.0 \pm 4.1$$

$$\sigma = 31.4 \pm 4.2$$

dE/dx (ADC counts/cm)

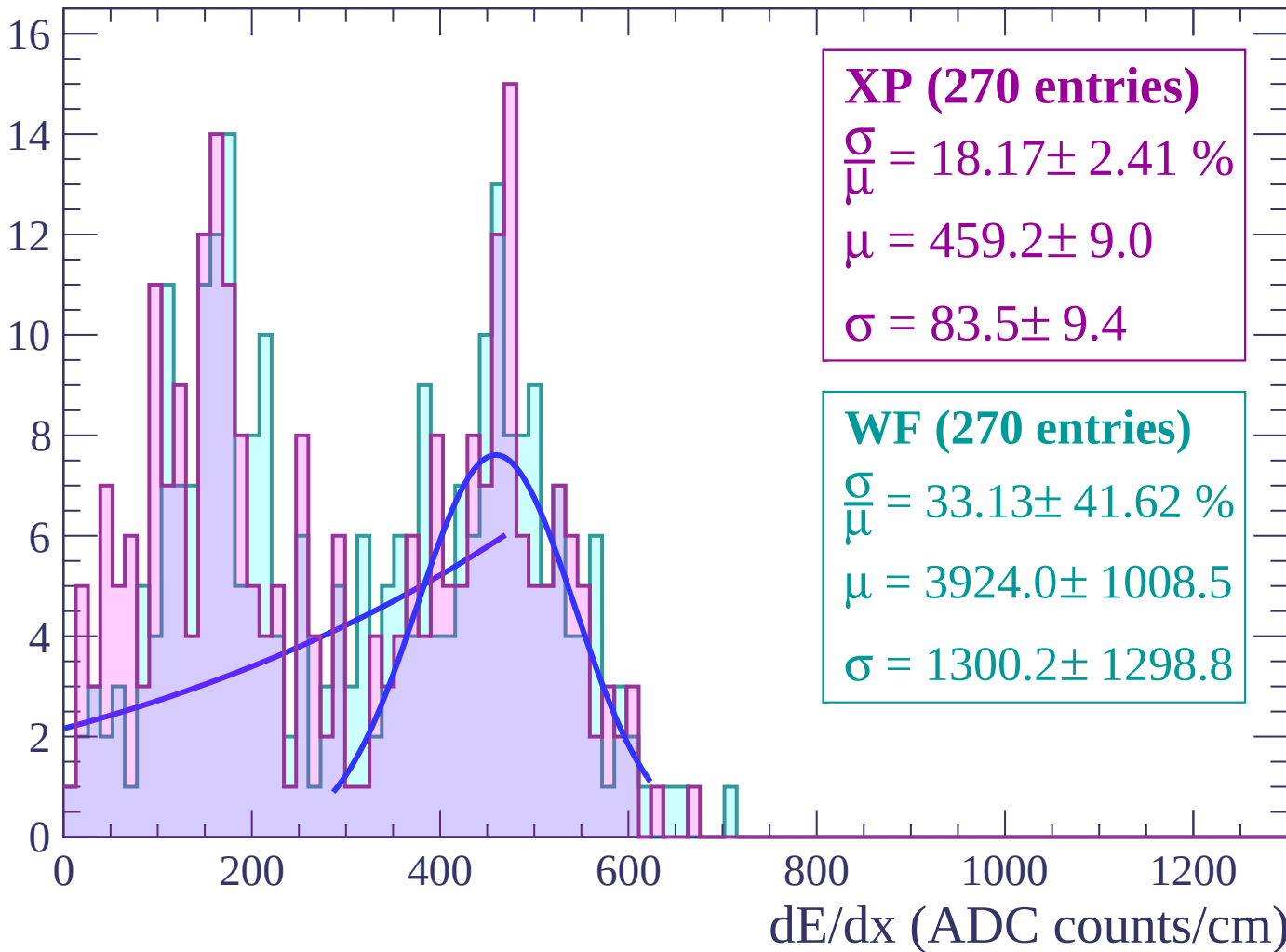
Energy loss | $-72 < \theta < -70$

Count



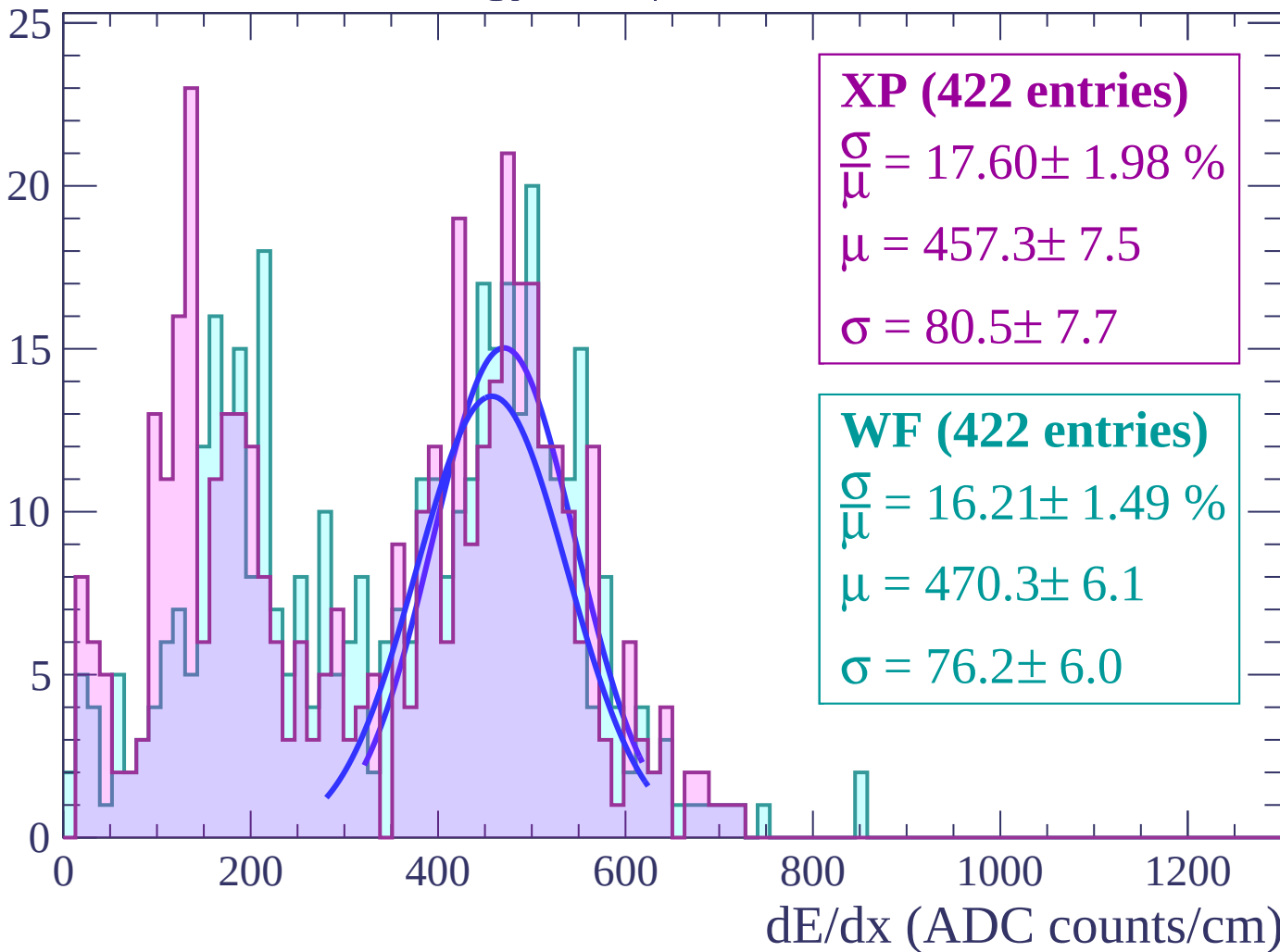
Energy loss | $-70 < \theta < -68$

Count



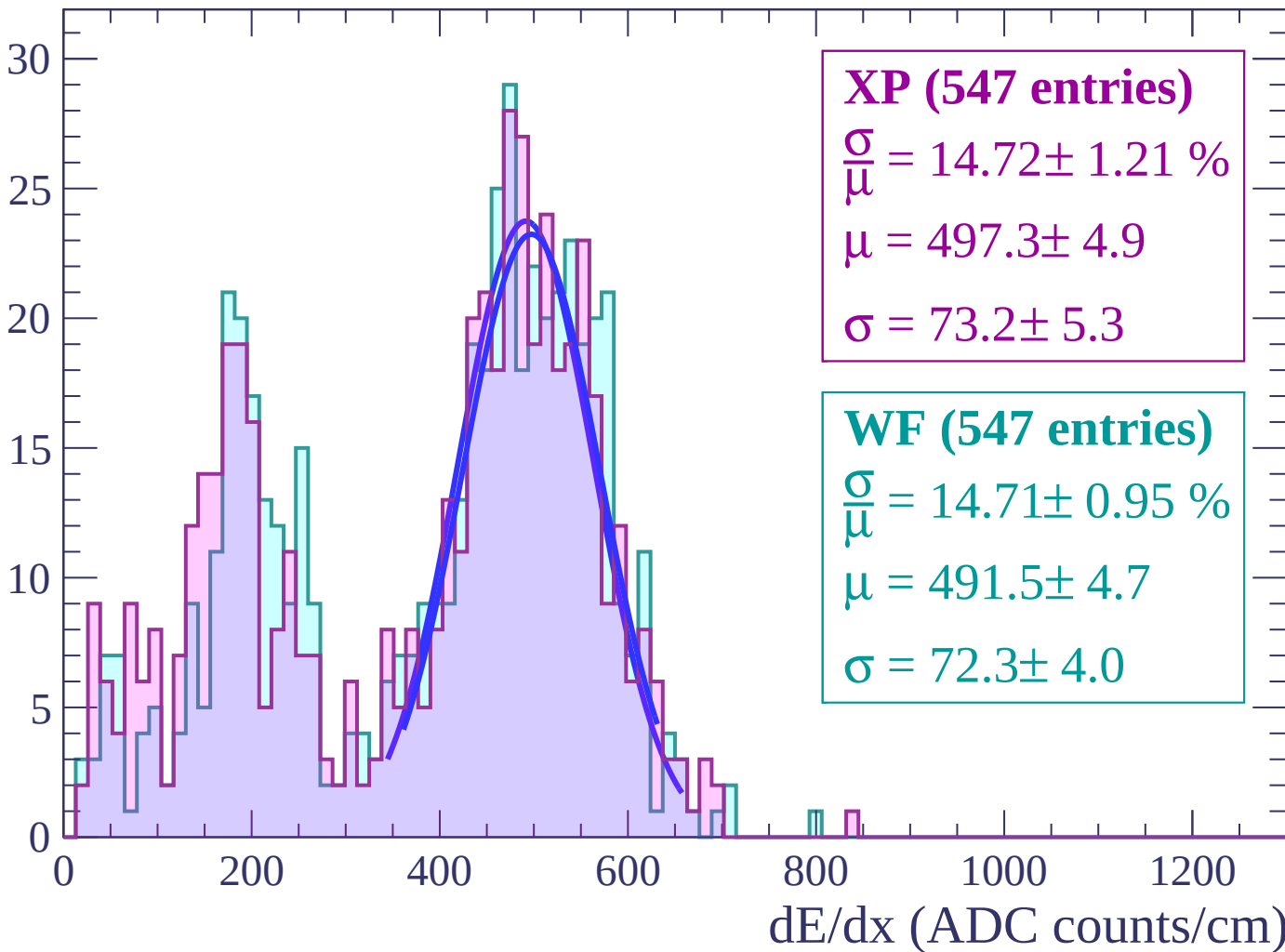
Energy loss | $-68 < \theta < -66$

Count



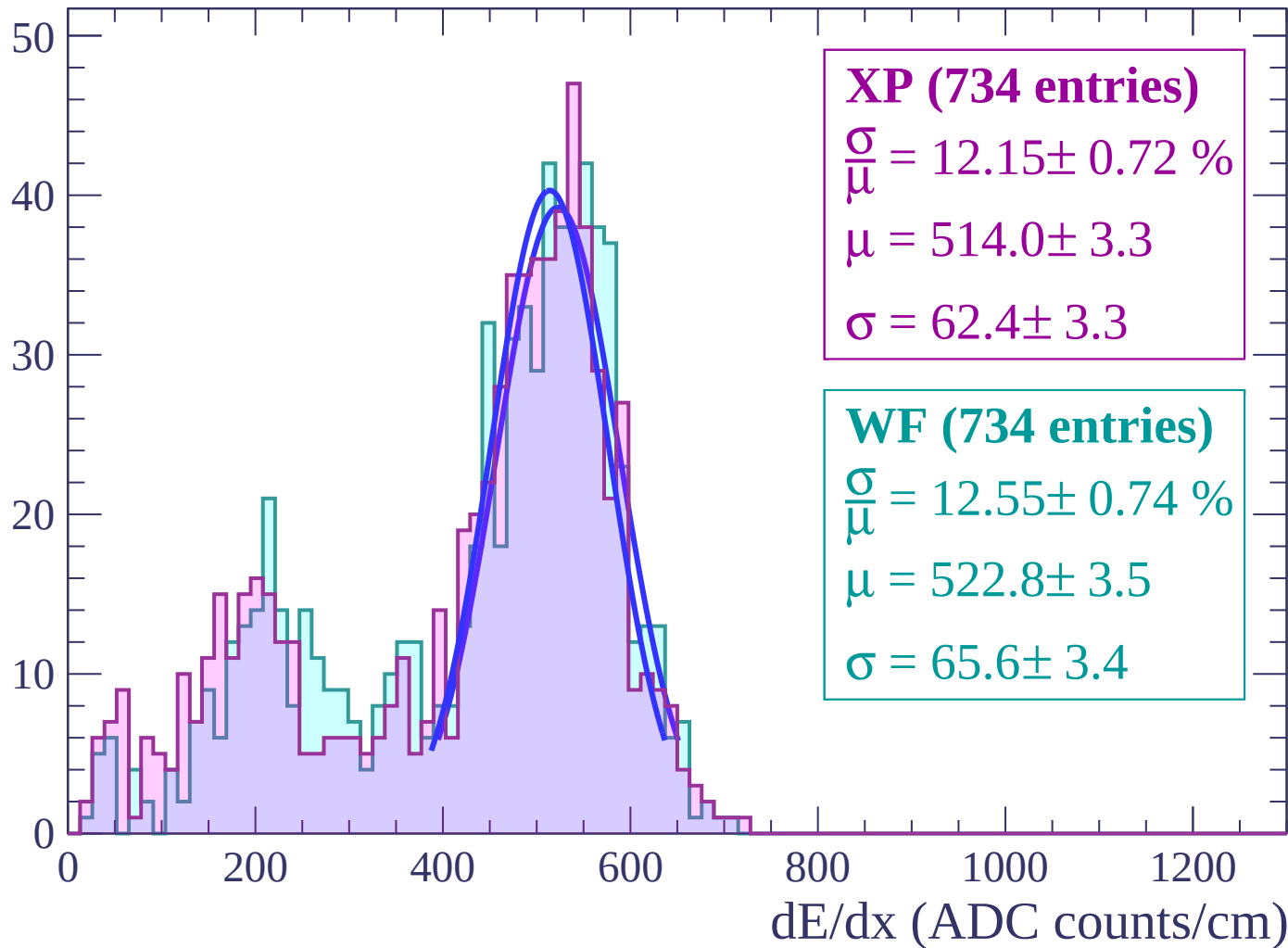
Energy loss | $-66 < \theta < -64$

Count



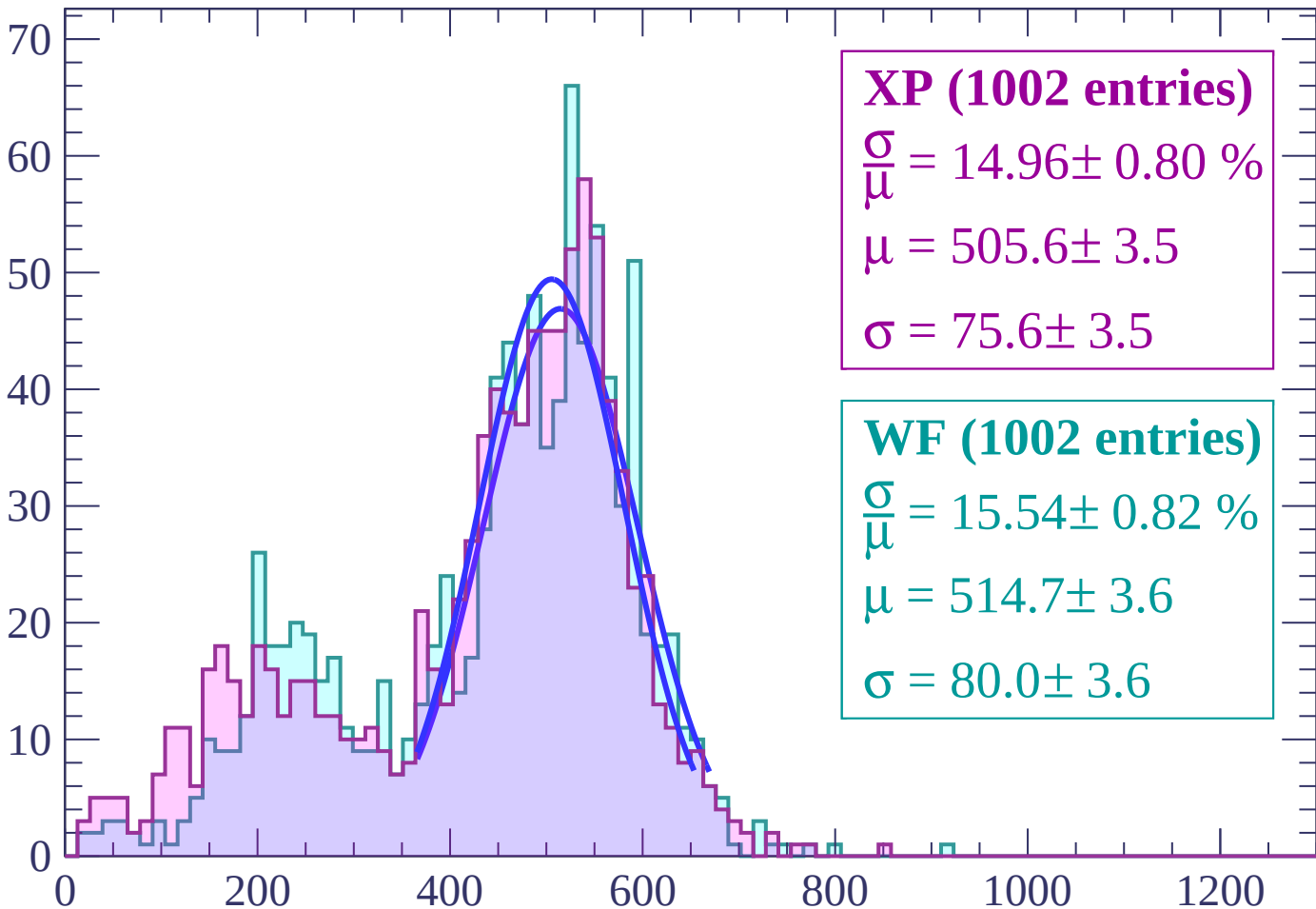
Energy loss | $-64 < \theta < -62$

Count



Energy loss | $-62 < \theta < -60$

Count



XP (1002 entries)

$$\frac{\sigma}{\mu} = 14.96 \pm 0.80 \%$$

$$\mu = 505.6 \pm 3.5$$

$$\sigma = 75.6 \pm 3.5$$

WF (1002 entries)

$$\frac{\sigma}{\mu} = 15.54 \pm 0.82 \%$$

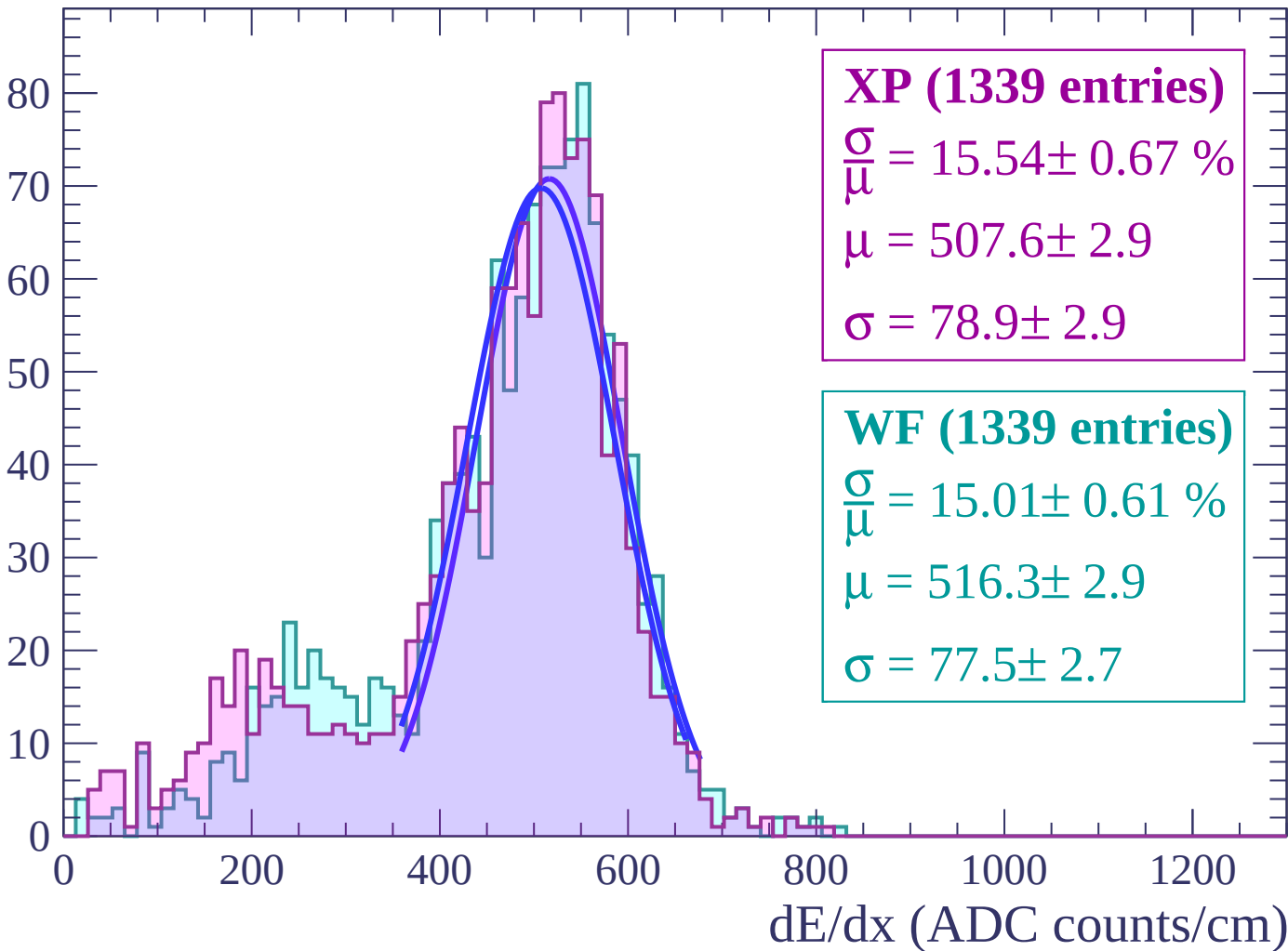
$$\mu = 514.7 \pm 3.6$$

$$\sigma = 80.0 \pm 3.6$$

dE/dx (ADC counts/cm)

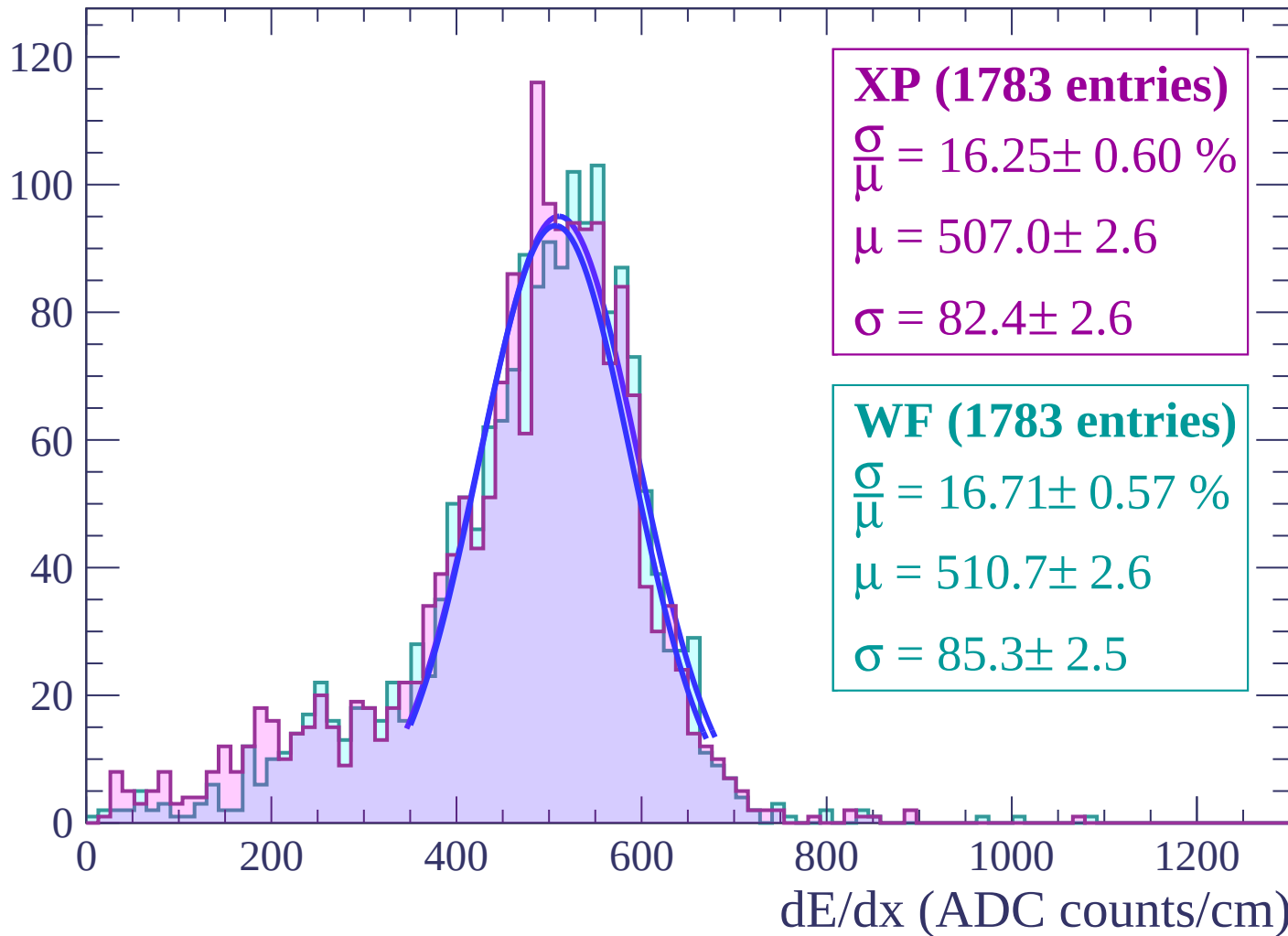
Energy loss | $-60 < \theta < -58$

Count



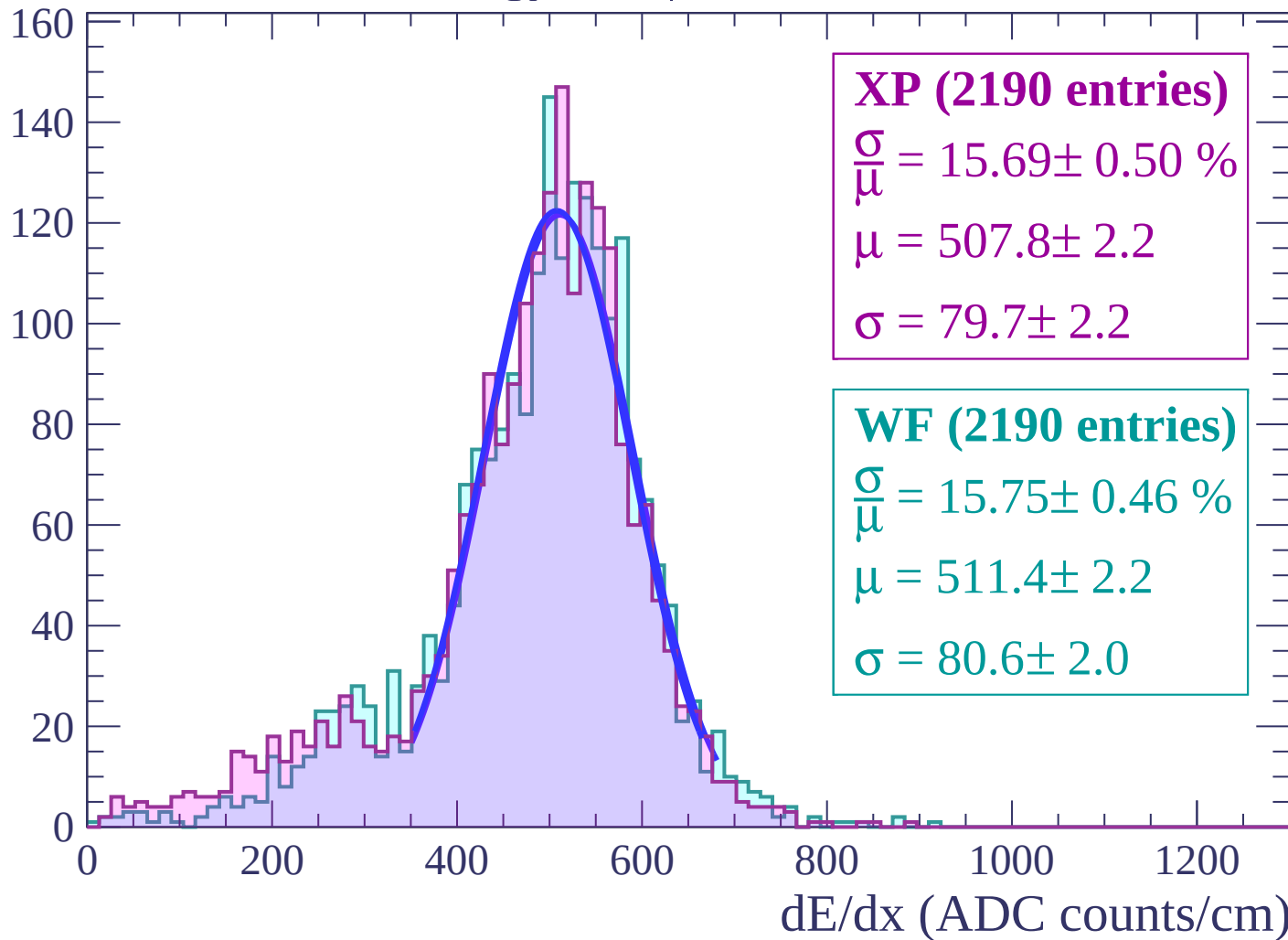
Energy loss | $-58 < \theta < -56$

Count



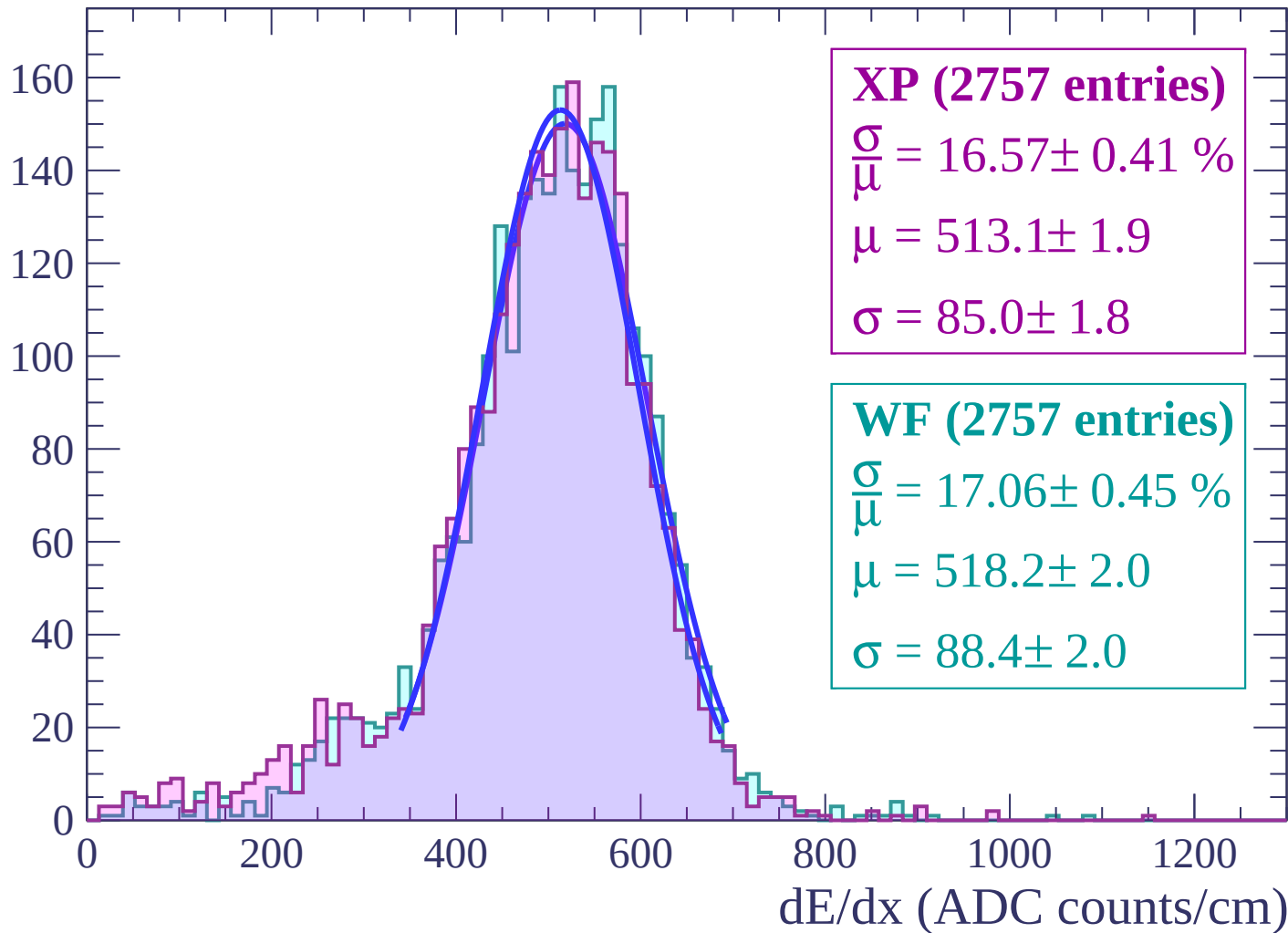
Energy loss | $-56 < \theta < -54$

Count



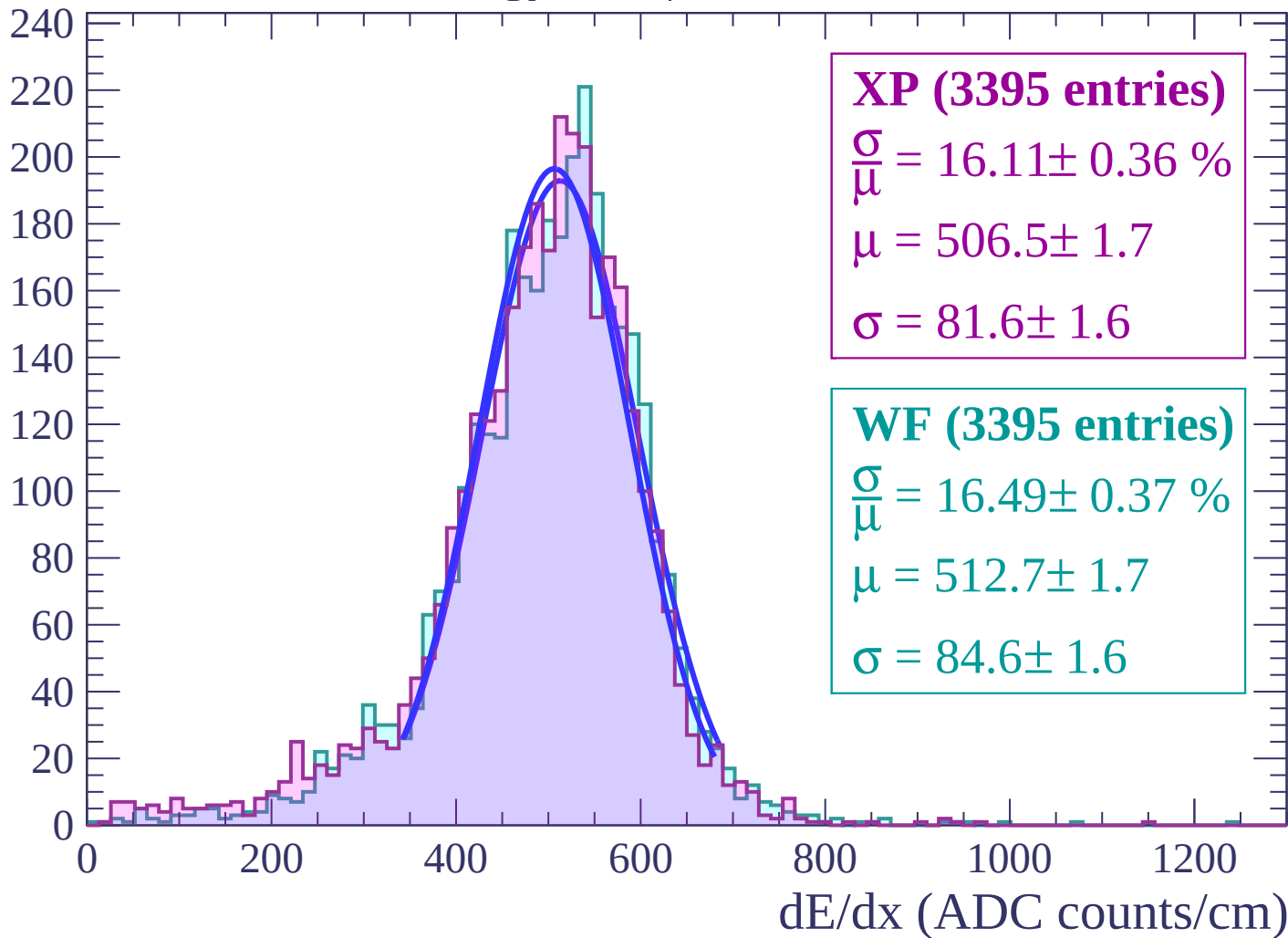
Energy loss | $-54 < \theta < -52$

Count



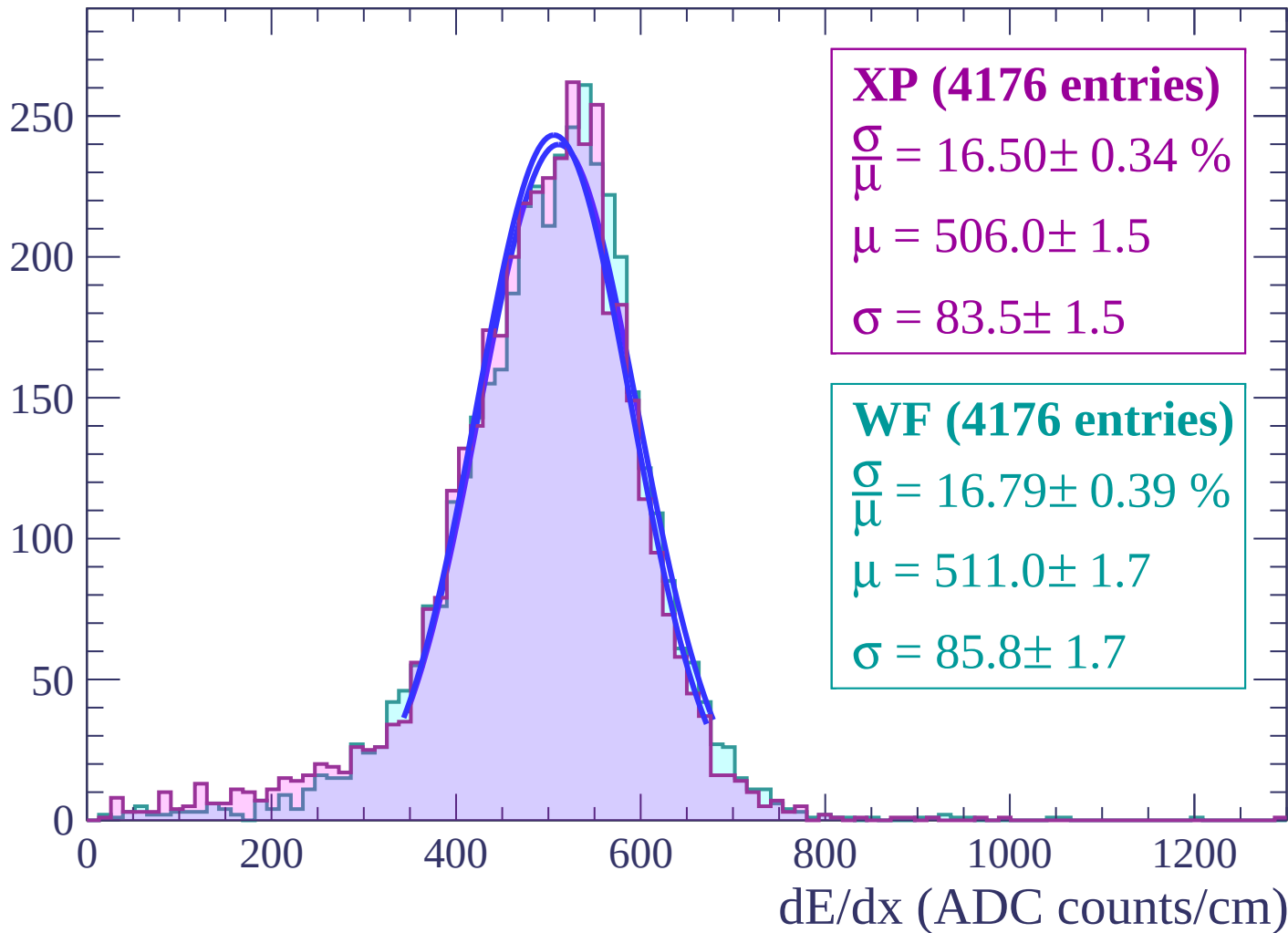
Energy loss | $-52 < \theta < -50$

Count



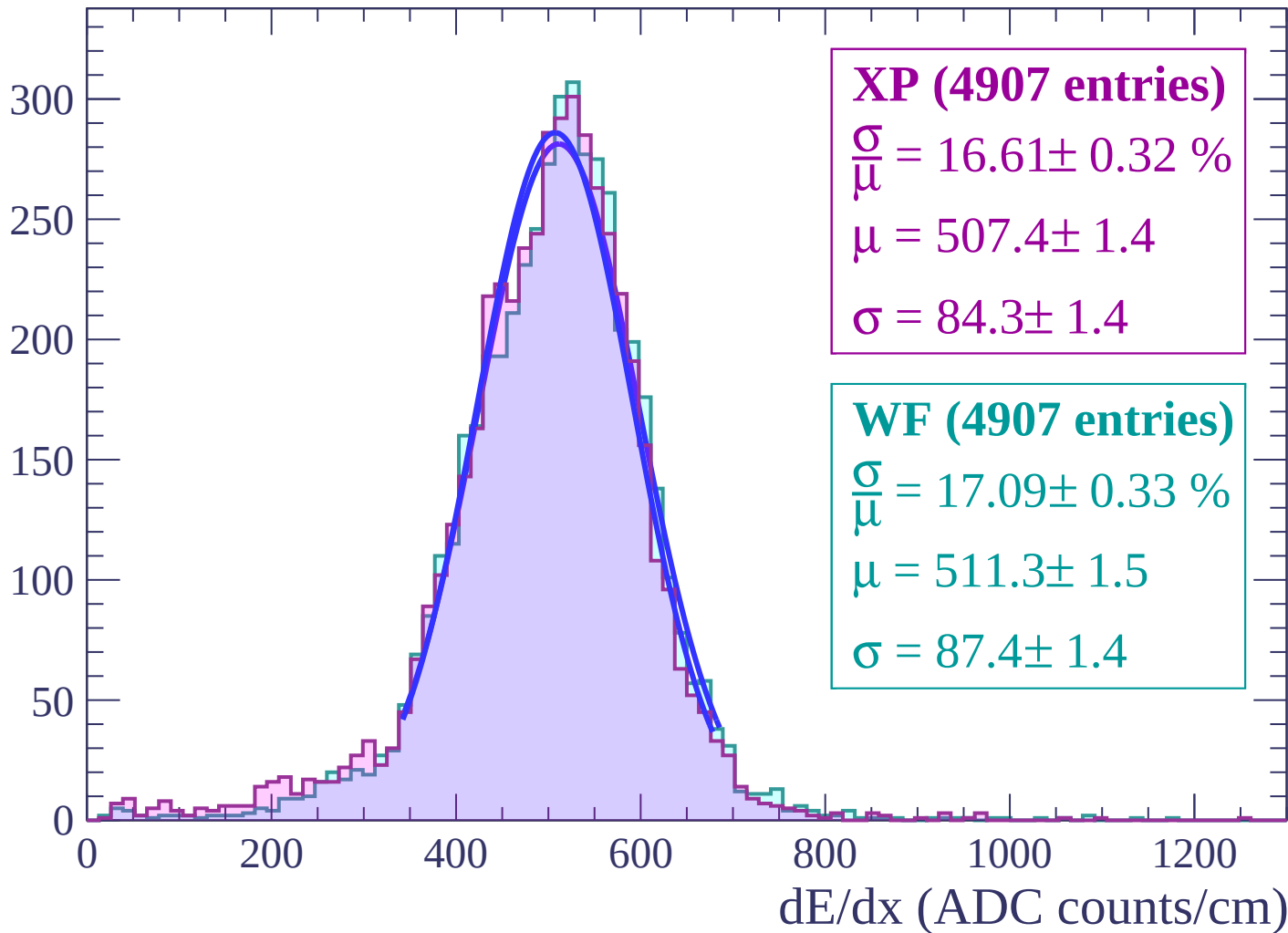
Energy loss | $-50 < \theta < -48$

Count



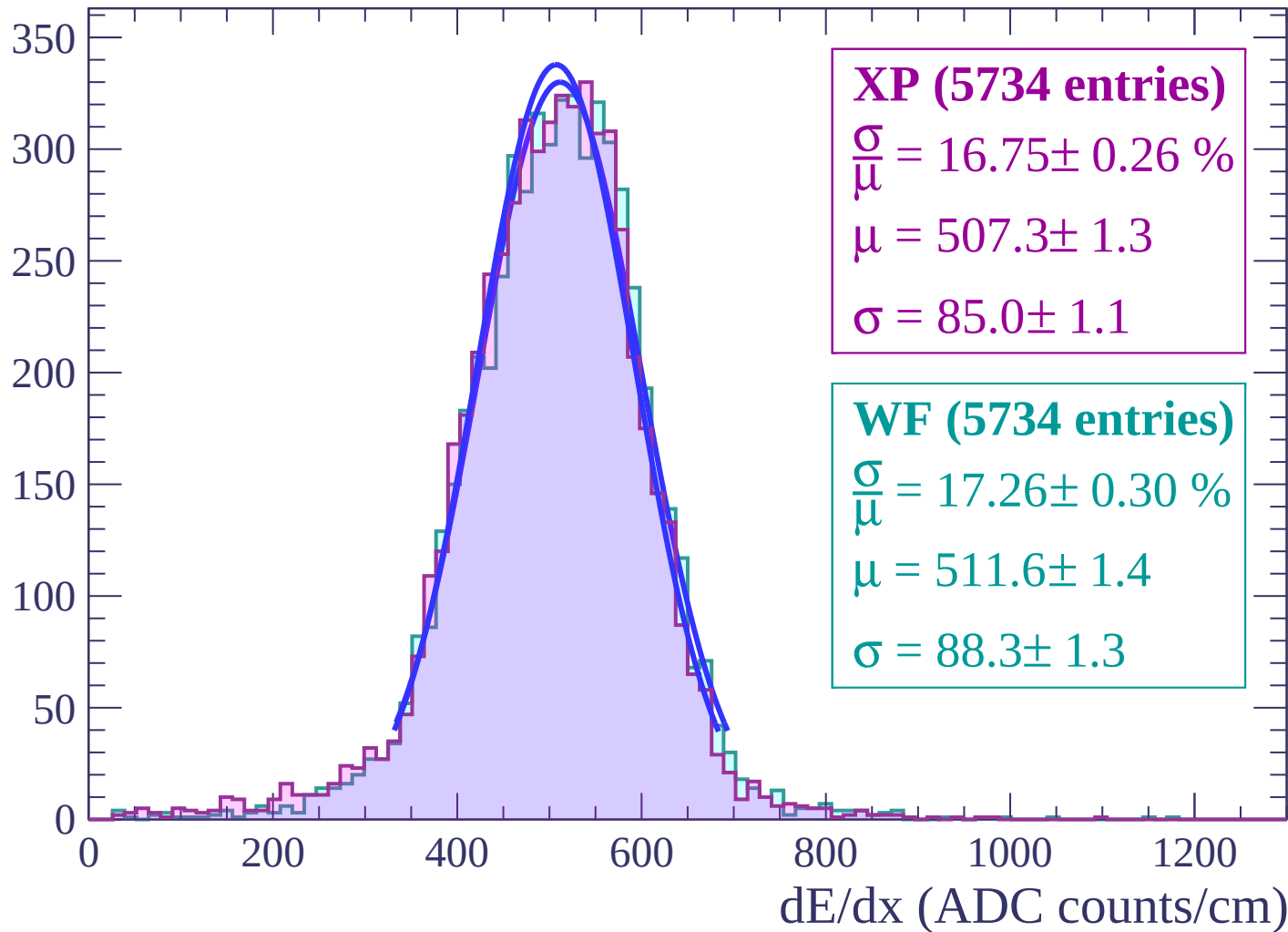
Energy loss | $-48 < \theta < -46$

Count



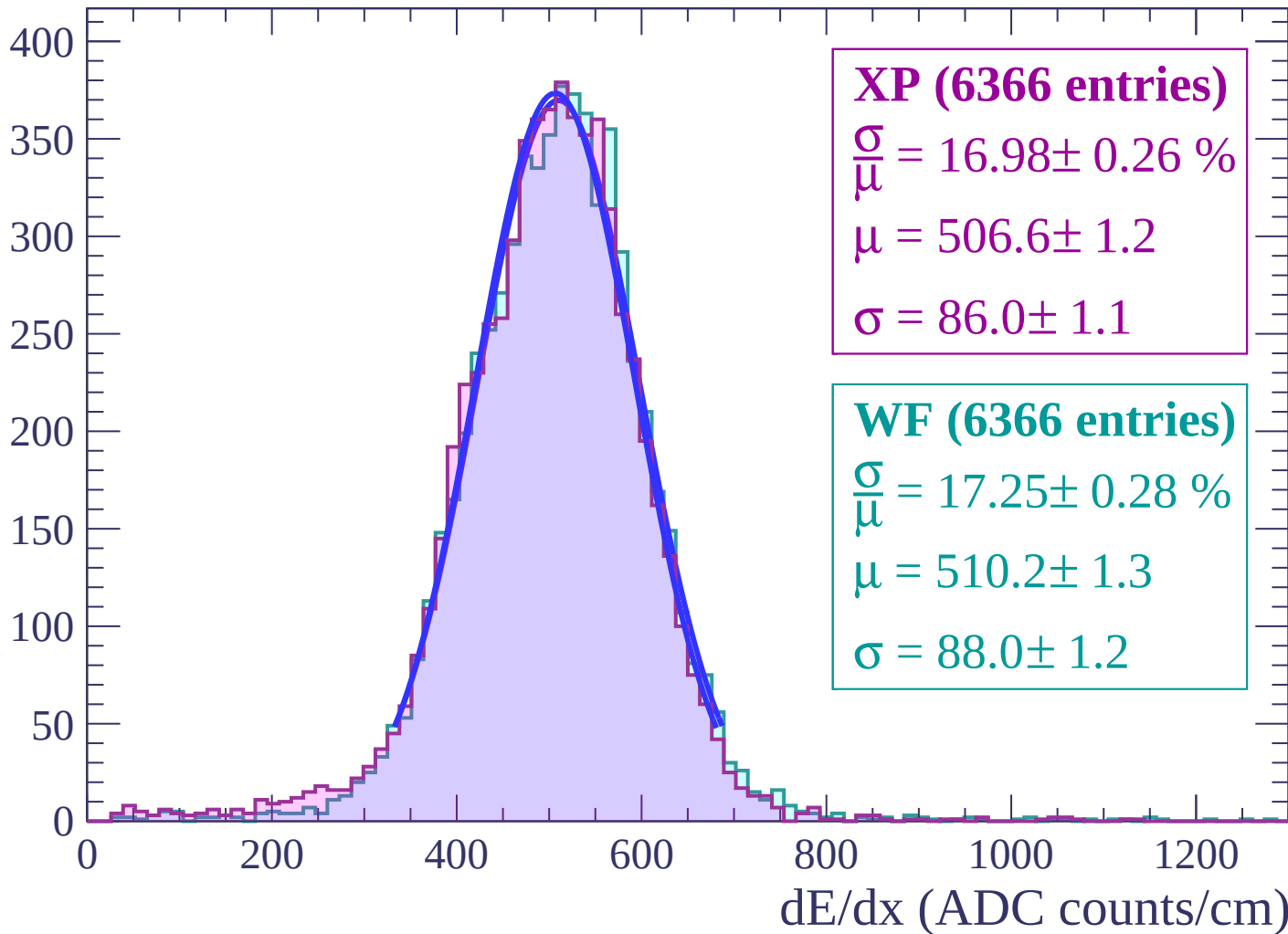
Energy loss | $-46 < \theta < -44$

Count



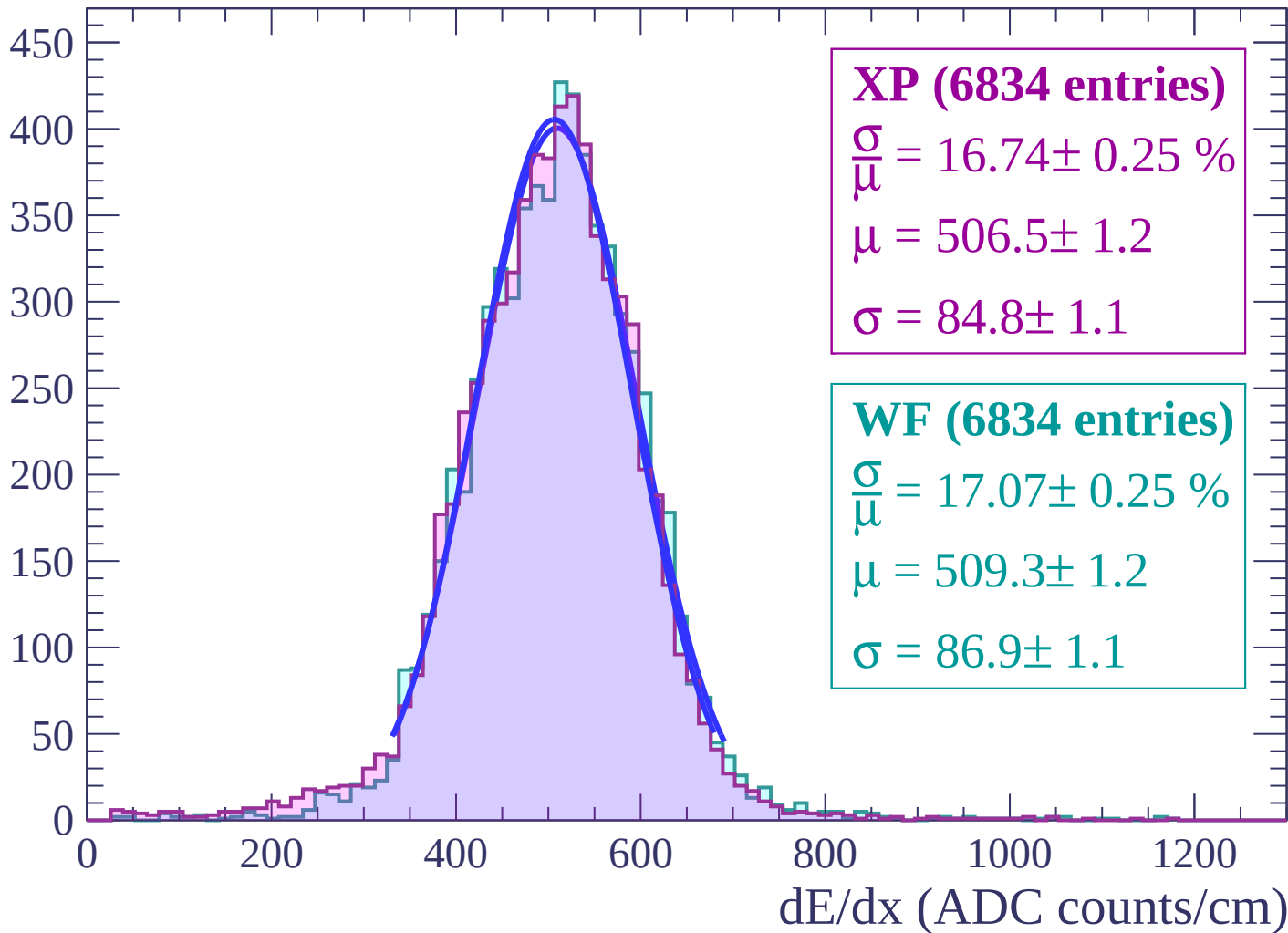
Energy loss | $-44 < \theta < -42$

Count



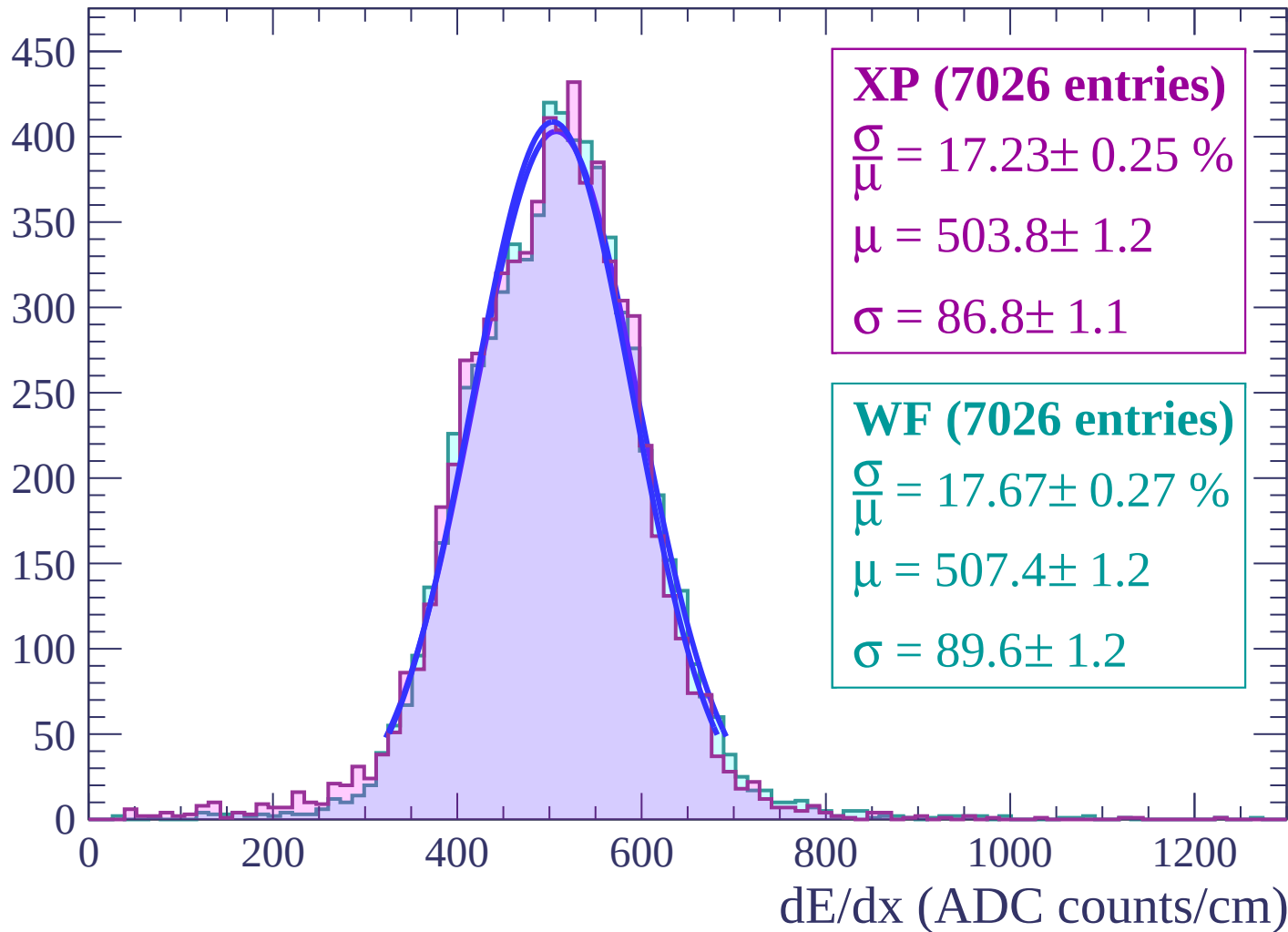
Energy loss | $-42 < \theta < -40$

Count



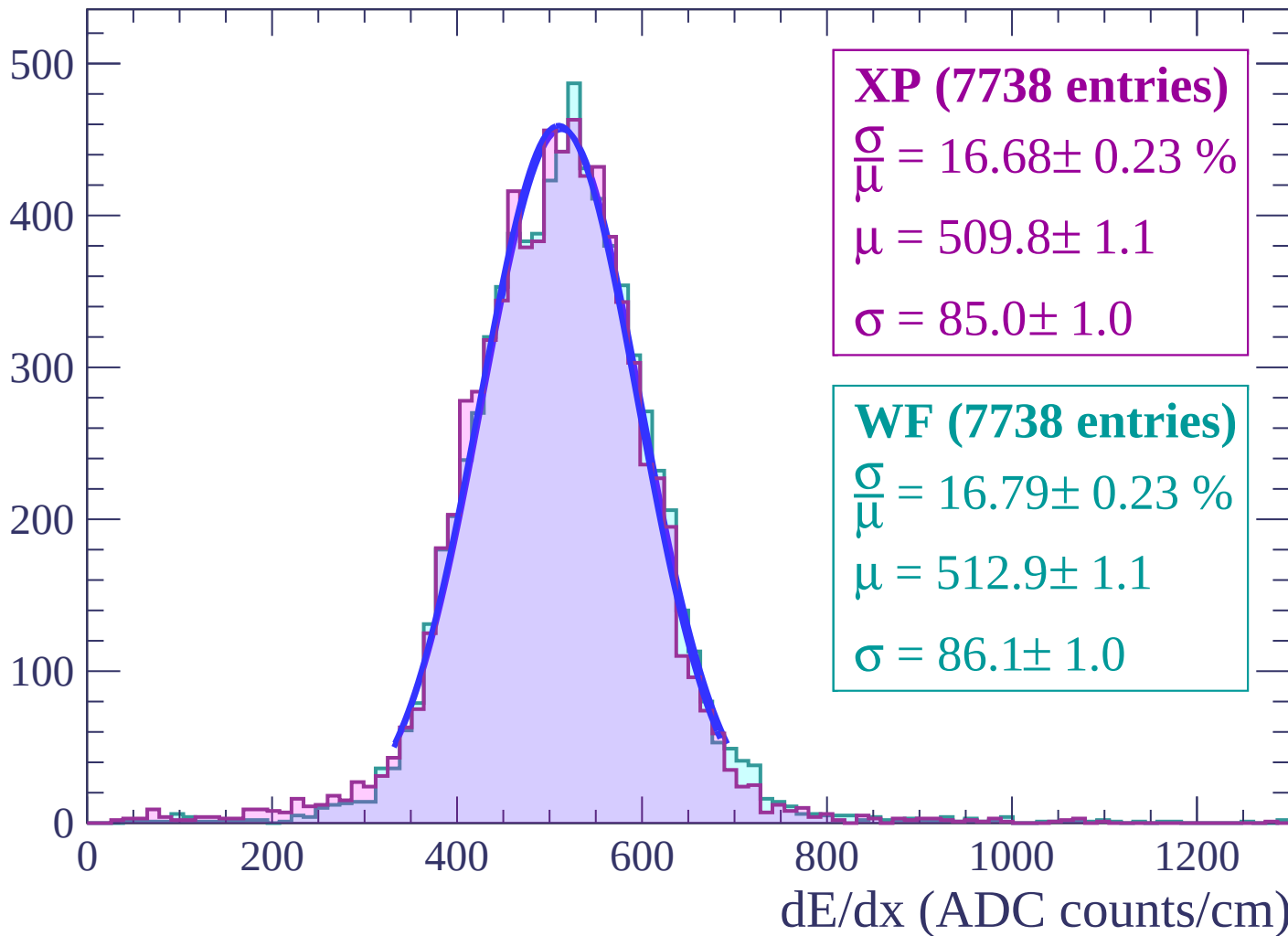
Energy loss | $-40 < \theta < -38$

Count



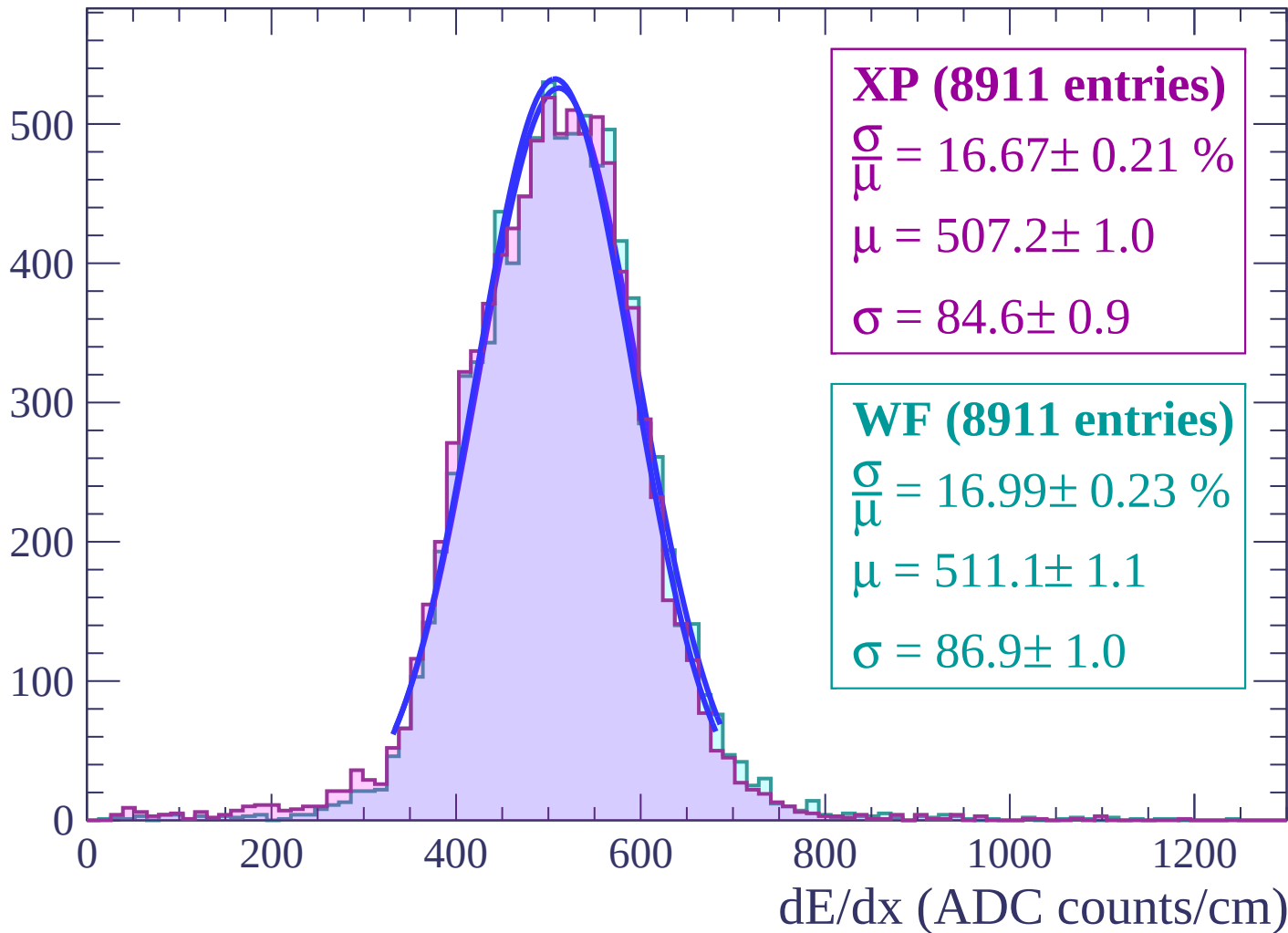
Energy loss | $-38 < \theta < -36$

Count



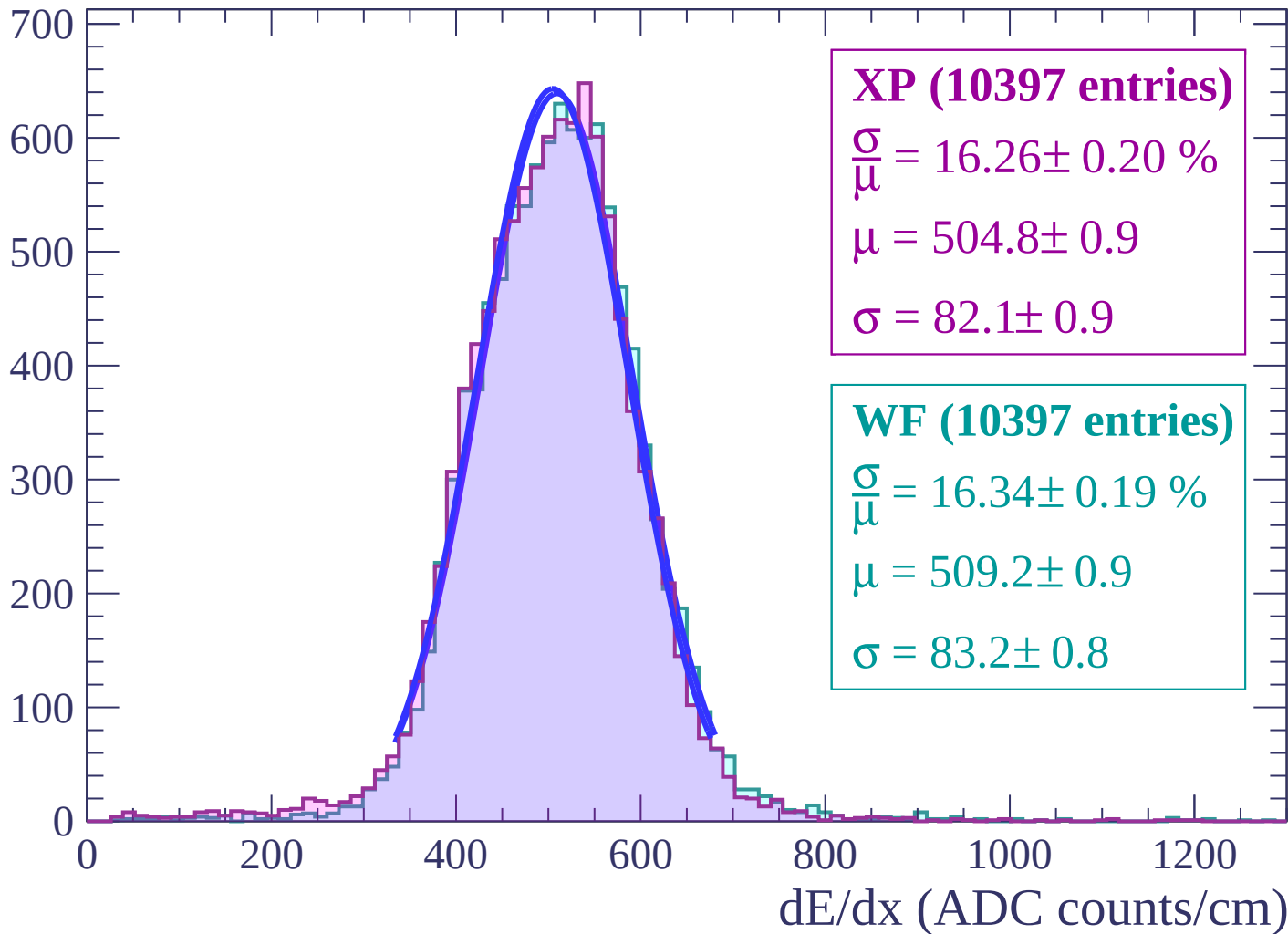
Energy loss | $-36 < \theta < -34$

Count



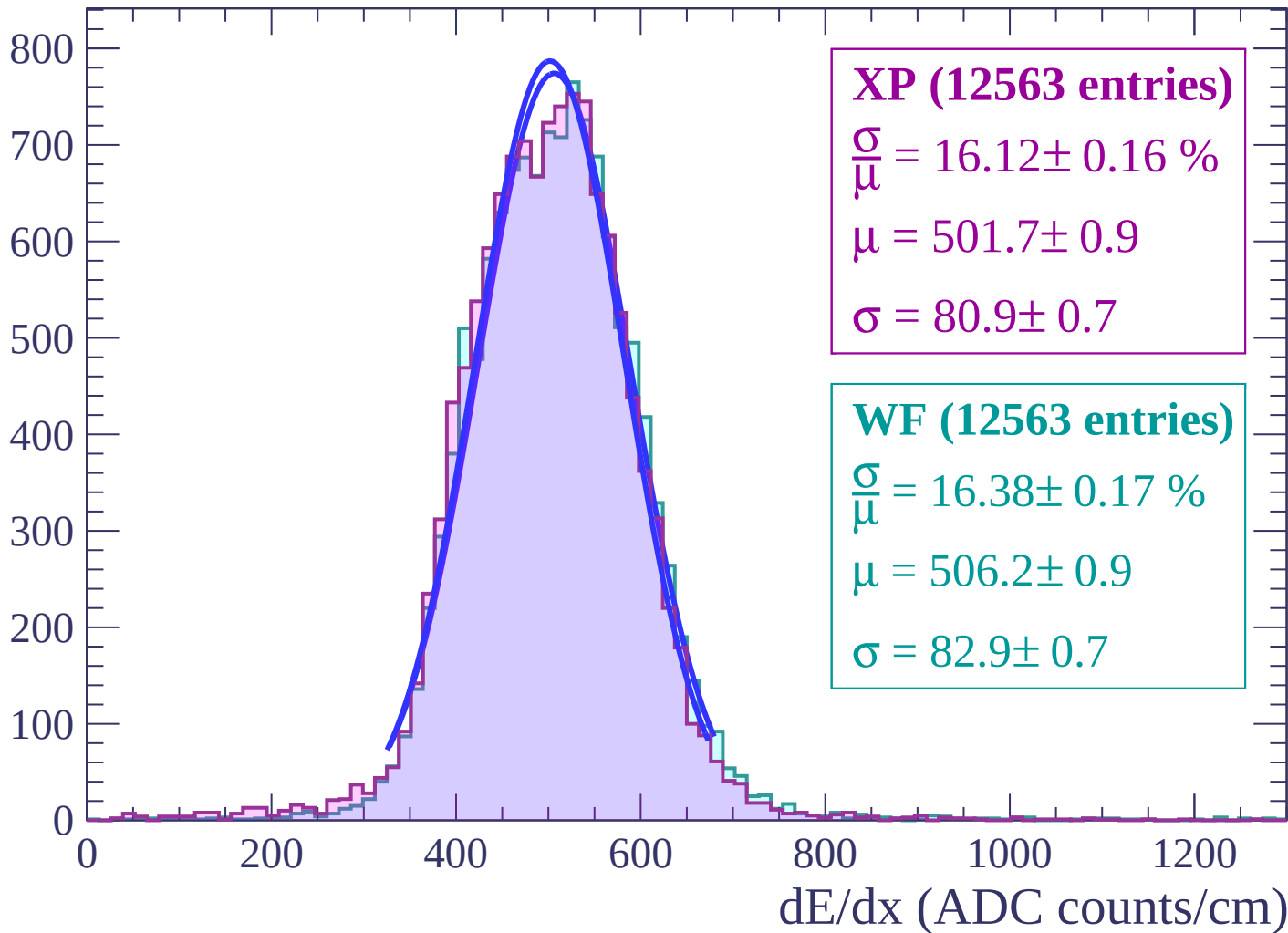
Energy loss | $-34 < \theta < -32$

Count



Energy loss | $-32 < \theta < -30$

Count



Energy loss | $-30 < \theta < -28$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15565 entries)

$\frac{\sigma}{\mu} = 16.13 \pm 0.15 \%$

$\mu = 496.4 \pm 0.7$

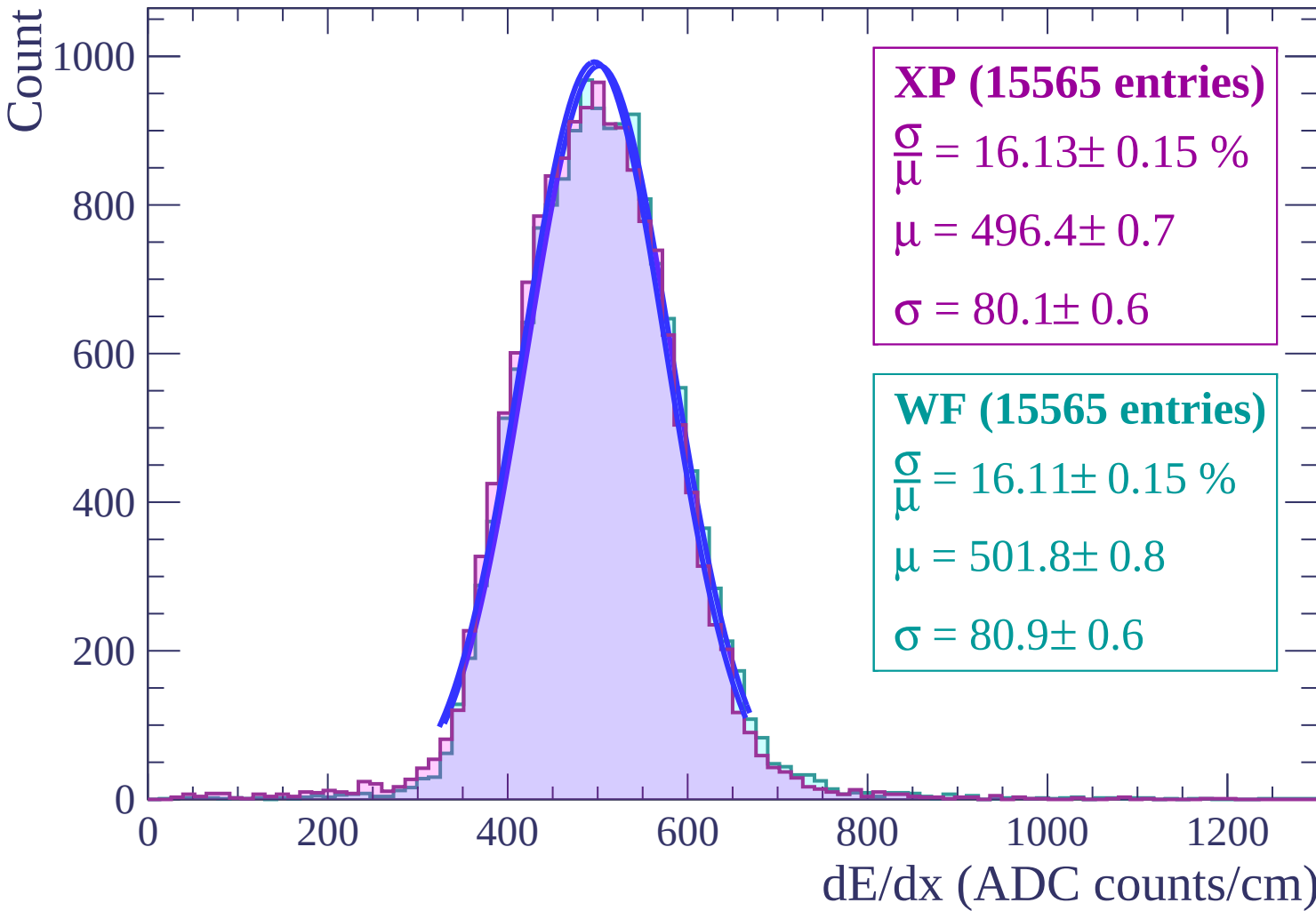
$\sigma = 80.1 \pm 0.6$

WF (15565 entries)

$\frac{\sigma}{\mu} = 16.11 \pm 0.15 \%$

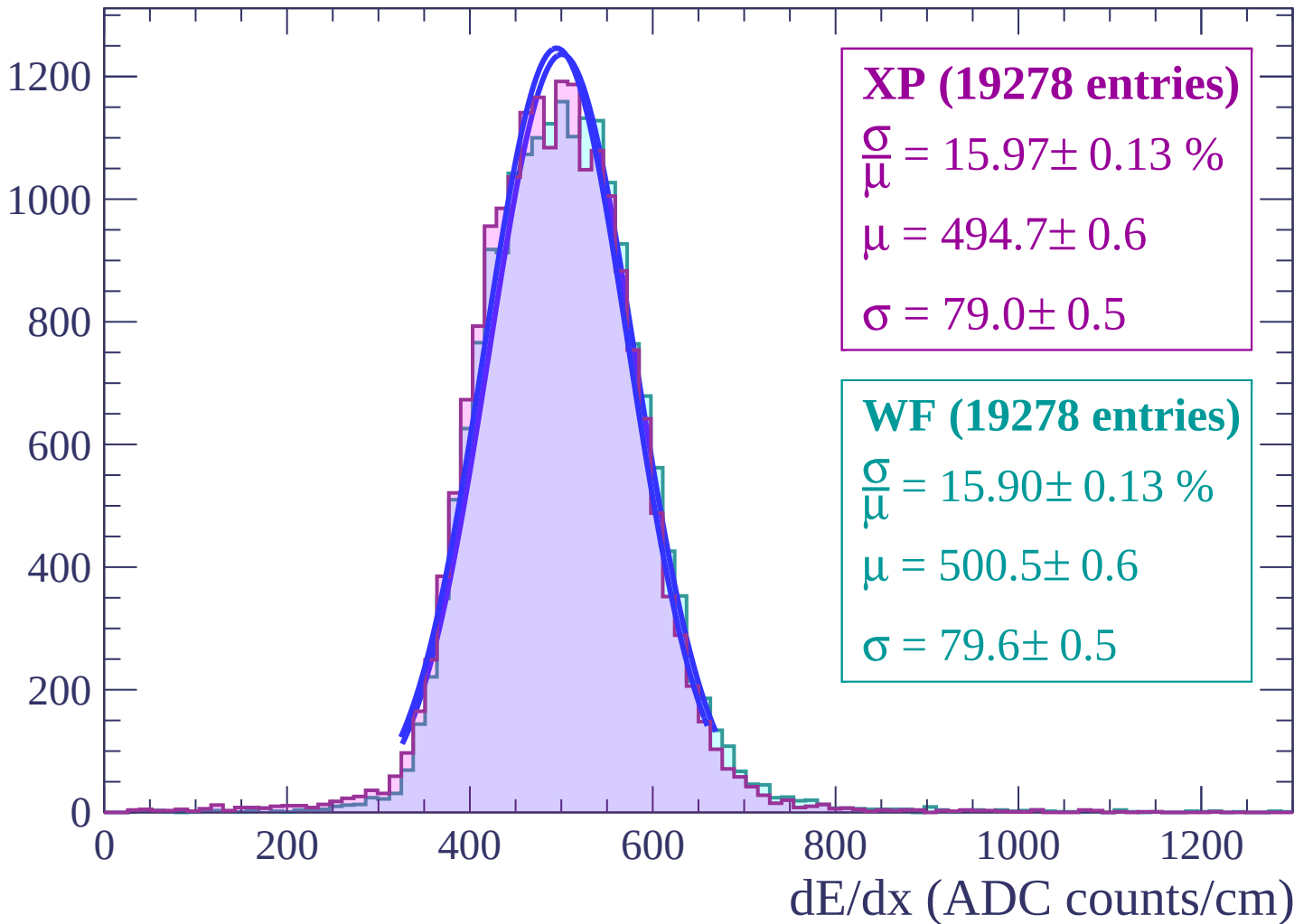
$\mu = 501.8 \pm 0.8$

$\sigma = 80.9 \pm 0.6$



Energy loss | $-28 < \theta < -26$

Count



Energy loss | $-26 < \theta < -24$

Count

1400

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (21927 entries)

$\frac{\sigma}{\mu} = 15.81 \pm 0.13 \%$

$\mu = 491.8 \pm 0.6$

$\sigma = 77.8 \pm 0.5$

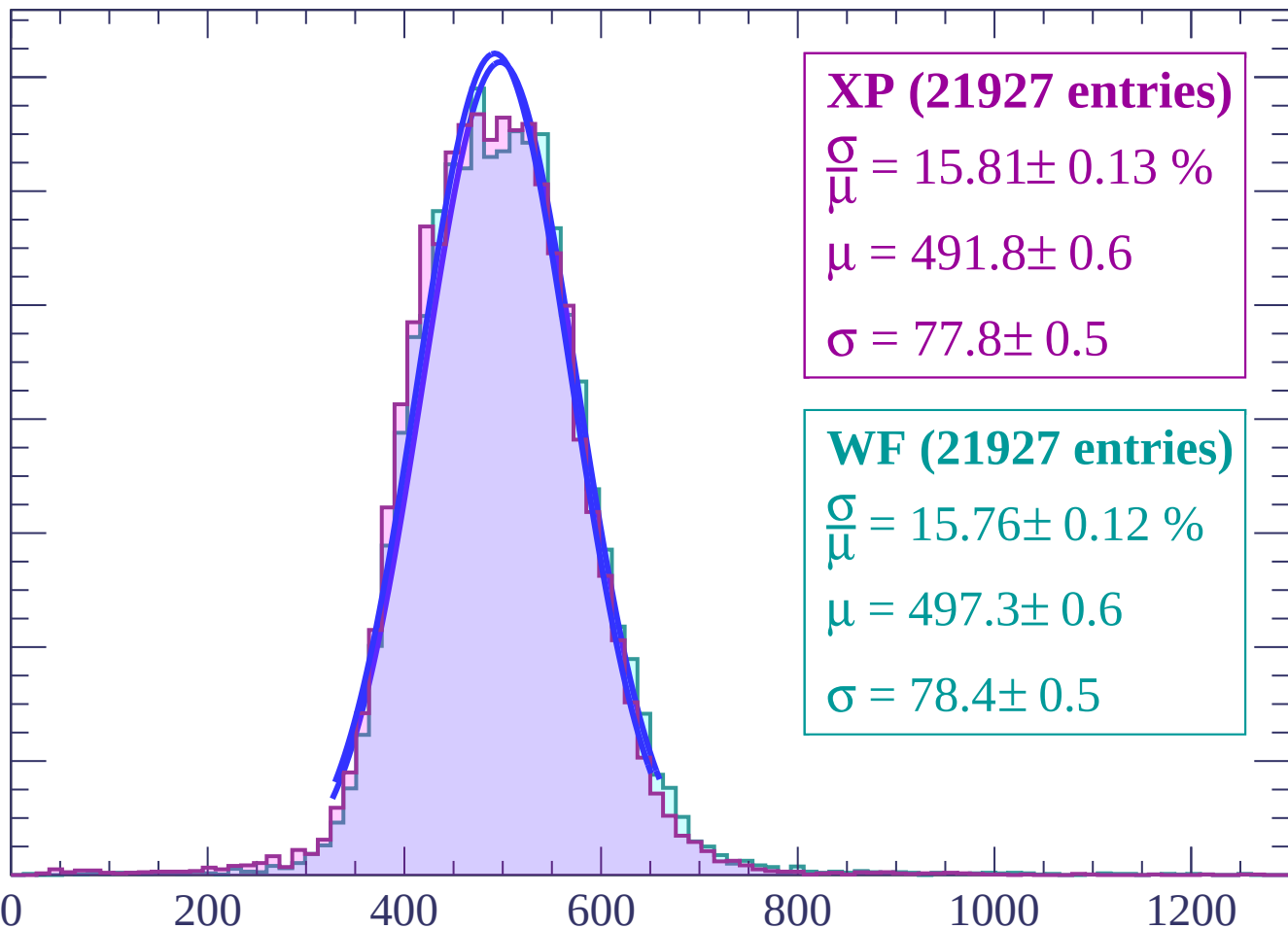
WF (21927 entries)

$\frac{\sigma}{\mu} = 15.76 \pm 0.12 \%$

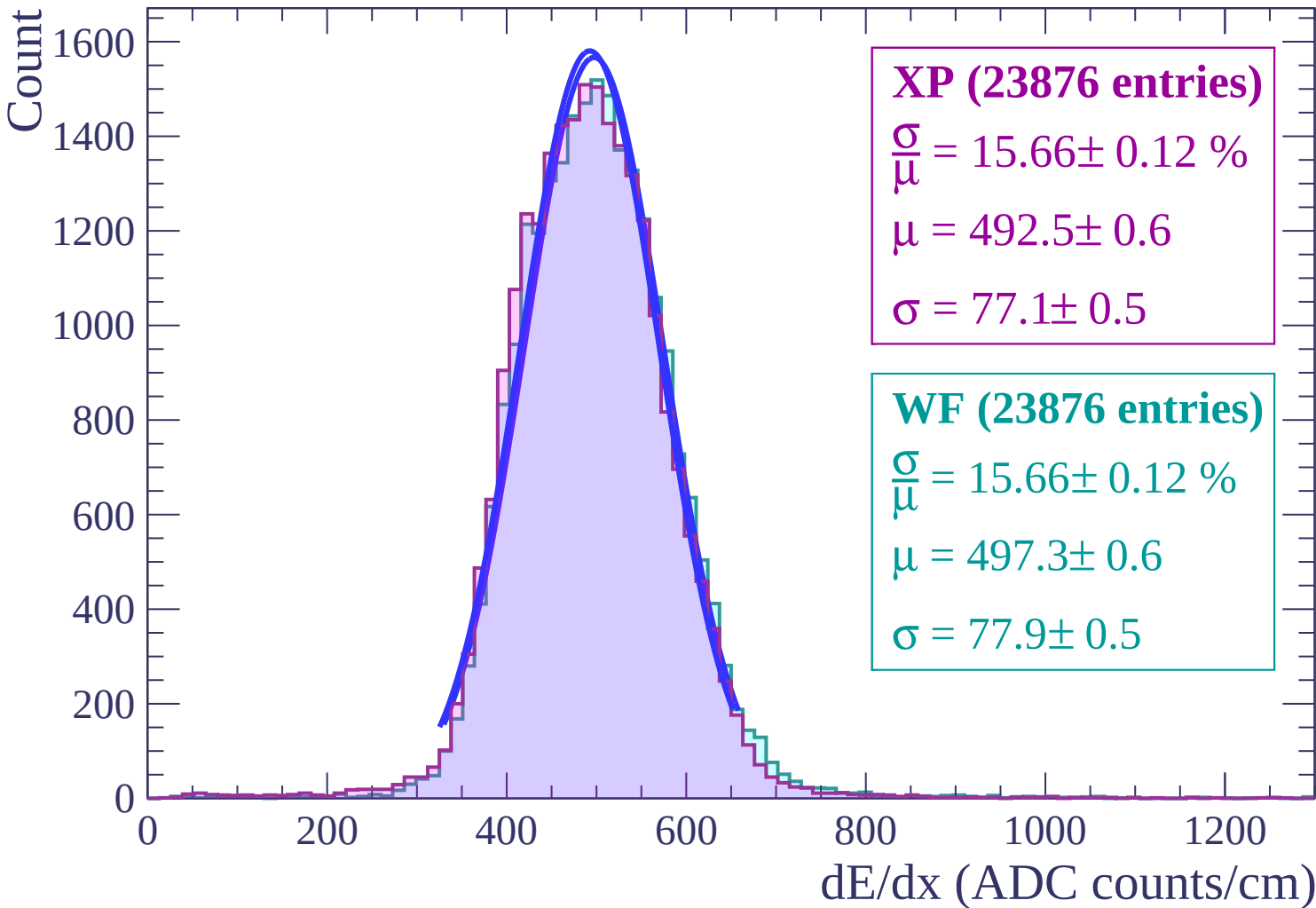
$\mu = 497.3 \pm 0.6$

$\sigma = 78.4 \pm 0.5$

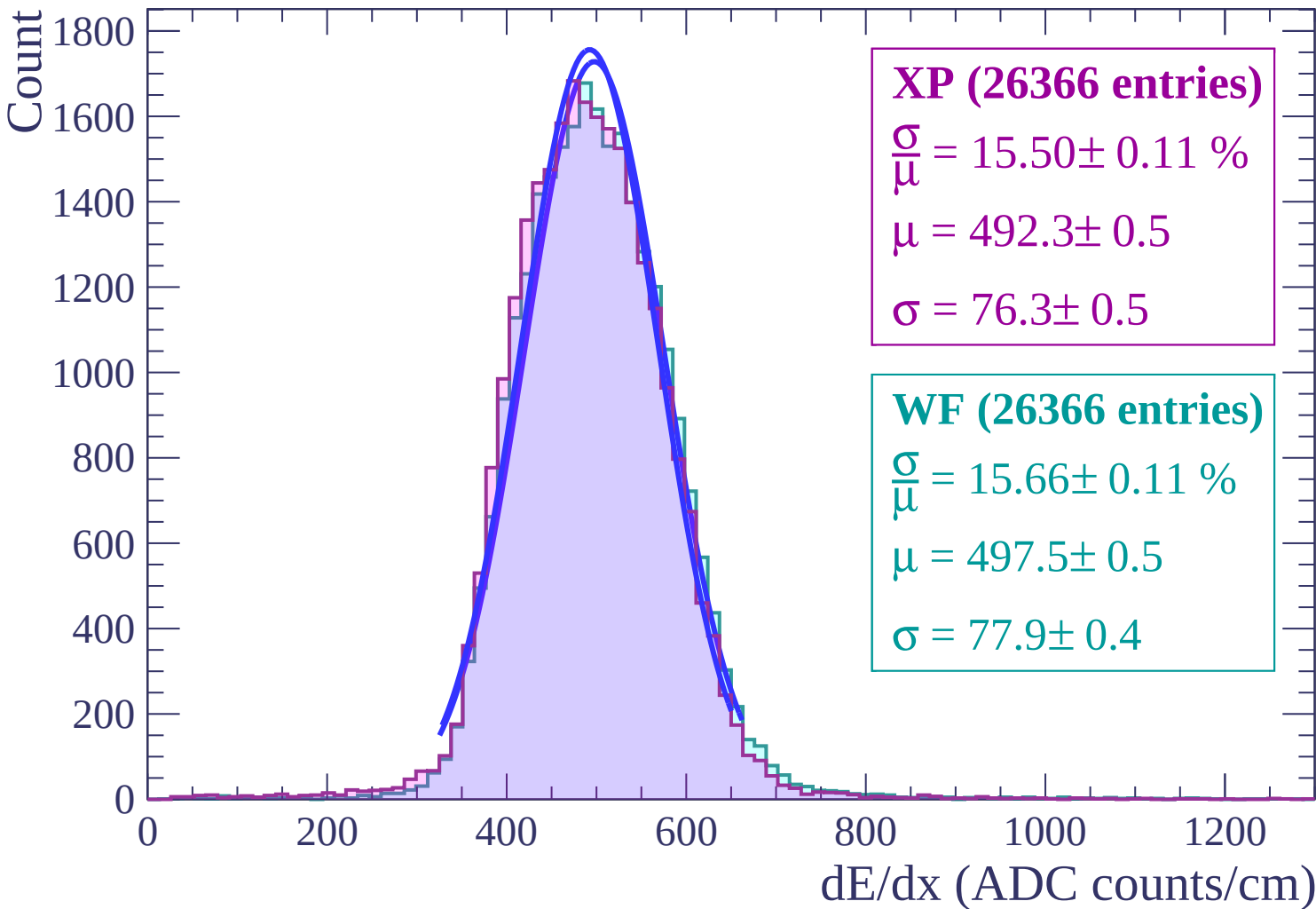
dE/dx (ADC counts/cm)



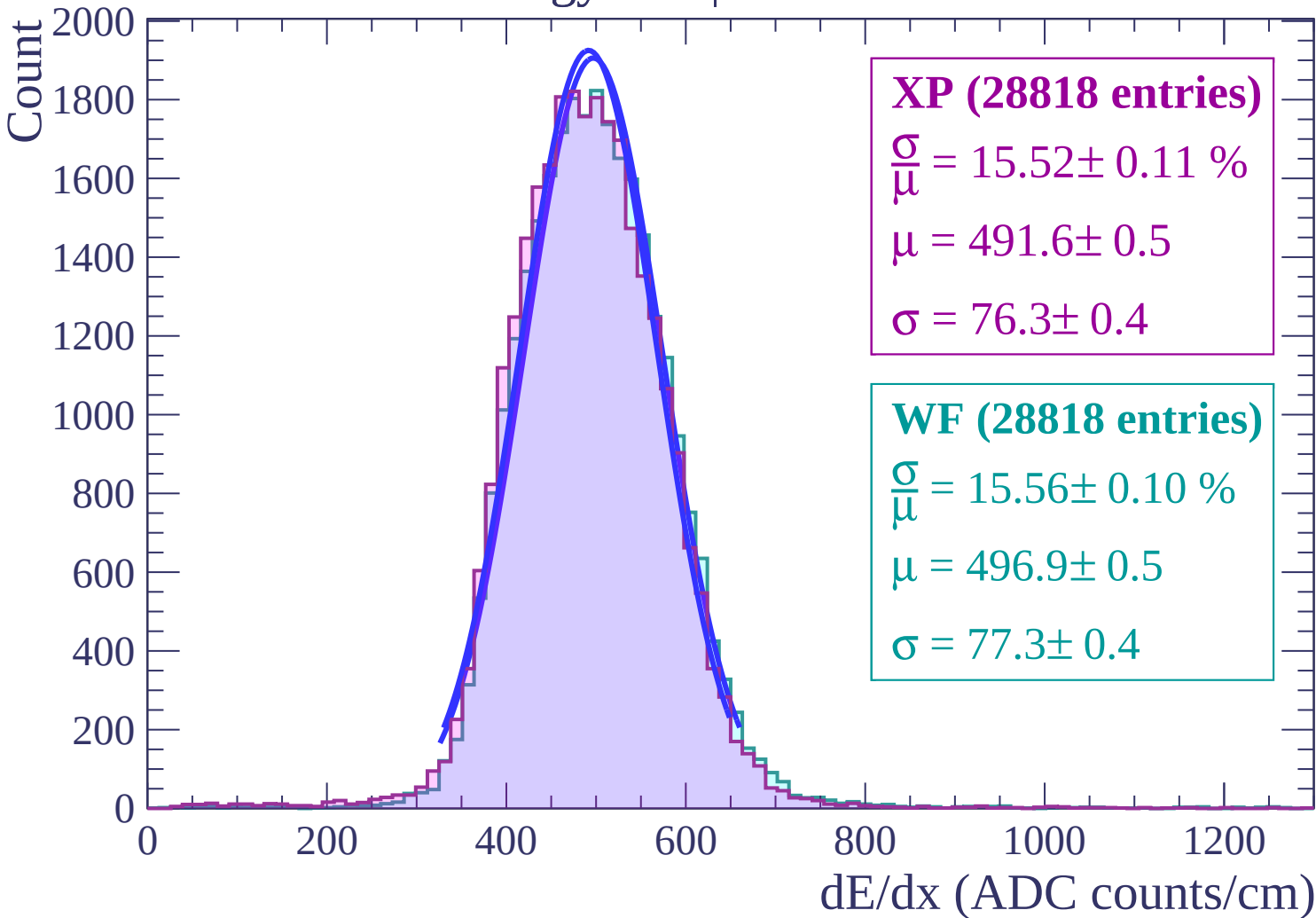
Energy loss | $-24 < \theta < -22$



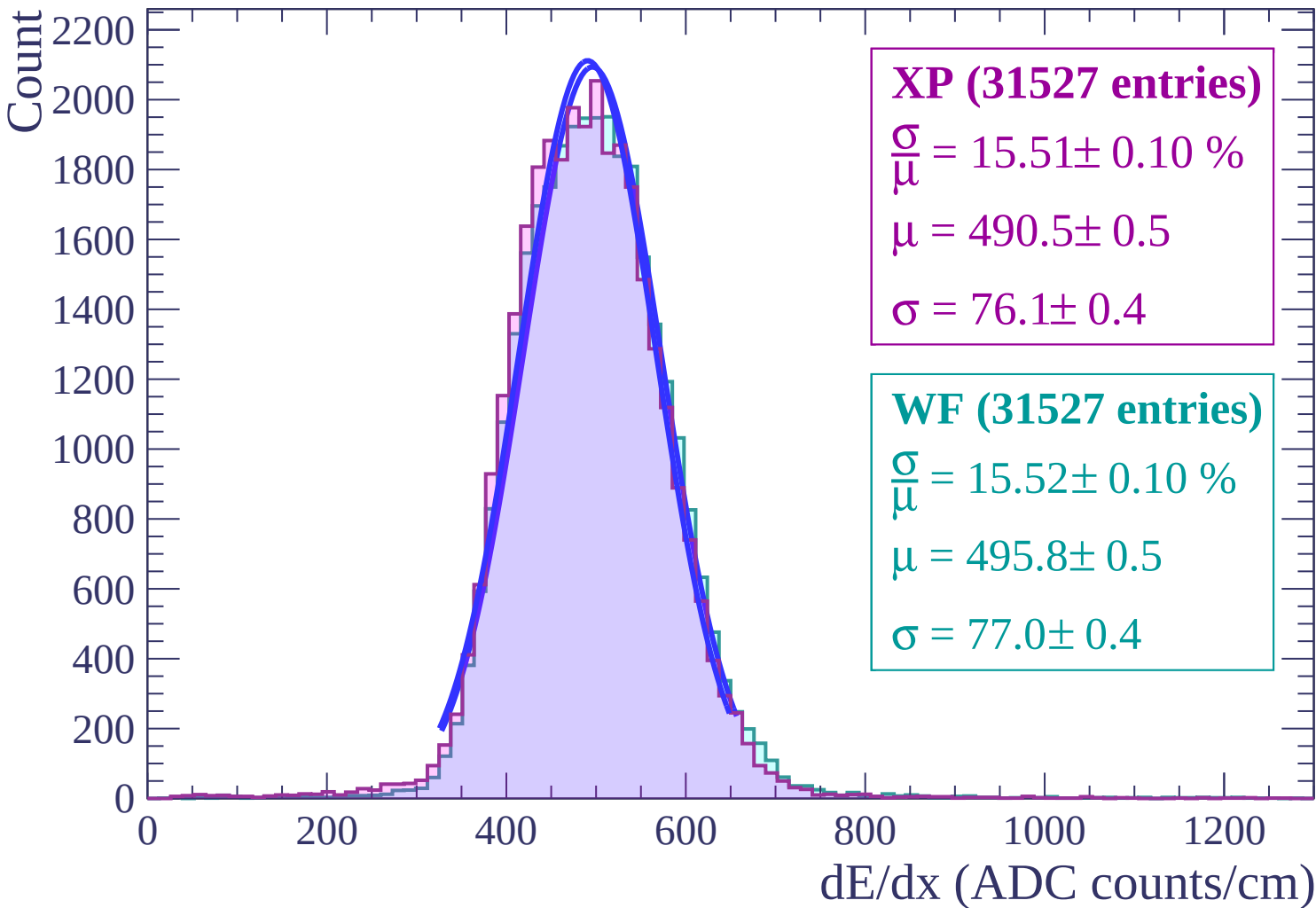
Energy loss | $-22 < \theta < -20$



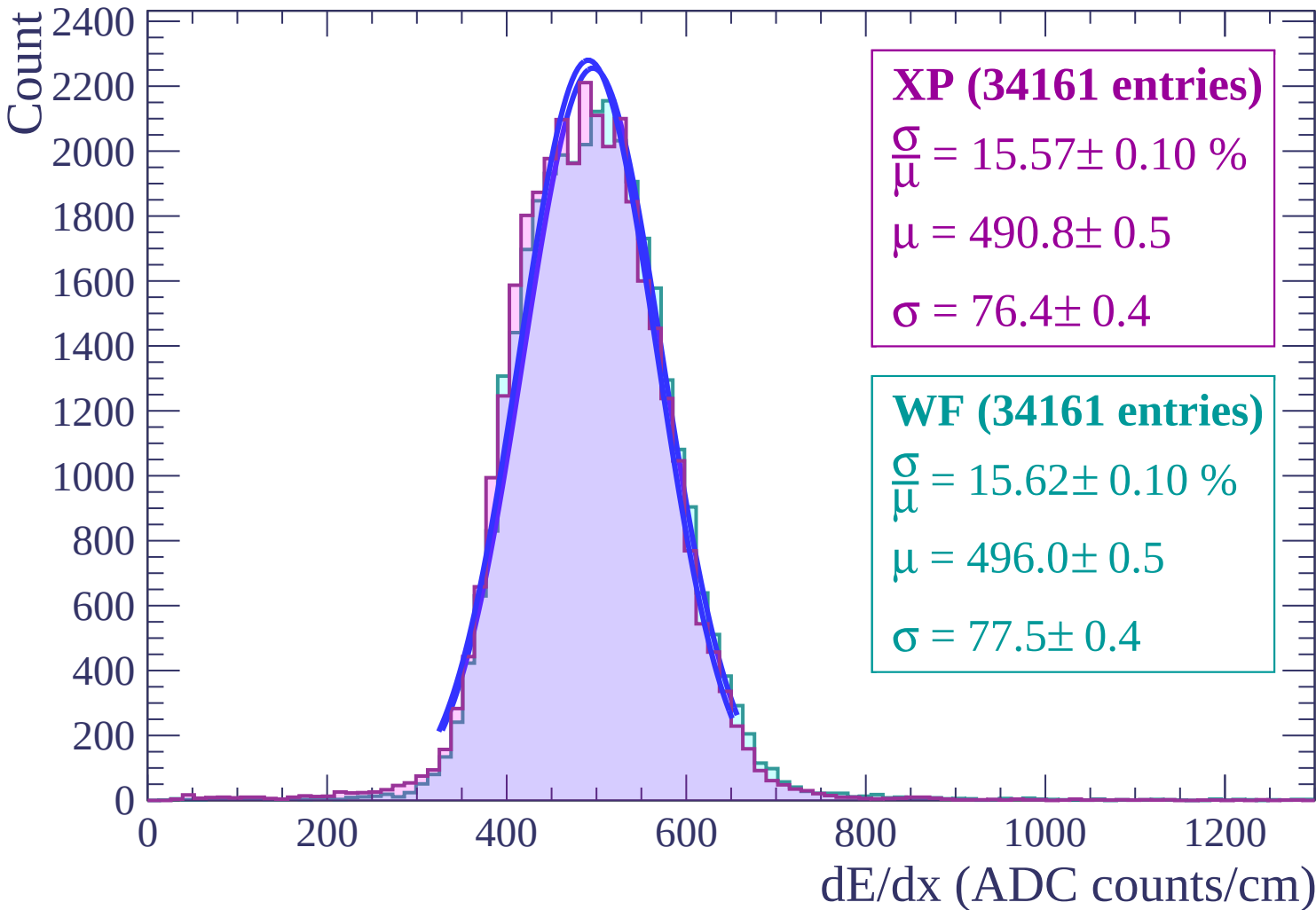
Energy loss | $-20 < \theta < -18$



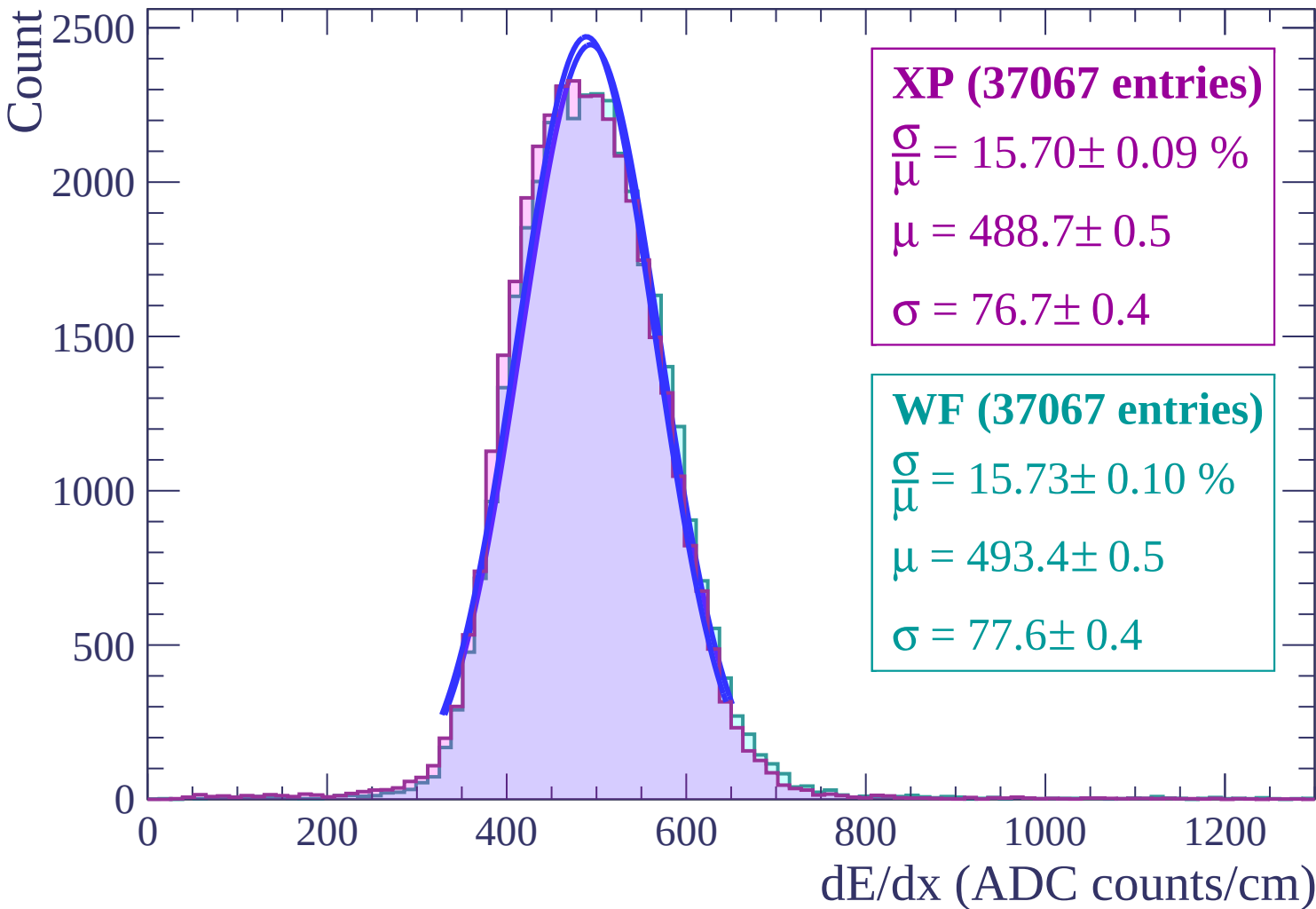
Energy loss | $-18 < \theta < -16$



Energy loss | $-16 < \theta < -14$

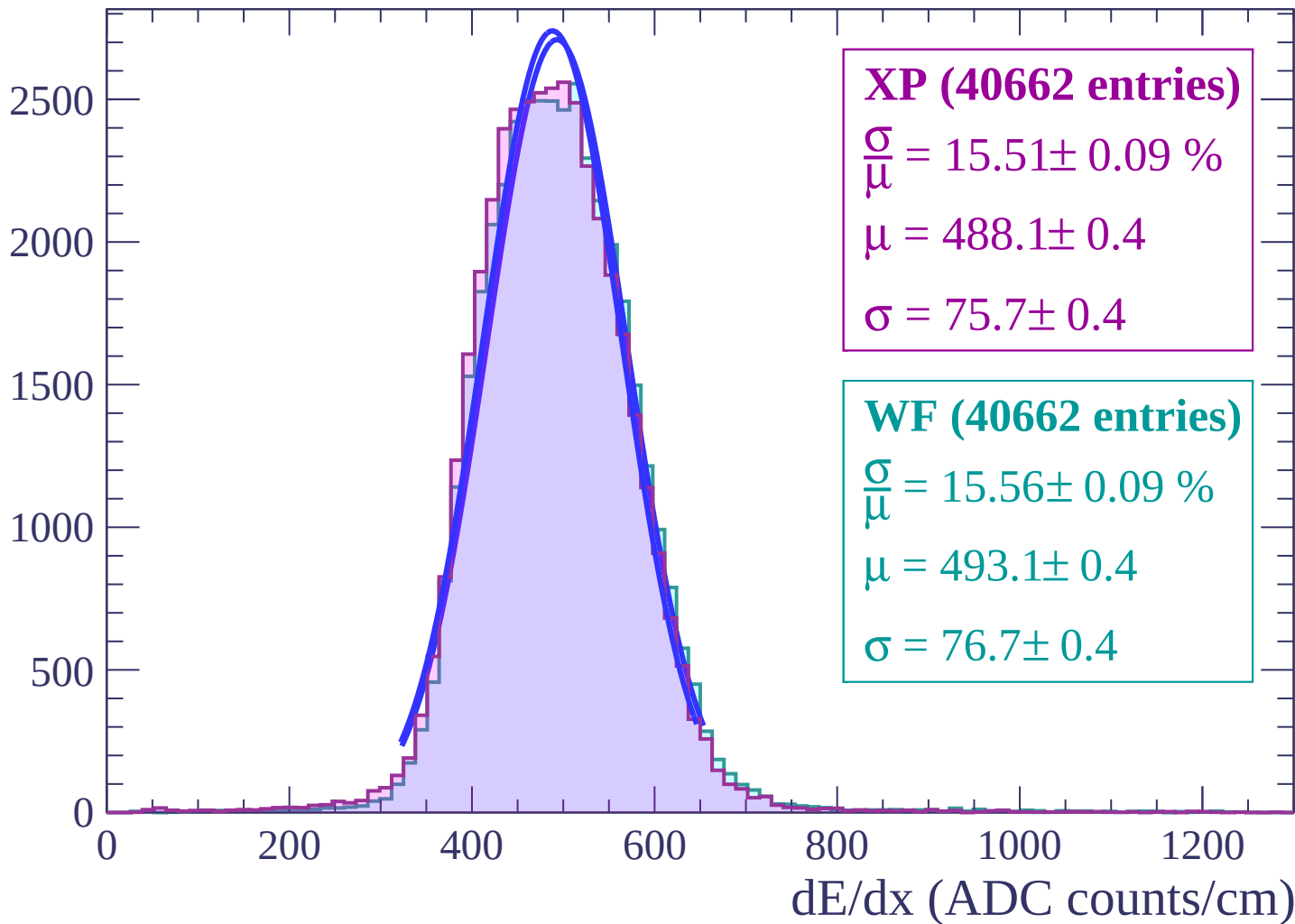


Energy loss | $-14 < \theta < -12$

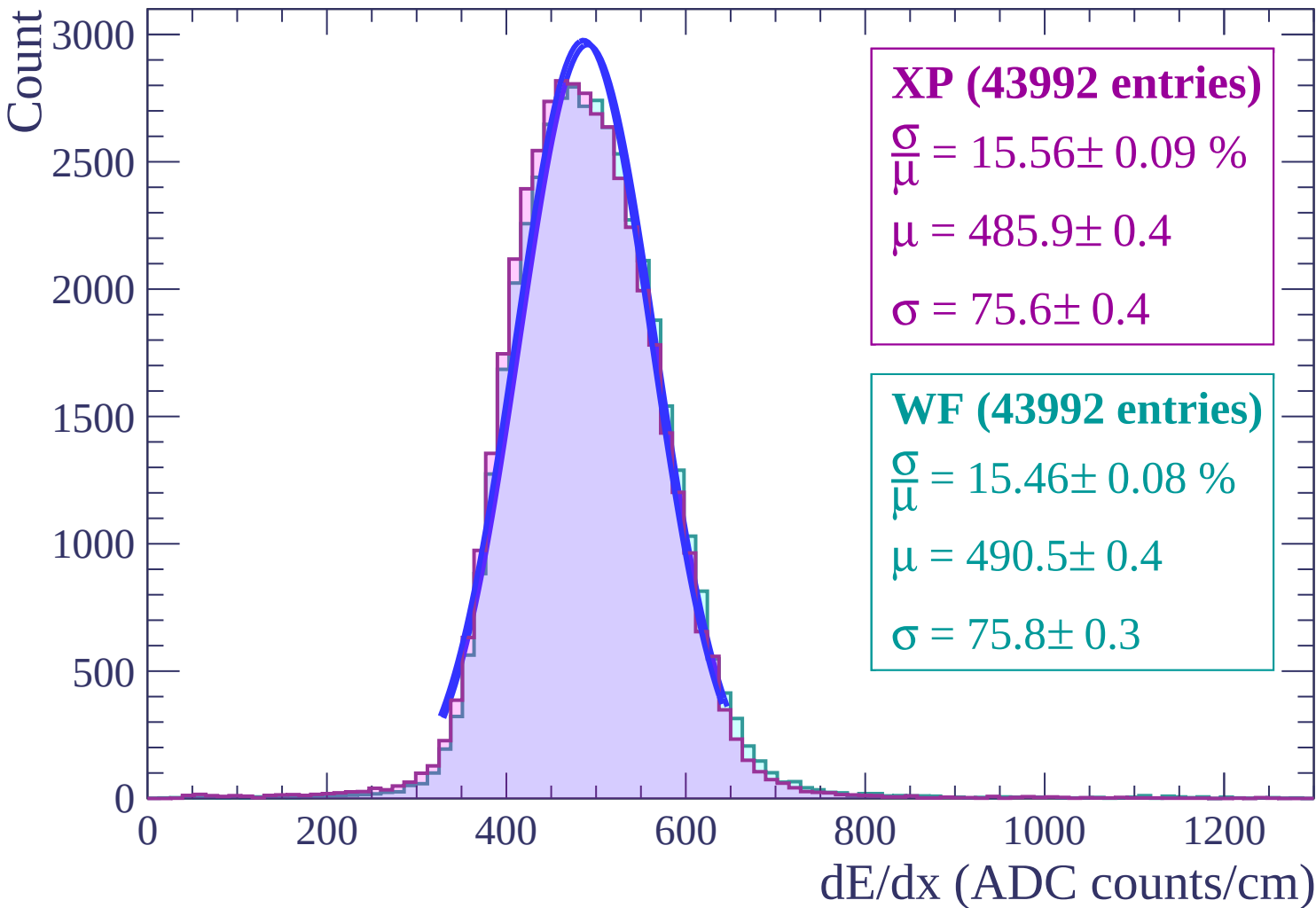


Energy loss | $-12 < \theta < -10$

Count



Energy loss | $-10 < \theta < -8$



Energy loss | $-8 < \theta < -6$

Count

3000

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (46393 entries)

$\frac{\sigma}{\mu} = 15.48 \pm 0.08 \%$

$\mu = 486.8 \pm 0.4$

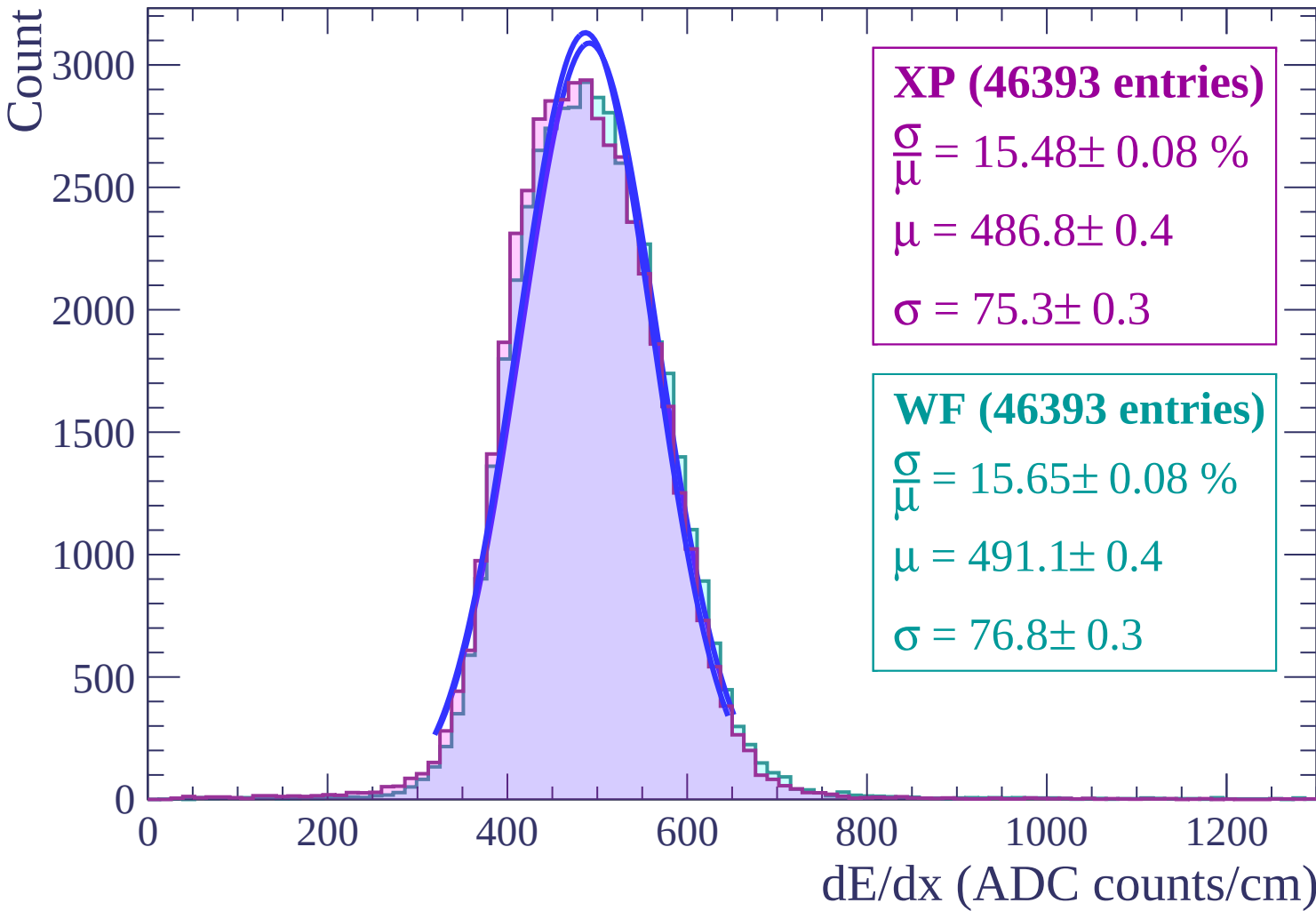
$\sigma = 75.3 \pm 0.3$

WF (46393 entries)

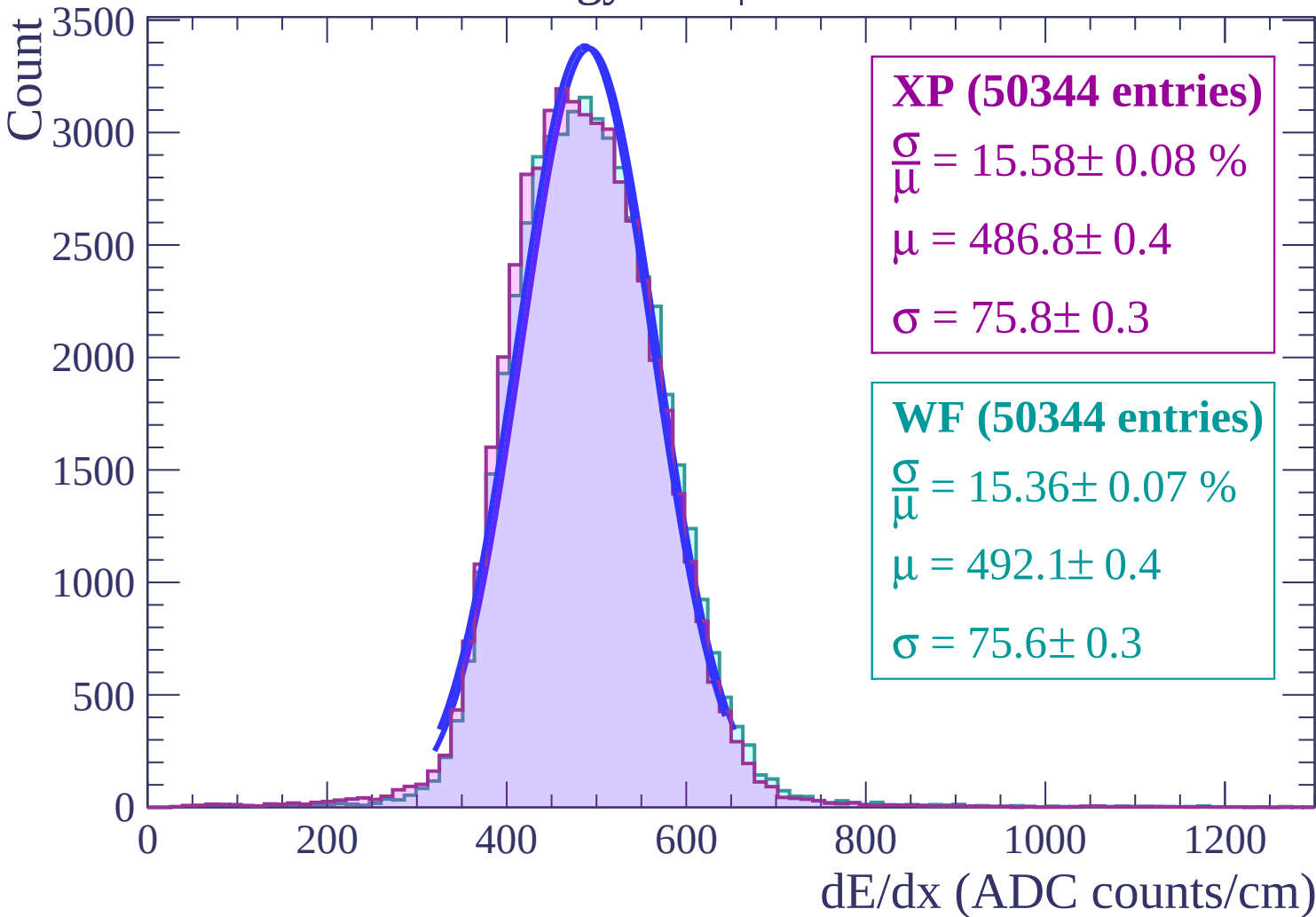
$\frac{\sigma}{\mu} = 15.65 \pm 0.08 \%$

$\mu = 491.1 \pm 0.4$

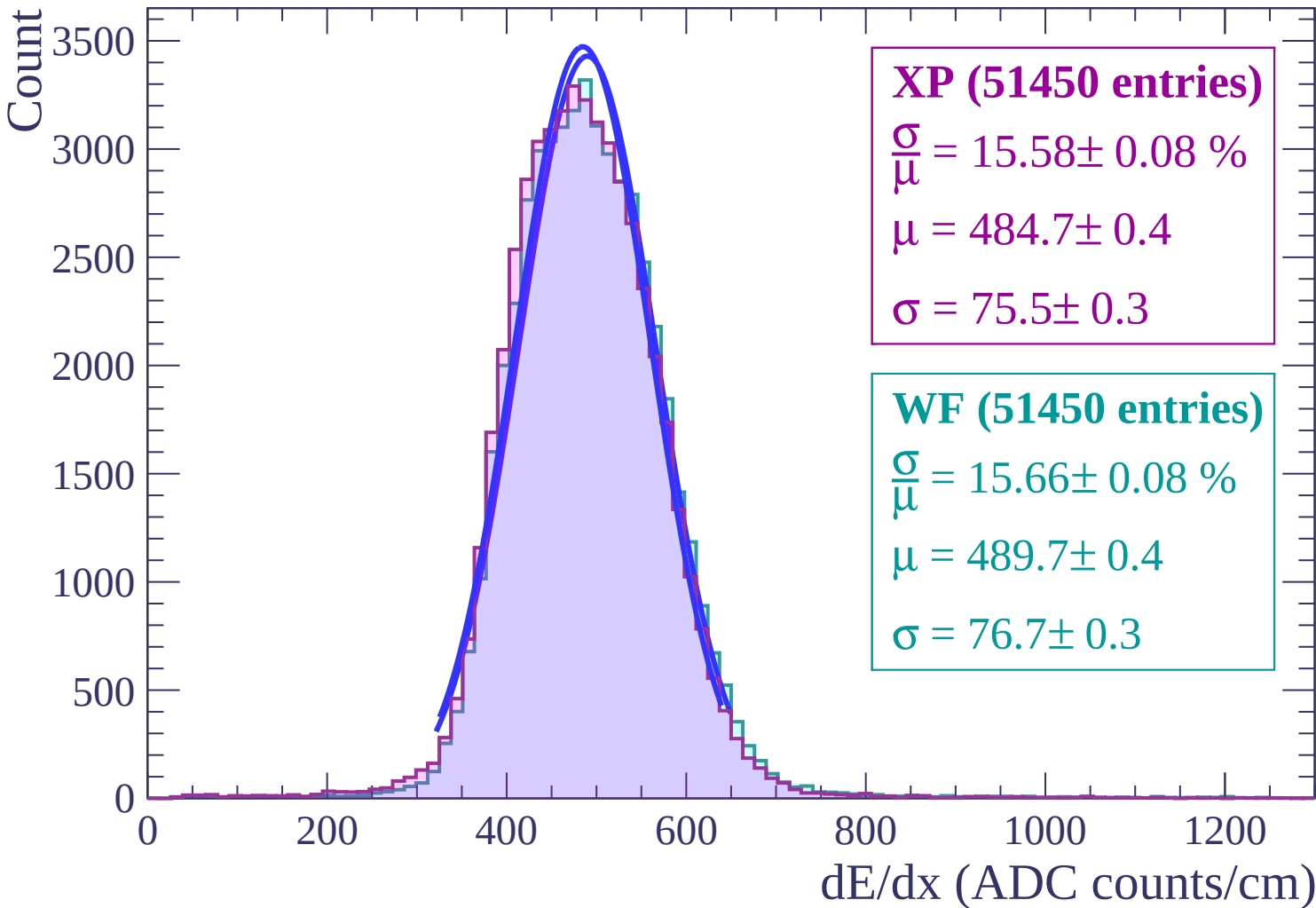
$\sigma = 76.8 \pm 0.3$



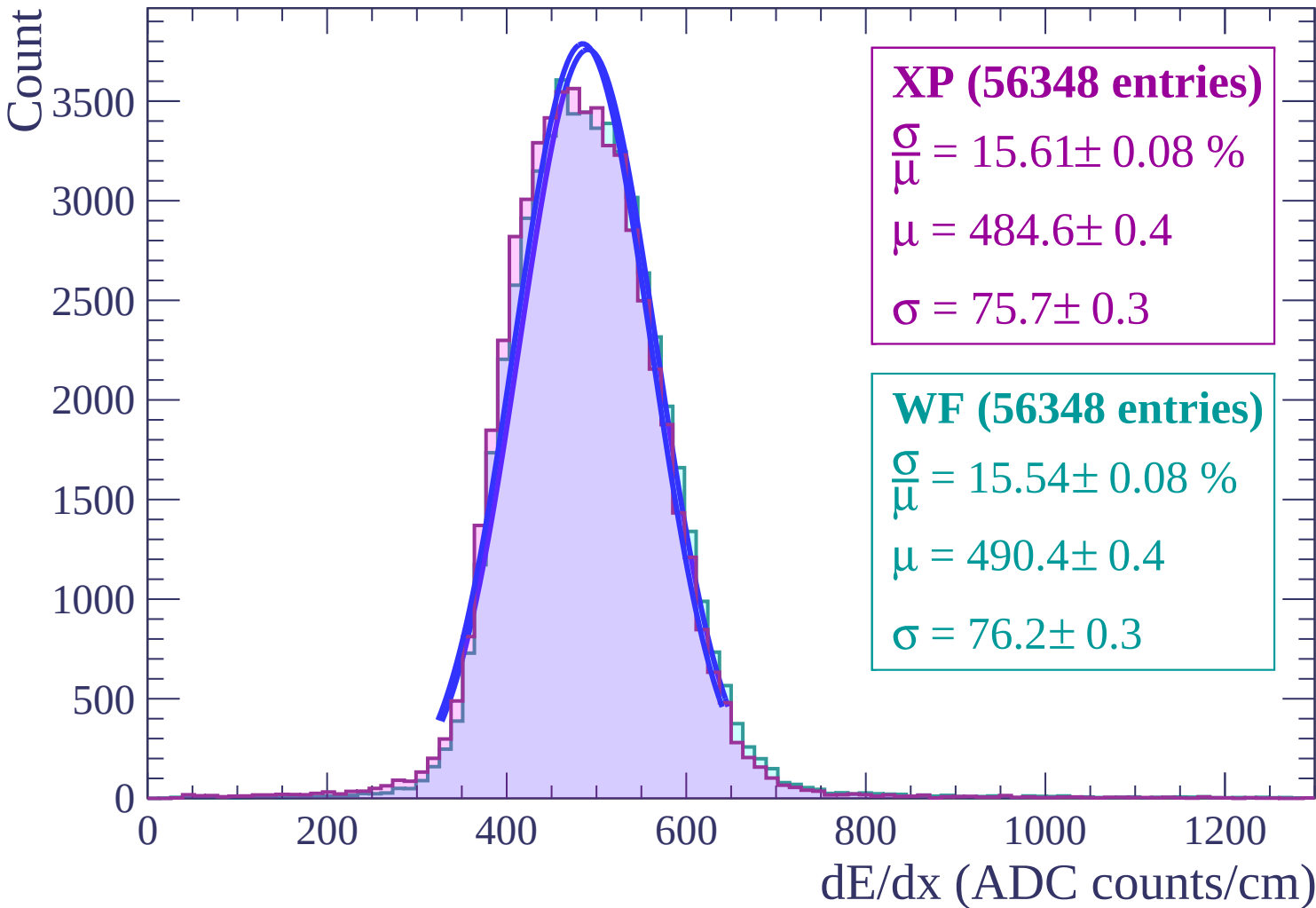
Energy loss | $-6 < \theta < -4$



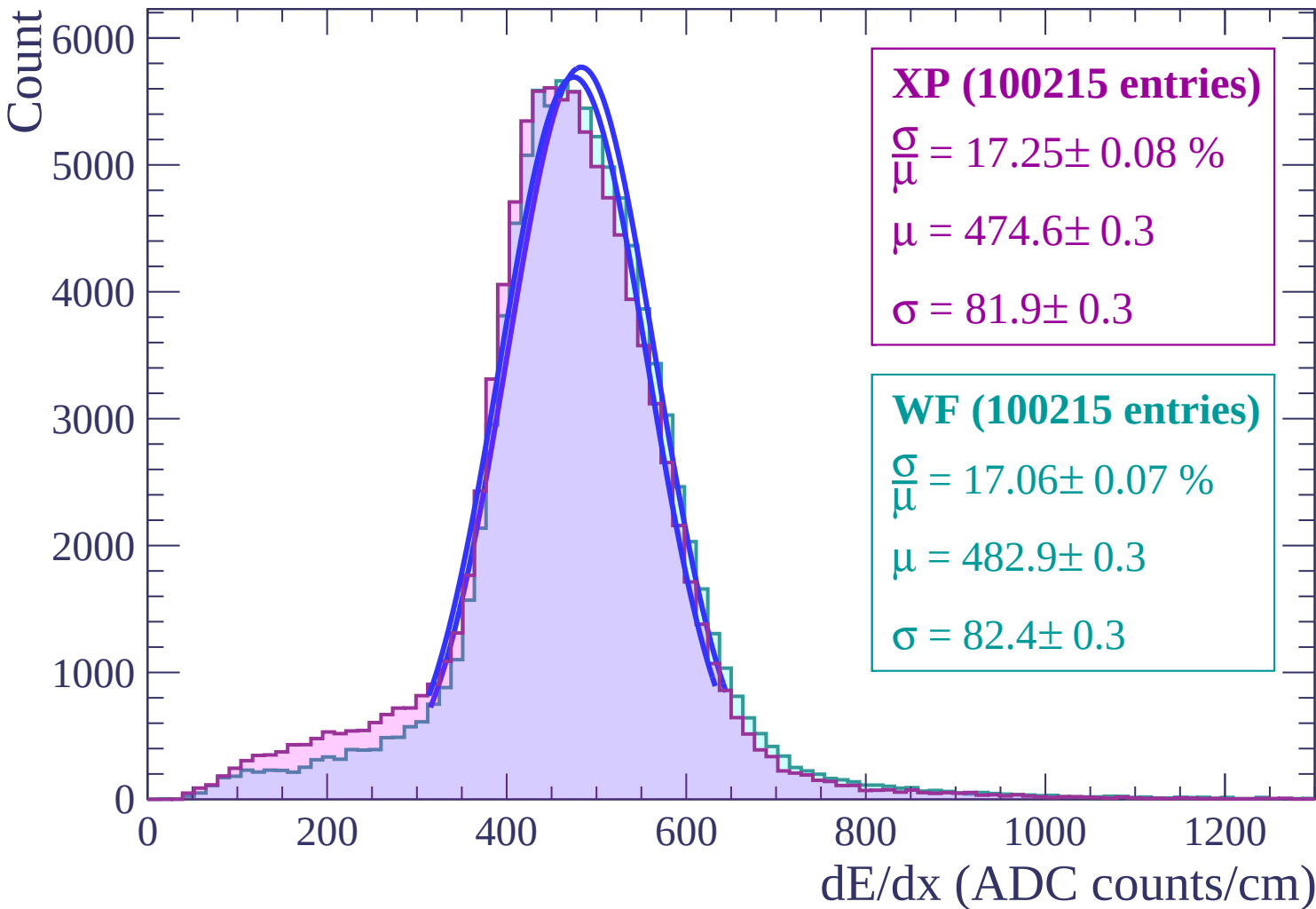
Energy loss | $-4 < \theta < -2$



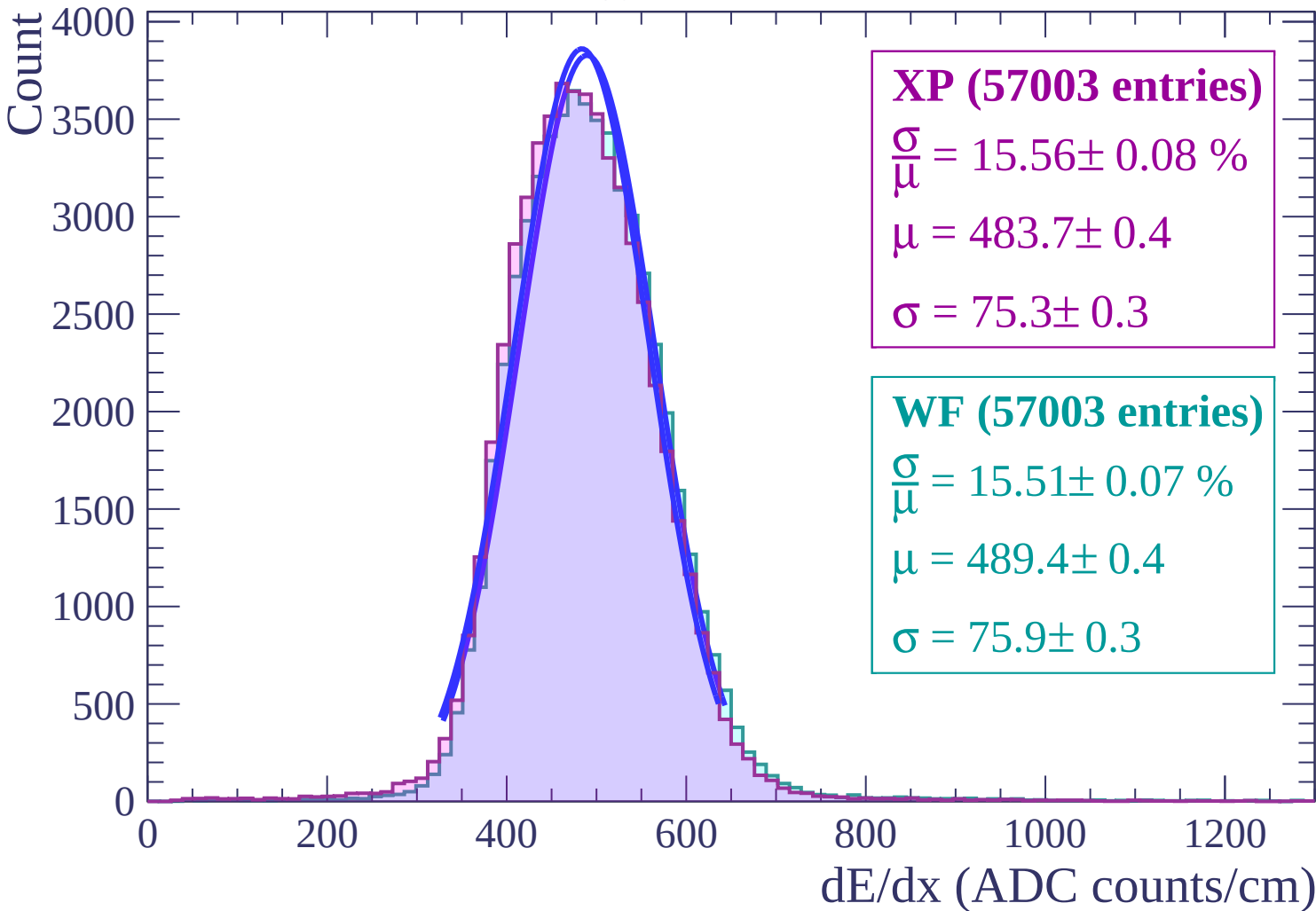
Energy loss | $-2 < \theta < 0$



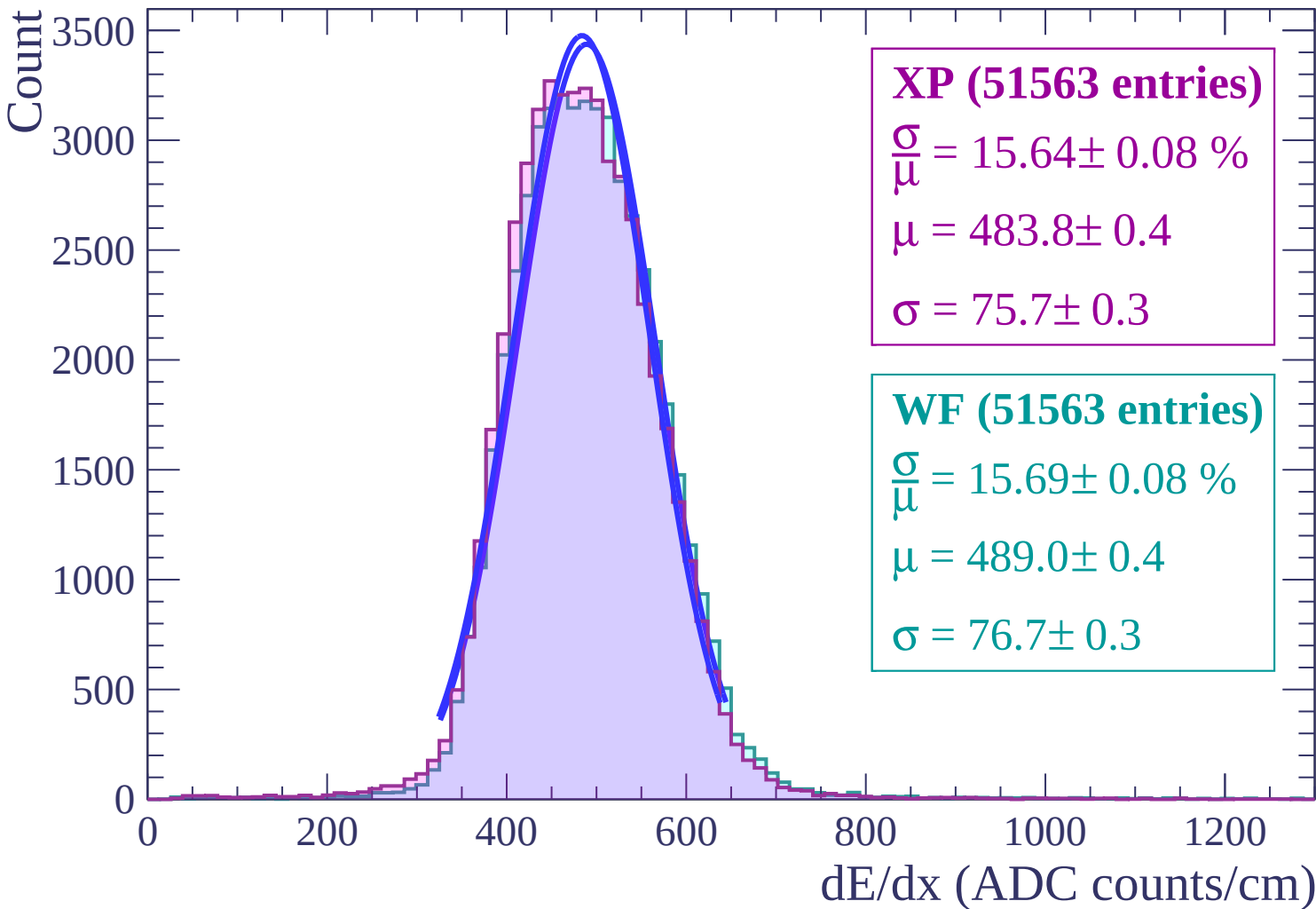
Energy loss | $0 < \theta < 2$



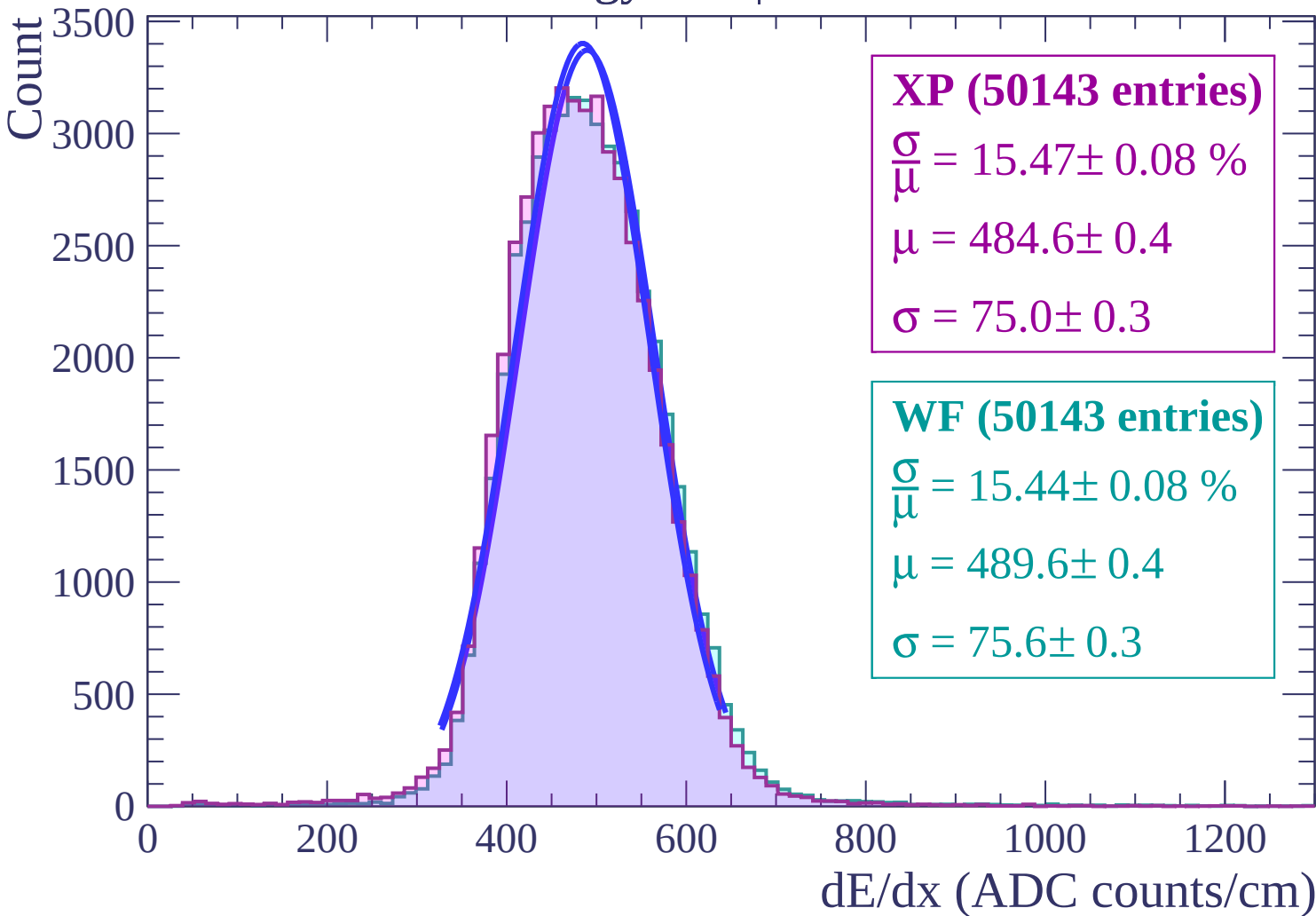
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$



Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

Count

3000

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (46568 entries)

$\frac{\sigma}{\mu} = 15.58 \pm 0.09 \%$

$\mu = 485.5 \pm 0.4$

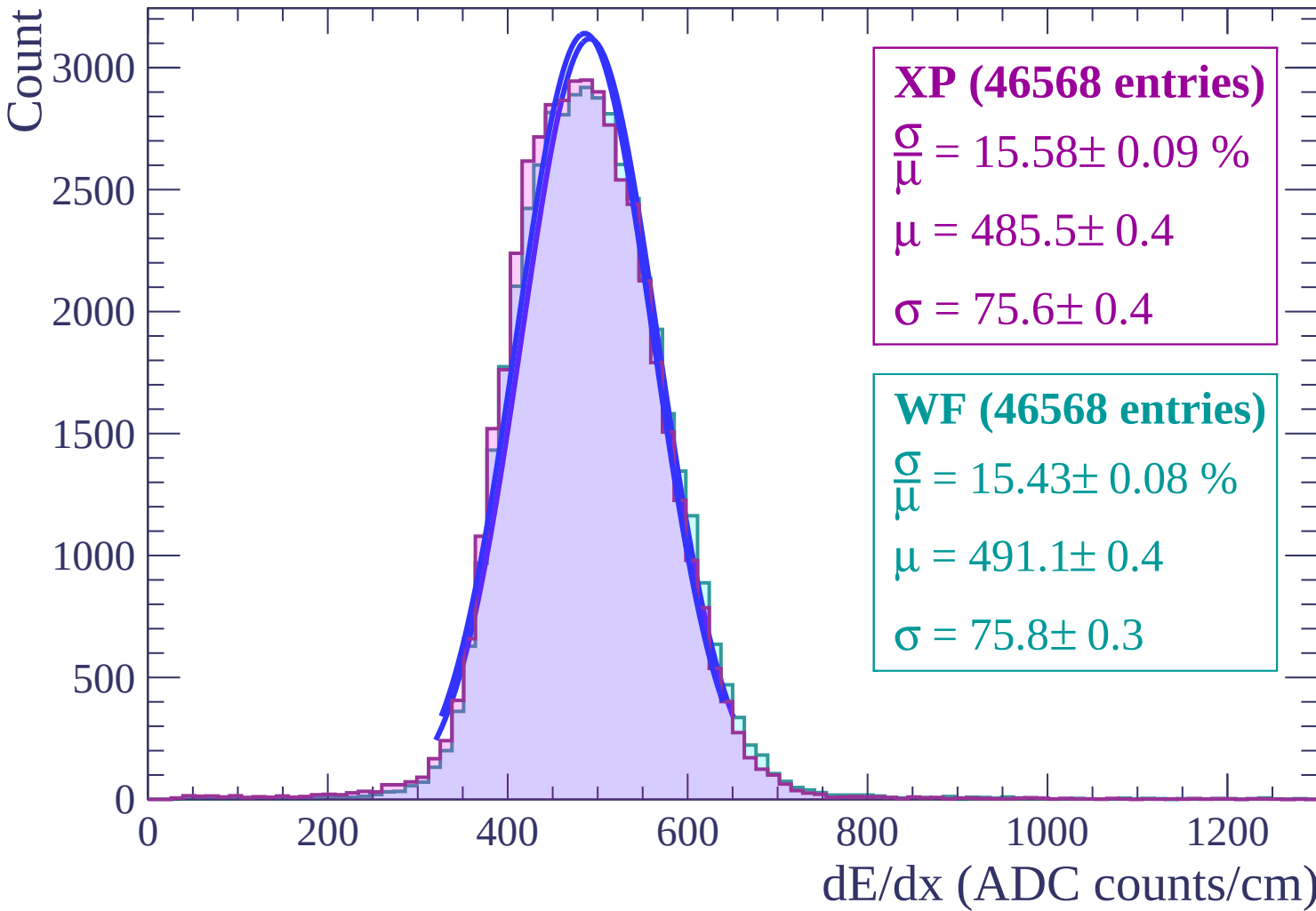
$\sigma = 75.6 \pm 0.4$

WF (46568 entries)

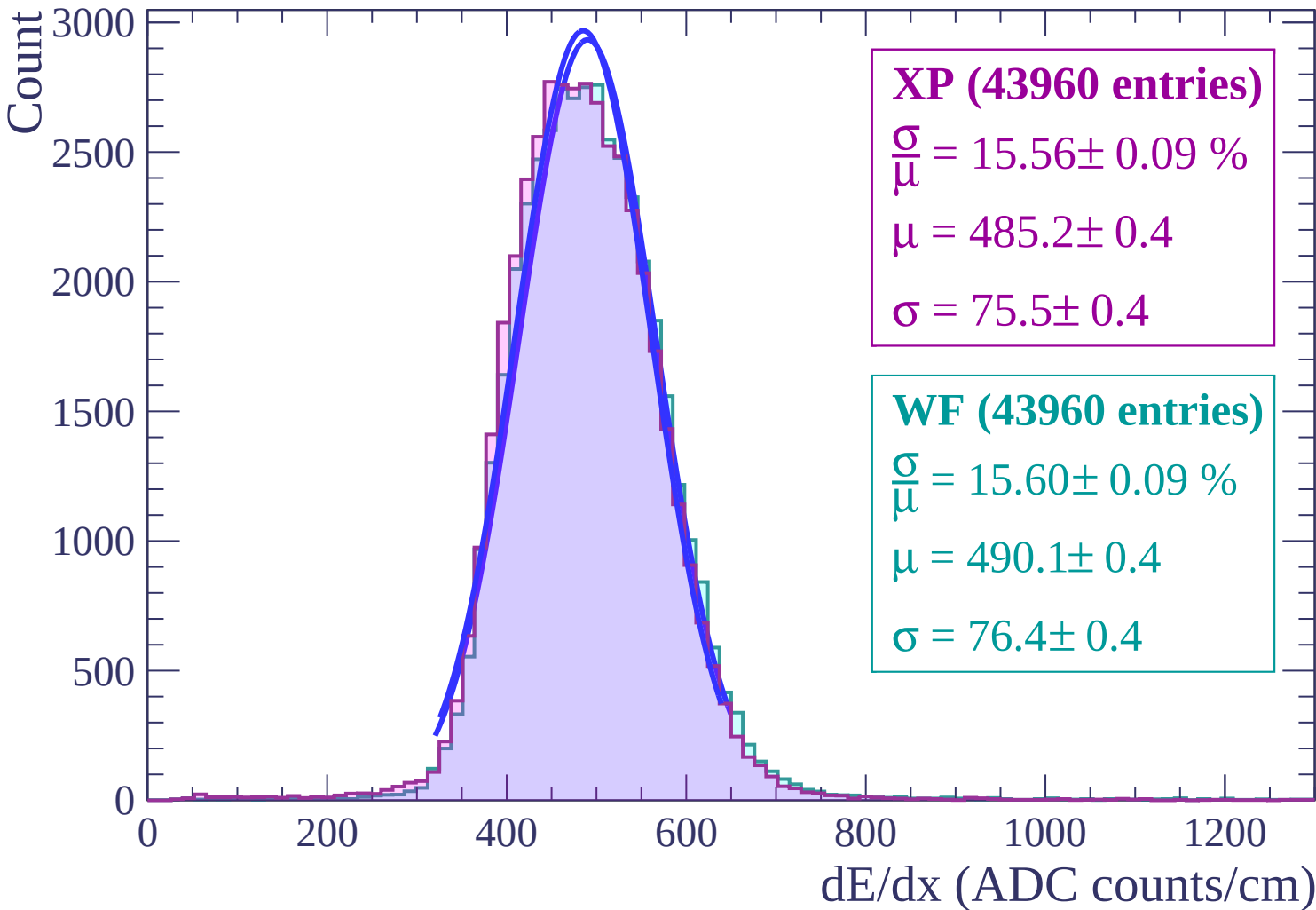
$\frac{\sigma}{\mu} = 15.43 \pm 0.08 \%$

$\mu = 491.1 \pm 0.4$

$\sigma = 75.8 \pm 0.3$

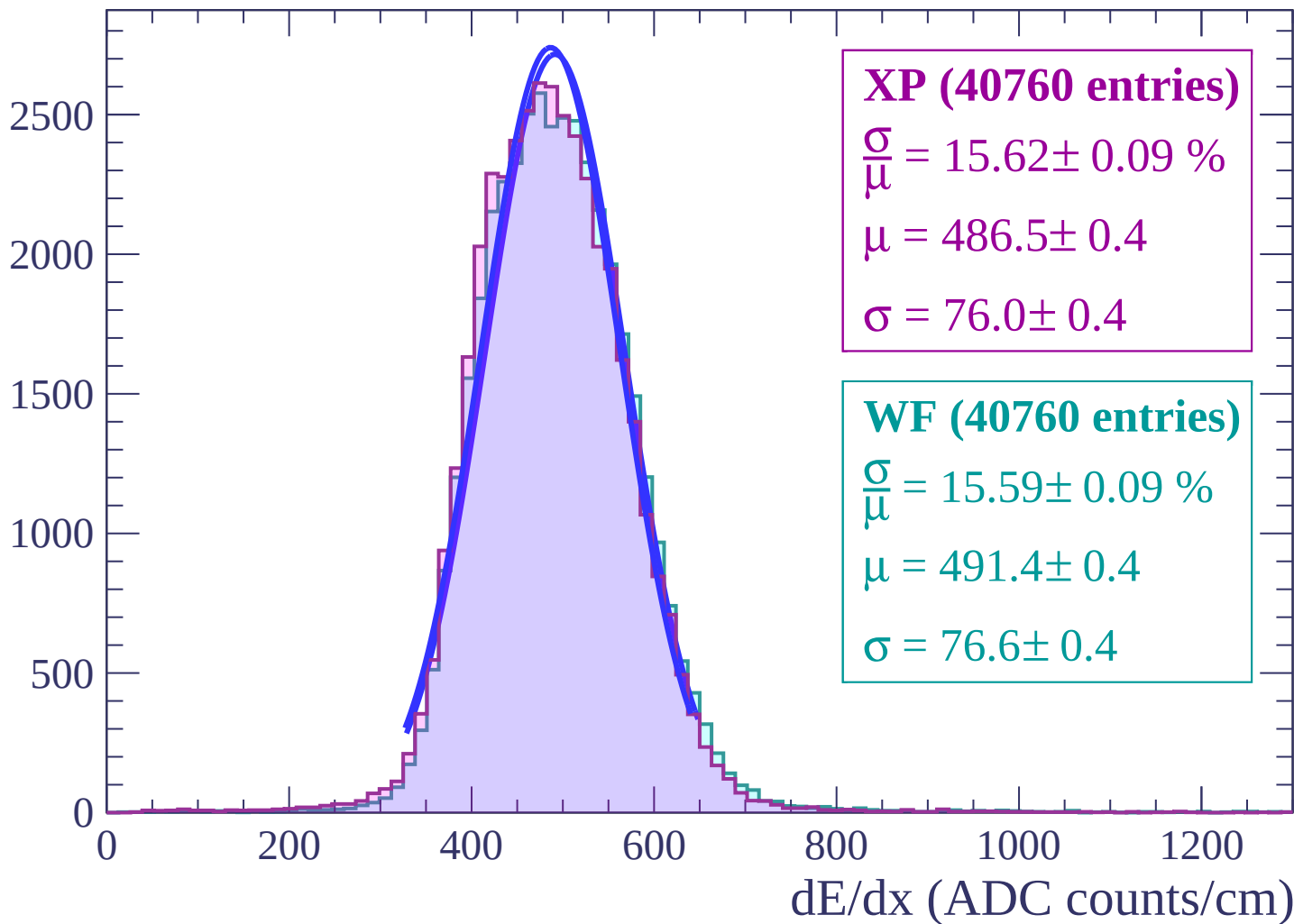


Energy loss | $10 < \theta < 12$

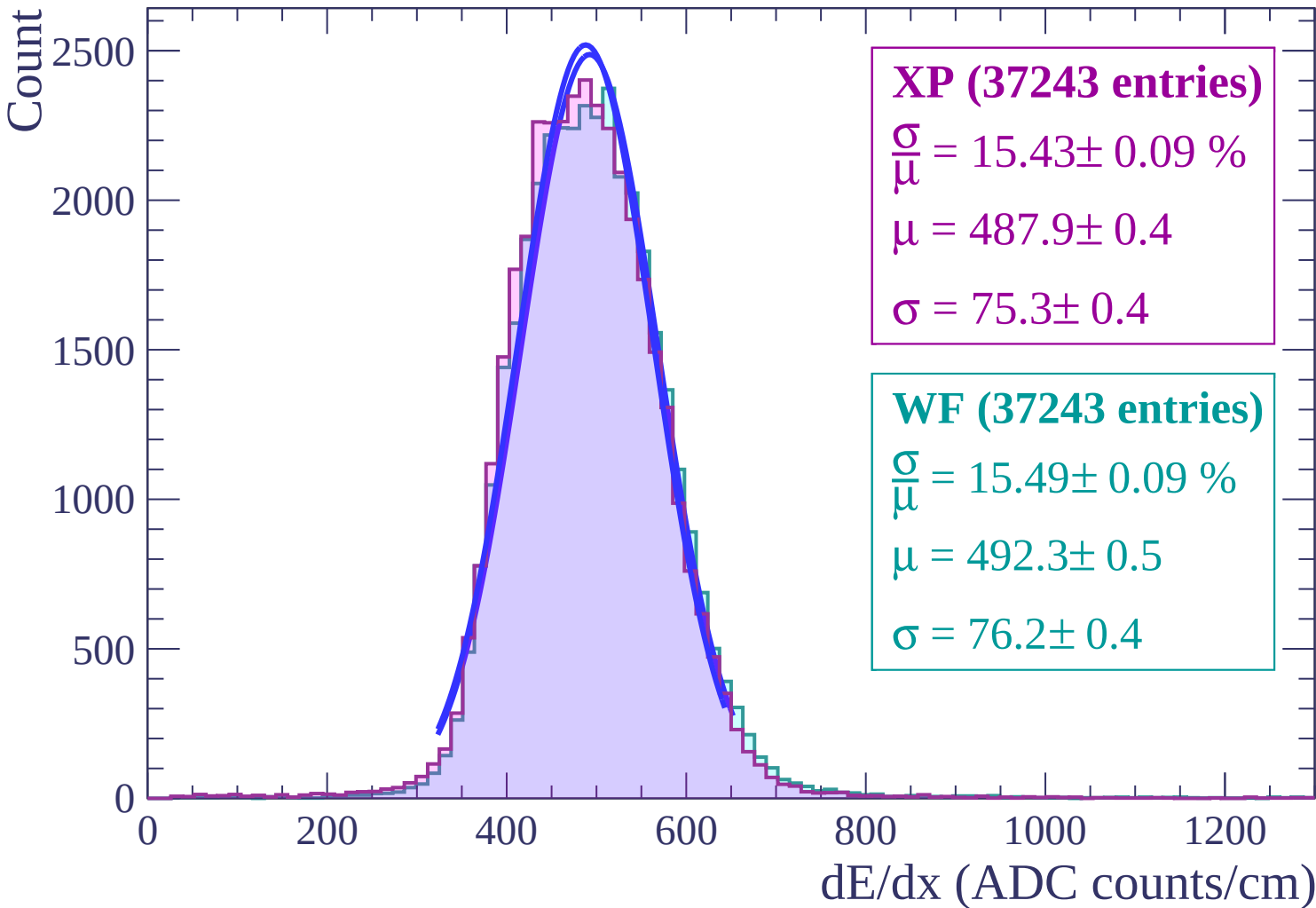


Energy loss | $12 < \theta < 14$

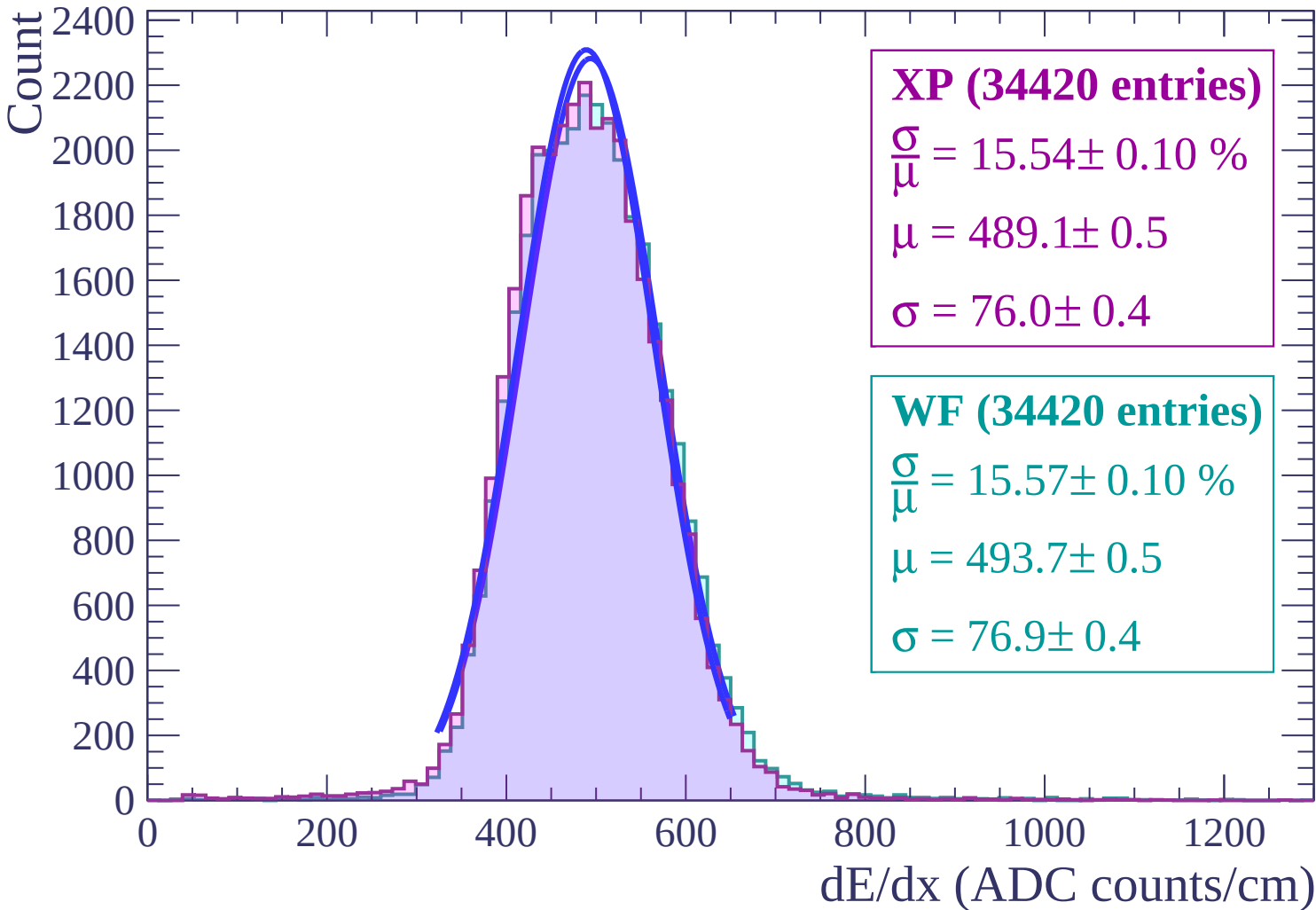
Count



Energy loss | $14 < \theta < 16$

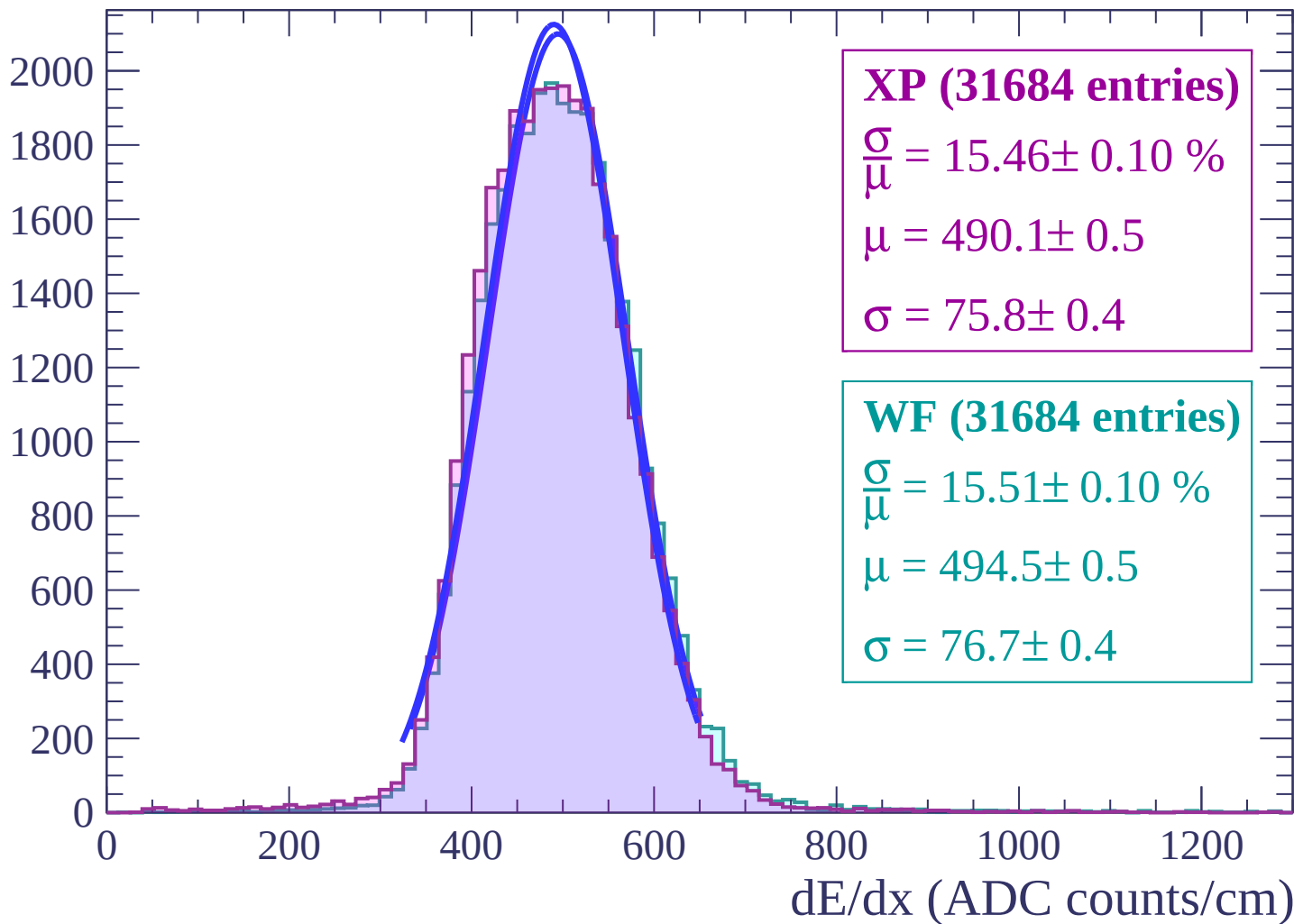


Energy loss | $16 < \theta < 18$

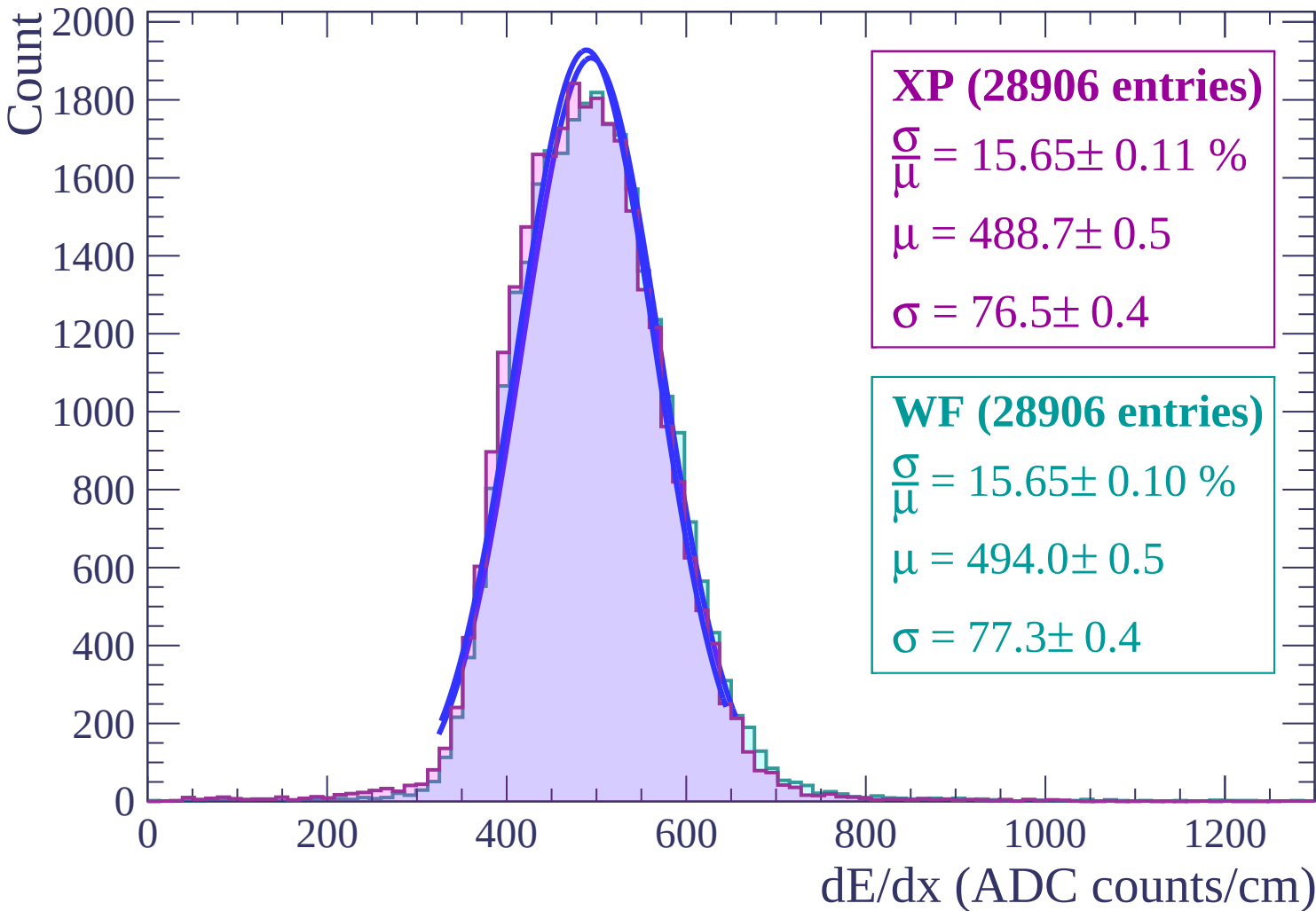


Energy loss | $18 < \theta < 20$

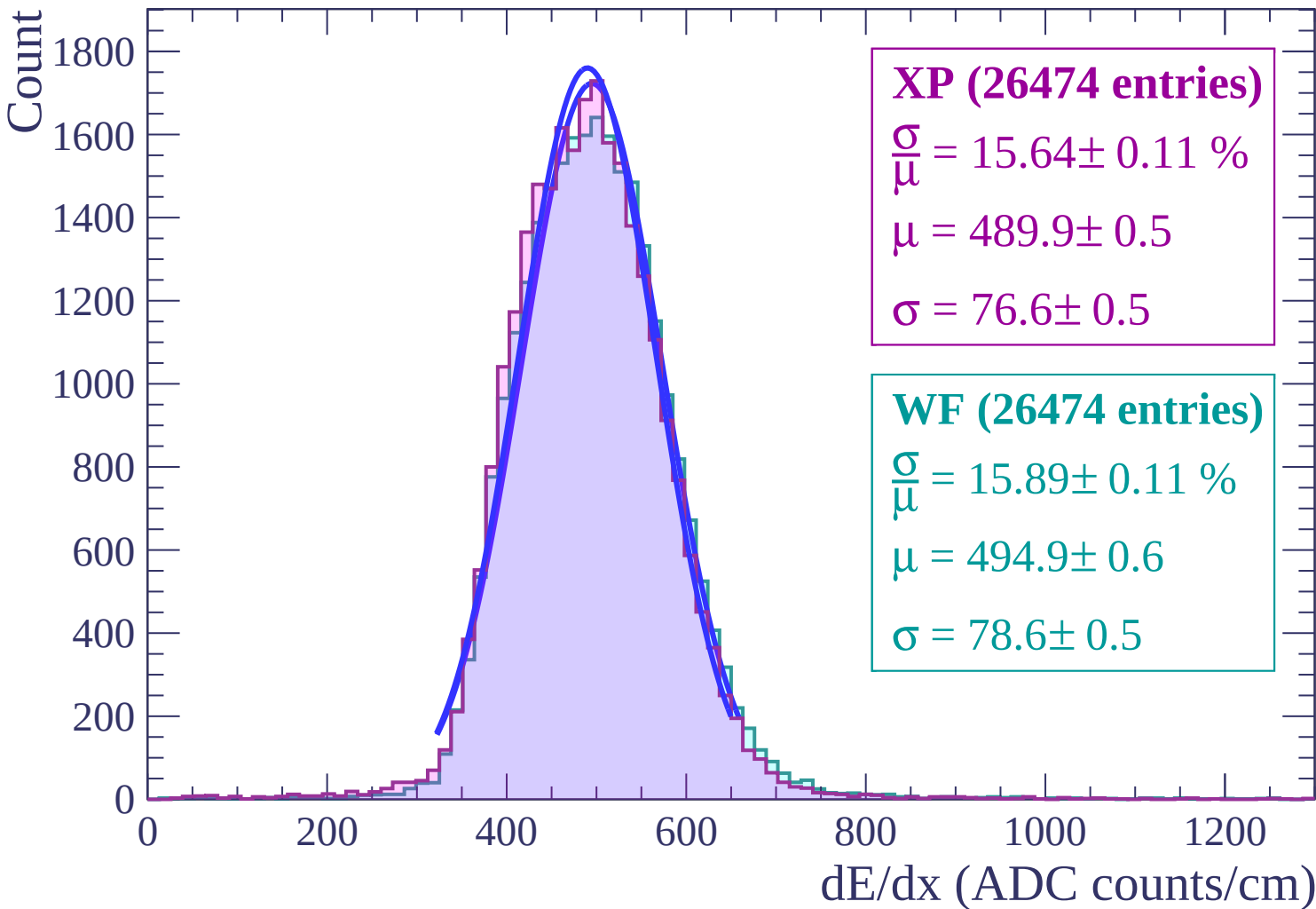
Count



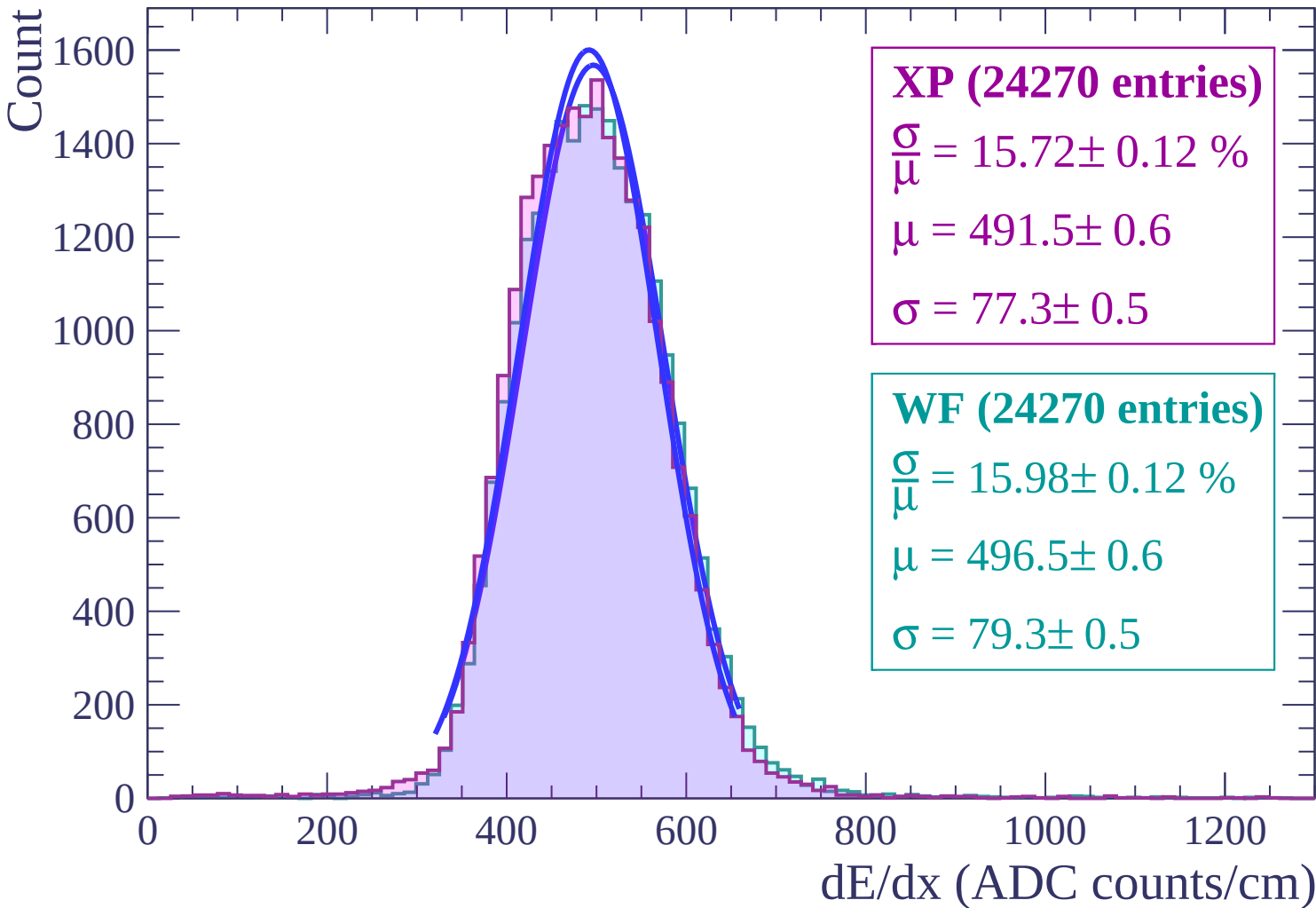
Energy loss | $20 < \theta < 22$



Energy loss | $22 < \theta < 24$



Energy loss | $24 < \theta < 26$



Energy loss | $26 < \theta < 28$

Count

1400
1200
1000
800
600
400
200
0

XP (22379 entries)

$\frac{\sigma}{\mu} = 15.97 \pm 0.12 \%$

$\mu = 492.5 \pm 0.6$

$\sigma = 78.7 \pm 0.5$

WF (22379 entries)

$\frac{\sigma}{\mu} = 16.02 \pm 0.12 \%$

$\mu = 497.3 \pm 0.6$

$\sigma = 79.7 \pm 0.5$

0

200

400

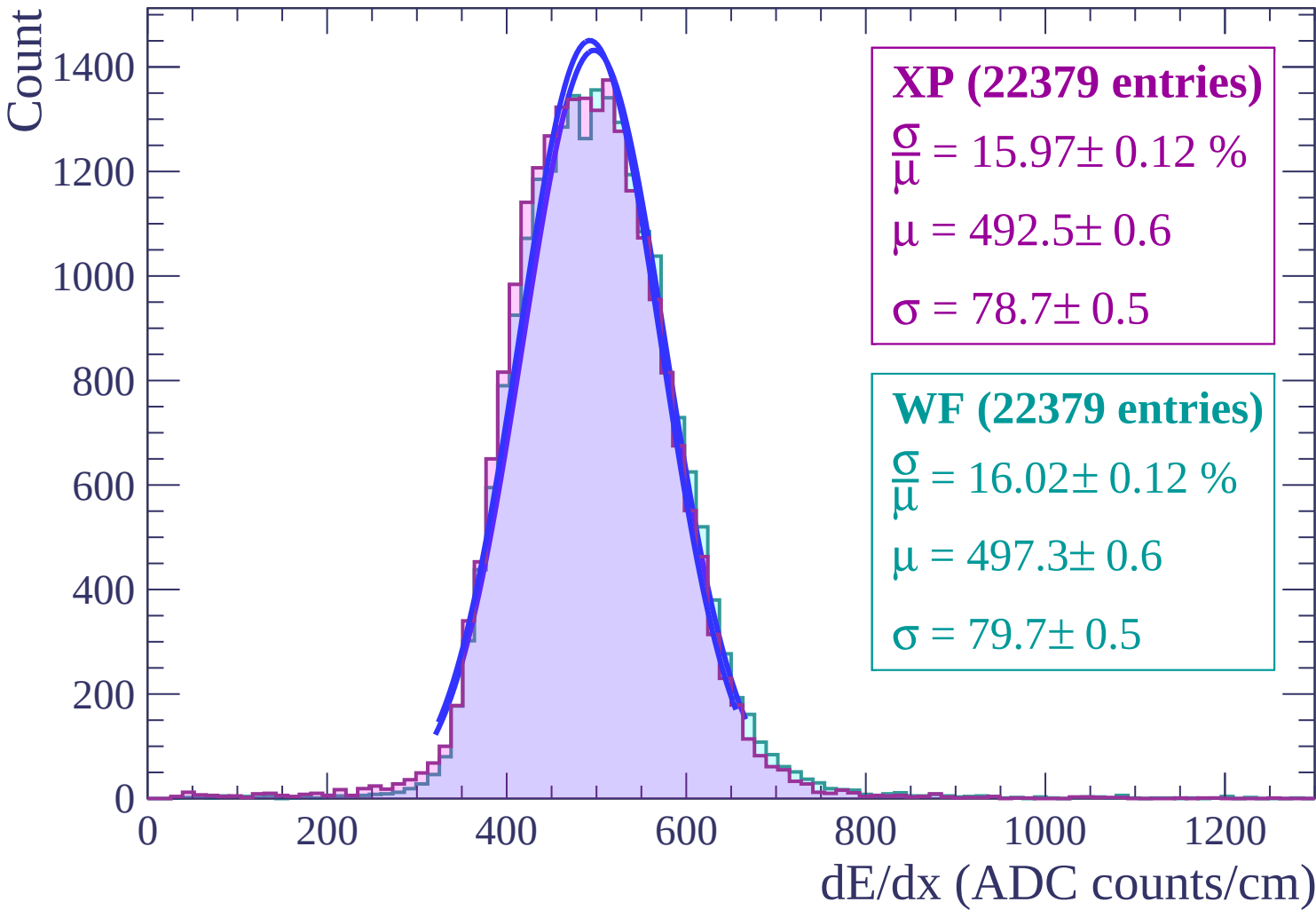
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $28 < \theta < 30$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (19634 entries)

$\frac{\sigma}{\mu} = 15.90 \pm 0.13 \%$

$\mu = 493.4 \pm 0.6$

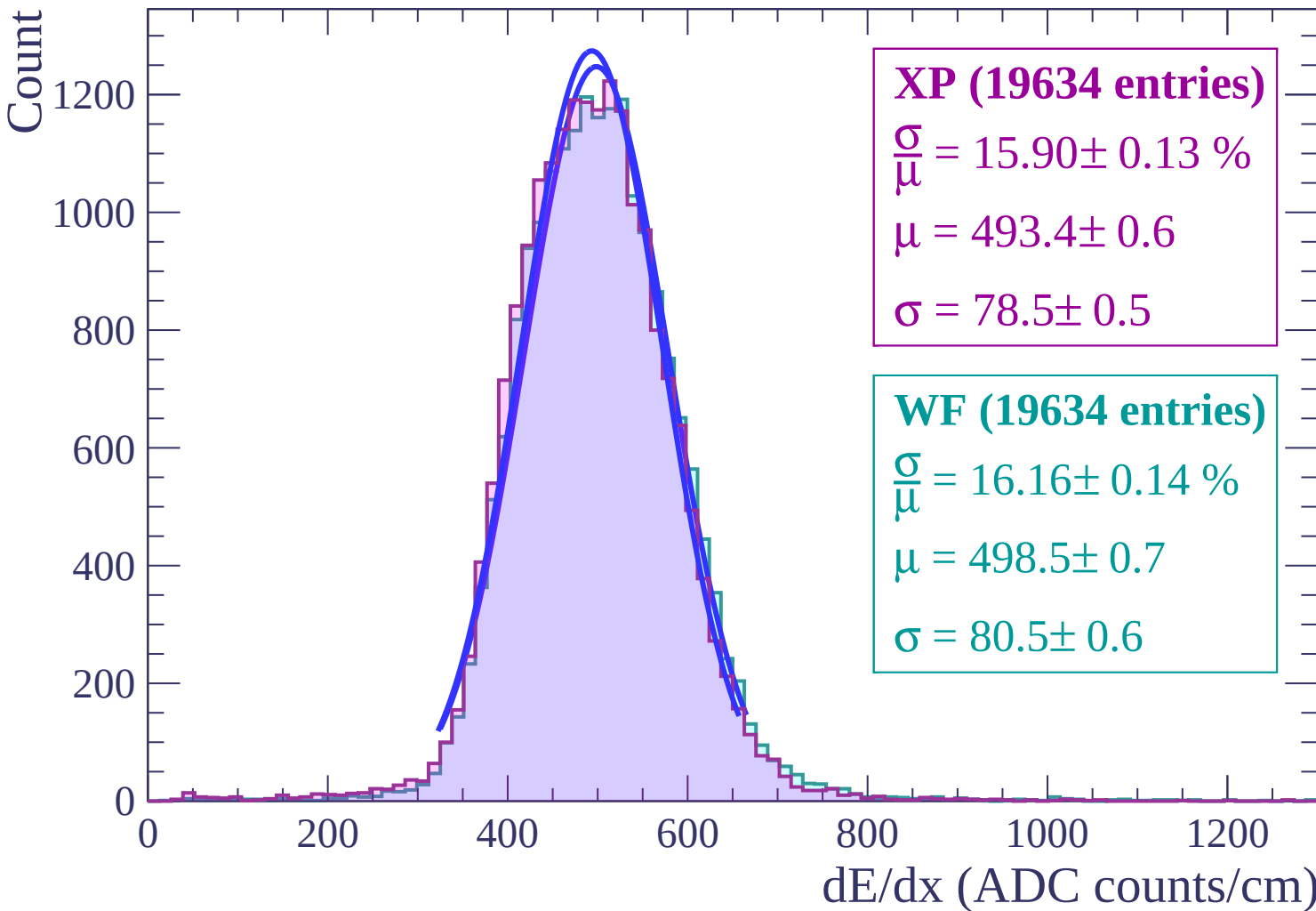
$\sigma = 78.5 \pm 0.5$

WF (19634 entries)

$\frac{\sigma}{\mu} = 16.16 \pm 0.14 \%$

$\mu = 498.5 \pm 0.7$

$\sigma = 80.5 \pm 0.6$



Energy loss | $30 < \theta < 32$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15894 entries)

$\frac{\sigma}{\mu} = 16.05 \pm 0.13 \%$

$\mu = 497.6 \pm 0.7$

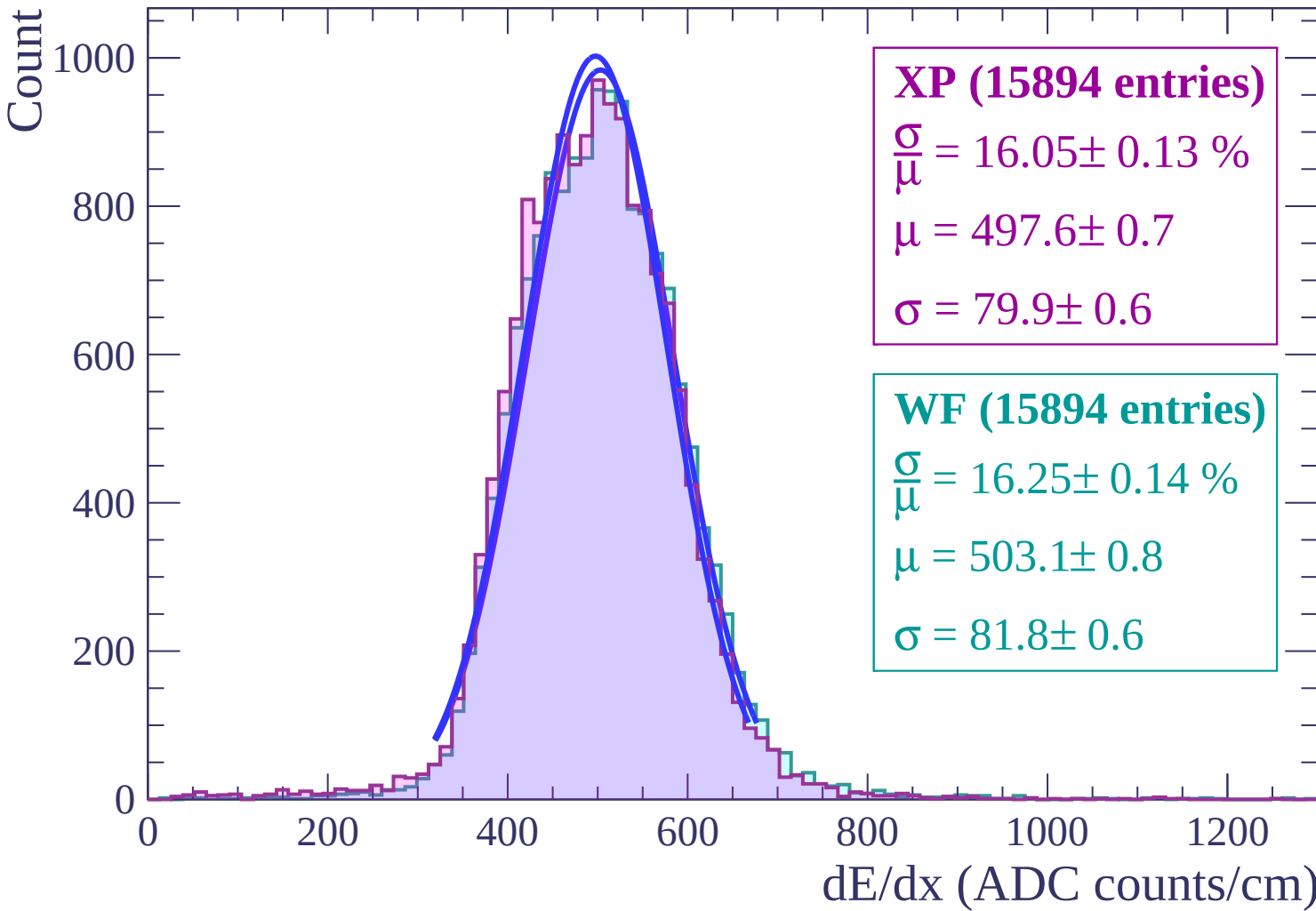
$\sigma = 79.9 \pm 0.6$

WF (15894 entries)

$\frac{\sigma}{\mu} = 16.25 \pm 0.14 \%$

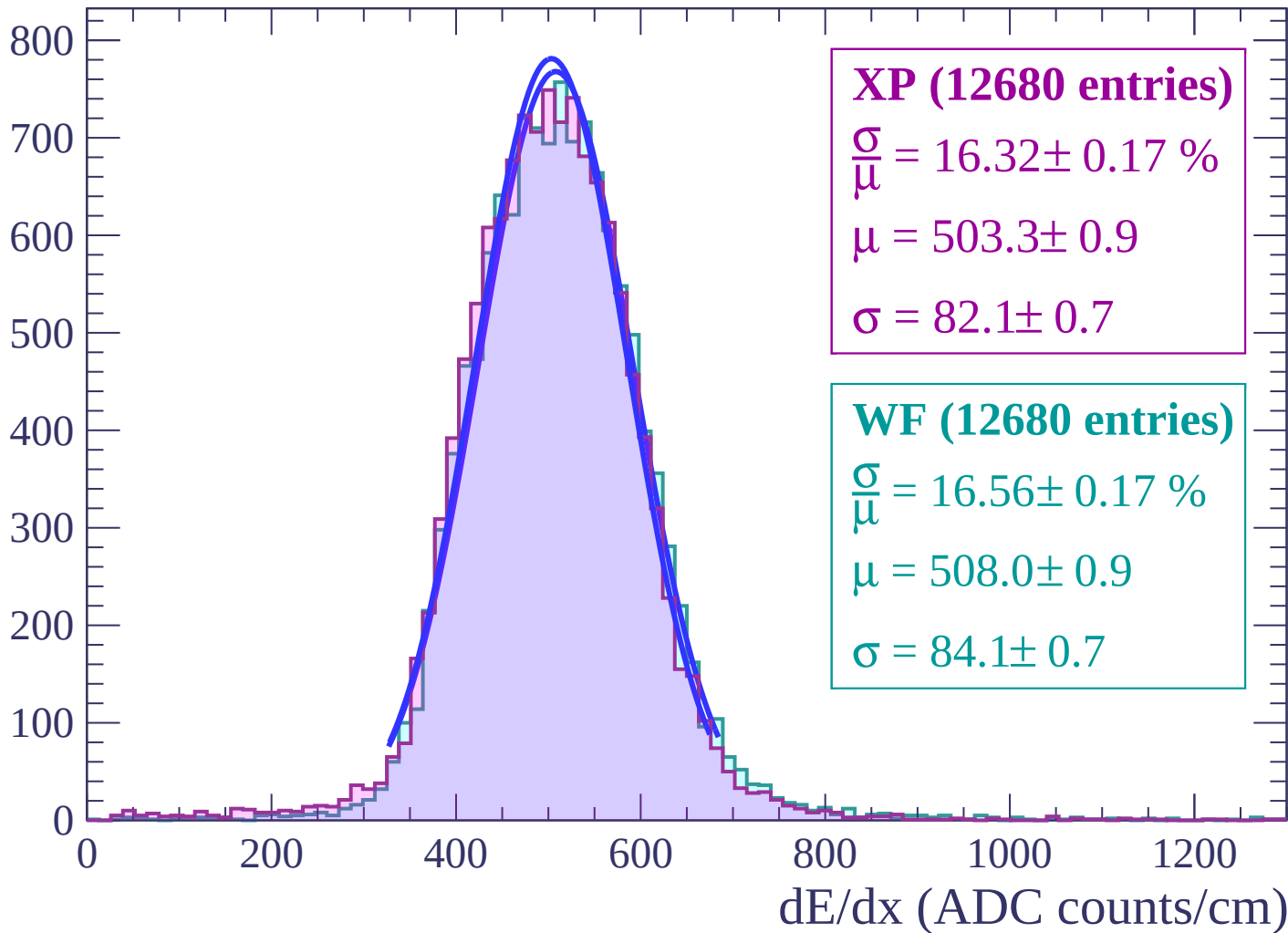
$\mu = 503.1 \pm 0.8$

$\sigma = 81.8 \pm 0.6$



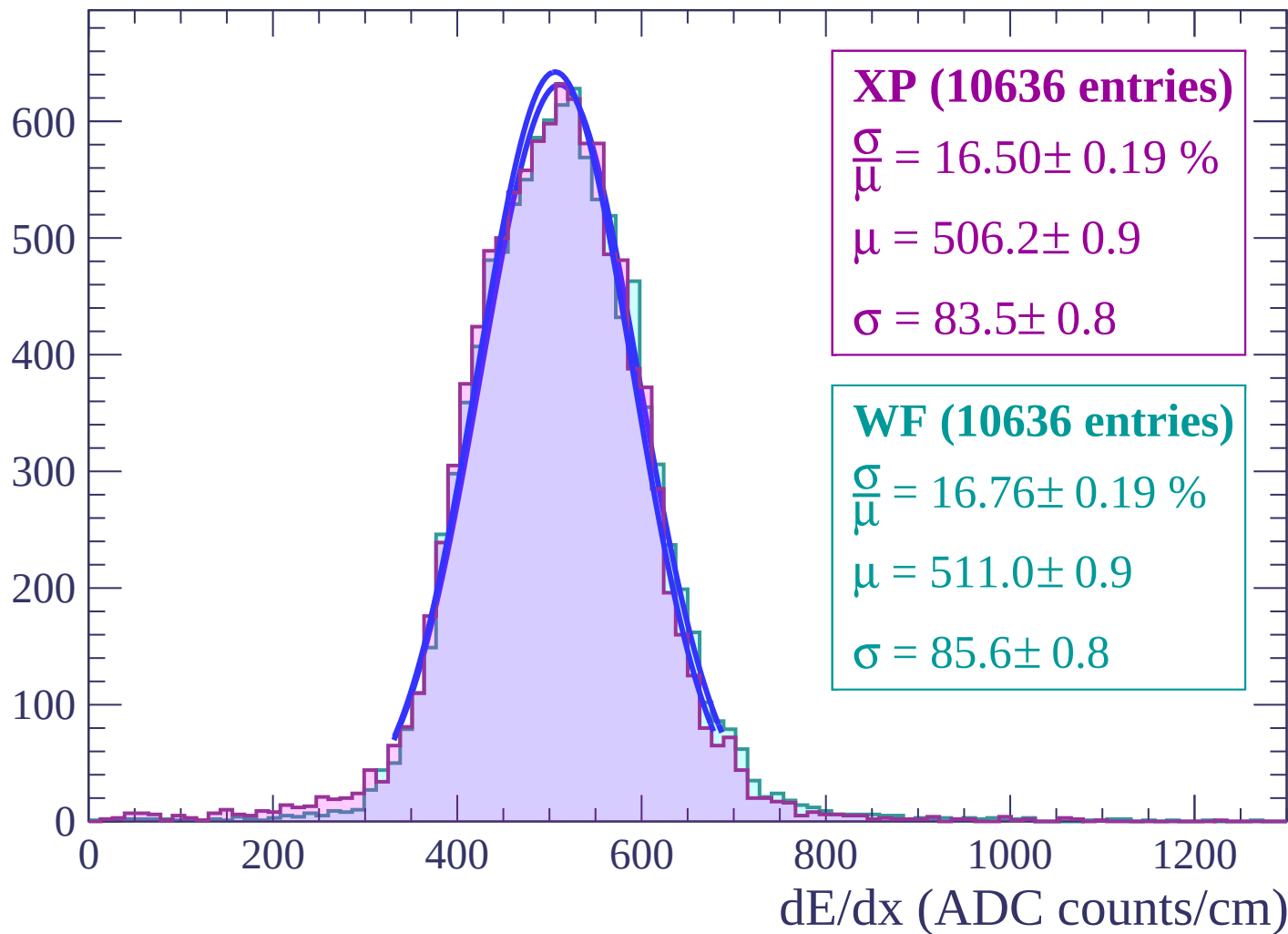
Energy loss | $32 < \theta < 34$

Count



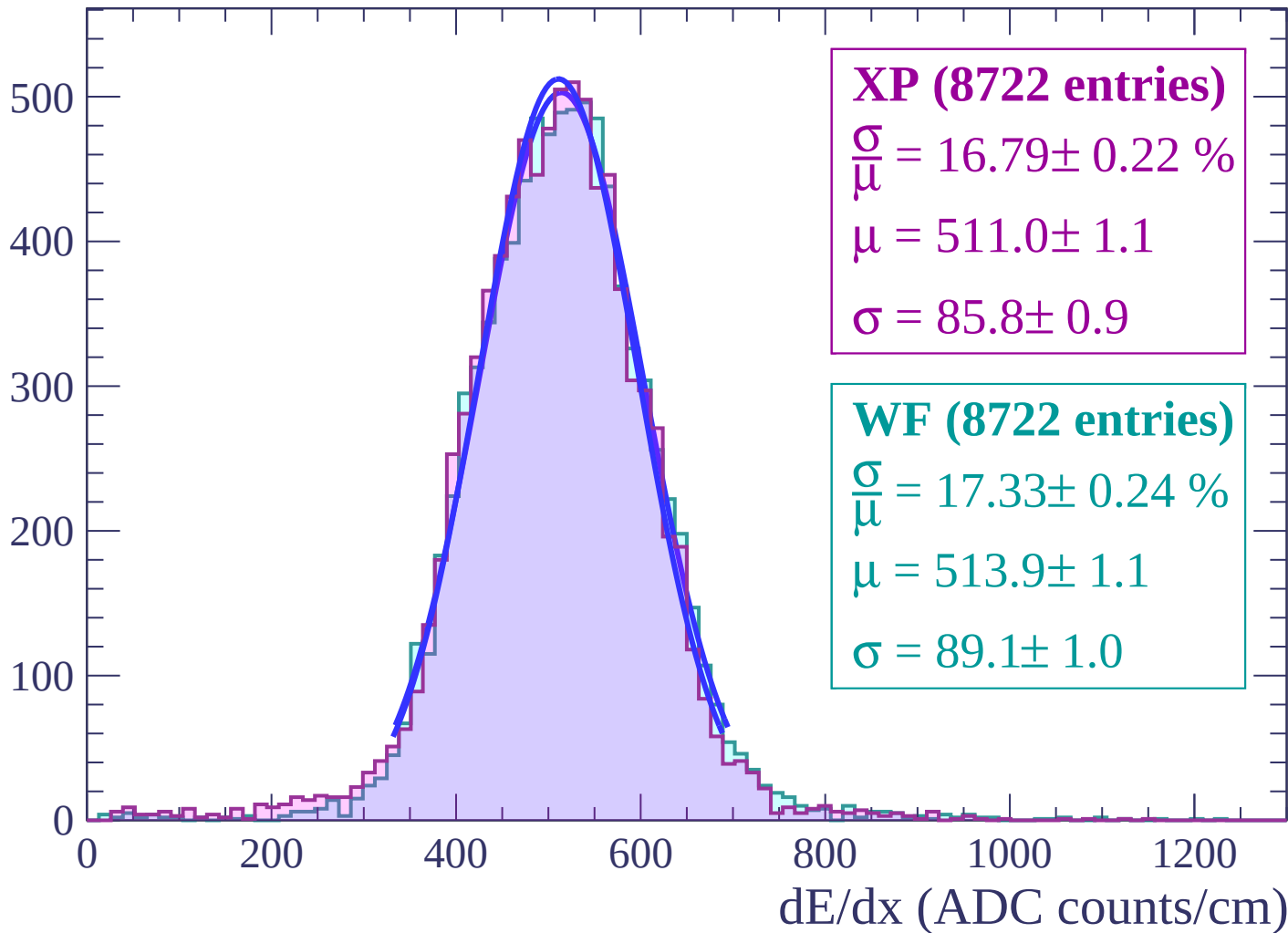
Energy loss | $34 < \theta < 36$

Count



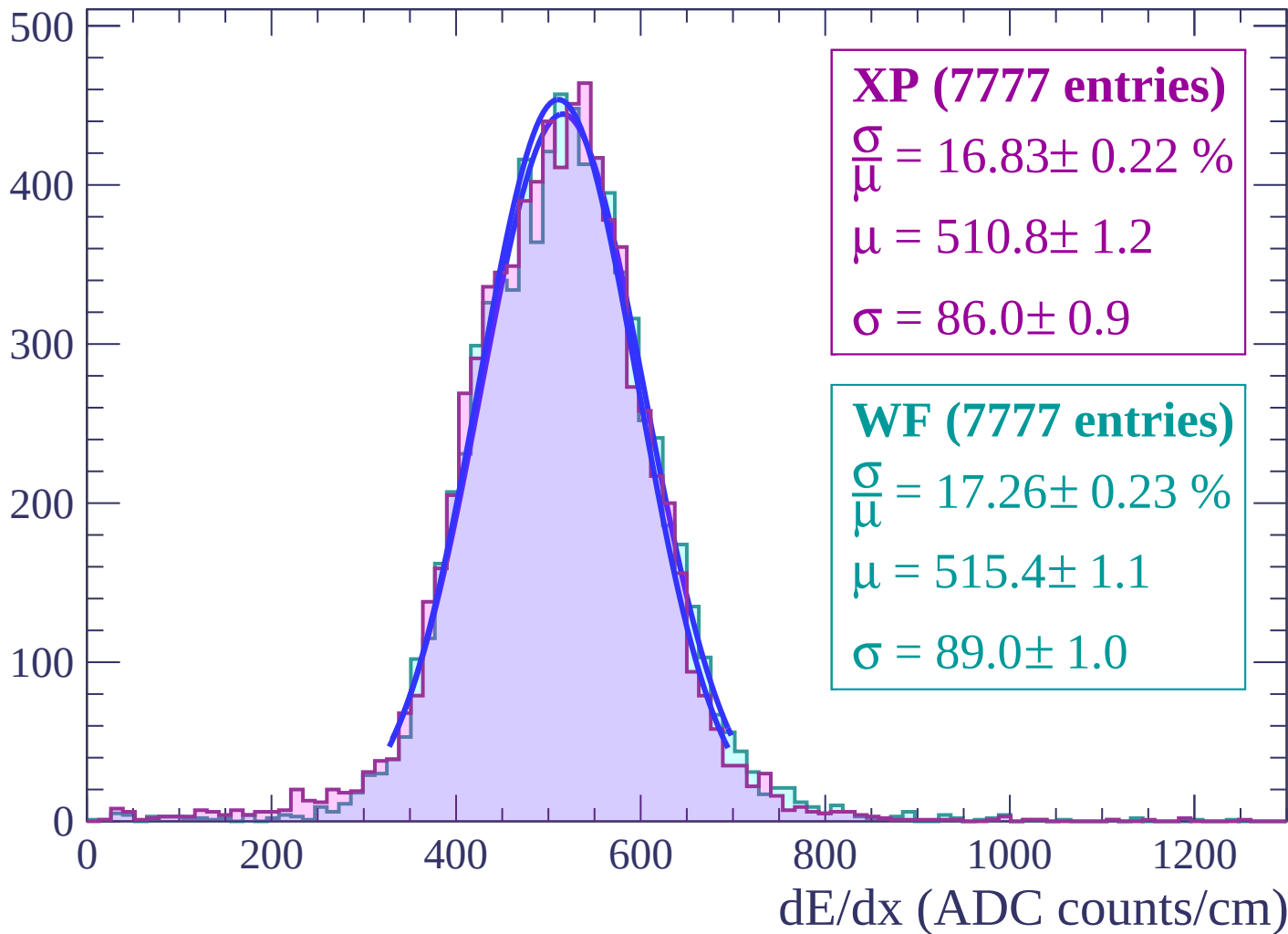
Energy loss | $36 < \theta < 38$

Count



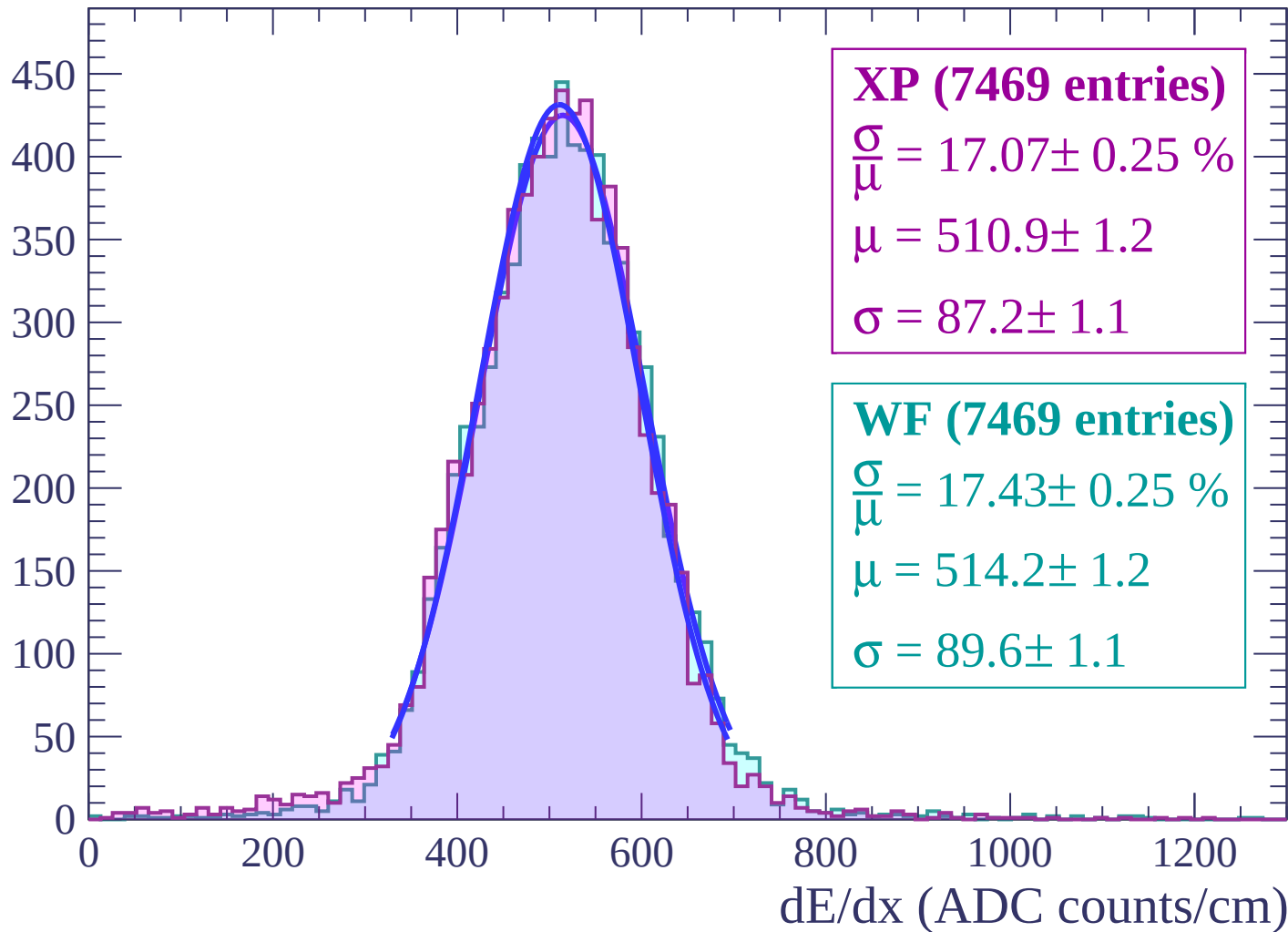
Energy loss | $38 < \theta < 40$

Count



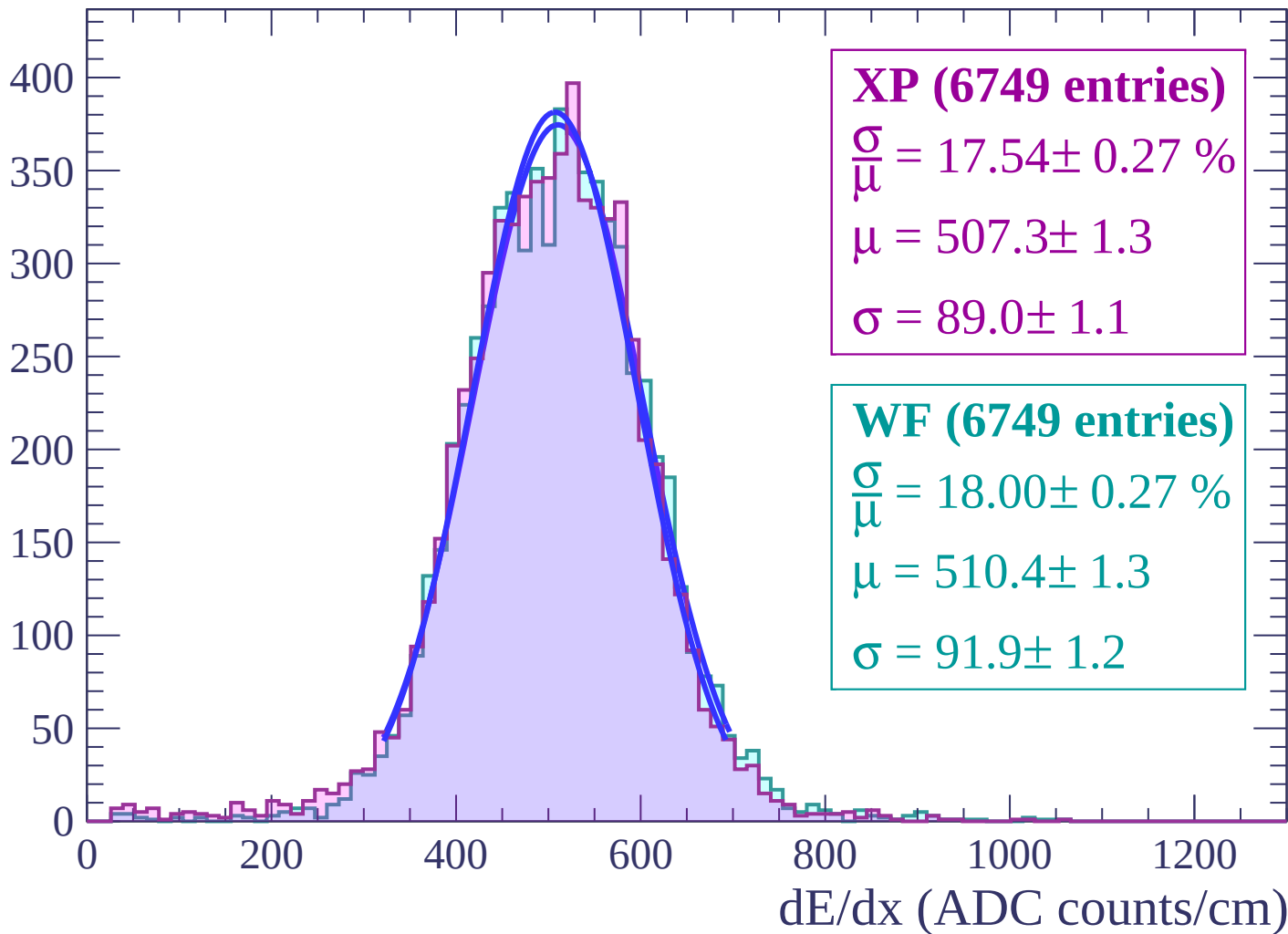
Energy loss | $40 < \theta < 42$

Count



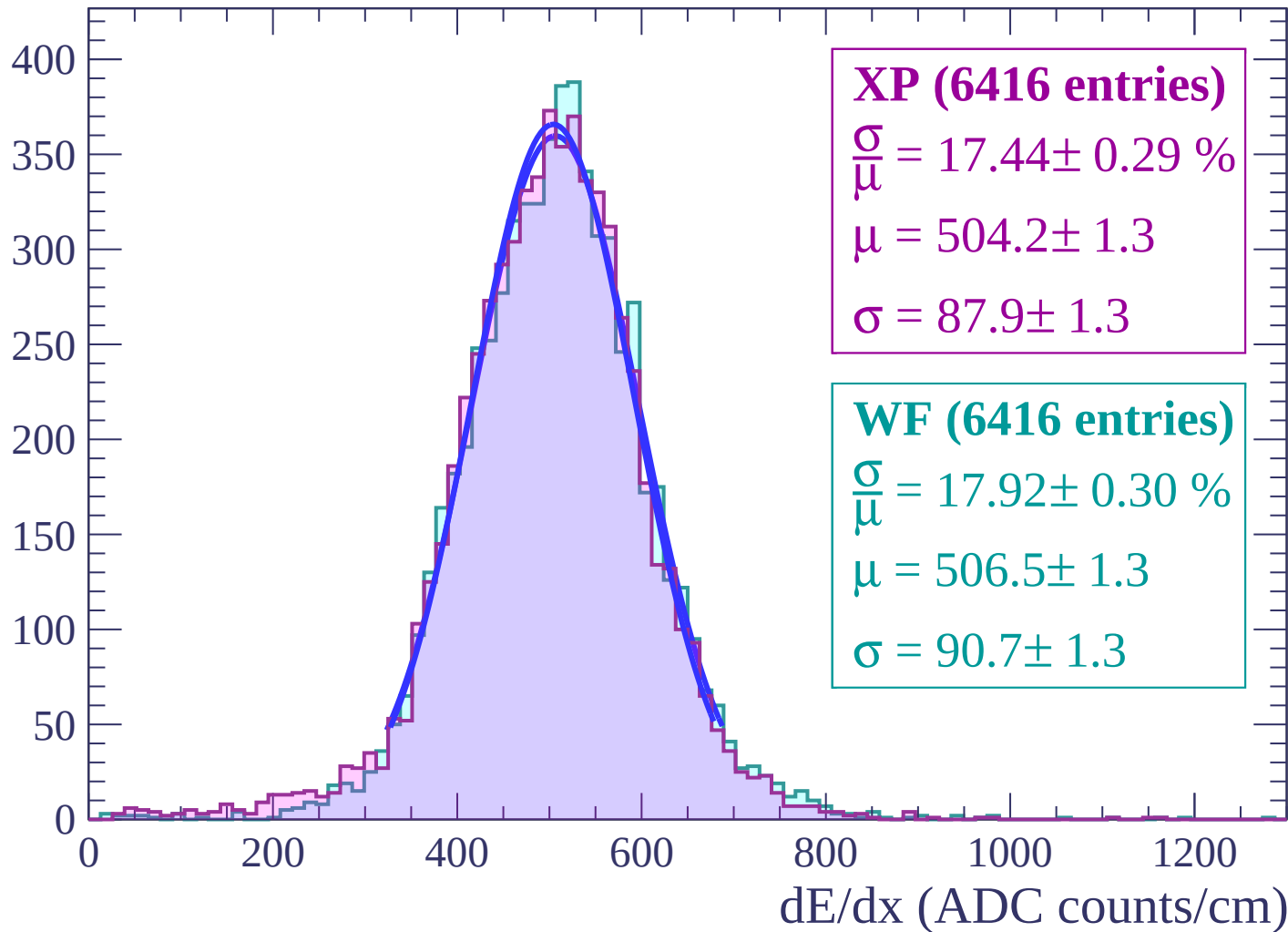
Energy loss | $42 < \theta < 44$

Count



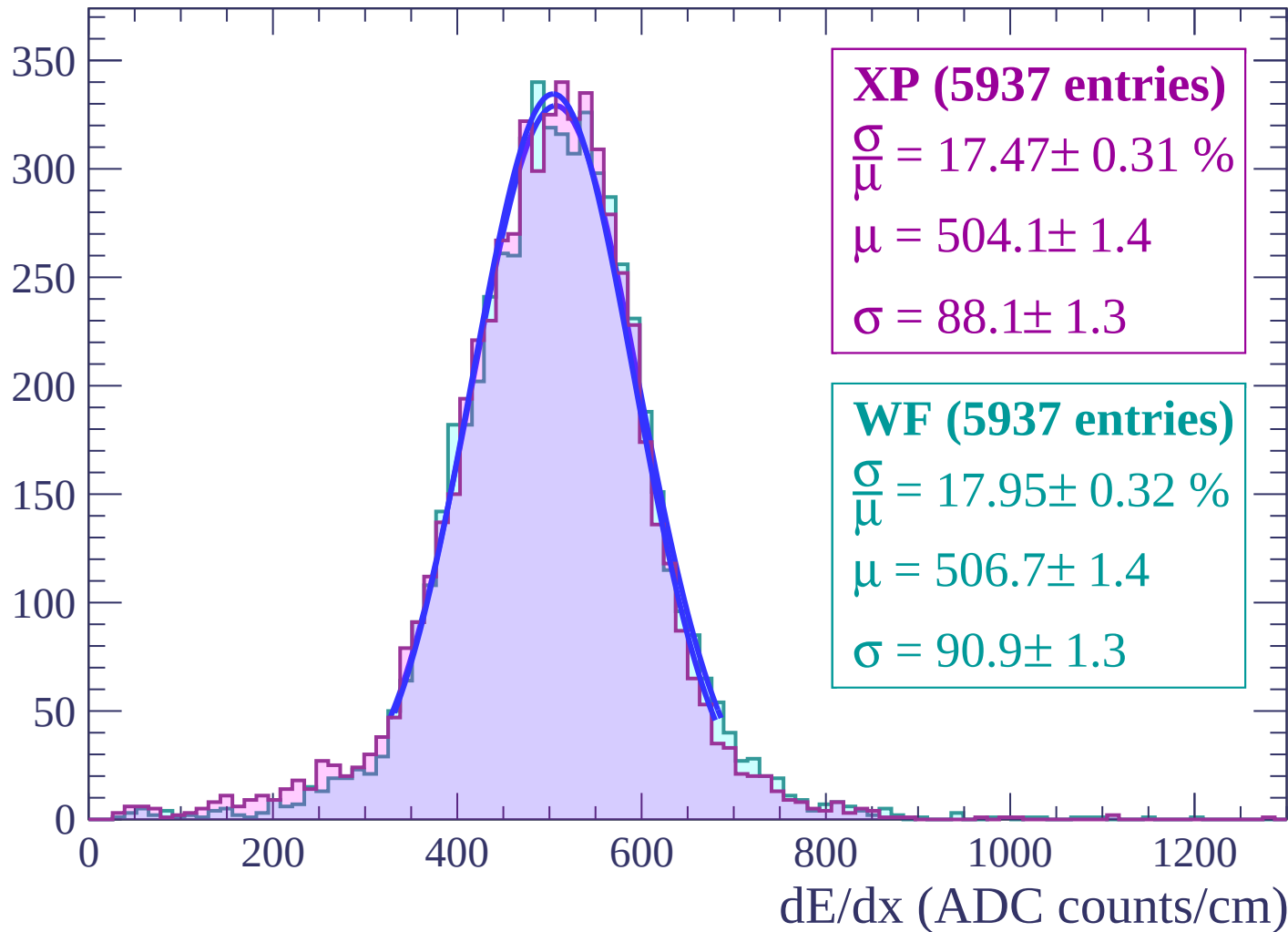
Energy loss | $44 < \theta < 46$

Count



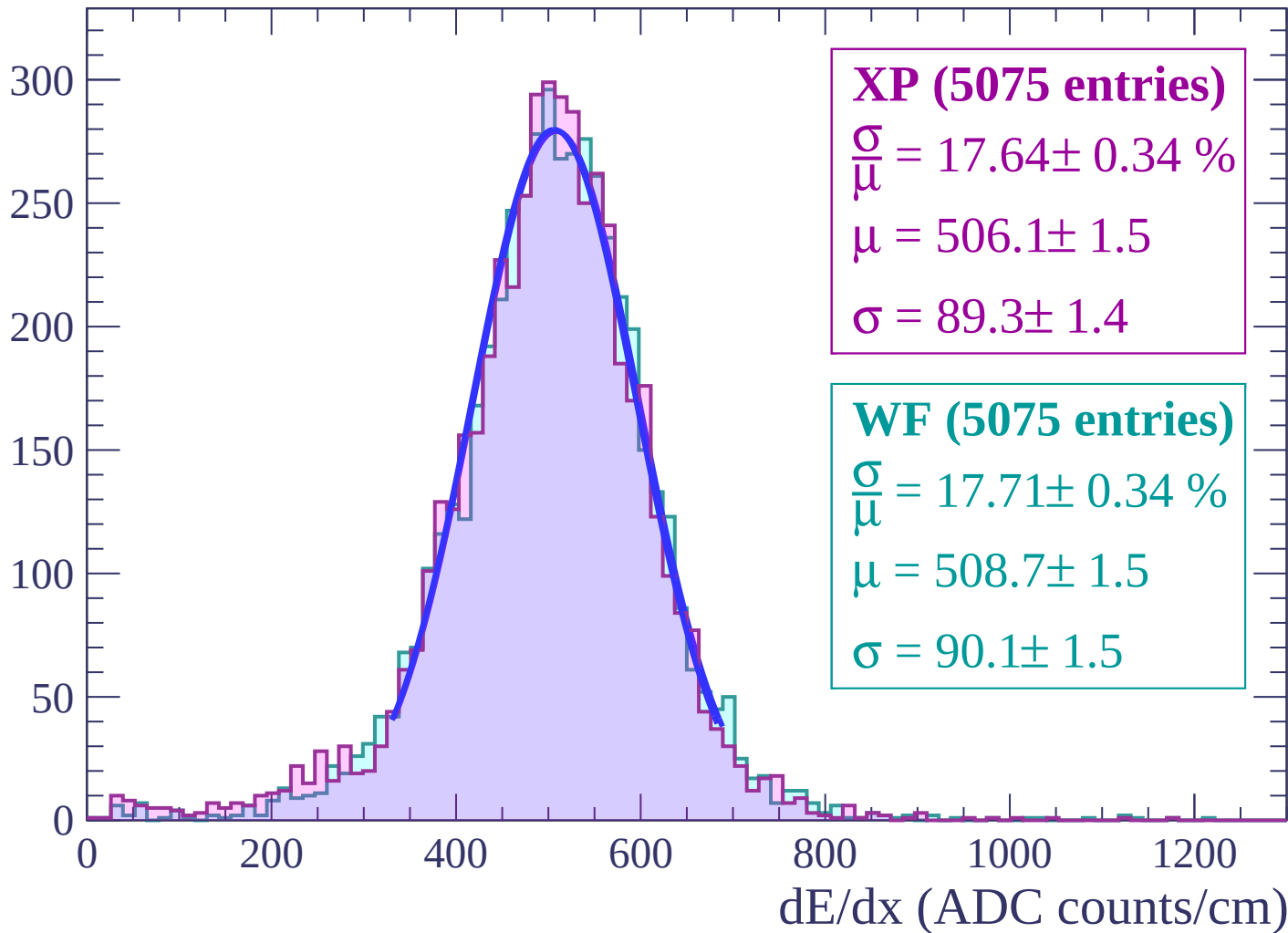
Energy loss | $46 < \theta < 48$

Count



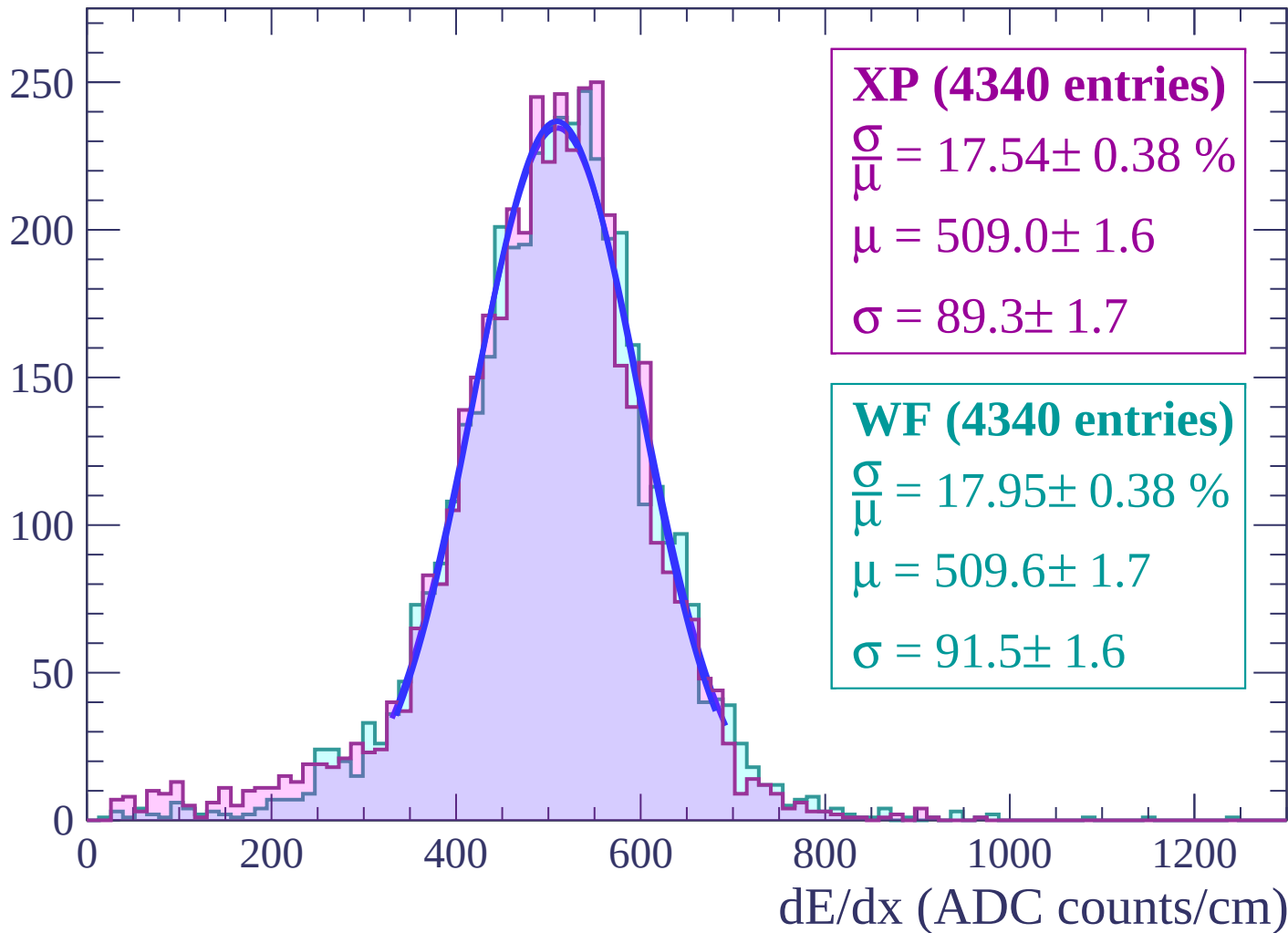
Energy loss | $48 < \theta < 50$

Count



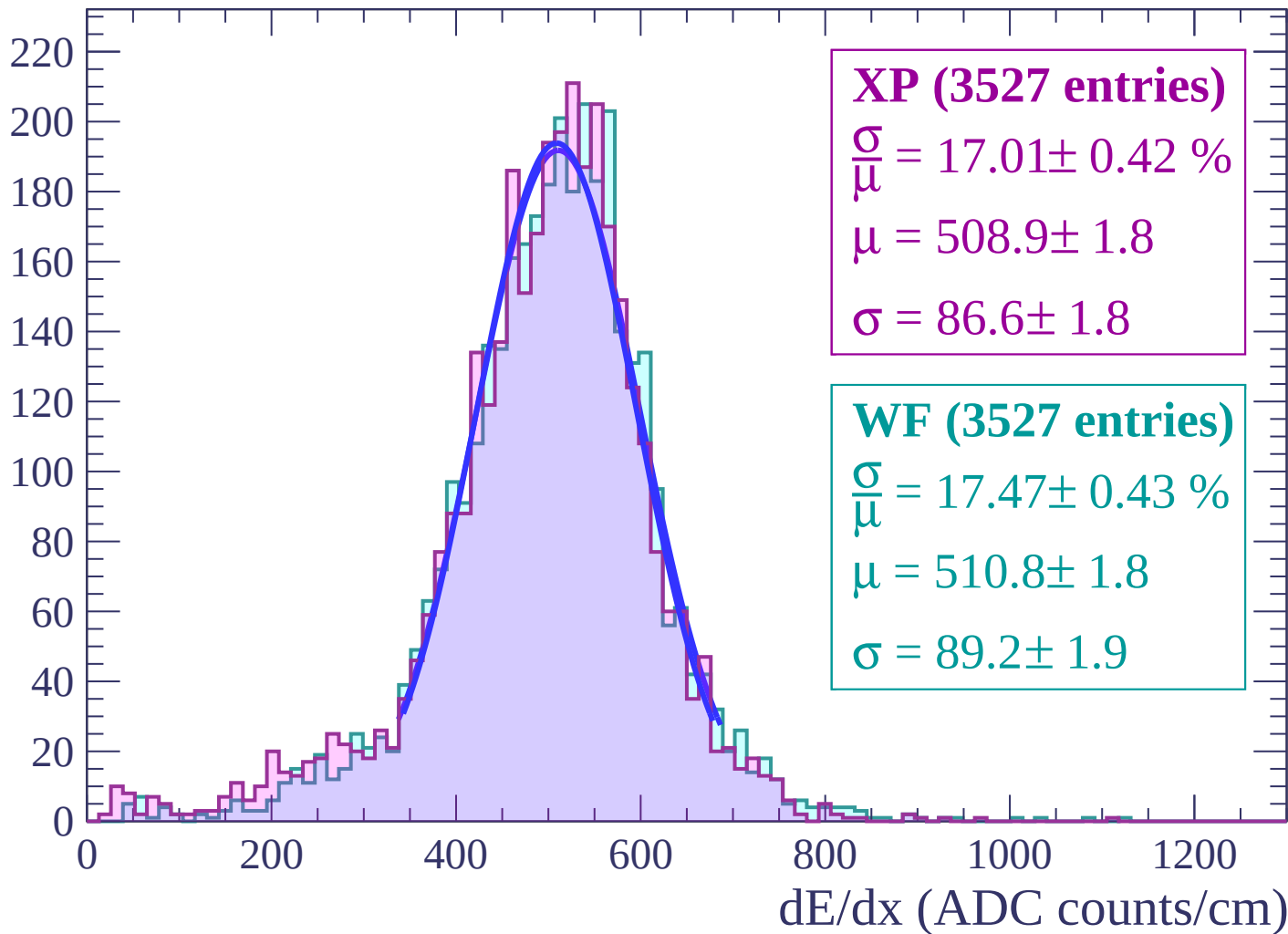
Energy loss | $50 < \theta < 52$

Count



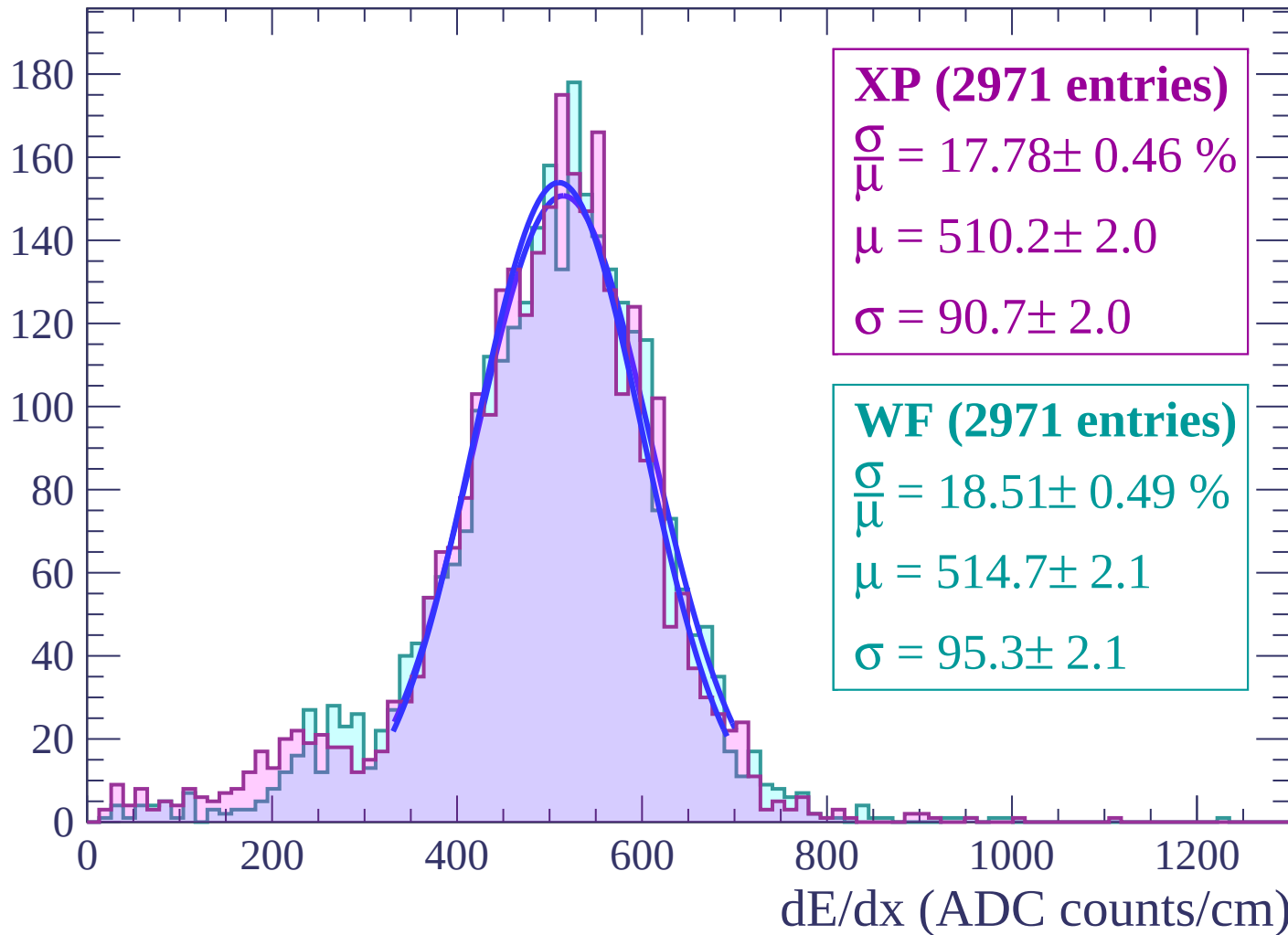
Energy loss | $52 < \theta < 54$

Count



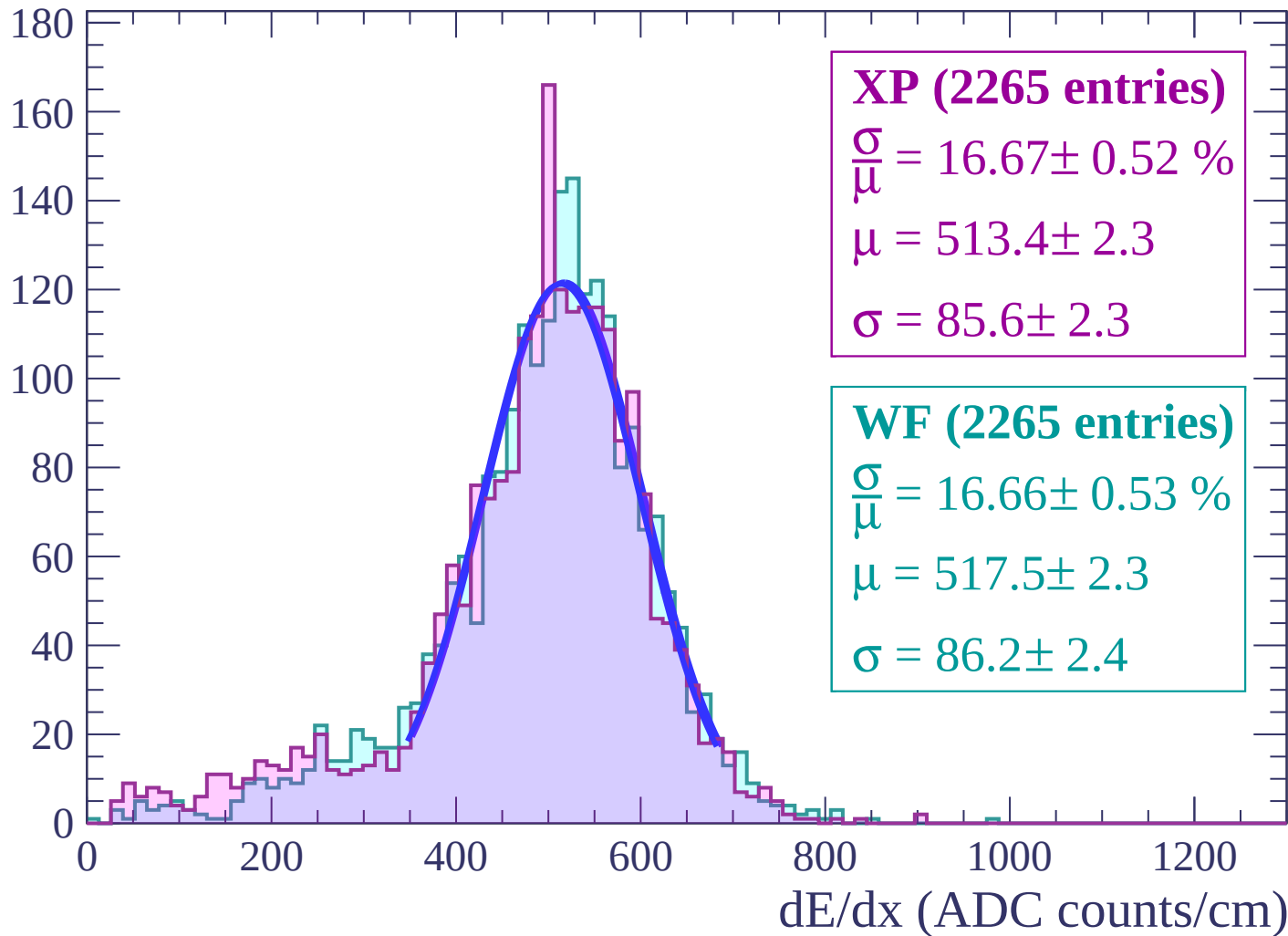
Energy loss | $54 < \theta < 56$

Count



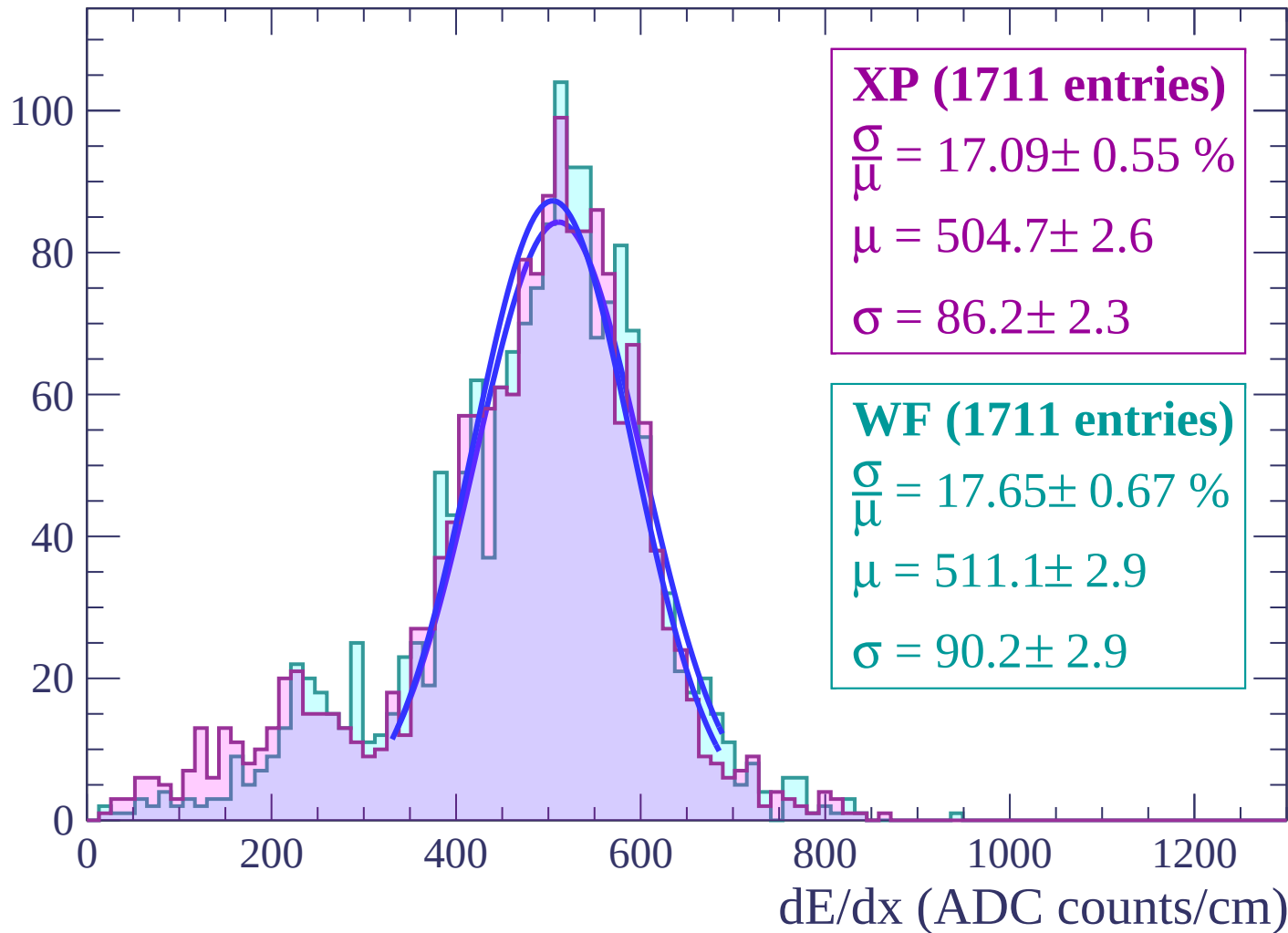
Energy loss | $56 < \theta < 58$

Count



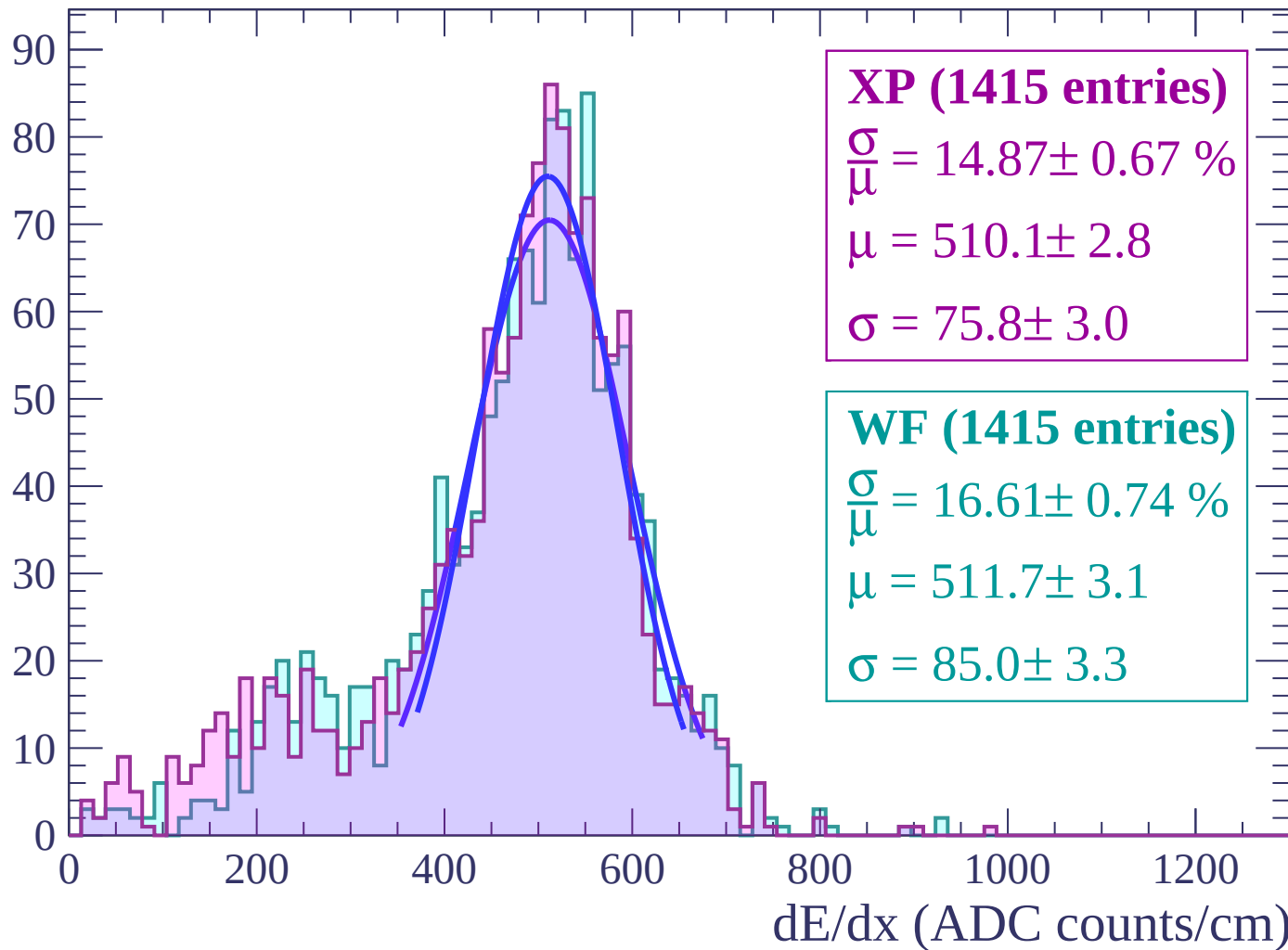
Energy loss | $58 < \theta < 60$

Count



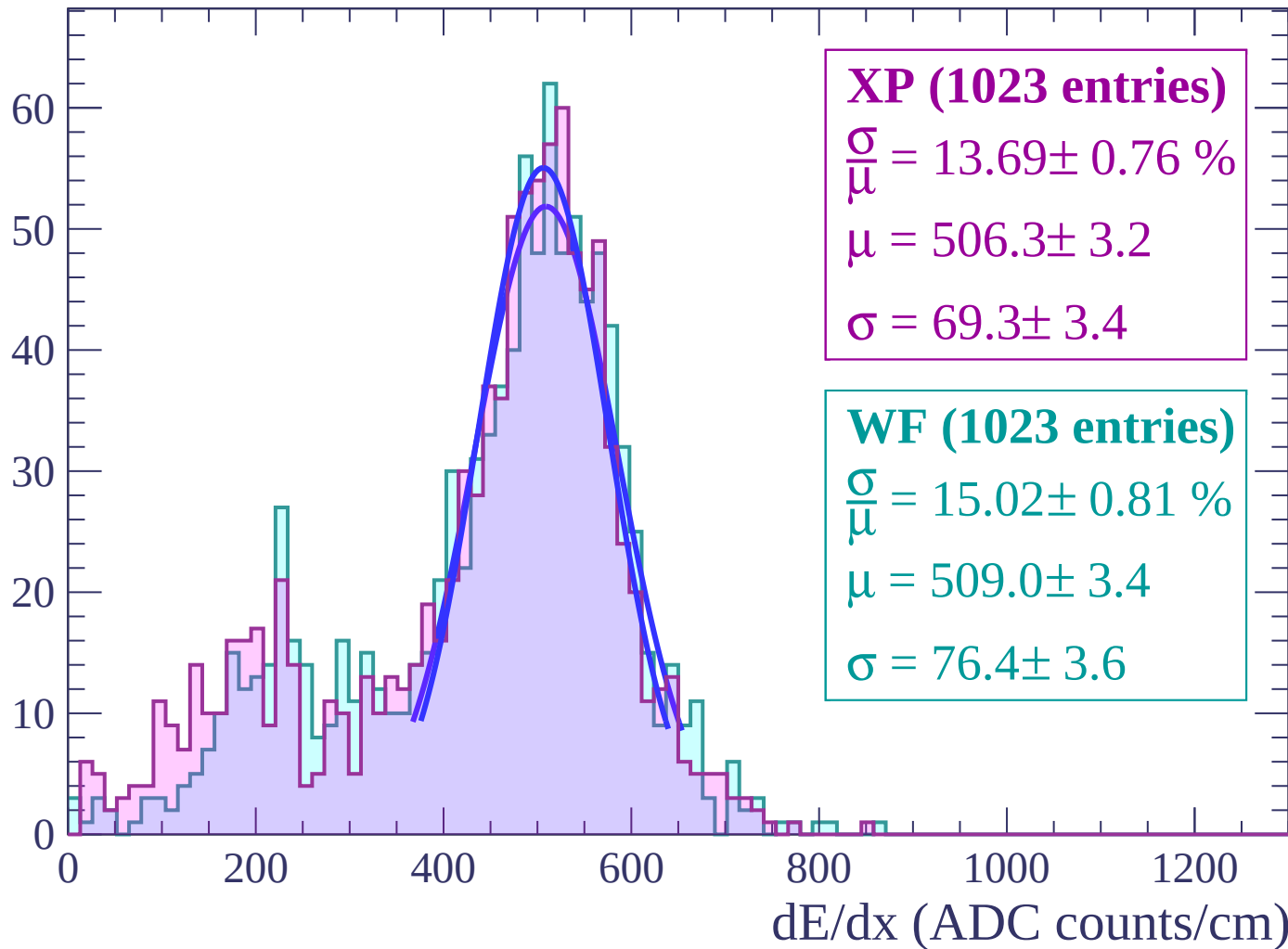
Energy loss | $60 < \theta < 62$

Count



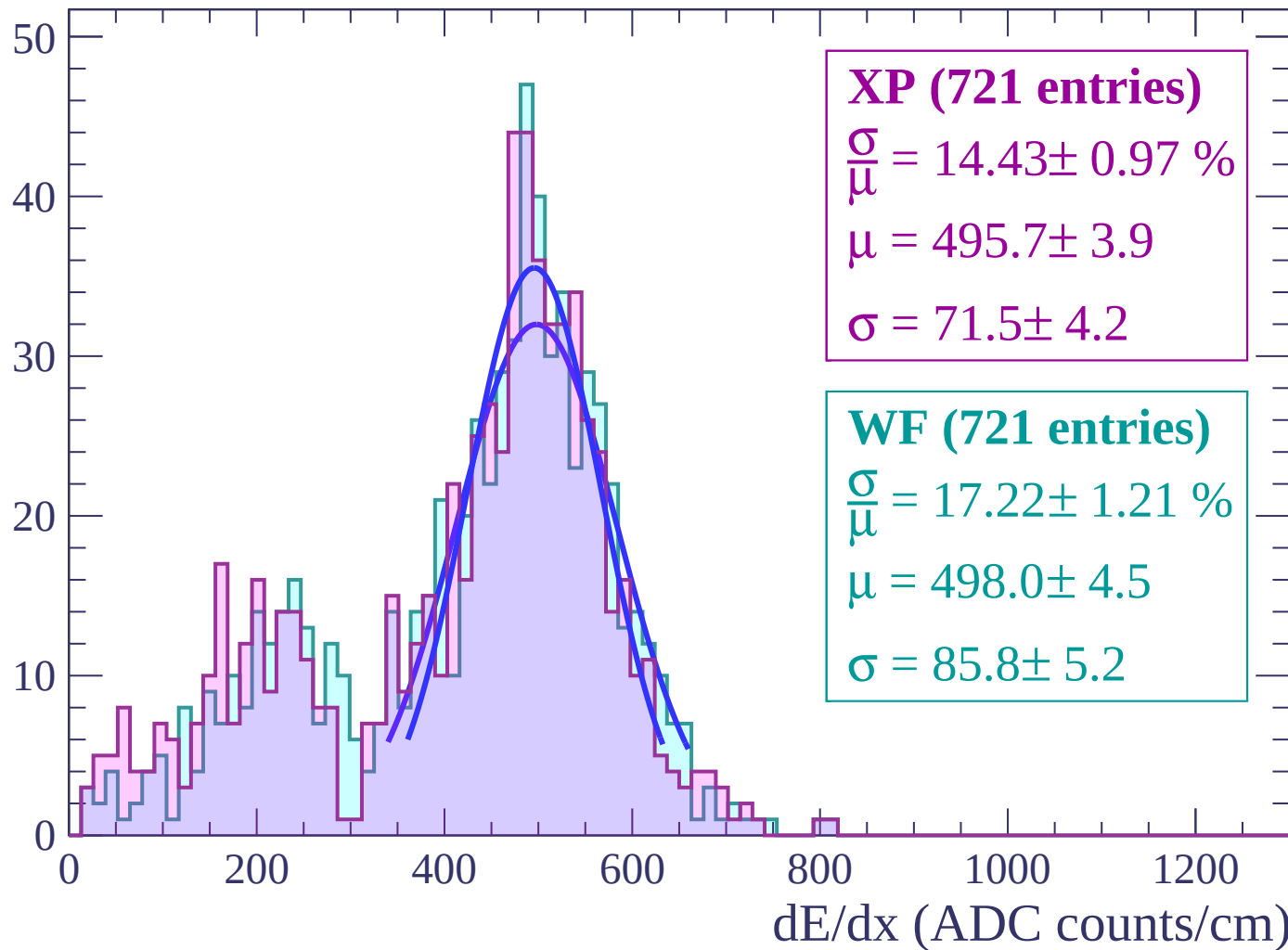
Energy loss | $62 < \theta < 64$

Count



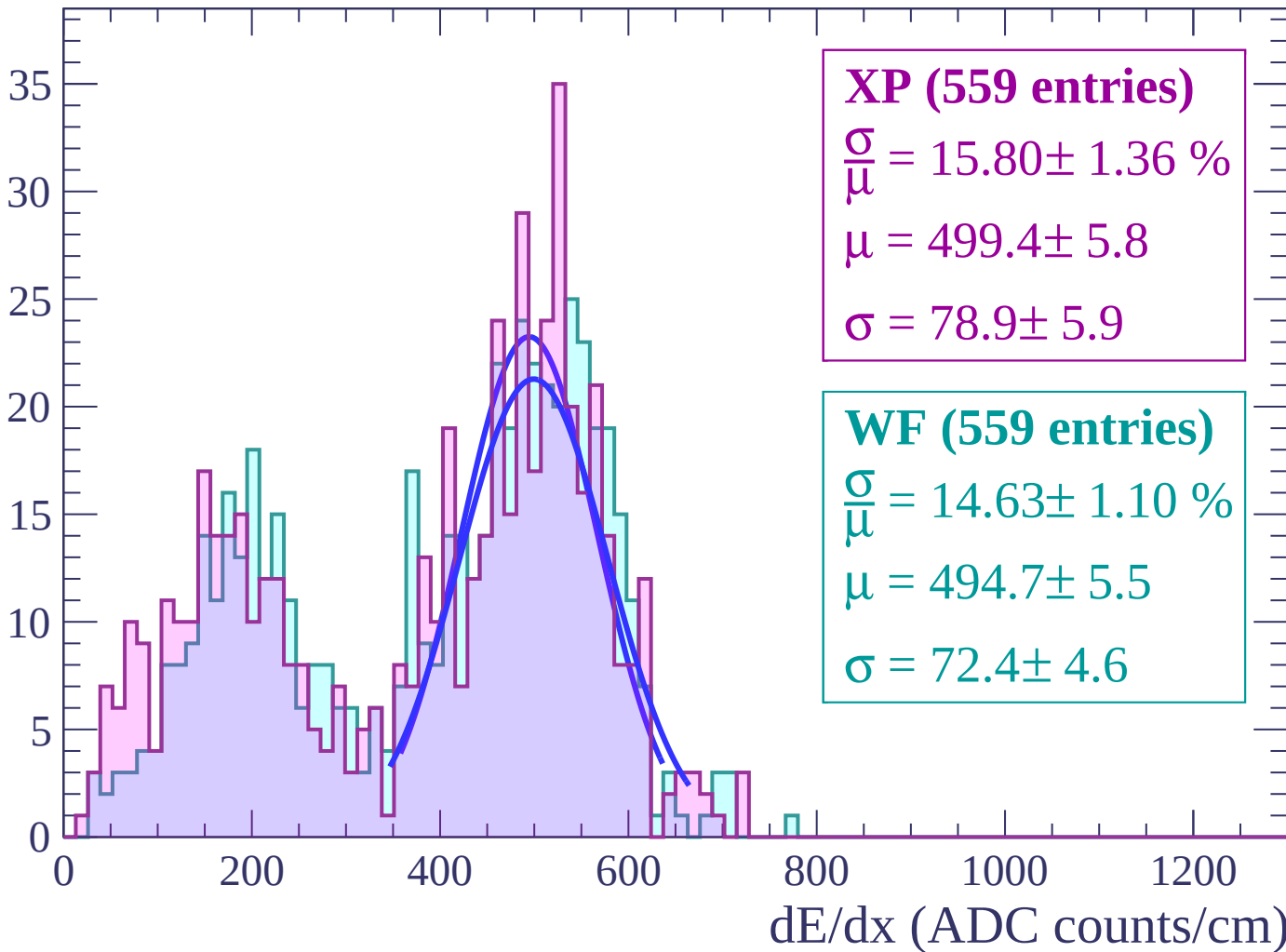
Energy loss | $64 < \theta < 66$

Count



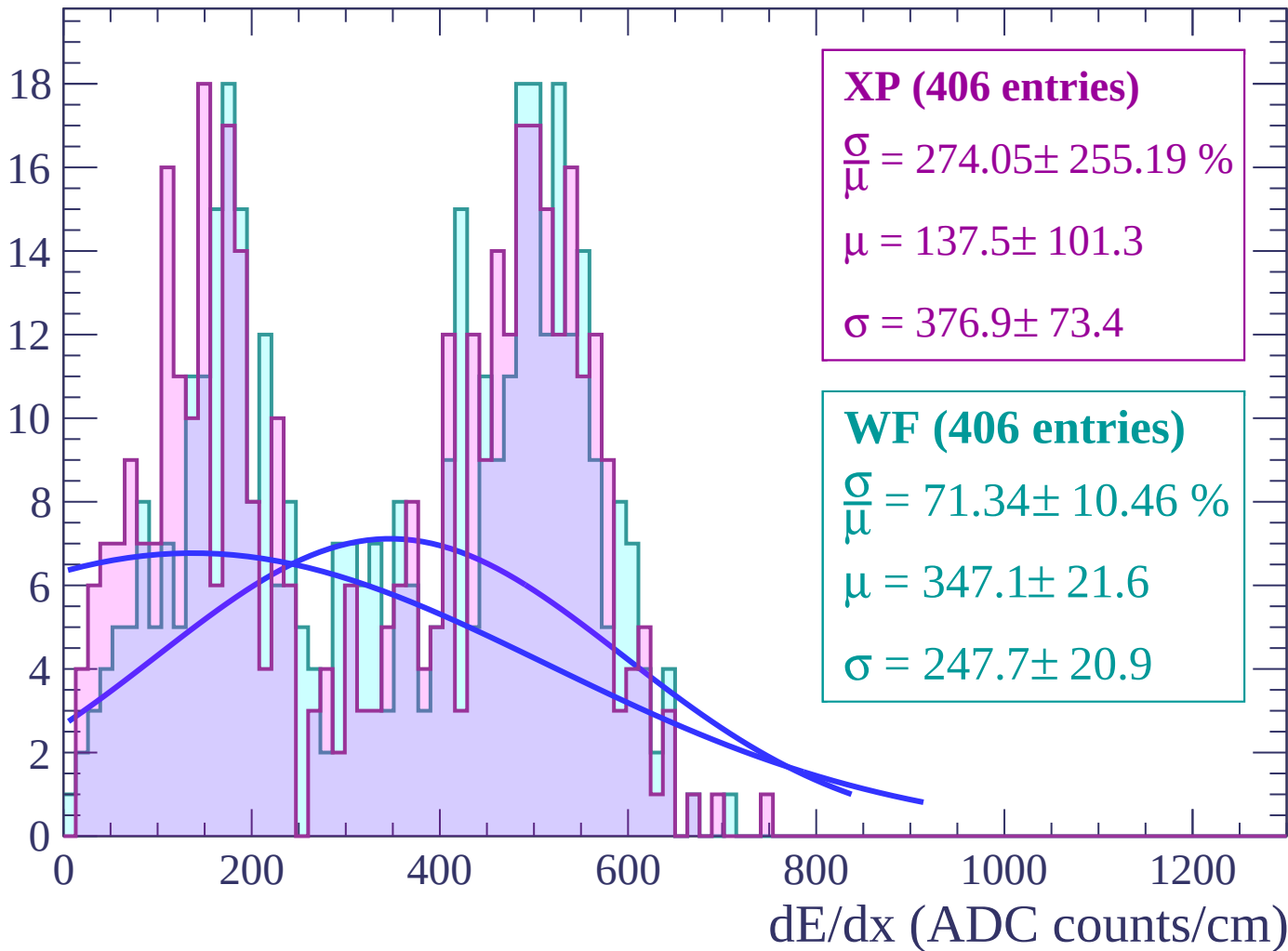
Energy loss | $66 < \theta < 68$

Count



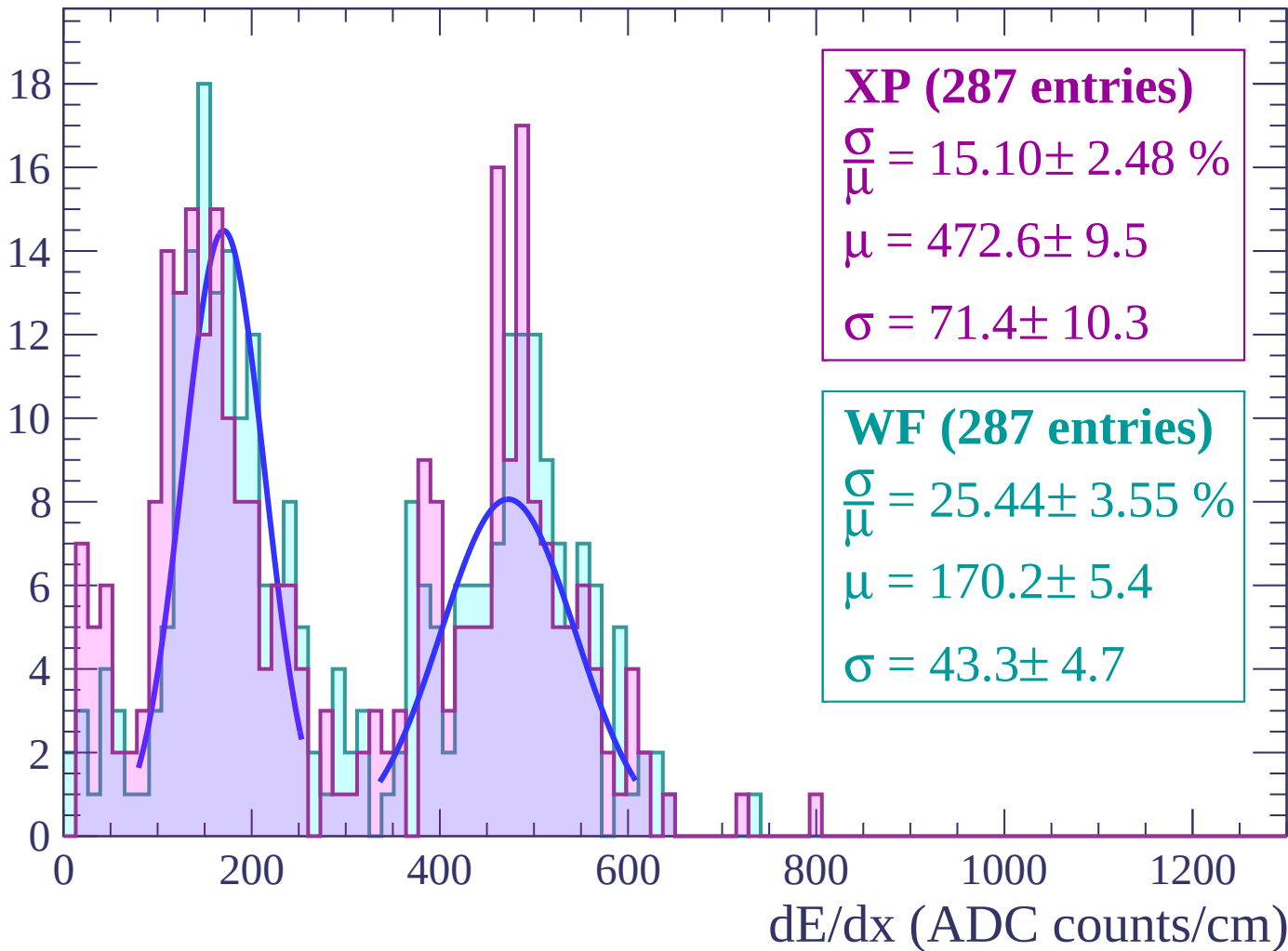
Energy loss | $68 < \theta < 70$

Count



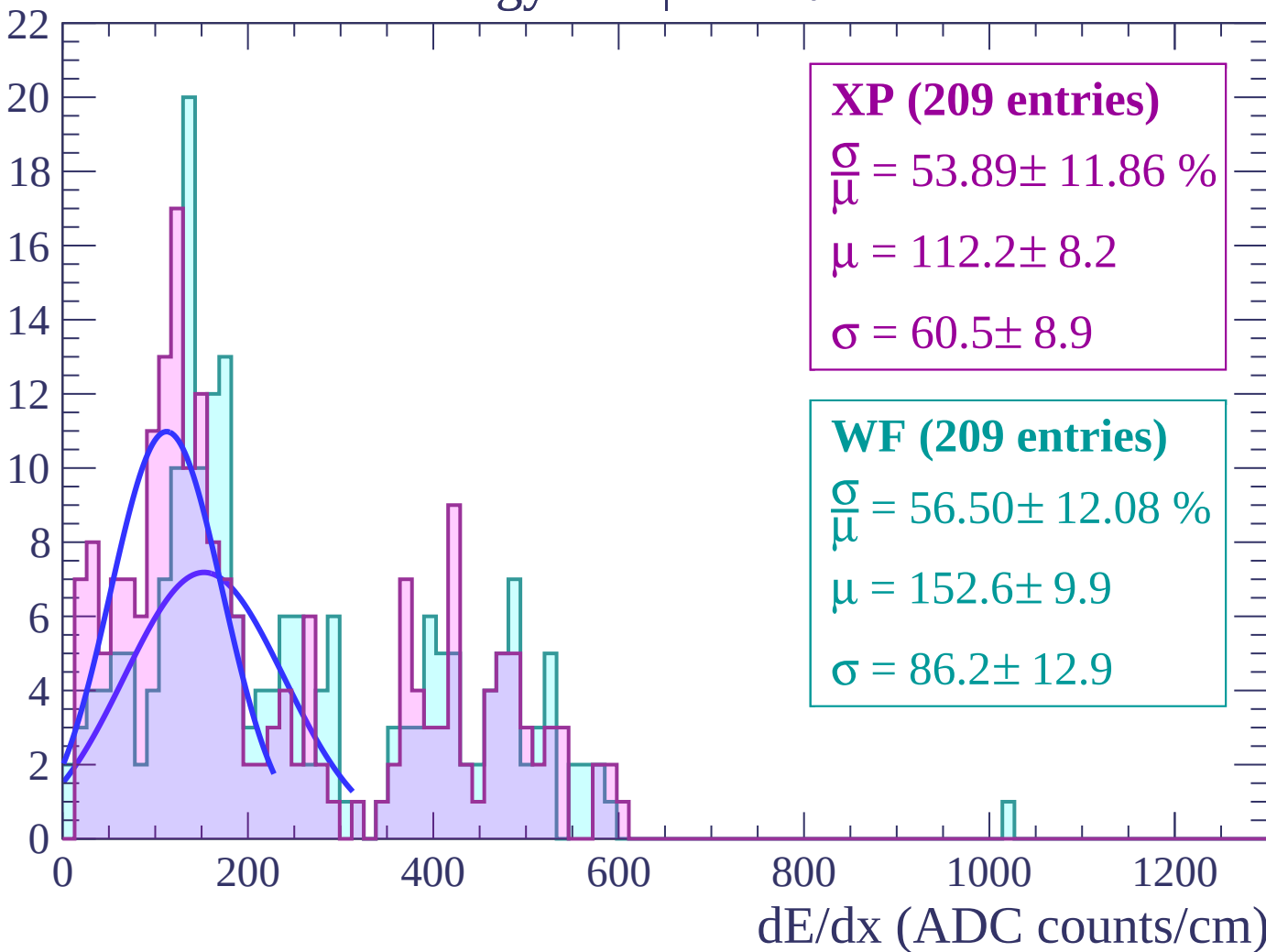
Energy loss | $70 < \theta < 72$

Count



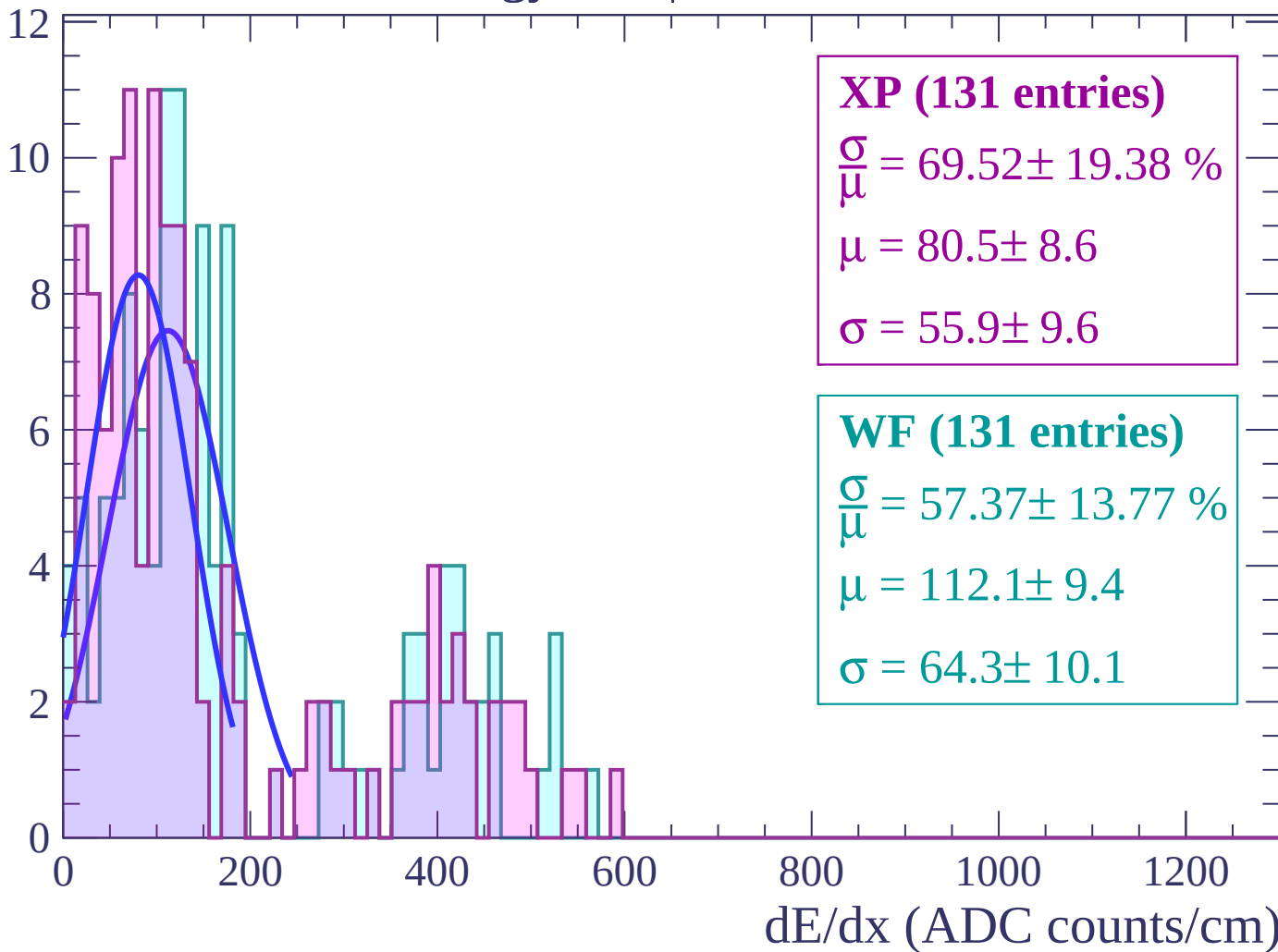
Energy loss | $72 < \theta < 74$

Count



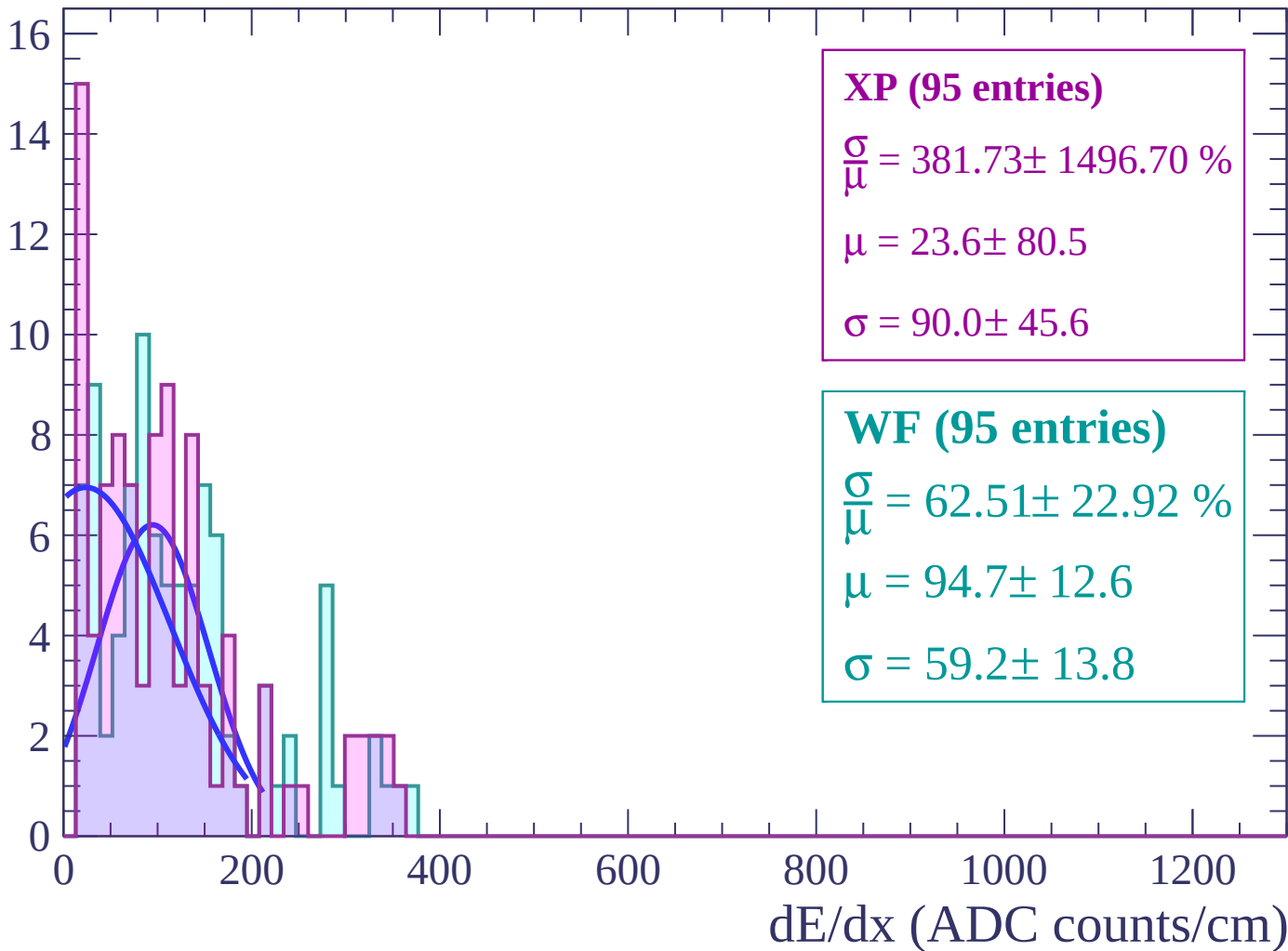
Energy loss | $74 < \theta < 76$

Count



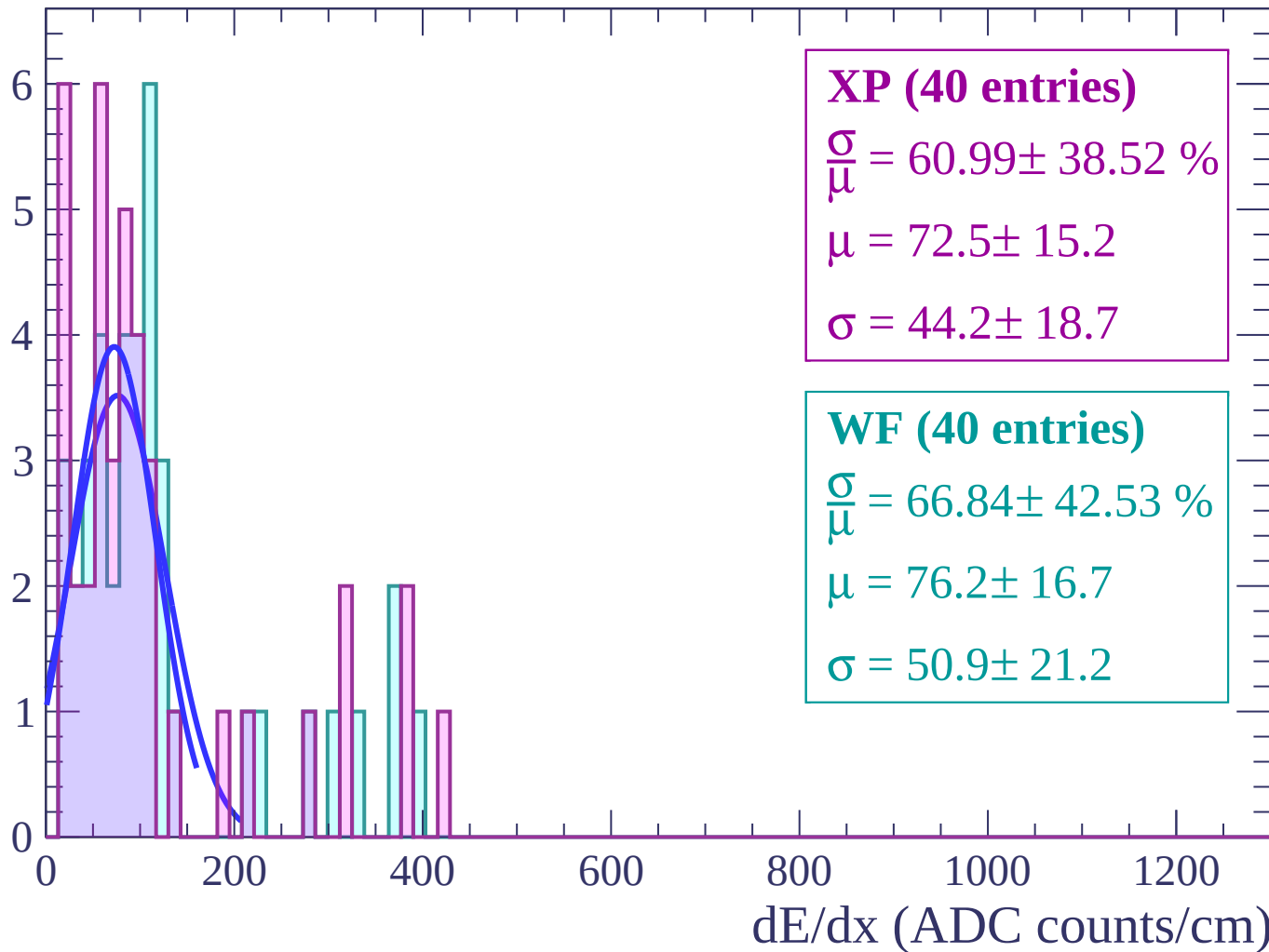
Energy loss | $76 < \theta < 78$

Count



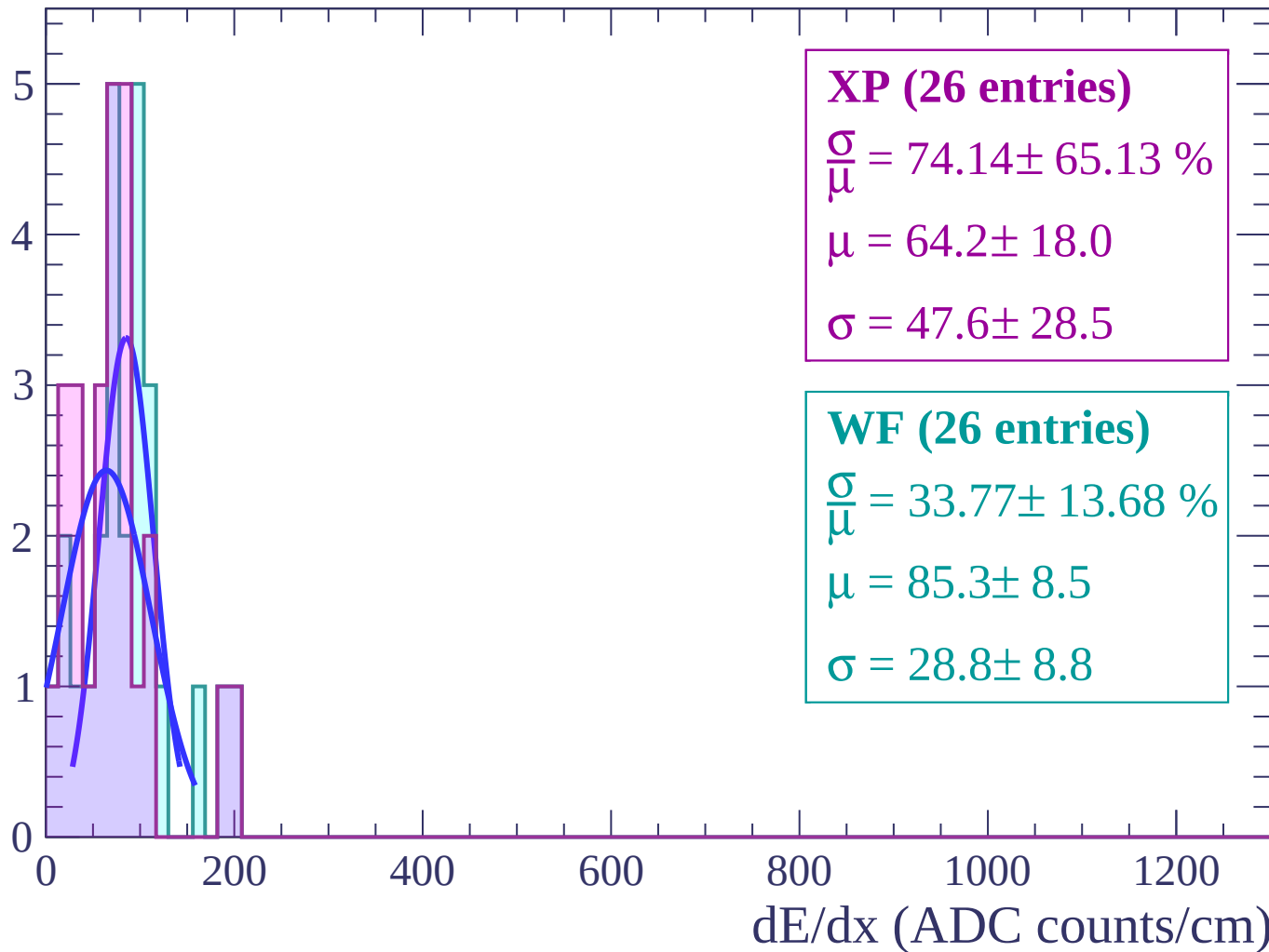
Energy loss | $78 < \theta < 80$

Count



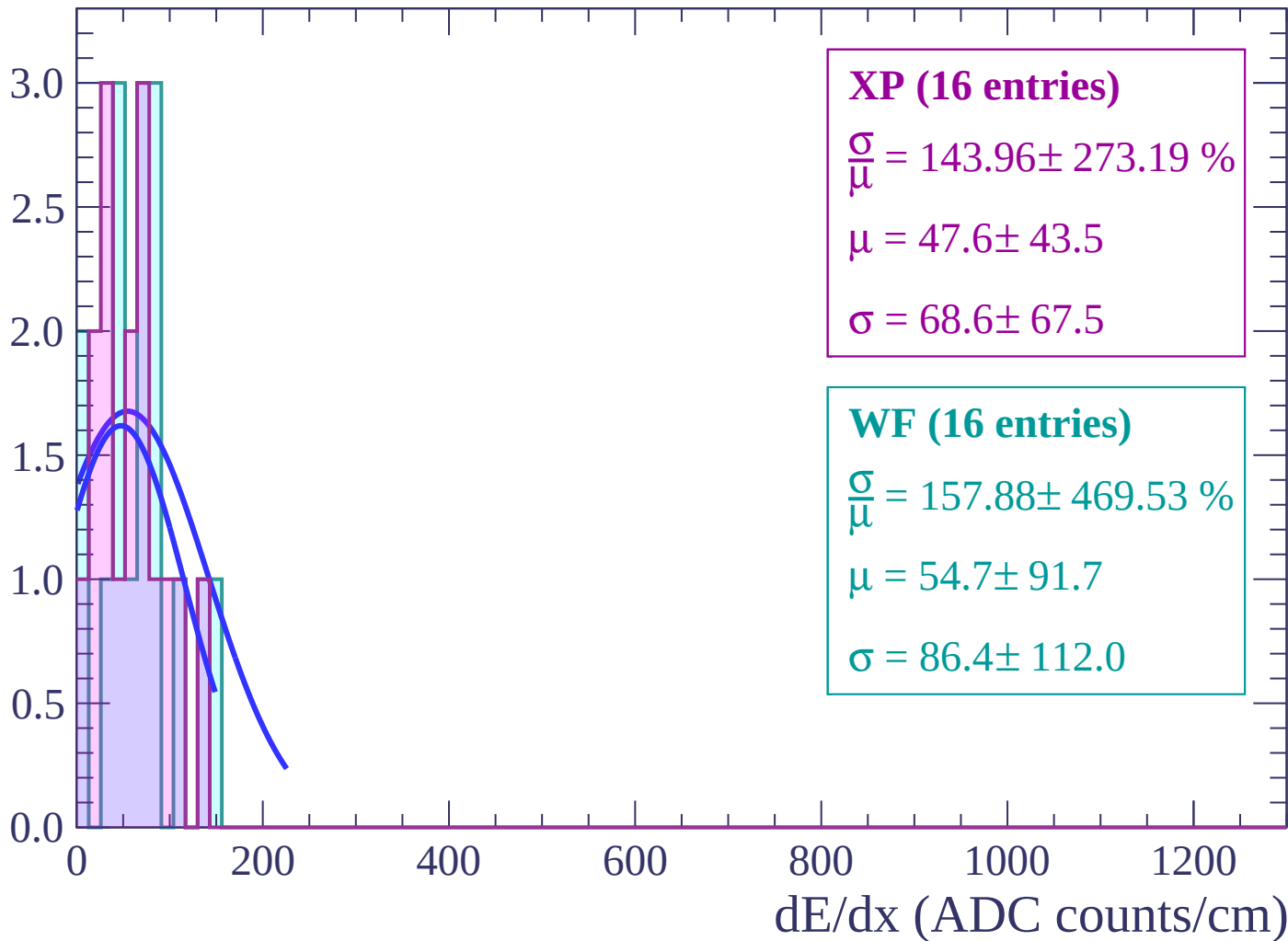
Energy loss | $80 < \theta < 82$

Count



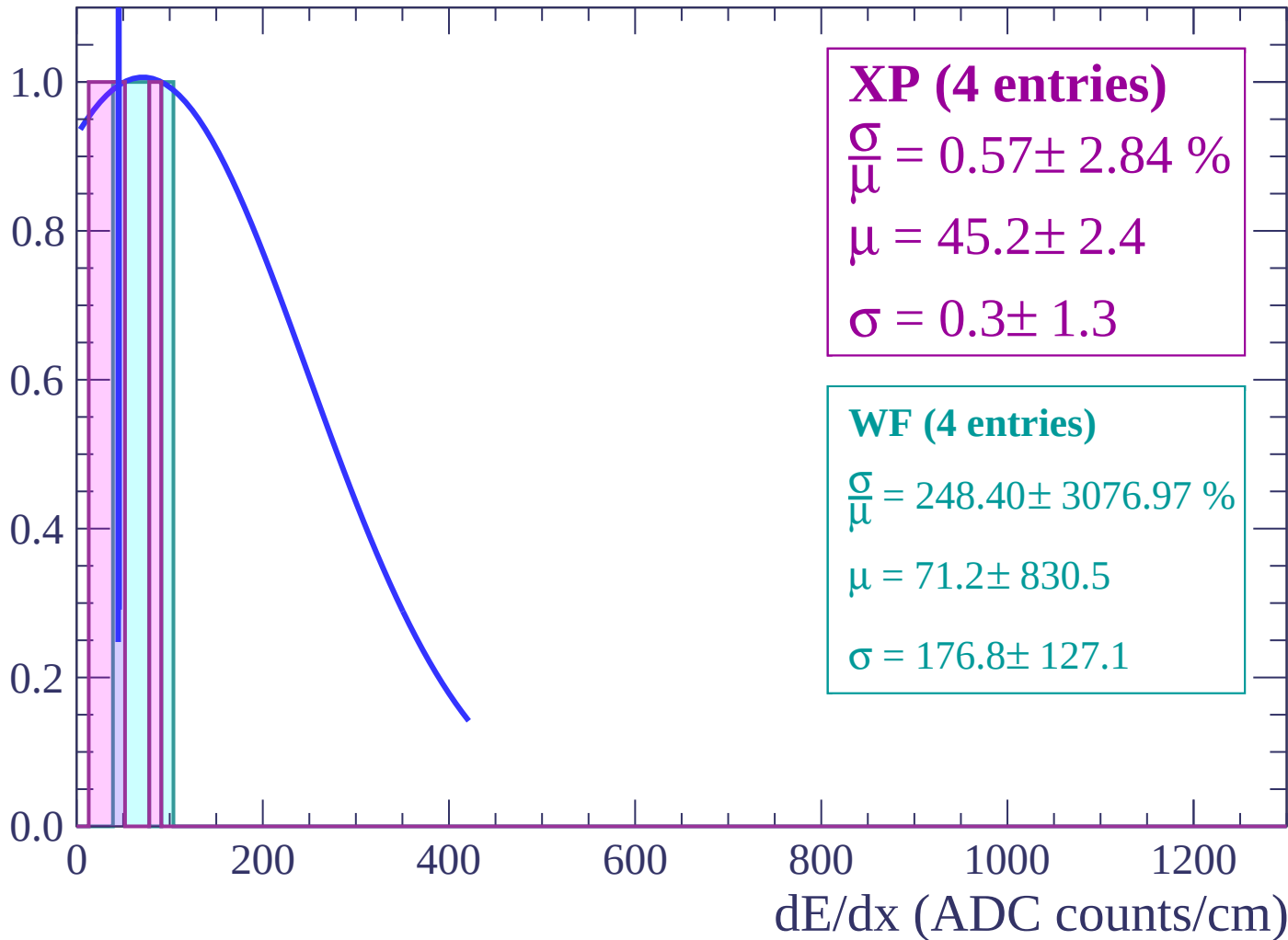
Energy loss | $82 < \theta < 84$

Count



Energy loss | $84 < \theta < 86$

Count



Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

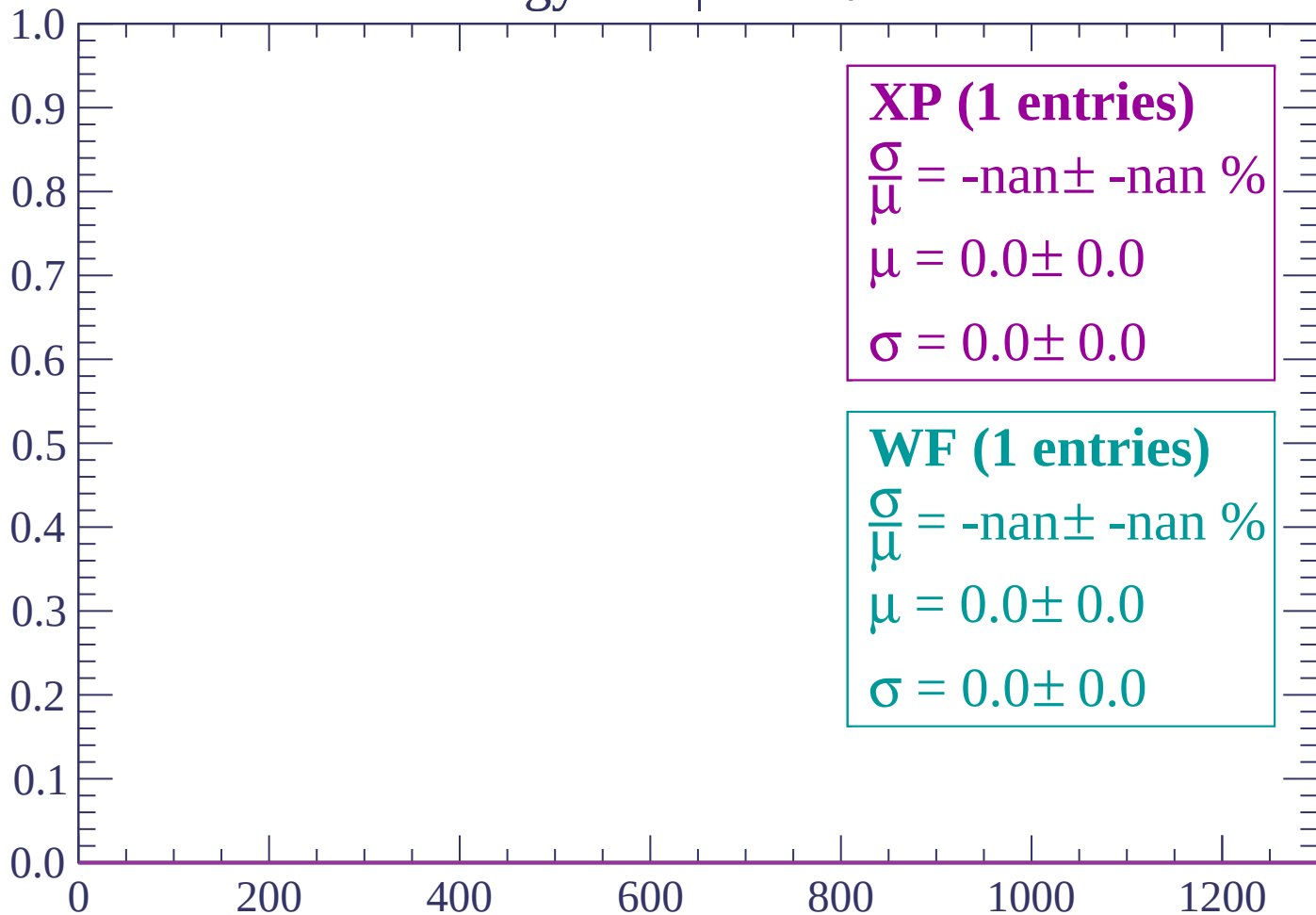
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

